MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3670 PRIMARY CYANIDE DESTRUCTION
DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to prepare design of foundation for structures of **Primary Cyanide destruction (3670) area**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Attachment 1. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 450 Kn/m^2 at 1.2 m below EGL. (Refer Table 5 - BH-27 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-X	Along-Z
2 , 669	0.60m	0.60m
1211 , 1273	0.60m	0.60m
Size of fdn. raft:	5.25m	7.75m
Thickness of raft:	0.70	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

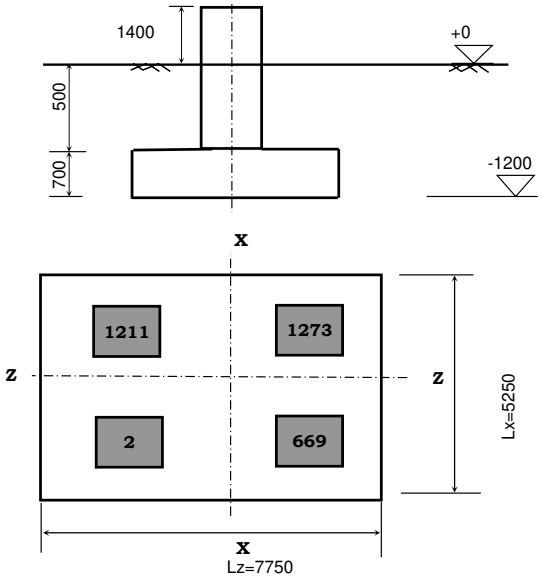
Weight of raft = (5.25*7.75*0.7*24) = 683.55KN

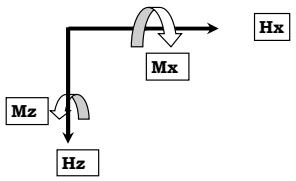
Weight of soil = {(5.25*7.75)-(0.6*0.6)*4}*0.5*20 = 392.48KN

Total = 1076.03KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of footing	Node 2 & 669				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	237.61	0.67	-1.74	-1.14	-2.96
202 DL+L+TL1	588.38	5.95	-15.47	-9.50	-24.70
203 DL+L+TL2	572.84	13.45	-34.97	1.60	4.16
204 DL+0.75L+0.75TL1	500.69	4.63	-12.04	-7.41	-19.27
205 DL+0.75L+0.75TL2	489.03	10.26	-26.66	0.92	2.38
206 DL+WL1	265.39	117.70	-306.02	-0.59	-1.53
207 DL+WL2	210.79	-117.83	306.36	-1.68	-4.37
208 DL+WL3	44.96	7.18	-18.67	23.90	62.14
209 DL+WL4	429.41	-5.15	13.39	-22.14	-57.56
210 DL+0.75WL1+0.75L	515.70	95.22	-247.56	-2.84	-7.37
211 DL+0.75WL2+0.75L	474.75	-81.43	211.73	-3.65	-9.50
212 DL+0.75WL3+0.75L	350.37	12.33	-32.05	15.53	40.39
213 DL+0.75WL4+0.75L	638.71	3.08	-8.00	-19.00	-49.39
214 0.6DL+WL1	170.35	117.43	-305.33	-0.13	-0.35
215 0.6DL+WL2	115.75	-118.10	307.06	-1.22	-3.18
216 0.6DL+WL3	-50.08	6.91	-17.97	24.36	63.33
217 0.6DL+WL4	334.37	-5.42	14.09	-21.68	-56.38
218 0.6DL+0.7SL1	140.72	2.22	-5.76	-0.52	-1.36
219 0.6DL+0.7SL2	144.41	-1.41	3.67	-0.85	-2.20

	Node 1211 & 1273				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
264.9	-0.1	0.2	2.5	6.6
762.6	-0.8	2.0	13.2	34.7
762.6	0.4	-1.0	13.2	34.7
638.2	-0.6	1.6	10.5	27.6
638.2	0.3	-0.7	10.5	27.6
59.4	10.6	-27.5	-16.7	-43.5
587.4	-12.4	32.3	21.3	55.6
481.2	-0.2	0.5	121.3	315.4
50.6	0.1	-0.2	-120.5	-313.4
475.4	7.9	-20.5	-3.4	-8.6
871.4	-9.4	24.4	25.2	65.7
791.7	-0.2	0.5	100.1	260.6
468.8	0.0	0.0	-81.2	-211.0
-46.6	10.6	-27.6	-17.7	-46.1
481.4	-12.4	32.2	20.3	53.0
375.3	-0.2	0.4	120.3	312.8
-55.3	0.1	-0.2	-121.5	-316.0
152.7	0.0	0.0	3.2	8.4
165.1	-0.1	0.2	-0.2	-0.5

Check for overturning and sliding:

	Resultant Load at bottom of footing								
						Overturning		Sliding	
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	1578.5	1.35	3.6	0.6	-1.6	2656.2	1698.2	1052.4	467.7
202 DL+L+TL1	2427.0	3.72	10.0	5.2	-13.4	474.0	944.8	187.8	261.0
203 DL+L+TL2	2411.5	14.82	38.8	13.8	-36.0	176.0	240.7	69.7	65.1
204 DL+0.75L+0.75TL1	2214.9	3.13	8.4	4.0	-10.5	555.3	1026.0	220.0	283.4
205 DL+0.75L+0.75TL2	2203.2	11.45	30.0	10.5	-27.4	211.4	284.5	83.7	76.9
206 DL+WL1	1400.8	-17.31	-45.0	128.3	-333.6	11.0	120.6	4.4	32.4
207 DL+WL2	1874.2	19.63	51.3	-130.3	338.7	14.5	141.7	5.8	38.2
208 DL+WL3	1602.2	145.15	377.6	7.0	-18.1	231.7	16.4	91.8	4.4
209 DL+WL4	1556.1	-142.68	-371.0	-5.1	13.2	308.7	16.3	122.3	4.4
210 DL+0.75WL1+0.75L	2067.1	-6.20	-15.9	103.1	-268.1	20.2	502.6	8.0	133.5
211 DL+0.75WL2+0.75L	2422.1	21.51	56.2	-90.8	236.1	26.9	166.9	10.7	45.0
212 DL+0.75WL3+0.75L	2218.1	115.65	301.0	12.1	-31.5	184.8	28.6	73.2	7.7
213 DL+0.75WL4+0.75L	2183.5	-100.22	-260.4	3.1	-8.0	718.7	32.5	284.8	8.7
214 0.6DL+WL1	1199.8	-17.85	-46.4	128.1	-332.9	9.5	100.1	3.7	26.9
215 0.6DL+WL2	1673.2	19.09	49.8	-130.5	339.3	12.9	130.2	5.1	35.1
216 0.6DL+WL3	1401.2	144.61	376.1	6.7	-17.5	209.9	14.4	83.2	3.9
217 0.6DL+WL4	1355.1	-143.22	-372.4	-5.3	13.9	256.7	14.1	101.7	3.8
218 0.6DL+0.7SL1	1369.5	2.69	7.0	2.2	-5.7	628.3	754.2	248.9	203.9
219 0.6DL+0.7SL2	1385.6	-1.07	-2.7	-1.5	3.9	939.8	1977.6	372.5	519.9

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	68.83	0
Min Px	2	201	0	68.83	0
Max Py	1326	211	0	198.03	0
Min Py	10	214	0	0.00	0
Max Pz	2	201	0	68.83	0
Min Pz	2	201	0	68.83	0

Gross Maximum Base pressure, p_{max} = 198.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.50 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 460 KN/m^2
> 198.0 KN/m^2
Hence OK

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Design of foundation raft F1

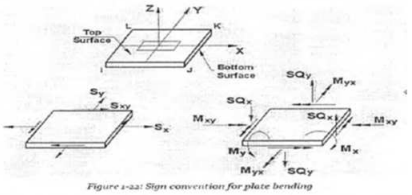
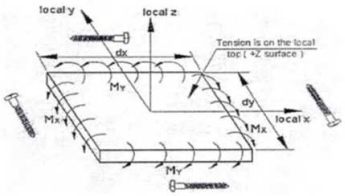


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	348	0.781	-0.006	0	0	0	-53.632	-5.716	45.35	-98.982	-51.066	106:1.2DL+1.6WL2+1.0L
Min Qx	659	-1.198	-0.109	0	0	0	-443.33	-59.111	-93.163	-536.488	-152.274	105:1.2DL+1.6WL1+1.0L
Max Qy	678	0.043	0.863	0	0	0	-236.83	-12.885	-50.153	-286.983	-63.038	105:1.2DL+1.6WL1+1.0L
Min Qy	1246	0.033	-0.579	0	0	0	-139.62	97.626	52.273	-191.895	149.899	106:1.2DL+1.6WL2+1.0L
Max Sx	1290	0.013	-0.063	0	0	0	-0.968	13.725	0.712	-1.68	14.437	101:1.4DL
Min Sx	1290	0.013	-0.063	0	0	0	-0.968	13.725	0.712	-1.68	14.437	101:1.4DL
Max Sy	1290	0.013	-0.063	0	0	0	-0.968	13.725	0.712	-1.68	14.437	101:1.4DL
Min Sy	1290	0.013	-0.063	0	0	0	-0.968	13.725	0.712	-1.68	14.437	101:1.4DL
Max Sxy	1290	0.013	-0.063	0	0	0	-0.968	13.725	0.712	-1.68	14.437	101:1.4DL
Min Sxy	1290	0.013	-0.063	0	0	0	-0.968	13.725	0.712	-1.68	14.437	101:1.4DL
Max Mx	659	0.706	0.134	0	0	0	343.468	31.805	63.914	407.382	95.719	112:0.9DL+1.6WL2
Min Mx	659	-1.198	-0.109	0	0	0	-443.33	-59.111	-93.163	-536.488	-152.274	105:1.2DL+1.6WL1+1.0L
Max My	1281	-0.005	-0.107	0	0	0	0.522	255.737	12.307	12.829	268.044	107:1.2DL+1.6WL3+1.0L
Min My	1246	0.105	-0.247	0	0	0	2.936	-187.498	11.039	13.975	-198.537	114:0.9DL+1.6WL4
Max Mxy	1238	-0.302	-0.388	0	0	0	-52.372	79.397	150.629	-203.001	230.026	106:1.2DL+1.6WL2+1.0L
Min Mxy	594	-0.208	0.103	0	0	0	-52.345	28.055	-154.365	-206.71	182.42	105:1.2DL+1.6WL1+1.0L

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	498	0.708	0.001	0	0	0	7.052	-0.663	55.125	62.177	-55.788	112:0.9DL+1.6WL2
Min Qx	575	-0.985	0.007	0	0	0	-112.03	-2.022	-67.51	-179.541	-69.532	105:1.2DL+1.6WL1+1.0L
Max Qy	789	-0.012	0.387	0	0	0	-2.077	120.851	29.007	-31.084	149.858	106:1.2DL+1.6WL2+1.0L
Min Qy	1283	0.014	-0.4	0	0	0	-3.159	179.311	26.302	-29.461	205.613	106:1.2DL+1.6WL2+1.0L
Max Sx	1285	0	-0.051	0	0	0	-0.087	54.587	2.706	-2.793	57.293	101:1.4DL
Min Sx	1285	0	-0.051	0	0	0	-0.087	54.587	2.706	-2.793	57.293	101:1.4DL
Max Sy	1285	0	-0.051	0	0	0	-0.087	54.587	2.706	-2.793	57.293	101:1.4DL
Min Sy	1285	0	-0.051	0	0	0	-0.087	54.587	2.706	-2.793	57.293	101:1.4DL
Max Sxy	1285	0	-0.051	0	0	0	-0.087	54.587	2.706	-2.793	57.293	101:1.4DL
Min Sxy	1285	0	-0.051	0	0	0	-0.087	54.587	2.706	-2.793	57.293	101:1.4DL
Max Mx	575	0.494	0.021	0	0	0	155.219	3.746	35.526	190.745	39.272	106:1.2DL+1.6WL2+1.0L
Min Mx	575	-0.816	-0.003	0	0	0	-134	-2.753	-57.112	-191.11	-59.865	111:0.9DL+1.6WL1
Max My	1281	-0.005	-0.107	0	0	0	0.522	255.737	12.307	12.829	268.044	107:1.2DL+1.6WL3+1.0L
Min My	1154	0.017	-0.053	0	0	0	-3.554	-88.896	-5.535	-9.089	-94.431	114:0.9DL+1.6WL4
Max Mxy	503	0.155	0.045	0	0	0	9.908	7.526	124.627	134.535	132.153	112:0.9DL+1.6WL2
Min Mxy	594	-0.208	0.103	0	0	0	-52.345	28.055	-154.365	-206.71	182.42	105:1.2DL+1.6WL1+1.0L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

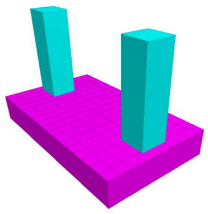
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
112:0.9D	407.38	top face	50	640	407.38	1.33	0.0032	2067.9	1152	20	151.98	150	2095.3	29.54	36.93	0.049	tension control
105:1.2D	-536.5	bottom face	50	640	536.49	1.75	0.0026	1677.5		20	187.35	150	2095.3	29.54	36.93	0.049	tension control
106:1.2D	-203.00	bottom face	50	640	203	0.66	0.0016	1018.4		20	272.82	150	2095.3	29.54	36.93	0.049	tension control
105:1.2D	-206.71	bottom face	50	640	206.71	0.67	0.0016	1037.2		20	272.82	150	2095.3	29.54	36.93	0.049	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.708	1	495.6	62.177	0.003	0.64	787.516	641.427	1098.02	641.43	2355	> 495.6	ok	481.1	φVc<Vu	required
	Min Qx	0.985	1	689.5	179.54			693.354			641.43	2355	> 689.5	ok	481.1	φVc<Vu	required

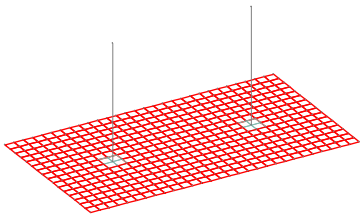
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107:1.2D	268.04	top face	50	640	268.04	0.87	0.0021	1349.7	1152	20	232.86	150	2095.3	29.54	36.93	0.049	tension control
114:0.9D	-198.54	bottom face	50	640	198.54	0.65	0.0016	995.7		20	272.82	150	2095.3	29.54	36.93	0.049	tension control
106:1.2D	230.03	bottom face	50	640	230.03	0.75	0.0018	1155.7		20	271.94	150	2095.3	29.54	36.93	0.049	tension control
105:1.2D	182.42	bottom face	50	640	182.42	0.59	0.0014	914.1		20	272.82	150	2095.3	29.54	36.93	0.049	tension control

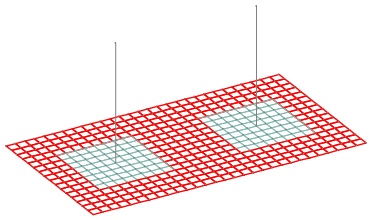
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qy	0.387	1	270.9	149.86	0.003	0.64	647.017	641.427	fc'1098	641.43	668	> 270.9	ok	481.1	φVc>Vu	not required
	Min Qy	0.4	1	280	205.61			636.851			636.85	665	> 280	ok	477.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	20ϕT	150	c/c
At Top	20ϕT	150	c/c
Along Z Direction :			
At Bottom	20ϕT	150	c/c
At Top	20ϕT	150	c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-X	Along-Z
1329	0.50m	0.50m
1332	0.50m	0.50m
Size of fdn. raft:	2.50m	4.25m
Thickness of raft:	0.70	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

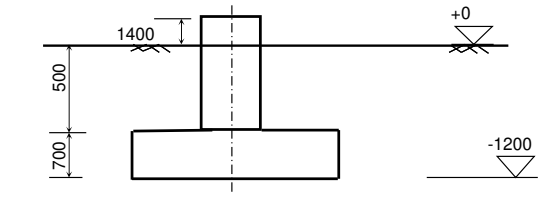
Weight of raft = (2.5*4.25*0.7*24) = 178.50KN

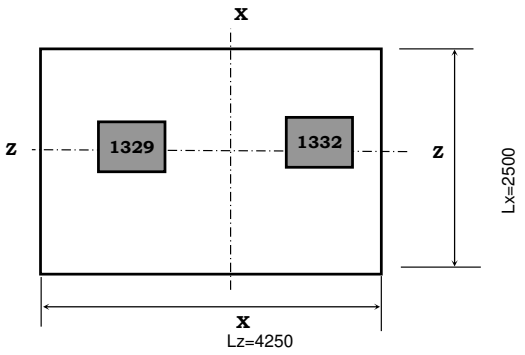
Weight of soil = {(2.5*4.25)-(0.5*0.5)*2}*0.5*20 = 101.25KN

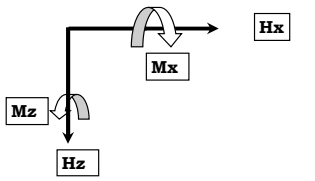
Total = 279.75KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2

Load at bottom of footing	Node 1329					Node 1332				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	113.40	0.35	-0.91	-0.39	-1.01	113.2	2.2	-5.6	-0.4	-1.0
202 DL+L+TL1	189.38	0.52	-1.35	-2.82	-7.33	86.1	3.3	-8.6	-4.9	-12.7
203 DL+L+TL2	123.30	-0.18	0.47	1.96	5.10	152.1	1.2	-3.2	-0.3	-0.8
204 DL+0.75L+0.75TL1	170.39	0.48	-1.24	-2.21	-5.75	92.9	3.0	-7.8	-3.8	-9.8
205 DL+0.75L+0.75TL2	120.83	-0.05	0.12	1.37	3.57	142.4	1.5	-3.8	-0.3	-0.9
206 DL+WL1	94.03	4.17	-10.84	-1.37	-3.56	390.0	-1.7	4.4	6.5	16.9
207 DL+WL2	94.03	-3.50	9.10	0.58	1.51	-162.3	6.0	-15.7	-7.1	-18.4
208 DL+WL3	-146.60	0.00	0.00	0.06	0.16	401.9	1.6	-4.1	60.6	157.6
209 DL+WL4	369.27	0.69	-1.79	-0.86	-2.24	-171.2	2.8	-7.2	-61.1	-159.0
210 DL+0.75WL1+0.75L	131.08	3.08	-8.01	-1.16	-3.00	325.2	-0.7	1.7	3.1	8.1
211 DL+0.75WL2+0.75L	131.08	-2.67	6.95	0.31	0.80	-89.0	5.1	-13.4	-7.1	-18.4
212 DL+0.75WL3+0.75L	-49.40	-0.05	0.12	-0.08	-0.22	334.1	1.8	-4.6	43.7	113.6
213 DL+0.75WL4+0.75L	337.51	0.47	-1.22	-0.77	-2.01	-95.7	2.7	-7.0	-47.6	-123.8
214 0.6DL+WL1	48.67	4.03	-10.48	-1.21	-3.16	344.7	-2.6	6.7	6.6	17.3
215 0.6DL+WL2	48.67	-3.64	9.46	0.74	1.91	-207.5	5.2	-13.4	-6.9	-18.0
216 0.6DL+WL3	-191.96	-0.14	0.36	0.22	0.56	356.6	0.7	-1.8	60.7	158.0
217 0.6DL+WL4	323.91	0.55	-1.43	-0.70	-1.83	-216.5	1.9	-4.9	-61.0	-158.6
218 0.6DL+0.7SL1	62.69	0.25	-0.64	-0.21	-0.55	62.3	1.4	-3.8	0.5	1.3
219 0.6DL+0.7SL2	73.40	0.18	-0.46	-0.26	-0.66	73.6	1.2	-3.0	-1.0	-2.5

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	506.4	-0.79	-2.0	2.5	-6.6	96.6	534.0	80.4	256.4
202 DL+L+TL1	555.3	-7.73	-20.1	3.8	-9.9	70.1	58.8	58.3	28.7
203 DL+L+TL2	555.1	1.63	4.3	1.1	-2.7	254.2	274.9	211.5	136.2
204 DL+0.75L+0.75TL1	543.0	-6.00	-15.6	3.5	-9.1	74.9	74.2	62.3	36.2
205 DL+0.75L+0.75TL2	542.9	1.03	2.7	1.4	-3.7	184.2	425.0	153.2	211.9
206 DL+WL1	763.8	5.09	13.4	2.5	-6.4	148.7	121.3	123.7	60.0
207 DL+WL2	211.5	-6.49	-16.9	2.5	-6.6	40.0	26.5	33.3	13.0
208 DL+WL3	535.1	60.62	157.8	1.6	-4.1	164.9	7.2	137.2	3.5
209 DL+WL4	477.8	-61.99	-161.2	3.5	-9.0	66.6	6.3	55.4	3.1
210 DL+0.75WL1+0.75L	736.1	1.93	5.1	2.4	-6.3	146.5	305.3	121.9	152.9
211 DL+0.75WL2+0.75L	321.9	-6.76	-17.6	2.5	-6.4	62.7	38.8	52.2	19.0
212 DL+0.75WL3+0.75L	564.5	43.57	113.4	1.7	-4.5	156.7	10.6	130.3	5.2
213 DL+0.75WL4+0.75L	521.5	-48.39	-125.8	3.2	-8.2	79.6	8.8	66.2	4.3
214 0.6DL+WL1	673.2	5.41	14.2	1.5	-3.8	221.4	100.8	184.2	49.8
215 0.6DL+WL2	120.9	-6.17	-16.1	1.5	-4.0	37.9	15.9	31.6	7.8
216 0.6DL+WL3	444.4	60.94	158.6	0.6	-1.4	387.1	6.0	322.0	2.9
217 0.6DL+WL4	387.1	-61.67	-160.4	2.4	-6.3	76.2	5.1	63.4	2.5
218 0.6DL+0.7SL1	404.7	0.26	0.7	1.7	-4.4	114.9	1230.3	95.6	620.2
219 0.6DL+0.7SL2	426.8	-1.21	-3.1	1.3	-3.5	154.3	290.8	128.3	141.2

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Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	492	201	0	74.70	0
Min Px	492	201	0	74.70	0
Max Py	1506	217	0	303.53	0
Min Py	494	208	0	0.00	0
Max Pz	492	201	0	74.70	0
Min Pz	492	201	0	74.70	0

Gross Maximum Base pressure, p_{max} = 303.5 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.50 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 460 KN/m^2
> 303.5 KN/m^2
Hence OK

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Design of foundation raft F2

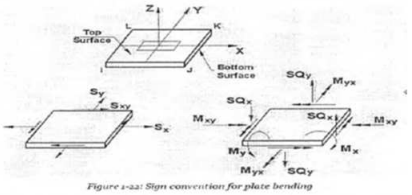
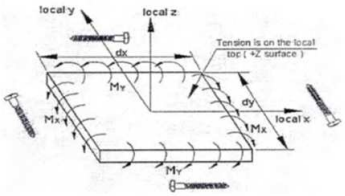


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F2_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	841	0.346	-0.014	0	0	0	-51.526	-34.167	-9.435	-60.961	-43.602	107:1.2DL+1.6WL3+1.0L
Min Qx	822	-0.355	-0.011	0	0	0	-54.659	-34.474	9.614	-64.273	-44.088	107:1.2DL+1.6WL3+1.0L
Max Qy	926	-0.143	0.57	0	0	0	-76.047	-123.69	12.774	-88.821	-136.464	113:0.9DL+1.6WL3
Min Qy	633	-0.05	-0.443	0	0	0	-58.915	-95.068	4.279	-63.194	-99.347	114:0.9DL+1.6WL4
Max Sx	1443	0.008	-0.003	0	0	0	-2.773	-0.607	-0.926	-3.699	-1.533	101:1.4DL
Min Sx	1443	0.008	-0.003	0	0	0	-2.773	-0.607	-0.926	-3.699	-1.533	101:1.4DL
Max Sy	1443	0.008	-0.003	0	0	0	-2.773	-0.607	-0.926	-3.699	-1.533	101:1.4DL
Min Sy	1443	0.008	-0.003	0	0	0	-2.773	-0.607	-0.926	-3.699	-1.533	101:1.4DL
Max Sxy	1443	0.008	-0.003	0	0	0	-2.773	-0.607	-0.926	-3.699	-1.533	101:1.4DL
Min Sxy	1443	0.008	-0.003	0	0	0	-2.773	-0.607	-0.926	-3.699	-1.533	101:1.4DL
Max Mx	926	0.128	-0.315	0	0	0	57.11	67.58	-15.257	72.367	82.837	114:0.9DL+1.6WL4
Min Mx	937	0.145	0.489	0	0	0	-83.695	-97.559	-11.071	-94.766	-108.63	107:1.2DL+1.6WL3+1.0L
Max My	831	-0.031	-0.056	0	0	0	-61.165	73.346	-29.321	-90.486	102.667	107:1.2DL+1.6WL3+1.0L
Min My	680	0.239	-0.074	0	0	0	-35.678	-183.729	10.217	-45.895	-193.946	114:0.9DL+1.6WL4
Max Mxy	810	-0.004	-0.022	0	0	0	-14.142	-36.432	46.487	-60.629	-82.919	113:0.9DL+1.6WL3
Min Mxy	832	0.017	-0.081	0	0	0	-13.953	29.298	-47.052	-61.005	76.35	107:1.2DL+1.6WL3+1.0L

PLATE CENTER STRESS : FDN_F2_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1298	0.032	0.008	0	0	0	-11.48	-103.802	13.95	-25.43	-117.752	108:1.2DL+1.6WL4+1.0L
Min Qx	606	-0.033	0.005	0	0	0	-11.713	-103.295	-16.982	-28.695	-120.277	108:1.2DL+1.6WL4+1.0L
Max Qy	819	-0.023	0.135	0	0	0	-0.516	-48.382	6.651	-7.167	-55.033	113:0.9DL+1.6WL3
Min Qy	572	-0.015	-0.283	0	0	0	-0.557	-81.498	-3.326	-3.883	-84.824	114:0.9DL+1.6WL4
Max Sx	1438	0.003	-0.002	0	0	0	-0.613	-0.566	-0.291	-0.904	-0.857	101:1.4DL
Min Sx	1438	0.003	-0.002	0	0	0	-0.613	-0.566	-0.291	-0.904	-0.857	101:1.4DL
Max Sy	1438	0.003	-0.002	0	0	0	-0.613	-0.566	-0.291	-0.904	-0.857	101:1.4DL
Min Sy	1438	0.003	-0.002	0	0	0	-0.613	-0.566	-0.291	-0.904	-0.857	101:1.4DL
Max Sxy	1438	0.003	-0.002	0	0	0	-0.613	-0.566	-0.291	-0.904	-0.857	101:1.4DL
Min Sxy	1438	0.003	-0.002	0	0	0	-0.613	-0.566	-0.291	-0.904	-0.857	101:1.4DL
Max Mx	685	-0.001	0.069	0	0	0	7.409	-59.408	-2.732	10.141	-62.14	112:0.9DL+1.6WL2
Min Mx	641	-0.008	-0.041	0	0	0	-26.156	17.736	4.587	-30.743	22.323	105:1.2DL+1.6WL1+1.0L
Max My	641	0.002	-0.013	0	0	0	-5.836	35.491	3.379	-9.215	38.87	107:1.2DL+1.6WL3+1.0L
Min My	588	-0.016	-0.076	0	0	0	-0.683	-168.953	-8.516	-9.199	-177.469	114:0.9DL+1.6WL4
Max Mxy	606	0.007	-0.05	0	0	0	-3.418	-35.724	14.191	-17.609	-49.915	113:0.9DL+1.6WL3
Min Mxy	606	-0.033	0.005	0	0	0	-11.713	-103.295	-16.982	-28.695	-120.277	108:1.2DL+1.6WL4+1.0L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

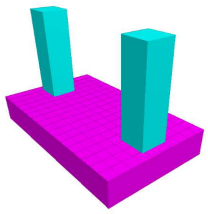
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
114:0.9D	72.37	top face	50	642	72.367	0.23	0.0006	359.3	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-94.8	bottom face	50	642	94.766	0.31	0.0007	471.0		16	174.06	150	1341	18.91	23.64	0.078	tension control
113:0.9D	-60.63	bottom face	50	642	60.629	0.20	0.0005	300.8		16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-61.01	bottom face	50	642	61.005	0.20	0.0005	302.7		16	174.06	150	1341	18.91	23.64	0.078	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			rks	KN	ks	
Load condition	Max Qx	0.032	1	22.4	25.43	0.002	0.642	620.592	630.497	1101.46	620.59	2346	> 22.4	ok	465.4	φVc>Vu	not required
	Min Qx	0.033	1	23.1	28.695			619.482			619.48	2345	> 23.1	ok	464.6	φVc>Vu	not required

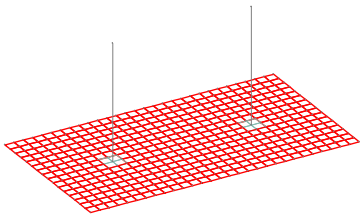
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107:1.2D	102.67	top face	50	642	102.67	0.33	0.0008	510.5	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
114:0.9D	-193.95	bottom face	50	642	193.95	0.63	0.0007	434.4		16	174.06	150	1341	18.91	23.64	0.078	tension control
113:0.9D	-82.92	bottom face	50	642	82.919	0.27	0.0006	411.9		16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	76.35	bottom face	50	642	76.35	0.25	0.0006	379.1		16	174.06	150	1341	18.91	23.64	0.078	tension control

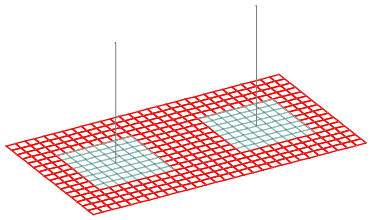
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			rks	KN	ks	
Load condition	Max Qy	0.135	1	94.5	55.033	0.002	0.642	632.831	630.497	fc'1101	630.5	661	> 94.5	ok	472.9	φVc>Vu	not required
	Min Qy	0.283	1	198.1	84.824			641.88			630.5	661	> 198.1	ok	472.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F3

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m
1538	0.80m	0.80m
Size of fdn. raft:	4.00m	4.50m
Thickness of raft:	0.70	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

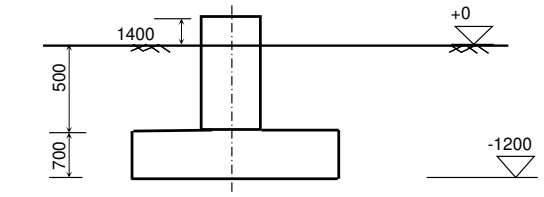
Weight of raft = (4*4.5*0.7*24) = 302.40KN

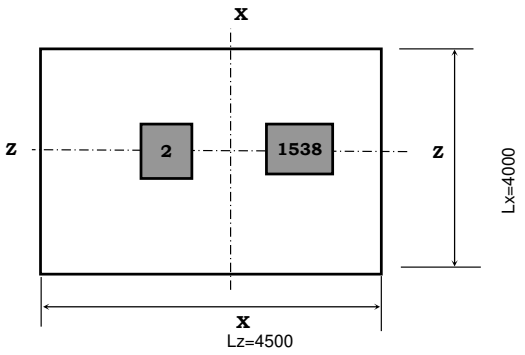
Weight of soil = {(4*4.5)-(0.6*0.6)-(0.8*0.8)}*0.5*20 = 170.00KN

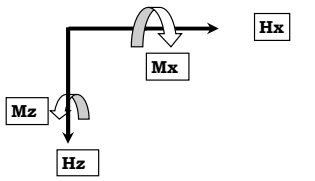
Total = 472.40KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F3

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	140.70	-0.38	0.99	-5.20	-13.52
202 DL+L+TL1	256.50	-10.38	26.99	-17.20	-44.72
203 DL+L+TL2	256.50	-10.38	26.99	-17.20	-44.72
204 DL+0.75L+0.75TL1	227.55	-7.88	20.49	-14.20	-36.92
205 DL+0.75L+0.75TL2	227.55	-7.88	20.49	-14.20	-36.92
206 DL+WL1	288.90	-70.58	183.51	-5.70	-14.82
207 DL+WL2	288.90	69.82	-181.53	-4.70	-12.22
208 DL+WL3	471.50	-0.68	1.77	-54.70	-142.22
209 DL+WL4	-190.10	-0.08	0.21	44.30	115.18
210 DL+0.75WL1+0.75L	338.70	-60.53	157.38	-14.58	-37.90
211 DL+0.75WL2+0.75L	338.70	44.77	-116.40	-13.83	-35.95
212 DL+0.75WL3+0.75L	475.65	-8.11	21.07	-51.33	-133.45
213 DL+0.75WL4+0.75L	-20.55	-7.66	19.90	22.93	59.61
214 0.6DL+WL1	232.62	-70.43	183.12	-3.62	-9.41
215 0.6DL+WL2	232.62	69.97	-181.93	-2.62	-6.81
216 0.6DL+WL3	415.22	-0.53	1.37	-52.62	-136.81
217 0.6DL+WL4	-246.38	0.07	-0.19	46.38	120.59
218 0.6DL+0.7SL1	84.42	-0.23	0.59	-3.12	-8.11
219 0.6DL+0.7SL2	84.42	-0.23	0.59	-3.12	-8.11

Node 1538				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
179.8	0.8	-2.0	4.9	39.7
246.2	-3.6	9.4	16.7	70.1
273.8	-4.2	10.9	16.8	89.8
229.6	-2.5	6.6	13.7	62.5
250.3	-2.9	7.7	13.9	77.3
409.9	-62.0	161.3	5.4	107.7
224.1	63.6	-165.3	4.4	-25.0
-171.7	1.8	-4.7	-38.5	-87.9
530.9	-0.3	0.7	48.4	167.1
412.6	-49.8	129.6	14.2	120.9
273.1	44.4	-115.4	13.4	21.3
-23.7	-2.0	5.1	-18.8	-25.8
503.3	-3.5	9.1	46.4	165.4
338.0	-62.4	162.1	3.4	91.8
152.2	63.3	-164.5	2.5	-40.9
-243.6	1.5	-3.9	-40.5	-103.8
459.0	-0.6	1.5	46.4	151.2
103.3	0.5	-1.2	4.1	23.7
112.4	0.4	-1.2	1.8	23.9

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	792.9	-0.29	26.2	0.4	-1.0	1584.1	68.1	823.7	1093.6
202 DL+L+TL1	975.1	-0.51	25.4	-14.0	36.4	53.6	86.4	27.8	764.8
203 DL+L+TL2	1002.7	-0.37	45.1	-14.6	37.9	53.0	50.1	27.5	1084.0
204 DL+0.75L+0.75TL1	929.5	-0.46	25.6	-10.4	27.1	68.7	81.7	35.7	817.2
205 DL+0.75L+0.75TL2	950.2	-0.35	40.3	-10.8	28.2	67.5	53.0	35.1	1086.0
206 DL+WL1	1171.2	-0.30	92.9	-132.6	344.8	6.8	28.4	3.5	1561.6
207 DL+WL2	985.4	-0.28	-37.3	133.4	-346.8	5.7	59.5	3.0	1407.6
208 DL+WL3	772.2	-93.24	-230.1	1.1	-2.9	532.7	7.6	277.0	3.3
209 DL+WL4	813.2	92.66	282.2	-0.3	0.9	1813.2	6.5	942.9	3.5
210 DL+0.75WL1+0.75L	1223.7	-0.41	83.0	-110.4	287.0	8.5	33.2	4.4	1193.8
211 DL+0.75WL2+0.75L	1084.2	-0.40	-14.6	89.1	-231.8	9.4	166.9	4.9	1098.0
212 DL+0.75WL3+0.75L	924.4	-70.12	-159.2	-10.1	26.2	70.6	13.1	36.7	5.3
213 DL+0.75WL4+0.75L	955.1	69.31	225.0	-11.2	29.0	65.8	9.6	34.2	5.5
214 0.6DL+WL1	1043.1	-0.18	82.4	-132.8	345.2	6.0	28.5	3.1	2267.5
215 0.6DL+WL2	857.2	-0.16	-47.7	133.2	-346.4	4.9	40.4	2.6	2090.7
216 0.6DL+WL3	644.0	-93.12	-240.6	1.0	-2.5	515.6	6.0	268.0	2.8
217 0.6DL+WL4	685.0	92.78	271.8	-0.5	1.3	1055.5	5.7	549.1	3.0
218 0.6DL+0.7SL1	660.1	1.02	15.6	0.2	-0.6	2072.6	95.0	1077.8	259.9
219 0.6DL+0.7SL2	669.2	-1.36	15.8	0.2	-0.6	2373.1	95.3	1233.6	196.2

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Base pressure check of footing F3

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	66.07	0
Min Px	2	201	0	66.07	0
Max Py	1537	217	0	290.77	0
Min Py	1445	208	0	0.00	0
Max Pz	2	201	0	66.07	0
Min Pz	2	201	0	66.07	0

Gross Maximum Base pressure, p_{max} = 290.8 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.50 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 460 KN/m^2
> 290.8 KN/m^2
Hence OK

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Design of foundation raft F3

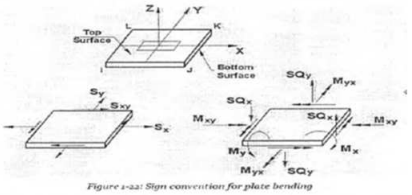
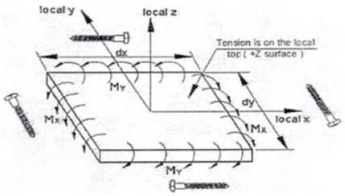


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F3_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1381	0.888	-0.094	0	0	0	-181.99	22.508	-58.703	-240.692	81.211	108:1.2DL+1.6WL4+1.0L
Min Qx	1378	-0.864	-0.102	0	0	0	-171.54	20.982	56.612	-228.151	77.594	108:1.2DL+1.6WL4+1.0L
Max Qy	417	0.27	0.431	0	0	0	-137.43	4.753	-11.07	-148.504	15.823	105:1.2DL+1.6WL1+1.0L
Min Qy	346	0.177	-0.645	0	0	0	-168.06	-127.33	12.204	-180.267	-139.534	107:1.2DL+1.6WL3+1.0L
Max Sx	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Min Sx	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Max Sy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Min Sy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Max Sxy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Min Sxy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Max Mx	1381	-0.448	0.075	0	0	0	85.456	-18.411	32.681	118.137	-51.092	113:0.9DL+1.6WL3
Min Mx	1381	0.848	-0.013	0	0	0	-269.94	32.062	-50.681	-320.617	82.743	105:1.2DL+1.6WL1+1.0L
Max My	884	0.015	-0.14	0	0	0	-144.87	151.084	23.006	-167.879	174.09	108:1.2DL+1.6WL4+1.0L
Min My	427	0.274	0.011	0	0	0	-60.651	-135.248	31.059	-91.71	-166.307	113:0.9DL+1.6WL3
Max Mxy	1364	-0.048	-0.106	0	0	0	-19.277	43.712	103.519	-122.796	147.231	105:1.2DL+1.6WL1+1.0L
Min Mxy	1368	0.224	-0.169	0	0	0	-77.048	54.328	-80.636	-157.684	134.964	108:1.2DL+1.6WL4+1.0L

PLATE CENTER STRESS : FDN_F3_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1384	0.378	-0.068	0	0	0	-48.546	8.572	-30.583	-79.129	39.155	105:1.2DL+1.6WL1+1.0L
Min Qx	1405	-0.355	-0.052	0	0	0	-8.624	8.38	42.662	-51.286	51.042	108:1.2DL+1.6WL4+1.0L
Max Qy	1533	-0.001	0.277	0	0	0	0.219	-17.478	-21.343	21.562	-38.821	114:0.9DL+1.6WL4
Min Qy	1543	-0.004	-0.241	0	0	0	0.264	-61.495	-13.717	13.981	-75.212	113:0.9DL+1.6WL3
Max Sx	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Min Sx	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Max Sy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Min Sy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Max Sxy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Min Sxy	1554	-0.005	0.003	0	0	0	-0.318	0.131	0.017	-0.335	0.148	101:1.4DL
Max Mx	1384	-0.035	0.01	0	0	0	16.576	-0.001	-2.525	19.101	-2.526	112:0.9DL+1.6WL2
Min Mx	770	0.013	-0.032	0	0	0	-74.209	135.25	-1.953	-76.162	137.203	105:1.2DL+1.6WL1+1.0L
Max My	770	0.013	-0.032	0	0	0	-74.209	135.25	-1.953	-76.162	137.203	105:1.2DL+1.6WL1+1.0L
Min My	1475	0.01	-0.026	0	0	0	-2.571	-107.994	26.302	-28.873	-134.296	113:0.9DL+1.6WL3
Max Mxy	1401	-0.102	-0.073	0	0	0	-10.188	42.666	65.64	-75.828	108.306	108:1.2DL+1.6WL4+1.0L
Min Mxy	1370	0.11	-0.077	0	0	0	-11.848	43.53	-67.133	-78.981	110.663	108:1.2DL+1.6WL4+1.0L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

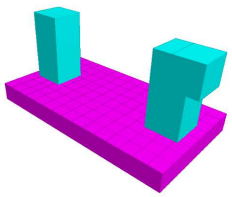
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
113:0.9D	118.14	top face	50	642	118.14	0.38	0.0009	588.0	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
105:1.2D	-320.6	bottom face	50	640	320.62	1.04	0.0025	1619.3		20	194.09	150	2095.3	29.54	36.93	0.049	tension control
105:1.2D	-122.80	bottom face	50	640	122.8	0.40	0.0010	613.3		20	271.97	150	2095.3	29.54	36.93	0.049	tension control
108:1.2D	-157.68	bottom face	50	640	157.68	0.51	0.0012	789.0		20	271.97	150	2095.3	29.54	36.93	0.049	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.378	1	264.6	79.129	0.002	0.642	656.64	630.497	1101.46	630.5	2353	> 264.6	ok	472.9	φVc>Vu	not required
	Min Qx	0.355	1	248.5	51.286			678.615			630.5	2353	> 248.5	ok	472.9	φVc>Vu	not required

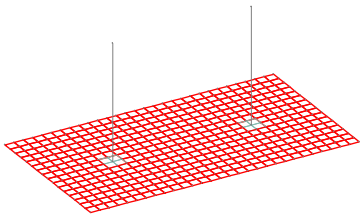
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:1.2D	174.09	top face	50	642	174.09	0.56	0.0014	869.2	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
113:0.9D	-166.31	bottom face	50	640	166.31	0.54	0.0013	832.6		20	271.97	150	2095.3	29.54	36.93	0.049	tension control
105:1.2D	147.23	top face	50	642	147.23	0.48	0.0011	734.0		16	174.06	150	1341	18.91	23.64	0.078	tension control
108:1.2D	134.96	top face	50	642	134.96	0.44	0.0010	672.3		16	174.06	150	1341	18.91	23.64	0.078	tension control

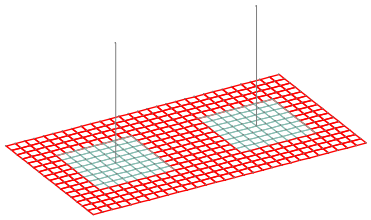
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qy	0.277	1	193.9	38.821	0.002	0.642	680.801	630.497	fc'1101	630.5	661	> 193.9	ok	472.9	φVc>Vu	not required
	Min Qy	0.241	1	168.7	75.212			640.527			630.5	661	> 168.7	ok	472.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **20ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **20ϕT** **150** c/c
At Top **16ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F4

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m
1561	0.60m	0.60m
Size of fdn. raft:	2.50m	4.25m
Thickness of raft:	0.70	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

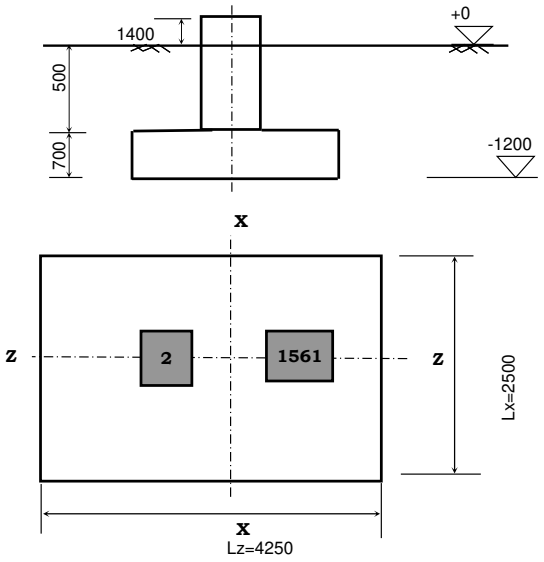
Weight of raft = (2.5*4.25*0.7*24) = 178.50KN

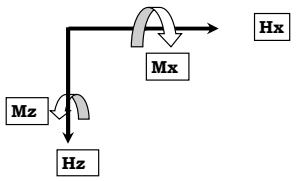
Weight of soil = {(2.5*4.25)-(0.6*0.6)-(0.6*0.6)}*0.5*20 = 99.05KN

Total = 277.55KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F4

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	115.42	-0.03	0.08	-0.35	-0.91
202 DL+L	211.12	0.18	-0.47	-0.45	-1.17
203 DL+0.75L	187.20	0.13	-0.33	-0.43	-1.11
204 DL+WL1	-32.78	-2.99	7.77	-0.35	-0.91
205 DL+WL2	263.62	2.93	-7.62	-0.35	-0.91
206 DL+WL3	-114.78	-0.47	1.22	-42.68	-110.97
207 DL+WL4	345.62	0.41	-1.07	41.98	109.15
208 DL+0.75WL1+0.75L	76.05	-2.09	5.44	-0.43	-1.11
209 DL+0.75WL2+0.75L	298.35	2.35	-6.10	-0.43	-1.11
210 DL+0.75WL3+0.75L	14.55	-0.20	0.53	-32.17	-83.65
211 DL+0.75WL4+0.75L	359.85	0.46	-1.19	31.32	81.44
212 0.6DL+WL1	-78.95	-2.98	7.74	-0.21	-0.55
213 0.6DL+WL2	217.45	2.94	-7.65	-0.21	-0.55
214 0.6DL+WL3	-160.95	-0.46	1.19	-42.54	-110.61
215 0.6DL+WL4	299.45	0.42	-1.10	42.12	109.51
216 0.6DL+WL3	415.22	-0.53	1.37	-52.62	-136.81
217 0.6DL+WL4	-246.38	0.07	-0.19	46.38	120.59
218 0.6DL+0.7SL1	84.42	-0.23	0.59	-3.12	-8.11
219 0.6DL+0.7SL2	84.42	-0.23	0.59	-3.12	-8.11

Node 1561				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
92.6	-0.3	0.8	0.6	1.5
142.7	-1.3	3.4	0.8	2.0
130.2	-1.0	2.7	0.7	1.9
227.6	0.0	0.1	0.5	1.5
-42.5	-0.6	1.6	0.6	1.6
322.8	-0.7	1.9	-41.6	-107.9
-137.6	0.1	-0.2	42.7	111.0
231.5	-0.8	2.2	0.7	1.9
28.9	-1.3	3.3	0.7	1.9
302.8	-1.4	3.5	-30.9	-80.2
-42.5	-0.7	1.9	32.3	84.0
190.6	0.1	-0.2	0.3	0.9
-79.5	-0.5	1.2	0.4	1.0
285.7	-0.6	1.6	-41.8	-108.6
-174.6	0.2	-0.6	42.5	110.4
-243.6	1.5	-3.9	-40.5	-103.8
459.0	-0.6	1.5	46.4	151.2
103.3	0.5	-1.2	4.1	23.7
112.4	0.4	-1.2	1.8	23.9

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	485.6	0.23	0.6	-0.4	0.9	667.0	1640.4	554.9	844.4
202 DL+L	631.4	0.31	0.9	-1.1	2.9	273.5	1567.4	227.5	814.7
203 DL+0.75L	594.9	0.29	0.8	-0.9	2.4	310.9	1582.3	258.7	820.6
204 DL+WL1	472.4	0.19	0.6	-3.0	7.9	75.0	1742.7	62.4	994.5
205 DL+WL2	498.7	0.27	0.7	2.3	-6.1	102.9	1556.2	85.6	738.8
206 DL+WL3	485.5	-84.24	-218.9	-1.2	3.1	194.5	4.7	161.8	2.3
207 DL+WL4	485.6	84.70	220.2	0.5	-1.3	466.9	4.7	388.5	2.3
208 DL+0.75WL1+0.75L	585.1	0.26	0.8	-2.9	7.6	96.0	1635.8	79.8	900.1
209 DL+0.75WL2+0.75L	604.8	0.32	0.8	1.1	-2.8	266.8	1531.8	221.9	756.0
210 DL+0.75WL3+0.75L	594.9	-63.06	-163.9	-1.6	4.1	183.6	7.7	152.7	3.8
211 DL+0.75WL4+0.75L	594.9	63.64	165.5	-0.3	0.7	1013.2	7.6	843.9	3.7
212 0.6DL+WL1	389.2	0.10	0.3	-2.9	7.5	64.7	2544.6	53.9	1588.5
213 0.6DL+WL2	415.5	0.18	0.4	2.5	-6.4	80.9	2053.4	67.3	933.8
214 0.6DL+WL3	402.3	-84.33	-219.2	-1.1	2.8	182.5	3.9	151.8	1.9
215 0.6DL+WL4	402.4	84.61	219.9	0.6	-1.7	302.3	3.9	251.5	1.9
216 0.6DL+WL3	449.1	-93.12	-240.6	1.0	-2.5	224.7	4.0	186.9	1.9
217 0.6DL+WL4	490.2	92.78	271.8	-0.5	1.3	472.1	3.8	392.9	2.1
218 0.6DL+0.7SL1	465.3	1.02	15.6	0.2	-0.6	913.0	63.3	759.6	183.2
219 0.6DL+0.7SL2	474.4	-1.36	15.8	0.2	-0.6	1051.3	63.8	874.4	139.1

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Base pressure check of footing F4

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	78.69	0
Min Px	2	201	0	78.69	0
Max Py	324	211	0	151.26	0
Min Py	2	212	0	0.00	0
Max Pz	2	201	0	78.69	0
Min Pz	2	201	0	78.69	0

Gross Maximum Base pressure, p_{max} = 151.3 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.50 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 460 KN/m^2
> 151.3 KN/m^2
Hence OK

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Design of foundation raft F4

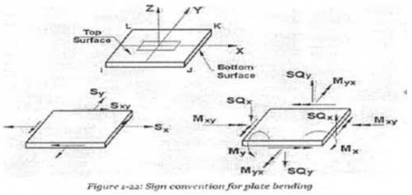
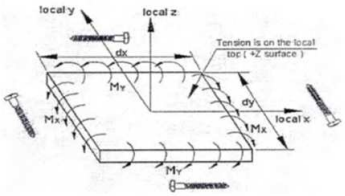


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F4_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	427	0.348	0.048	0	0	0	-54.057	-50.749	2.234	-56.291	-52.983	107:1.2DL+1.6WL4+1.0L
Min Qx	424	-0.352	0.05	0	0	0	-55.736	-50.82	-1.194	-56.93	-52.014	107:1.2DL+1.6WL4+1.0L
Max Qy	415	-0.122	0.487	0	0	0	-79.916	-85.143	3.299	-83.215	-88.442	107:1.2DL+1.6WL4+1.0L
Min Qy	1366	0.115	-0.471	0	0	0	-74.469	-80.497	4.722	-79.191	-85.219	106:1.2DL+1.6WL3+1.0L
Max Sx	1427	-0.042	-0.004	0	0	0	-6.981	-0.51	0.529	-7.51	-1.039	101:1.4DL
Min Sx	1427	-0.042	-0.004	0	0	0	-6.981	-0.51	0.529	-7.51	-1.039	101:1.4DL
Max Sy	1427	-0.042	-0.004	0	0	0	-6.981	-0.51	0.529	-7.51	-1.039	101:1.4DL
Min Sy	1427	-0.042	-0.004	0	0	0	-6.981	-0.51	0.529	-7.51	-1.039	101:1.4DL
Max Sxy	1427	-0.042	-0.004	0	0	0	-6.981	-0.51	0.529	-7.51	-1.039	101:1.4DL
Min Sxy	1427	-0.042	-0.004	0	0	0	-6.981	-0.51	0.529	-7.51	-1.039	101:1.4DL
Max Mx	1420	0.068	0.05	0	0	0	39.19	-15.266	-17.765	56.955	-33.031	111:0.9DL+1.6WL4
Min Mx	345	0.022	-0.021	0	0	0	-97.13	12.062	19.447	-116.577	31.509	107:1.2DL+1.6WL4+1.0L
Max My	415	0.073	-0.179	0	0	0	8.739	114.566	-12.231	20.97	126.797	106:1.2DL+1.6WL3+1.0L
Min My	415	-0.109	0.389	0	0	0	-61.55	-100.38	5.287	-66.837	-105.667	111:0.9DL+1.6WL4
Max Mxy	344	-0.025	-0.038	0	0	0	-43.815	-7.904	28.181	-71.996	-36.085	107:1.2DL+1.6WL4+1.0L
Min Mxy	347	0.026	-0.036	0	0	0	-42.877	-7.746	-28.204	-71.081	-35.95	107:1.2DL+1.6WL4+1.0L

PLATE CENTER STRESS : FDN_F4_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	446	0.034	0.08	0	0	0	-13.262	21.104	14.952	-28.214	36.056	107:1.2DL+1.6WL4+1.0L
Min Qx	1339	-0.034	0.08	0	0	0	-13.343	21.077	-14.137	-27.48	35.214	107:1.2DL+1.6WL4+1.0L
Max Qy	443	-0.012	0.122	0	0	0	-26.511	22.78	-3.575	-30.086	26.355	107:1.2DL+1.6WL4+1.0L
Min Qy	761	0.011	-0.127	0	0	0	-25.795	30.212	-2.719	-28.514	32.931	106:1.2DL+1.6WL3+1.0L
Max Sx	1425	-0.008	-0.004	0	0	0	-0.306	0.362	0.263	-0.569	0.625	101:1.4DL
Min Sx	1425	-0.008	-0.004	0	0	0	-0.306	0.362	0.263	-0.569	0.625	101:1.4DL
Max Sy	1425	-0.008	-0.004	0	0	0	-0.306	0.362	0.263	-0.569	0.625	101:1.4DL
Min Sy	1425	-0.008	-0.004	0	0	0	-0.306	0.362	0.263	-0.569	0.625	101:1.4DL
Max Sxy	1425	-0.008	-0.004	0	0	0	-0.306	0.362	0.263	-0.569	0.625	101:1.4DL
Min Sxy	1425	-0.008	-0.004	0	0	0	-0.306	0.362	0.263	-0.569	0.625	101:1.4DL
Max Mx	761	-0.007	0.089	0	0	0	9.645	7.418	2.463	12.108	9.881	111:0.9DL+1.6WL4
Min Mx	443	-0.012	0.122	0	0	0	-26.511	22.78	-3.575	-30.086	26.355	107:1.2DL+1.6WL4+1.0L
Max My	444	-0.004	-0.05	0	0	0	2.366	63.574	-1.225	3.591	64.799	106:1.2DL+1.6WL3+1.0L
Min My	503	0.021	-0.058	0	0	0	-0.587	-29.46	4.613	-5.2	-34.073	111:0.9DL+1.6WL4
Max Mxy	446	0.034	0.08	0	0	0	-13.262	21.104	14.952	-28.214	36.056	107:1.2DL+1.6WL4+1.0L
Min Mxy	1339	-0.034	0.08	0	0	0	-13.343	21.077	-14.137	-27.48	35.214	107:1.2DL+1.6WL4+1.0L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

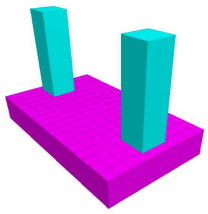
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
111:0.9D	56.96	top face	50	642	56.955	0.18	0.0004	282.5	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-116.6	bottom face	50	642	116.58	0.38	0.0009	580.2		16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-72.00	bottom face	50	642	71.996	0.23	0.0006	357.4		16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-71.08	bottom face	50	642	71.081	0.23	0.0005	352.9		16	174.06	150	1341	18.91	23.64	0.078	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			rks	KN	ks	
Load condition	Max Qx	0.034	1	23.8	28.214	0.002	0.642	620.046	630.497	1101.46	620.05	2345	> 23.8	ok	465	φVc>Vu	not required
	Min Qx	0.034	1	23.8	27.48			620.375			620.38	2345	> 23.8	ok	465.3	φVc>Vu	not required

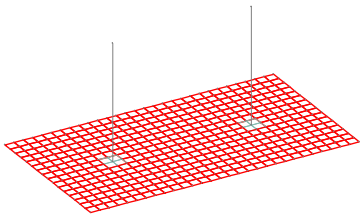
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:1.2D	126.80	top face	50	642	126.8	0.41	0.0010	631.4	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
111:0.9D	-105.67	bottom face	50	642	105.67	0.34	0.0008	525.5		16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-36.09	bottom face	50	642	36.085	0.12	0.0003	178.8		16	174.06	150	1341	18.91	23.64	0.078	tension control
107:1.2D	-35.95	bottom face	50	642	35.95	0.12	0.0003	178.1		16	174.06	150	1341	18.91	23.64	0.078	tension control

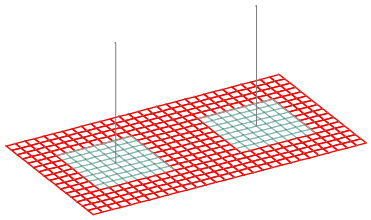
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			rks	KN	ks	
Load condition	Max Qy	0.122	1	85.4	26.355	0.002	0.642	655.125	630.497	fc'1101	630.5	661	> 85.4	ok	472.9	φVc>Vu	not required
	Min Qy	0.127	1	88.9	32.931			647.21			630.5	661	> 88.9	ok	472.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F5

Size of Columns at node	Along-X	Along-Z
1562	0.50m	0.50m
1878 , 1998	0.50m	0.50m
Size of fdn. raft:	2.75m	9.88m
Thickness of raft:	0.70	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

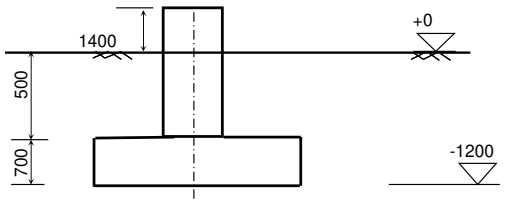
Weight of raft = (2.75*9.875*0.7*24) = 456.23KN

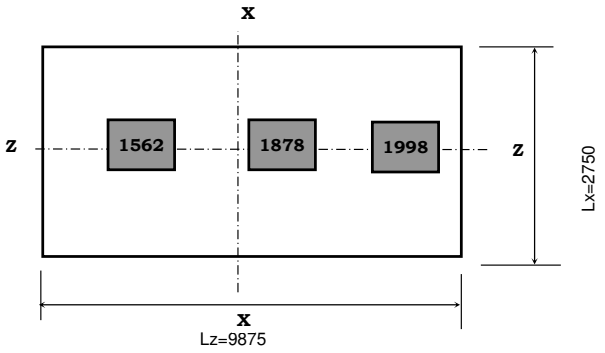
Weight of soil = {(2.75*9.875)-(0.5*0.5)*3}*0.5*20 = 264.06KN

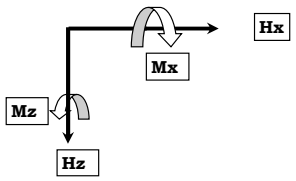
Total = 720.29KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F5

Load at bottom of footing	Node			1562 & 1878	
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	383.86	-1.60	4.16	-1.52	-3.81
202 DL+L+TL1	721.54	-7.34	19.08	1.23	3.47
203 DL+L+TL2	715.24	-14.28	37.13	-16.97	-43.86
204 DL+0.75L+0.75TL1	637.12	-5.91	15.35	0.54	1.65
205 DL+0.75L+0.75TL2	632.40	-11.11	28.89	-13.11	-33.85
206 DL+WL1	649.33	177.66	-461.92	10.97	28.76
207 DL+WL2	116.41	-181.74	472.53	-13.86	-36.00
208 DL+WL3	329.27	-13.86	36.04	160.24	416.75
209 DL+WL4	436.94	9.95	-25.87	-161.92	-420.84
210 DL+0.75WL1+0.75L	833.86	125.94	-327.44	3.09	8.33
211 DL+0.75WL2+0.75L	434.17	-143.61	373.40	-15.54	-40.24
212 DL+0.75WL3+0.75L	593.82	-17.70	46.03	115.04	299.32
213 DL+0.75WL4+0.75L	674.57	0.16	-0.40	-126.58	-328.87
214 0.6DL+WL1	495.79	178.30	-463.59	11.58	30.29
215 0.6DL+WL2	-37.13	-181.10	470.87	-13.25	-34.47
216 0.6DL+WL3	175.73	-13.22	34.37	160.85	418.27
217 0.6DL+WL4	283.40	10.59	-27.53	-161.31	-419.31
218 0.6DL+0.7SL1	222.34	1.78	-4.64	0.26	0.75
219 0.6DL+0.7SL2	238.30	-3.70	9.63	-2.08	-5.32

Node	1998			
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
124.2	-2.2	5.6	-0.1	-0.2
154.5	-3.3	8.6	0.3	0.7
152.1	-1.2	3.2	-0.6	-1.6
146.9	-3.0	7.8	0.2	0.5
145.1	-1.5	3.8	-0.5	-1.2
-105.9	60.4	-157.1	-0.1	-0.2
354.2	-64.8	168.4	-0.1	-0.1
126.6	4.3	-11.1	2.4	6.3
122.0	-8.6	22.4	-2.6	-6.7
-26.6	44.7	-116.3	-0.1	-0.4
318.6	-49.2	127.9	-0.2	-0.3
147.8	2.6	-6.7	1.7	4.5
144.3	-7.1	18.4	-2.0	-5.2
-155.6	61.3	-159.4	0.0	-0.1
304.6	-63.9	166.1	-0.1	0.0
76.9	5.1	-13.4	2.4	6.4
72.3	-7.7	20.1	-2.5	-6.6
66.9	0.2	-0.6	0.0	-0.1
82.1	-2.8	7.3	-0.1	-0.1

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	1228.3	-1.60	-4.0	-3.8	9.8	172.3	1525.0	130.3	307.1
202 DL+L+TL1	1596.3	1.49	4.2	-10.6	27.6	79.4	1877.5	60.1	428.5
203 DL+L+TL2	1587.6	-17.59	-45.4	-15.5	40.3	54.1	172.6	40.9	36.1
204 DL+0.75L+0.75TL1	1504.3	0.72	2.2	-8.9	23.2	89.2	3448.2	67.5	839.2
205 DL+0.75L+0.75TL2	1497.8	-13.59	-35.1	-12.6	32.7	63.0	211.0	47.6	44.1
206 DL+WL1	1263.7	10.91	28.6	238.1	-619.0	2.8	218.4	2.1	46.3
207 DL+WL2	1190.9	-13.95	-36.1	-246.5	640.9	2.6	162.9	1.9	34.1
208 DL+WL3	1176.2	162.64	423.0	-9.6	24.9	64.9	13.7	49.1	2.9
209 DL+WL4	1279.2	-164.50	-427.5	1.3	-3.5	504.8	14.8	381.8	3.1
210 DL+0.75WL1+0.75L	1527.6	2.95	8.0	170.7	-443.7	4.7	948.3	3.6	207.5
211 DL+0.75WL2+0.75L	1473.0	-15.70	-40.5	-192.8	501.3	4.0	179.4	3.1	37.5
212 DL+0.75WL3+0.75L	1461.9	116.74	303.8	-15.1	39.3	51.2	23.8	38.7	5.0
213 DL+0.75WL4+0.75L	1539.2	-128.61	-334.1	-6.9	18.0	117.8	22.7	89.1	4.8
214 0.6DL+WL1	1060.5	11.55	30.2	239.6	-623.0	2.3	173.6	1.8	36.7
215 0.6DL+WL2	987.7	-13.31	-34.5	-245.0	637.0	2.1	141.3	1.6	29.7
216 0.6DL+WL3	972.9	163.28	424.6	-8.1	21.0	63.7	11.3	48.2	2.4
217 0.6DL+WL4	1076.0	-163.86	-425.9	2.8	-7.4	199.8	12.5	151.1	2.6
218 0.6DL+0.7SL1	1009.5	0.22	0.7	2.0	-5.2	266.8	7298.2	201.8	1810.8
219 0.6DL+0.7SL2	1040.7	-2.14	-5.5	-6.5	17.0	84.3	941.8	63.8	194.2

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Base pressure check of footing F5

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	366	201	0	82.68	0
Min Px	366	201	0	82.68	0
Max Py	1652	213	0	208.40	0
Min Py	1971	206	0	0.00	0
Max Pz	366	201	0	82.68	0
Min Pz	366	201	0	82.68	0

Gross Maximum Base pressure, p_{max} = 208.4 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.50 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 460 KN/m^2
> 208.4 KN/m^2
Hence OK

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Design of foundation raft F5

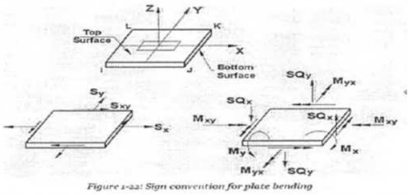
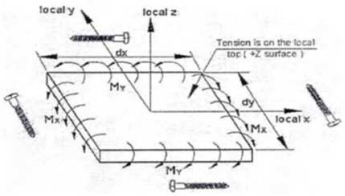


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F5_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1546	1.306	0.025	0	0	0	-102.89	-1.362	25.412	-128.297	-26.774	108:1.2DL+1.6WL4+1.0L
Min Qx	1727	-1.056	0.354	0	0	0	-316.38	-224.337	-7.616	-323.999	-231.953	105:1.2DL+1.6WL1+1.0L
Max Qy	1746	0.27	1.031	0	0	0	-105.76	-464.704	-27.504	-133.265	-492.208	113:0.9DL+1.6WL3
Min Qy	1746	-0.235	-0.814	0	0	0	65.379	373.047	31.941	97.32	404.988	114:0.9DL+1.6WL4
Max Sx	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Min Sx	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Max Sy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Min Sy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Max Sxy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Min Sxy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Max Mx	1727	0.84	-0.352	0	0	0	286.675	227.339	-40.809	327.484	268.148	106:1.2DL+1.6WL2+1.0L
Min Mx	1545	0.832	-0.141	0	0	0	-352.94	-2.801	-5.753	-358.697	-8.554	108:1.2DL+1.6WL4+1.0L
Max My	1746	-0.226	-0.768	0	0	0	51.113	381.303	22.406	73.519	403.709	108:1.2DL+1.6WL4+1.0L
Min My	1746	0.27	1.031	0	0	0	-105.76	-464.704	-27.504	-133.265	-492.208	113:0.9DL+1.6WL3
Max Mxy	1711	0.076	-0.323	0	0	0	-53.126	119.639	123.702	-176.828	243.341	112:0.9DL+1.6WL2
Min Mxy	775	-0.357	0.428	0	0	0	-74.136	-47.915	-162.884	-237.02	-210.799	108:1.2DL+1.6WL4+1.0L

PLATE CENTER STRESS : FDN_F5_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1548	0.822	-0.026	0	0	0	61.317	-0.253	19.212	80.529	-19.465	108:1.2DL+1.6WL4+1.0L
Min Qx	1548	-0.514	0.022	0	0	0	-79.415	1.058	-12.45	-91.865	13.508	113:0.9DL+1.6WL3
Max Qy	1534	0.462	0.827	0	0	0	-24.493	48.699	-4.142	-28.635	52.841	108:1.2DL+1.6WL4+1.0L
Min Qy	1692	-0.026	-0.499	0	0	0	-3.357	-59.165	-29.864	-33.221	-89.029	105:1.2DL+1.6WL1+1.0L
Max Sx	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Min Sx	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Max Sy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Min Sy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Max Sxy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Min Sxy	1858	0.012	-0.003	0	0	0	-0.665	0.14	-0.353	-1.018	0.493	101:1.4DL
Max Mx	1548	0.494	-0.018	0	0	0	106.463	-1.068	11.07	117.533	-12.138	106:1.2DL+1.6WL2+1.0L
Min Mx	1472	0.04	0.345	0	0	0	-103.89	98.308	-102.073	-205.963	200.381	108:1.2DL+1.6WL4+1.0L
Max My	1729	0.27	-0.067	0	0	0	58.842	254.573	35.385	94.227	289.958	112:0.9DL+1.6WL2
Min My	1743	0.164	0.062	0	0	0	-13.928	-256.581	5.125	-19.053	-261.706	113:0.9DL+1.6WL3
Max Mxy	1691	-0.062	0.027	0	0	0	5.576	165.079	84.269	89.845	249.348	112:0.9DL+1.6WL2
Min Mxy	759	-0.064	0.256	0	0	0	-65.617	72.99	-130.666	-196.283	203.656	108:1.2DL+1.6WL4+1.0L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

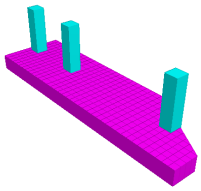
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:1.2D	327.48	top face	50	640	327.48	1.07	0.0026	1654.6	1152	20	189.94	150	2095.3	29.54	36.93	0.049	tension control
108:1.2D	-358.7	bottom face	50	640	358.7	1.17	0.0028	1815.6		20	173.1	150	2095.3	29.54	36.93	0.049	tension control
112:0.9D	-176.83	bottom face	50	640	176.83	0.58	0.0014	885.8		20	272.82	150	2095.3	29.54	36.93	0.049	tension control
108:1.2D	-237.02	bottom face	50	640	237.02	0.77	0.0019	1191.3		20	263.81	150	2095.3	29.54	36.93	0.049	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$	Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN	rks	KN	ks	
Load condition	Max Qx	0.462	1	323.4	80.529	0.003	0.64	697.358	641.427	1098.02	641.43	2355 > 323.4	ok	481.1	φVc>Vu	not required
	Min Qx	0.514	1	359.8	91.865			695.093			641.43	2355 > 359.8	ok	481.1	φVc>Vu	not required

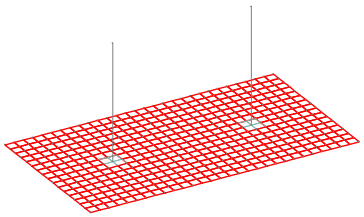
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:1.2D	403.71	top face	50	640	403.71	1.31	0.0032	2048.8	1152	20	153.4	150	2095.3	29.54	36.93	0.049	tension control
113:0.9D	-492.21	bottom face	50	640	492.21	1.60	0.0031	1977.7		20	158.91	150	2095.3	29.54	36.93	0.049	tension control
112:0.9D	243.34	top face	50	640	243.34	0.79	0.0019	1223.6		20	256.86	150	2095.3	29.54	36.93	0.049	tension control
108:1.2D	-210.80	bottom face	50	640	210.8	0.69	0.0017	1058.0		20	272.82	150	2095.3	29.54	36.93	0.049	tension control

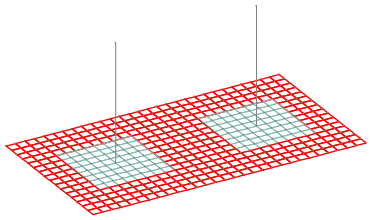
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$	Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN	rks	KN	ks	
Load condition	Max Qy	0.462	1	323.4	52.841	0.003	0.64	745.329	641.427	fc'1098	641.43	668 > 323.4	ok	481.1	φVc>Vu	not required
	Min Qy	0.499	1	349.3	89.029			695.249			641.43	668 > 349.3	ok	481.1	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **20ϕT** **150** c/c
At Top **20ϕT** **150** c/c
Along Z Direction :
At Bottom **20ϕT** **150** c/c
At Top **20ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F6

Size of Columns at node	Along-X	Along-Z
2	0.50m	0.50m
1562	0.50m	0.50m
Size of fdn. raft:	2.00m	3.60m
Thickness of raft:	0.70	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

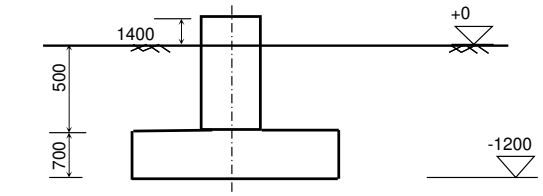
Weight of raft = (2*3.6*0.7*24) = 120.96KN

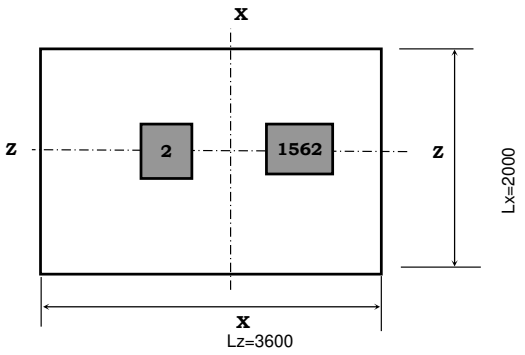
Weight of soil = {(2*3.6)-(0.5*0.5)-(0.5*0.5)}*0.5*20 = 67.00KN

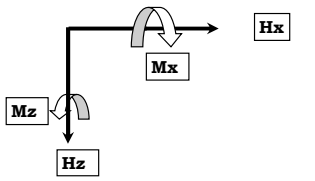
Total = 187.96KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F6

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	60.95	0.21	-0.55	-0.42	-1.09
202 DL+L	89.51	0.06	-0.16	-0.57	-1.48
203 DL+0.75L	82.37	0.10	-0.25	-0.53	-1.39
204 DL+WL1	-31.76	28.66	-74.52	-1.43	-3.72
205 DL+WL2	149.70	-27.52	71.55	0.46	1.20
206 DL+WL3	-41.11	-2.98	7.75	32.92	85.59
207 DL+WL4	163.29	3.33	-8.66	-33.75	-87.75
208 DL+0.75WL1+0.75L	12.84	21.44	-55.73	-1.29	-3.35
209 DL+0.75WL2+0.75L	148.93	-20.70	53.82	0.13	0.33
210 DL+0.75WL3+0.75L	5.83	-2.30	5.97	24.47	63.63
211 DL+0.75WL4+0.75L	159.13	2.44	-6.34	-25.53	-66.38
212 0.6DL+WL1	-56.14	28.58	-74.30	-1.26	-3.28
213 0.6DL+WL2	125.32	-27.60	71.77	0.63	1.63
214 0.6DL+WL3	-65.49	-3.06	7.97	33.09	86.03
215 0.6DL+WL4	138.91	3.25	-8.44	-33.58	-87.31
216 DL+L+TL1	115.09	-3.56	9.26	-2.62	-6.81
217 DL+L+TL2	63.93	3.68	-9.57	1.48	3.85
218 DL+0.75L+0.75TL1	101.56	-2.62	6.81	-2.07	-5.38
219 DL+0.75L+0.75TL2	63.19	2.81	-7.31	1.01	2.61

Node 1562				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
95.4	0.0	0.1	0.1	0.2
153.9	-0.1	0.3	0.2	0.6
139.3	-0.1	0.2	0.2	0.5
63.6	6.1	-15.8	0.0	0.0
125.0	-6.1	15.9	0.2	0.5
213.2	0.0	-0.1	0.7	1.9
-22.5	-0.1	0.2	-0.6	-1.5
115.4	4.5	-11.6	0.1	0.3
161.5	-4.7	12.2	0.2	0.7
227.6	-0.1	0.1	0.6	1.8
50.8	-0.1	0.4	-0.3	-0.8
25.5	6.1	-15.8	0.0	-0.1
86.8	-6.1	15.9	0.1	0.4
175.0	0.0	-0.1	0.7	1.8
-60.7	-0.1	0.2	-0.6	-1.6
150.6	-0.2	0.6	0.5	1.4
157.2	0.0	0.1	-0.1	-0.2
136.8	-0.2	0.4	0.4	1.1
141.7	0.0	0.1	0.0	-0.1

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	344.3	-0.35	-0.9	0.2	-0.5	697.0	705.9	724.9	393.5
202 DL+L	431.4	-0.36	-0.9	-0.1	0.2	2765.1	880.3	2875.7	479.3
203 DL+0.75L	409.6	-0.36	-0.9	0.003	0.0	58514.0	835.9	54613.1	457.7
204 DL+WL1	219.8	-1.45	-3.8	34.7	-90.3	2.4	105.5	2.5	60.6
205 DL+WL2	462.7	0.62	1.7	-33.7	87.5	5.3	503.2	5.5	298.5
206 DL+WL3	360.0	33.62	87.5	-2.9	7.6	47.1	7.4	49.0	4.3
207 DL+WL4	328.7	-34.31	-89.2	3.3	-8.5	38.9	6.6	40.5	3.8
208 DL+0.75WL1+0.75L	316.2	-1.18	-3.0	25.9	-67.4	4.7	187.5	4.9	107.0
209 DL+0.75WL2+0.75L	498.4	0.37	1.0	-25.4	66.0	7.6	880.3	7.9	537.3
210 DL+0.75WL3+0.75L	421.4	25.12	65.4	-2.3	6.1	69.1	11.6	71.9	6.7
211 DL+0.75WL4+0.75L	397.9	-25.83	-67.1	2.3	-6.0	66.6	10.7	69.3	6.2
212 0.6DL+WL1	157.3	-1.31	-3.4	34.7	-90.1	1.7	83.3	1.8	48.0
213 0.6DL+WL2	400.1	0.76	2.0	-33.7	87.7	4.6	358.9	4.7	210.6
214 0.6DL+WL3	297.5	33.76	87.8	-3.0	7.8	37.9	6.1	39.5	3.5
215 0.6DL+WL4	266.2	-34.17	-88.9	3.2	-8.3	32.3	5.4	33.5	3.1
216 DL+L+TL1	453.6	-2.12	-5.5	-3.8	9.8	46.2	149.6	48.0	85.6
217 DL+L+TL2	409.1	1.40	3.7	3.7	-9.5	43.0	199.3	44.7	116.9
218 DL+0.75L+0.75TL1	426.3	-1.68	-4.3	-2.8	7.2	58.8	177.9	61.2	101.7
219 DL+0.75L+0.75TL2	392.9	0.96	2.6	2.8	-7.3	54.1	277.1	56.3	163.4

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Base pressure check of footing F6

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	64.87	0
Min Px	2	201	0	64.87	0
Max Py	1466	214	0	246.64	0
Min Py	308	204	0	0.00	0
Max Pz	2	201	0	64.87	0
Min Pz	2	201	0	64.87	0

Gross Maximum Base pressure, p_{max} = 246.6 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.50 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 460 KN/m^2
> 246.6 KN/m^2
Hence OK

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Design of foundation raft F6

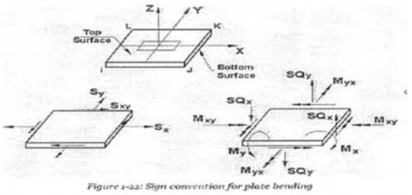
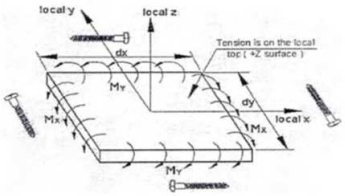


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F6_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	0.342	0.106	0	0	0	-32.237	28.852	24.26	-56.497	53.112	107:1.2DL+1.6WL4+1.0L
Min Qx	327	-0.364	0.104	0	0	0	-42.355	30.977	-23.783	-66.138	54.76	107:1.2DL+1.6WL4+1.0L
Max Qy	416	0.017	0.208	0	0	0	-27.467	25.358	-21.374	-48.841	46.732	105:1.2DL+1.6WL2+1.0L
Min Qy	885	0.034	-0.177	0	0	0	-29.457	30.54	-6.394	-35.851	36.934	105:1.2DL+1.6WL2+1.0L
Max Sx	1407	-0.044	-0.01	0	0	0	-3.208	0.924	2.267	-5.475	3.191	101:1.4DL
Min Sx	1407	-0.044	-0.01	0	0	0	-3.208	0.924	2.267	-5.475	3.191	101:1.4DL
Max Sy	1407	-0.044	-0.01	0	0	0	-3.208	0.924	2.267	-5.475	3.191	101:1.4DL
Min Sy	1407	-0.044	-0.01	0	0	0	-3.208	0.924	2.267	-5.475	3.191	101:1.4DL
Max Sxy	1407	-0.044	-0.01	0	0	0	-3.208	0.924	2.267	-5.475	3.191	101:1.4DL
Min Sxy	1407	-0.044	-0.01	0	0	0	-3.208	0.924	2.267	-5.475	3.191	101:1.4DL
Max Mx	333	-0.131	0.001	0	0	0	52.099	-7.058	3.737	55.836	-10.795	108:0.9DL+1.6WL1
Min Mx	333	0.259	0.029	0	0	0	-66.48	6.934	1.462	-67.942	8.396	105:1.2DL+1.6WL2+1.0L
Max My	416	-0.028	0.011	0	0	0	-32.574	116.647	20.659	-53.233	137.306	107:1.2DL+1.6WL4+1.0L
Min My	416	0.039	0.036	0	0	0	15.253	-113.068	-18.502	33.755	-131.57	110:0.9DL+1.6WL3
Max Mxy	414	-0.029	0.038	0	0	0	-14.703	-13.769	39.421	-54.124	-53.19	108:0.9DL+1.6WL1
Min Mxy	414	-0.014	0.05	0	0	0	-0.261	34.24	-38.274	-38.535	72.514	105:1.2DL+1.6WL2+1.0L

PLATE CENTER STRESS : FDN_F6_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	762	0.027	-0.067	0	0	0	-8.533	62.315	-10.222	-18.755	72.537	105:1.2DL+1.6WL2+1.0L
Min Qx	768	-0.027	0.014	0	0	0	-2.341	-8.412	17.879	-20.22	-26.291	108:0.9DL+1.6WL1
Max Qy	764	0.008	0.151	0	0	0	-0.864	-49.57	-7.285	-8.149	-56.855	110:0.9DL+1.6WL3
Min Qy	447	0	-0.107	0	0	0	0.128	-19.311	9.771	9.899	-29.082	108:0.9DL+1.6WL1
Max Sx	1355	-0.002	0.008	0	0	0	-1.059	28.267	0.432	-1.491	28.699	101:1.4DL
Min Sx	1355	-0.002	0.008	0	0	0	-1.059	28.267	0.432	-1.491	28.699	101:1.4DL
Max Sy	1355	-0.002	0.008	0	0	0	-1.059	28.267	0.432	-1.491	28.699	101:1.4DL
Min Sy	1355	-0.002	0.008	0	0	0	-1.059	28.267	0.432	-1.491	28.699	101:1.4DL
Max Sxy	1355	-0.002	0.008	0	0	0	-1.059	28.267	0.432	-1.491	28.699	101:1.4DL
Min Sxy	1355	-0.002	0.008	0	0	0	-1.059	28.267	0.432	-1.491	28.699	101:1.4DL
Max Mx	760	0.001	0.009	0	0	0	2.694	-15.999	-1.132	3.826	-17.131	111:0.9DL+1.6WL4
Min Mx	760	-0.004	0.003	0	0	0	-16.717	22.387	1.617	-18.334	24.004	106:1.2DL+1.6WL3+1.0L
Max My	443	0	-0.039	0	0	0	-3.528	80	0.401	-3.929	80.401	107:1.2DL+1.6WL4+1.0L
Min My	444	0.003	0.013	0	0	0	-3.957	-85.147	-4.936	-8.893	-90.083	110:0.9DL+1.6WL3
Max Mxy	443	-0.014	0	0	0	0	-1.157	-14.159	26.397	-27.554	-40.556	108:0.9DL+1.6WL1
Min Mxy	442	0.006	0.027	0	0	0	-6.484	62.398	-17.968	-24.452	80.366	105:1.2DL+1.6WL2+1.0L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

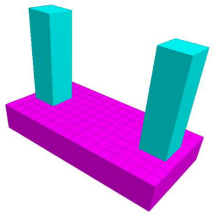
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:0.9D	55.84	top face	50	642	55.836	0.18	0.0004	276.9	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
105:1.2D	-67.9	bottom face	50	642	67.942	0.22	0.0005	337.2		16	174.06	150	1341	18.91	23.64	0.078	tension control
108:0.9D	-54.12	bottom face	50	642	54.124	0.18	0.0004	268.4		16	174.06	150	1341	18.91	23.64	0.078	tension control
105:1.2D	-38.54	bottom face	50	642	38.535	0.12	0.0003	191.0		16	174.06	150	1341	18.91	23.64	0.078	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN			
Load condition	Max Qx	0.027	1	18.9	18.755	0.002	0.642	622.449	630.497	1101.46	622.45	2347	> 18.9	ok	466.8	φVc>Vu	not required	
	Min Qx	0.027	1	18.9	20.22			621.38			621.38	2346	> 18.9	ok	466	φVc>Vu	not required	

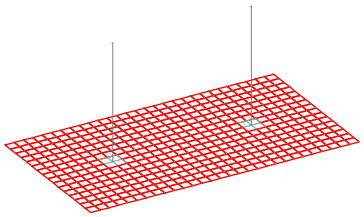
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107:1.2D	137.31	top face	50	642	137.31	0.44	0.0011	684.1	1155.6	16	174.06	150	1341	18.91	23.64	0.078	tension control
110:0.9D	-131.57	bottom face	50	642	131.57	0.43	0.0010	655.3		16	174.06	150	1341	18.91	23.64	0.078	tension control
108:0.9D	-53.19	bottom face	50	642	53.19	0.17	0.0004	263.8		16	174.06	150	1341	18.91	23.64	0.078	tension control
105:1.2D	72.51	top face	50	642	72.514	0.23	0.0006	360.0		16	174.06	150	1341	18.91	23.64	0.078	tension control

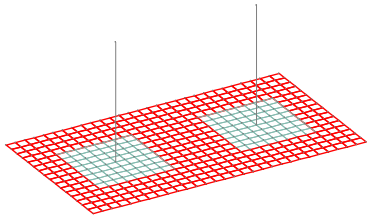
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN			
Load condition	Max Qy	0.151	1	105.7	56.855	0.002	0.642	634.909	630.497	fc'1101	630.5	661	> 105.7	ok	472.9	φVc>Vu	not required	
	Min Qy	0.107	1	74.9	29.082			645.394			630.5	661	> 74.9	ok	472.9	φVc>Vu	not required	



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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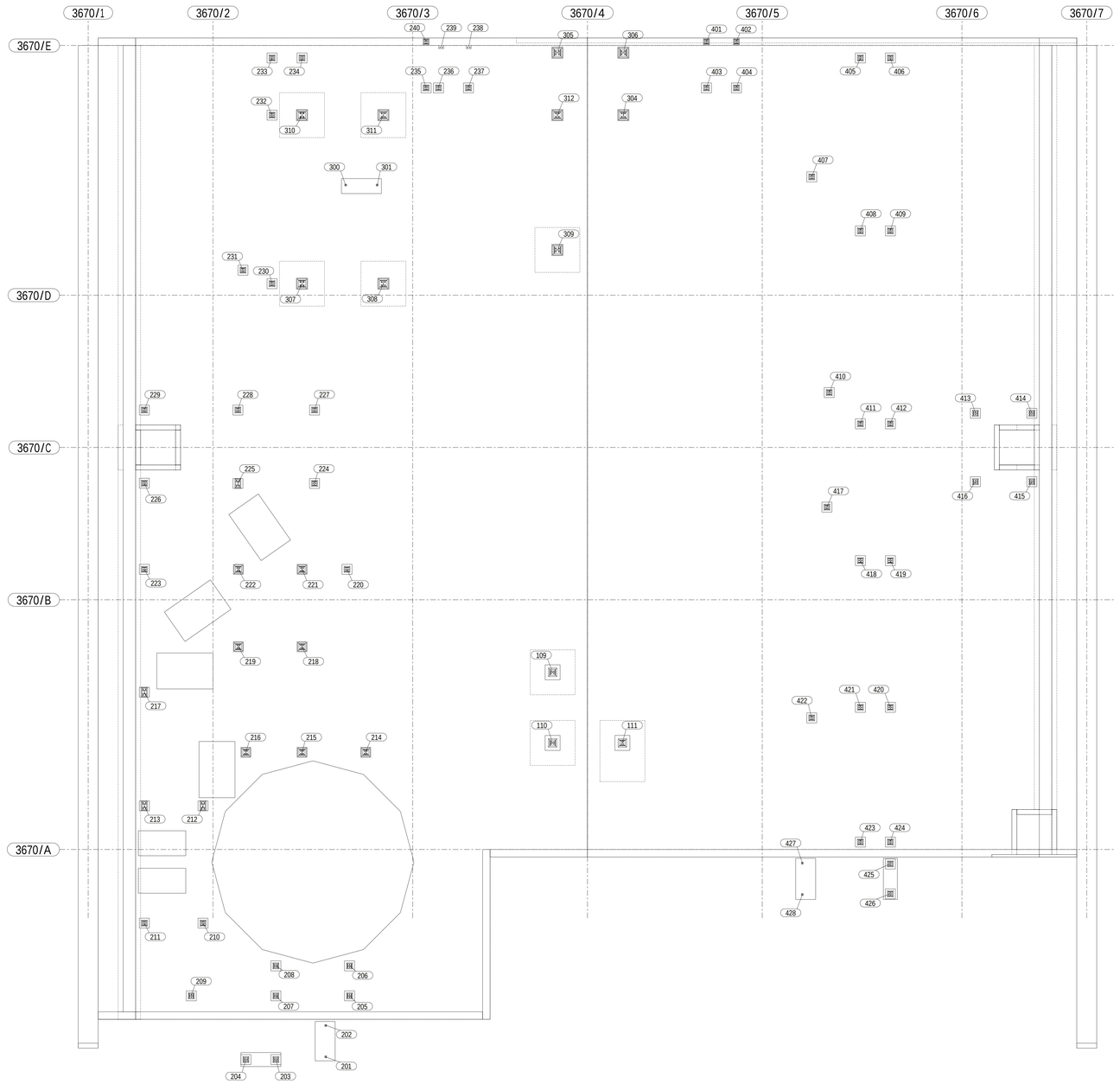
ATTACHMENT 1 INPUT DRAWINGS & DOCUMENTS

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REV	DRAWING NUMBER FOUNDATION IDENTIFICATION

LEVEL +912.200



SCHEDULE OF MEMBERS		
Section	Profile	Material
h	HEA120	S275JR
h	HEA160	S275JR
h	HEA220	S275JR
h	HEA260	S275JR
i	IPE240	S275JR
l	L130*10	S275JR
s	PB0*4	S275JR
s	P150*6	S275JR
u	UNP200	S275JR
u	UNP240	S275JR

WELDS:

All welds shall be in accordance with the project specification
MD-101-SMEM-EG-ST-SPC-001 and AWS D1.1

THE SIZE OF A FILLET WELD REFERS TO THE LEG LENGTH

BOLTS:

HIGH STRENGTH BOLTS ASTM A325/A325M TYPE 1, UNLESS SPECIFIED OTHERWISE

MATERIALS:

- HOT ROLLED SECTIONS AND PLATES IN CONNECTIONS S275JR EN 10025
- RAILING AND LADDERS S235JR EN 10025

PAINT SYSTEMS:

ANY ONE OF THE FOLLOWING PAINT SYSTEMS
CORRESPONDING TO STRUCTURAL STEEL EXPOSED TO WEATHER IN A MILD CORROSIVE
ENVIRONMENT SHALL BE ADOPTED :

i) Primer: 1 coat 125 - 150 µm NDFT Amercoat 385PA (or Amercoat 370 for winter application),
Finish: 1 coat 125 -150 µm NDFT PSX700 Engineering Epoxy Polysiloxane

ii) Primer: 1 coat 150 µm NDFT Hempadur Mastic 45881,
Finish: 1 coat 125 µm NDFT Hempaxane 55000

iii) Primer: 1 coat 75 µm NDFT Interzinc 22 / Interzinc 52,
Finish: 1 coat 125 µm NDFT Interfine 979

COLOR CODING:

Steelwork: RAL 7040 Window Grey
Hoist beams: RAL 2004 Pure Orange
Railing and ladders: RAL 1004 Golden Yellow

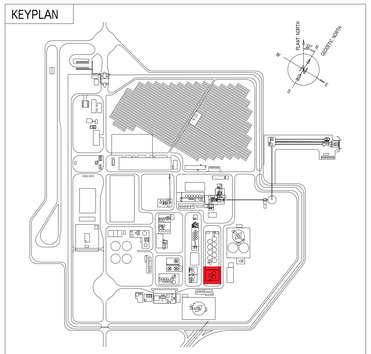
LEGEND:

TOS TOP OF STEEL
TYP TYPICAL
TOC TOP OF CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN METRES
2. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH CALCULATION REPORT
3. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER — REVISION 0

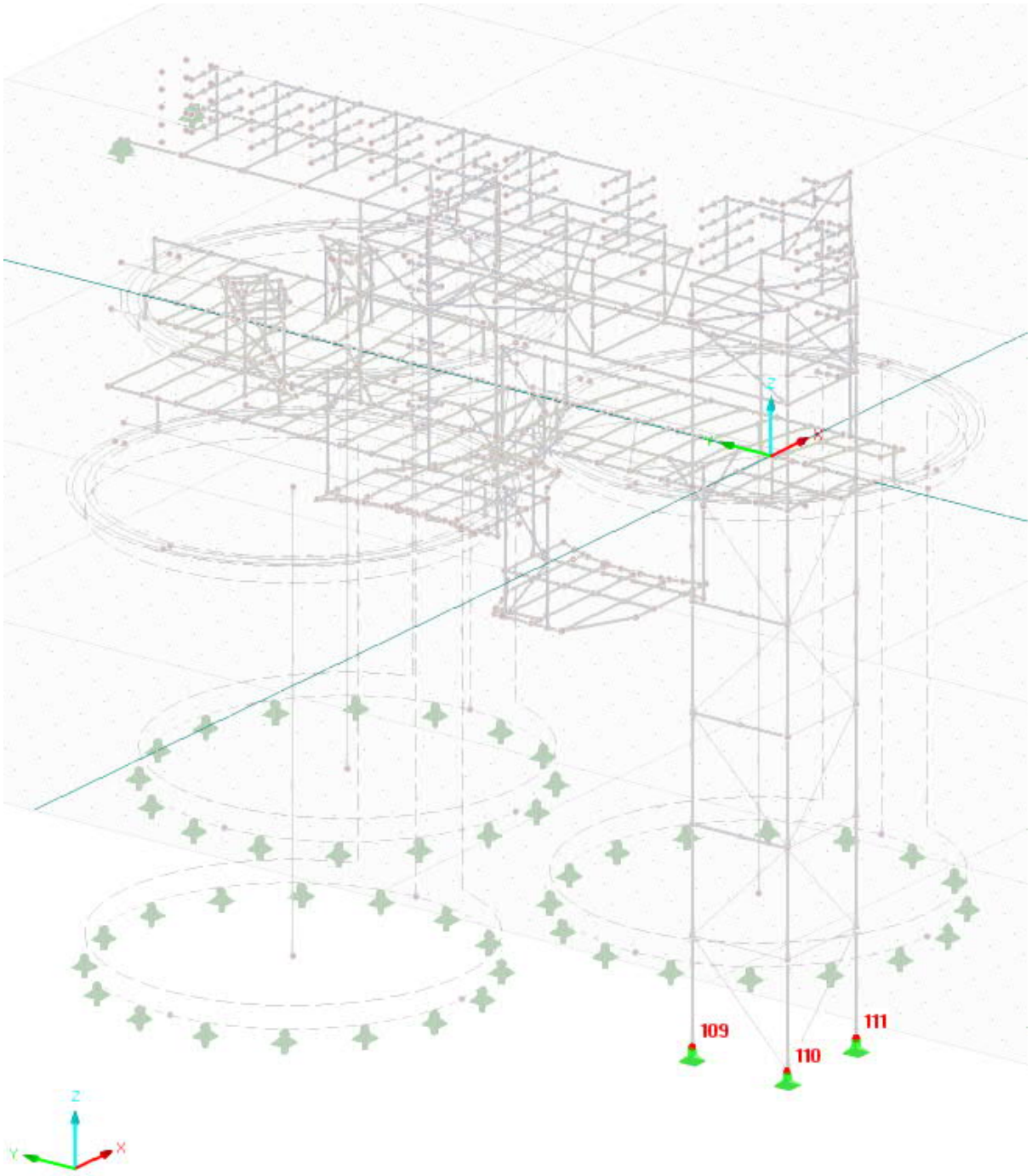
OUTOTEC DRAWING NUMBER — DOCUMENT ID — REVISION 0

DRAWN	NikJan	22.10.2020
CHECKED	MauNot	22.10.2020
ENGINEER	LuiPiz	22.10.2020
APPROVED	SouCha	22.10.2020
EMPLOYER		
SCALE	1:75	A0

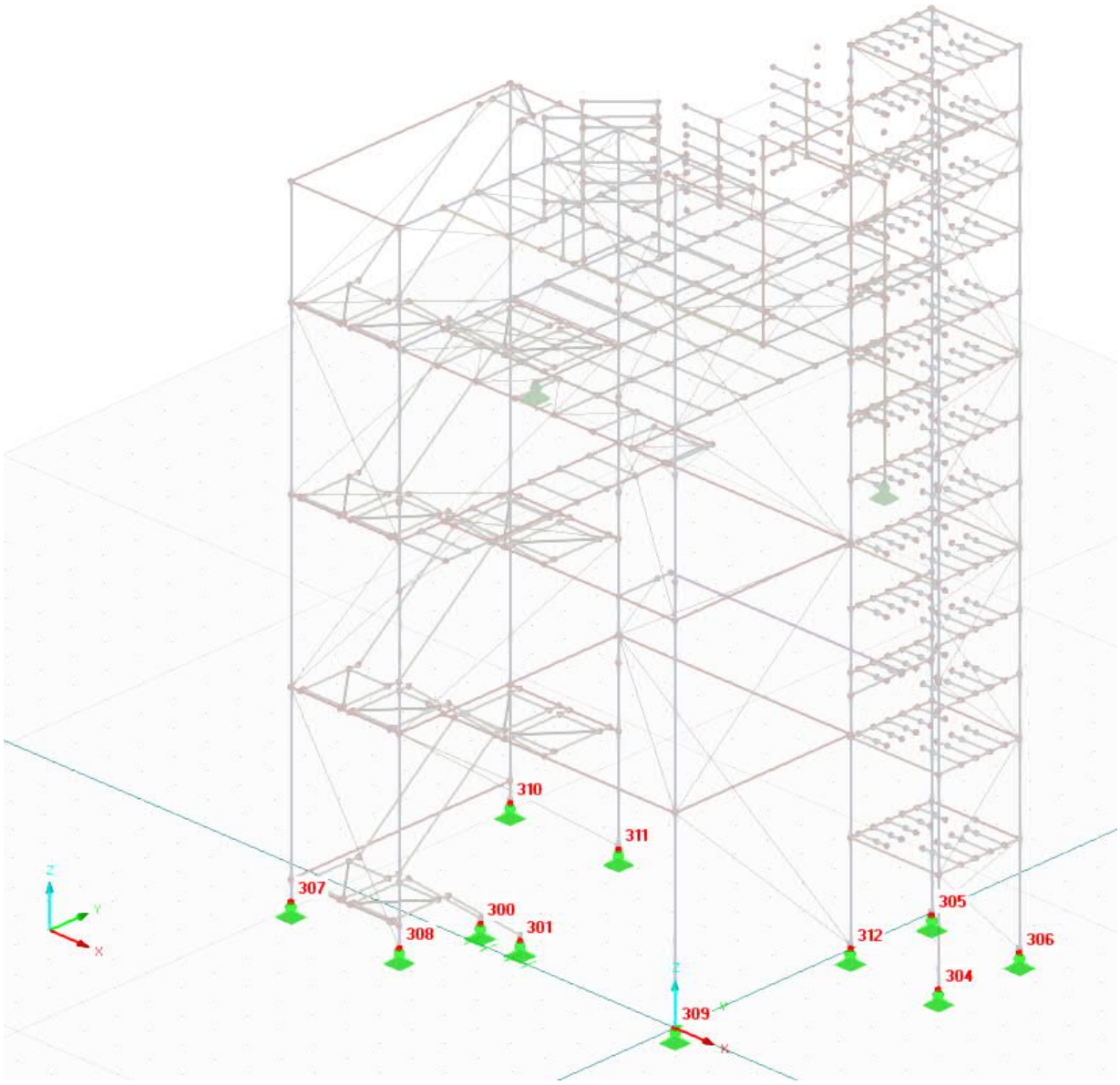
MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION LOADS 3670
PRIMARY CYANIDE DESTRUCTION
PLAN AT EL. +912.200

DRAWING NUMBER — REV
FOUNDATION IDENTIFICATION

Tekla Structures


Z
X
109
110
111
Z
Y
X

Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
109	LC1 - Self- Weight + Secondary structures	-0,02	0,07	-72,00
	LC2 - Live + Technical Load	-0,10	0,14	-58,49
	LC3 - Wind in +X	6,09	-0,09	31,79
	LC4 - Wind in -X	-6,11	0,09	-29,60
	LC5 - Wind in +Y	0,06	0,63	-117,78
	LC6 - Wind in -Y	-0,06	-0,63	117,92
	LC7 - Temperature +25	-0,10	0,29	3,30
	LC8 - Temperature -25	0,10	-0,29	-3,30
110	LC1 - Self- Weight + Secondary structures	0,21	-0,42	-37,55
	LC2 - Live + Technical Load	-0,15	-0,15	-28,56
	LC3 - Wind in +X	28,45	-1,01	92,71
	LC4 - Wind in -X	-27,73	0,88	-88,75
	LC5 - Wind in +Y	-3,19	33,34	102,06
	LC6 - Wind in -Y	3,12	-33,33	-102,34
	LC7 - Temperature +25	-3,62	-2,05	-25,58
	LC8 - Temperature -25	3,62	2,05	25,58
111	LC1 - Self- Weight + Secondary structures	0,04	-0,09	-33,15
	LC2 - Live + Technical Load	0,03	-0,15	-13,43
	LC3 - Wind in +X	0,69	-0,01	-95,18
	LC4 - Wind in -X	-0,69	0,01	90,70
	LC5 - Wind in +Y	-0,09	6,15	20,68
	LC6 - Wind in -Y	0,09	-6,15	-20,36
	LC7 - Temperature +25	0,28	-0,07	13,79
	LC8 - Temperature -25	-0,28	0,07	-13,79


310
311
307
308
300
301
312
305
306
304
309
Z
Y
X

Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
300	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,64
	LC2 - Live + Technical Load	0,00	0,00	-6,70
	LC3 - Wind in +X	0,00	0,00	0,00
	LC4 - Wind in -X	0,00	0,00	0,00
	LC5 - Wind in +Y	0,01	0,00	0,00
	LC6 - Wind in -Y	-0,01	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
301	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,60
	LC2 - Live + Technical Load	0,00	0,00	-6,42
	LC3 - Wind in +X	0,00	0,00	0,00
	LC4 - Wind in -X	0,00	0,00	0,00
	LC5 - Wind in +Y	0,01	0,00	0,00
	LC6 - Wind in -Y	-0,01	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
304	LC1 - Self- Weight + Secondary structures	0,35	-0,39	-90,00
	LC2 - Live + Technical Load	-0,18	-0,04	-42,94
	LC3 - Wind in +X	3,82	-0,98	19,37
	LC4 - Wind in -X	-3,85	0,97	-18,46
	LC5 - Wind in +Y	-0,35	0,45	260,00
	LC6 - Wind in -Y	0,34	-0,47	-255,87
	LC7 - Temperature +25	0,35	-2,39	-33,04
	LC8 - Temperature -25	-0,35	2,39	33,04
305	LC1 - Self- Weight + Secondary structures	-2,17	-0,08	-107,92
	LC2 - Live + Technical Load	-0,09	-0,10	-29,11
	LC3 - Wind in +X	62,59	0,02	230,10
	LC4 - Wind in -X	-62,59	-0,01	-230,07
	LC5 - Wind in +Y	6,44	2,48	-2,43
	LC6 - Wind in -Y	-6,44	-2,50	2,21
	LC7 - Temperature +25	-1,03	0,44	-1,17
	LC8 - Temperature -25	1,03	-0,44	1,17
306	LC1 - Self- Weight + Secondary structures	2,17	-0,40	-89,83
	LC2 - Live + Technical Load	0,09	-2,22	-5,87
	LC3 - Wind in +X	-3,87	6,86	-276,80
	LC4 - Wind in -X	3,87	-6,67	275,48
	LC5 - Wind in +Y	-0,61	60,96	-288,67
	LC6 - Wind in -Y	0,59	-60,73	284,46
	LC7 - Temperature +25	1,03	-2,29	32,98
	LC8 - Temperature -25	-1,03	2,29	-32,98
307	LC1 - Self- Weight + Secondary structures	0,01	-0,88	-52,59
	LC2 - Live + Technical Load	0,14	-1,99	-101,70
	LC3 - Wind in +X	2,54	0,24	393,02

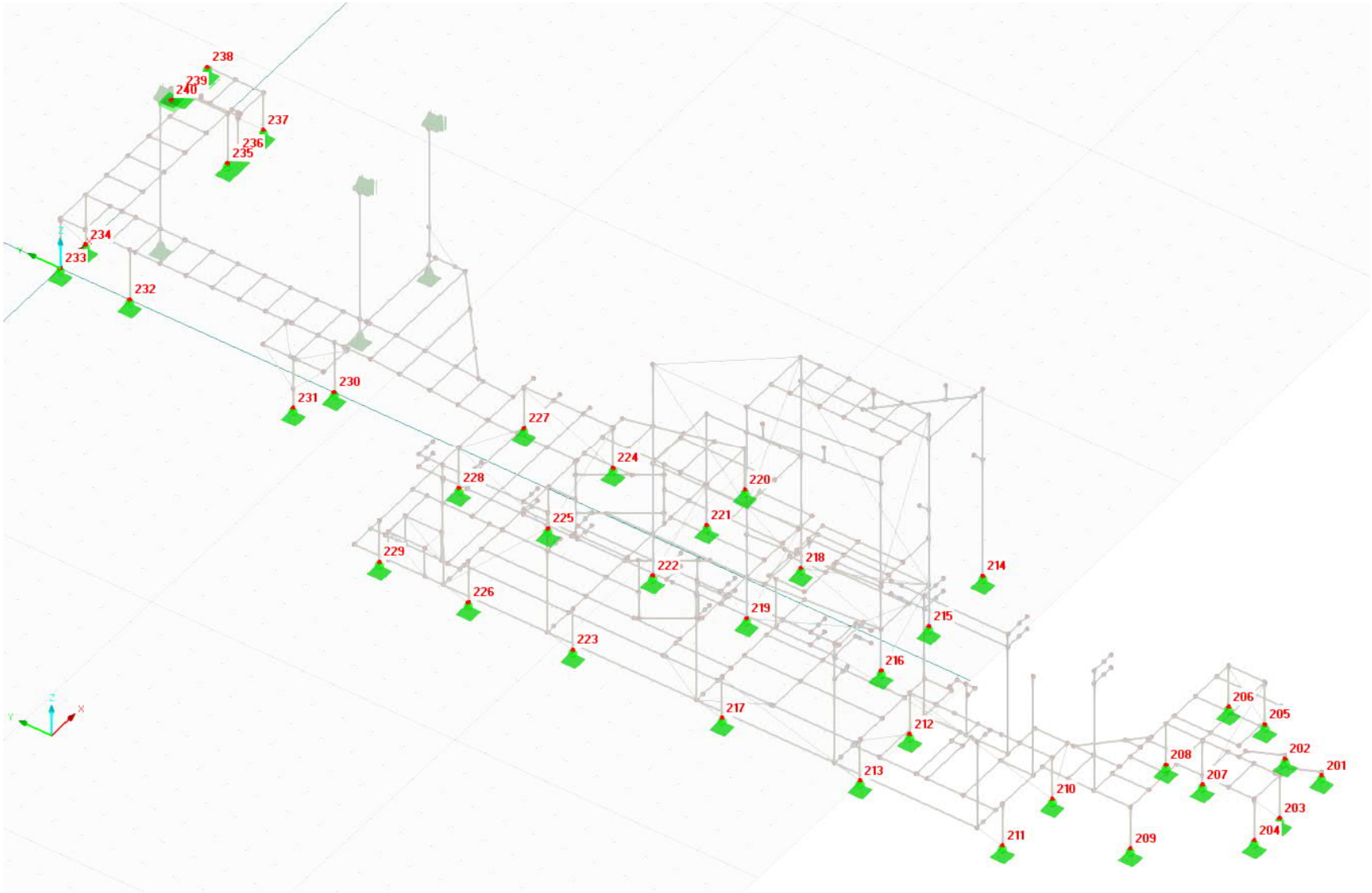
	LC4 - Wind in -X	-11,98	-0,23	-390,10
	LC5 - Wind in +Y	0,04	15,99	111,68
	LC6 - Wind in -Y	-0,04	-15,39	-108,97
	LC7 - Temperature +25	-8,90	-4,05	-9,77
	LC8 - Temperature -25	8,90	4,05	9,77
308	LC1 - Self- Weight + Secondary structures	0,66	-0,26	-138,22
	LC2 - Live + Technical Load	8,89	-0,81	-241,29
	LC3 - Wind in +X	114,49	0,31	-420,79
	LC4 - Wind in -X	-106,51	-0,31	416,92
	LC5 - Wind in +Y	6,47	9,04	80,95
	LC6 - Wind in -Y	-5,78	-4,99	-82,79
	LC7 - Temperature +25	5,15	-1,49	1,99
	LC8 - Temperature -25	-5,15	1,49	-1,99
309	LC1 - Self- Weight + Secondary structures	-0,50	-0,02	-203,10
	LC2 - Live + Technical Load	-0,31	0,07	-245,85
	LC3 - Wind in +X	0,73	-0,07	13,53
	LC4 - Wind in -X	-0,74	0,07	-12,99
	LC5 - Wind in +Y	0,46	1,21	399,87
	LC6 - Wind in -Y	-0,45	-1,21	-394,16
	LC7 - Temperature +25	-0,06	-0,18	23,97
	LC8 - Temperature -25	0,06	0,18	-23,97
310	LC1 - Self- Weight + Secondary structures	-0,07	0,42	-74,10
	LC2 - Live + Technical Load	-0,09	1,06	-139,24
	LC3 - Wind in +X	5,79	-1,21	58,16
	LC4 - Wind in -X	-4,49	1,24	-56,45
	LC5 - Wind in +Y	-0,10	45,02	-85,93
	LC6 - Wind in -Y	0,10	-45,91	85,55
	LC7 - Temperature +25	-0,58	3,75	0,30
	LC8 - Temperature -25	0,58	-3,75	-0,30
311	LC1 - Self- Weight + Secondary structures	0,00	2,07	-143,96
	LC2 - Live + Technical Load	0,04	10,35	-346,91
	LC3 - Wind in +X	4,87	-17,97	263,66
	LC4 - Wind in -X	-7,87	17,58	-266,01
	LC5 - Wind in +Y	-0,03	73,74	-130,38
	LC6 - Wind in -Y	0,03	-77,13	128,72
	LC7 - Temperature +25	0,08	-3,07	11,26
	LC8 - Temperature -25	-0,08	3,07	-11,26
312	LC1 - Self- Weight + Secondary structures	-1,09	-1,49	-148,26
	LC2 - Live + Technical Load	-8,91	-6,42	-88,68
	LC3 - Wind in +X	178,53	12,56	-279,00
	LC4 - Wind in -X	-179,41	-12,41	280,45
	LC5 - Wind in +Y	-12,72	160,55	-345,28
	LC6 - Wind in -Y	12,00	-159,19	341,08
	LC7 - Temperature +25	3,53	9,28	-27,12
	LC8 - Temperature -25	-3,53	-9,28	27,12

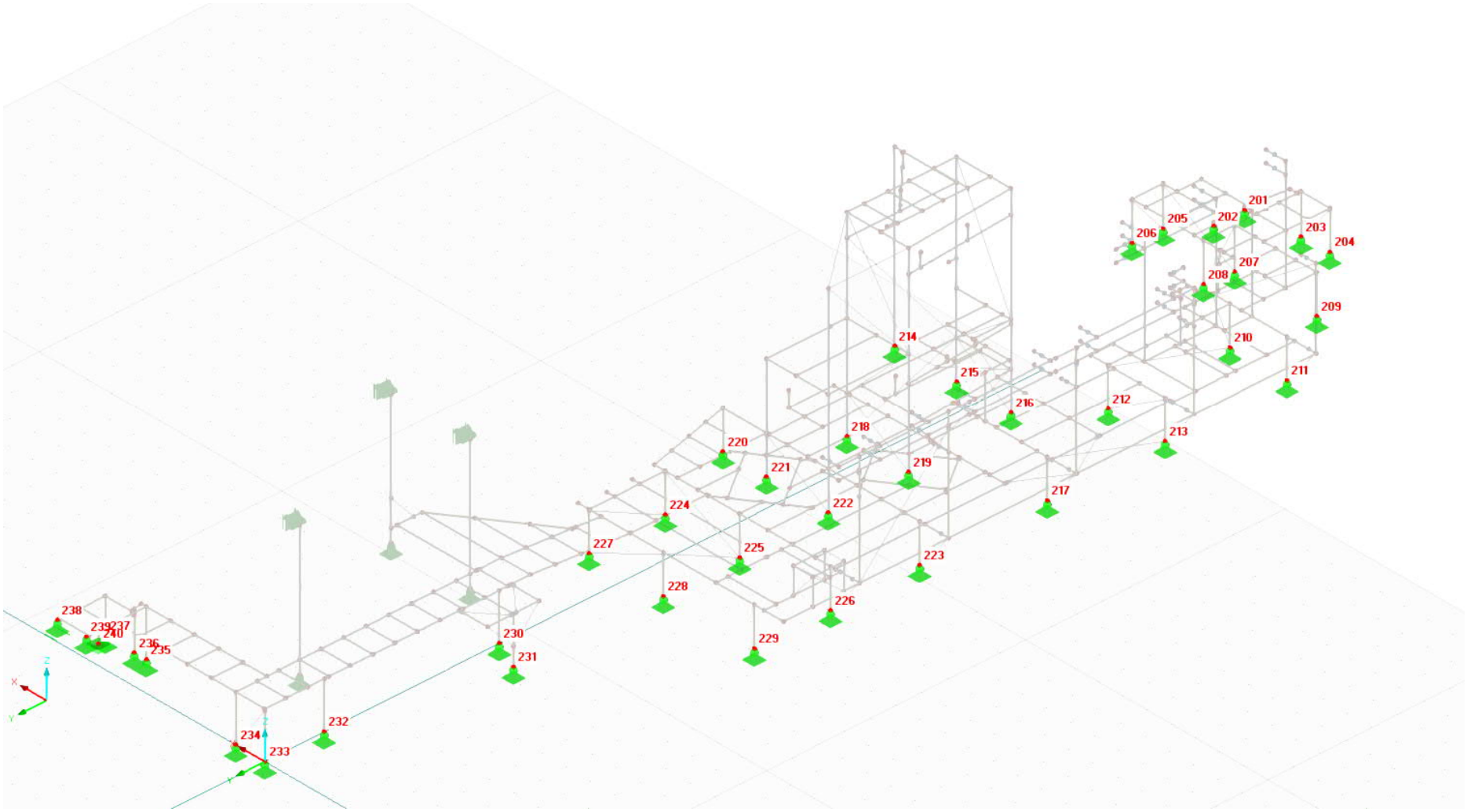
Node No.	LC/CO		Support Forces [kN]		
			$P_{X'}$	$P_{Y'}$	$P_{Z'}$
300	Seismic Envelope +X	Max	0,00	0,00	0,00
	Seismic Envelope -X	Min	0,00	0,00	0,00
301	Seismic Envelope +X	Max	0,00	0,00	0,00
	Seismic Envelope -X	Min	0,00	0,00	0,00
304	Seismic Envelope +X	Max	0,05	0,03	7,65
	Seismic Envelope -X	Min	-0,05	-0,03	-7,65
305	Seismic Envelope +X	Max	2,18	0,02	10,83
	Seismic Envelope -X	Min	-2,18	-0,02	-10,83
306	Seismic Envelope +X	Max	0,21	1,02	8,10
	Seismic Envelope -X	Min	-0,21	-1,02	-8,10
307	Seismic Envelope +X	Max	0,02	0,20	11,69
	Seismic Envelope -X	Min	-0,02	-0,20	-11,69
308	Seismic Envelope +X	Max	2,57	0,03	14,34
	Seismic Envelope -X	Min	-2,57	-0,03	-14,34
309	Seismic Envelope +X	Max	0,03	0,00	5,07
	Seismic Envelope -X	Min	-0,03	0,00	-5,07
310	Seismic Envelope +X	Max	0,02	0,75	2,37
	Seismic Envelope -X	Min	-0,02	-0,75	-2,37
311	Seismic Envelope +X	Max	0,02	1,70	6,48
	Seismic Envelope -X	Min	-0,02	-1,70	-6,48
312	Seismic Envelope +X	Max	3,89	1,67	6,33
	Seismic Envelope -X	Min	-3,89	-1,67	-6,33

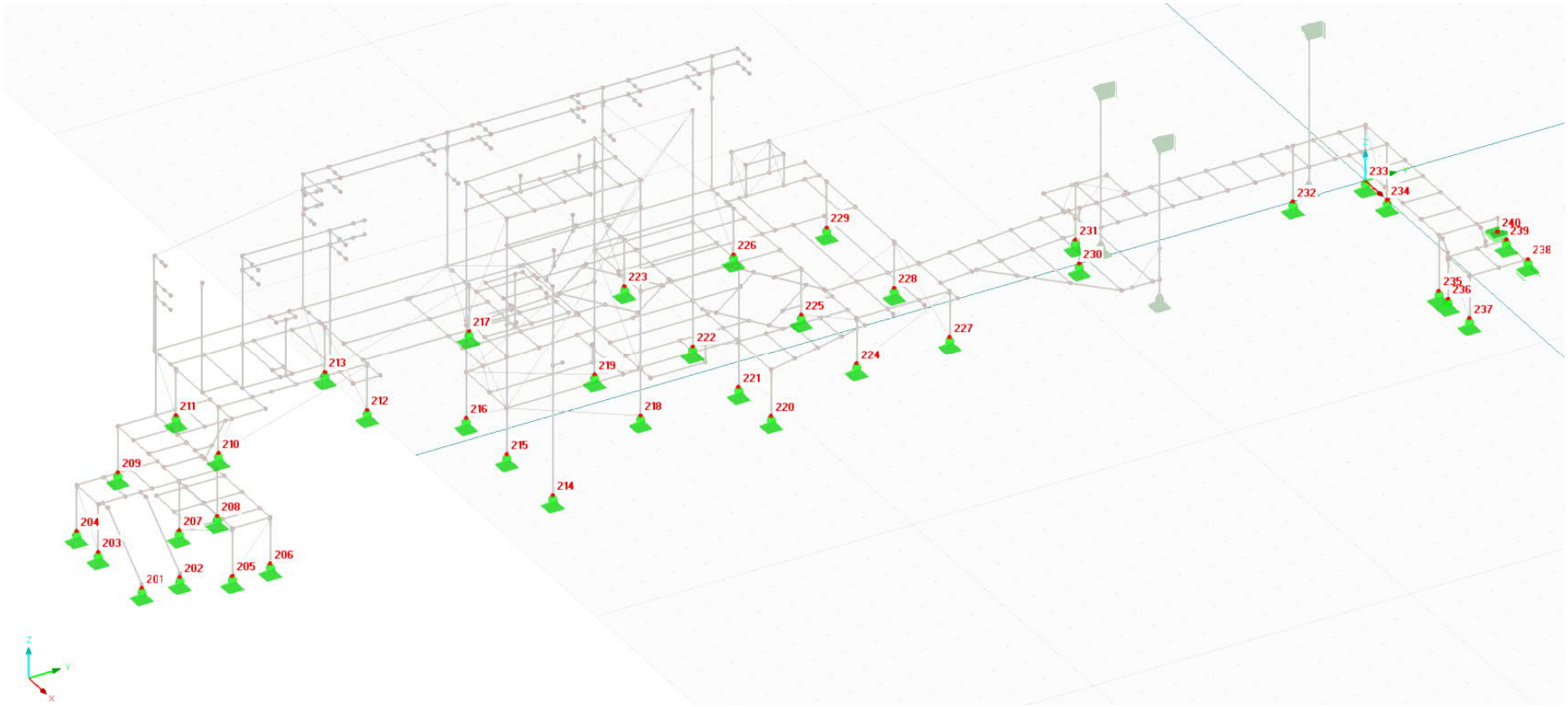
Node No.	LC/CO		Support Forces [kN]		
			$P_{X'}$	$P_{Y'}$	$P_{Z'}$
300	Seismic Envelope +Y	Max	0,00	0,00	0,00
	Seismic Envelope -Y	Min	0,00	0,00	0,00
301	Seismic Envelope +Y	Max	0,00	0,00	0,00
	Seismic Envelope -Y	Min	0,00	0,00	0,00
304	Seismic Envelope +Y	Max	0,02	0,05	8,31
	Seismic Envelope -Y	Min	-0,02	-0,05	-8,31
305	Seismic Envelope +Y	Max	1,76	0,03	7,58
	Seismic Envelope -Y	Min	-1,76	-0,03	-7,58
306	Seismic Envelope +Y	Max	0,18	1,26	13,56
	Seismic Envelope -Y	Min	-0,18	-1,26	-13,56
307	Seismic Envelope +Y	Max	0,02	0,51	4,73
	Seismic Envelope -Y	Min	-0,02	-0,51	-4,73
308	Seismic Envelope +Y	Max	0,80	0,11	4,25
	Seismic Envelope -Y	Min	-0,80	-0,11	-4,25
309	Seismic Envelope +Y	Max	0,03	0,00	13,89
	Seismic Envelope -Y	Min	-0,03	0,00	-13,89
310	Seismic Envelope +Y	Max	0,03	1,65	3,03
	Seismic Envelope -Y	Min	-0,03	-1,65	-3,03
311	Seismic Envelope +Y	Max	0,05	2,07	4,14
	Seismic Envelope -Y	Min	-0,05	-2,07	-4,14
312	Seismic Envelope +Y	Max	1,49	4,62	13,94
	Seismic Envelope -Y	Min	-1,49	-4,62	-13,94

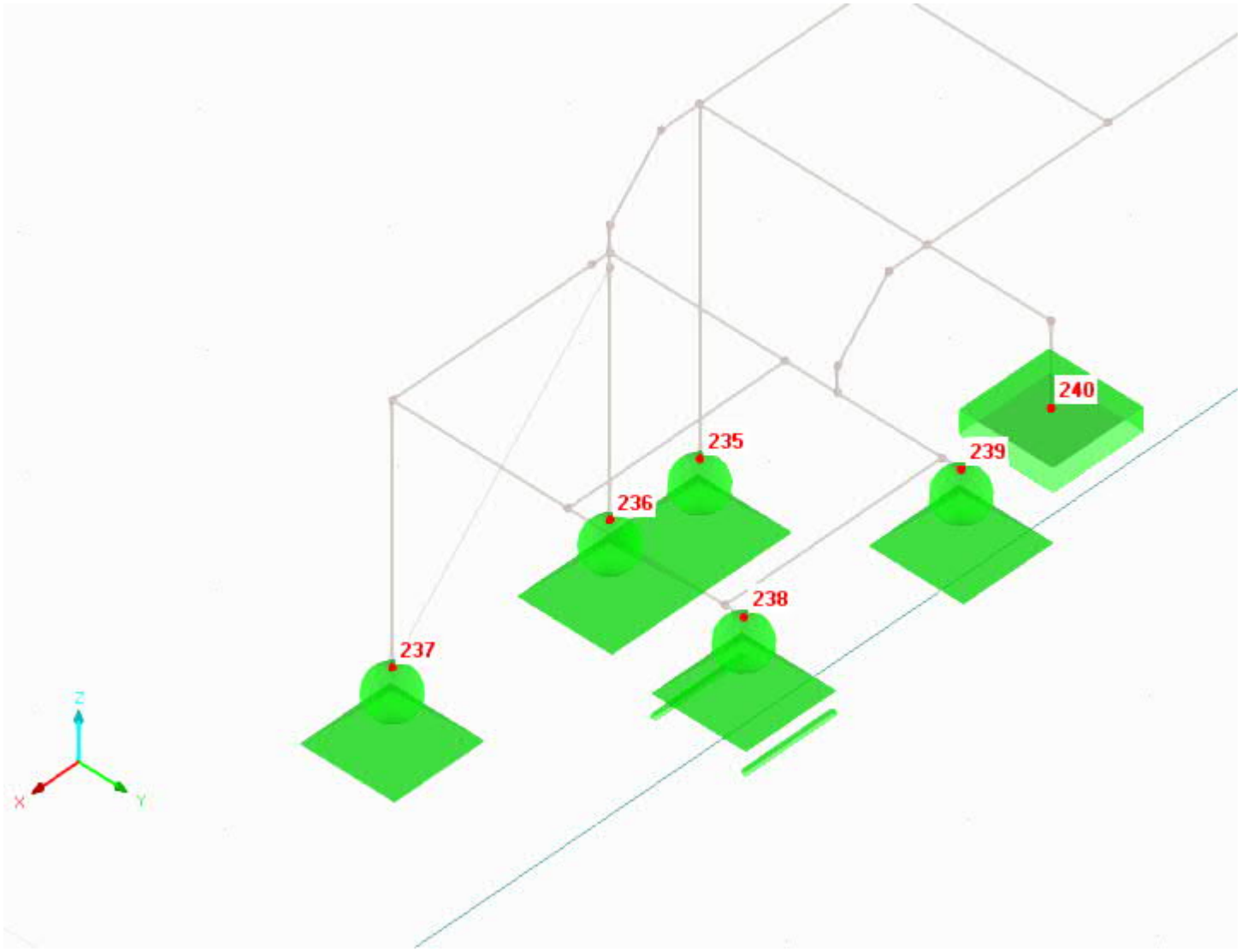
Node No.	LC/CO		Support Forces [kN]		
			$P_{X'}$	$P_{Y'}$	$P_{Z'}$
300	Seismic Envelope +Z	Max	0,00	0,00	0,00
	Seismic Envelope -Z	Min	0,00	0,00	0,00
301	Seismic Envelope +Z	Max	0,00	0,00	0,00
	Seismic Envelope -Z	Min	0,00	0,00	0,00
304	Seismic Envelope +Z	Max	0,02	0,02	2,34
	Seismic Envelope -Z	Min	-0,02	-0,02	-2,34
305	Seismic Envelope +Z	Max	0,53	0,02	2,27
	Seismic Envelope -Z	Min	-0,53	-0,02	-2,27
306	Seismic Envelope +Z	Max	0,05	0,62	2,13
	Seismic Envelope -Z	Min	-0,05	-0,62	-2,13
307	Seismic Envelope +Z	Max	0,02	0,18	1,31
	Seismic Envelope -Z	Min	-0,02	-0,18	-1,31
308	Seismic Envelope +Z	Max	0,27	0,06	1,29
	Seismic Envelope -Z	Min	-0,27	-0,06	-1,29
309	Seismic Envelope +Z	Max	0,02	0,00	2,07
	Seismic Envelope -Z	Min	-0,02	0,00	-2,07
310	Seismic Envelope +Z	Max	0,05	0,59	1,19
	Seismic Envelope -Z	Min	-0,05	-0,59	-1,19
311	Seismic Envelope +Z	Max	0,03	0,47	1,50
	Seismic Envelope -Z	Min	-0,03	-0,47	-1,50
312	Seismic Envelope +Z	Max	0,57	0,63	1,61
	Seismic Envelope -Z	Min	-0,57	-0,63	-1,61

PRELIMINARY SUPPOR FORCES FOR WALKWAY WEST




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Node No.	LC/CO	Support Forces [kN]			Support Moments [kNm]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$	$M_{X'}$	$M_{Y'}$	$M_{Z'}$
201	LC1 - Self- Weight + Secondary structures	0,00	0,00	-1,20	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-3,30	0,00	0,00	0,01
	LC3 - Wind in +X	0,00	0,00	0,00	0,00	0,00	0,00
	LC4 - Wind in -X	0,00	0,00	0,00	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,19	0,00	0,00	0,00	0,03
	LC6 - Wind in -Y	0,00	-0,19	0,00	0,00	0,00	-0,03
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
202	LC1 - Self- Weight + Secondary structures	0,00	0,00	-1,20	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-3,30	0,00	0,00	0,00
	LC3 - Wind in +X	0,00	0,00	0,00	0,00	0,00	0,00
	LC4 - Wind in -X	0,00	0,00	0,00	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,19	0,00	0,00	0,00	0,03
	LC6 - Wind in -Y	0,00	-0,19	0,00	0,00	0,00	-0,03
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
203	LC1 - Self- Weight + Secondary structures	0,00	0,00	-3,53	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-8,28	0,00	0,00	0,00
	LC3 - Wind in +X	1,13	0,00	-1,16	0,00	0,00	0,00
	LC4 - Wind in -X	-1,13	0,00	1,16	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,31	0,03	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,31	-0,03	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
204	LC1 - Self- Weight + Secondary structures	0,00	0,00	-1,94	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-3,83	0,00	0,00	0,00
	LC3 - Wind in +X	0,14	0,00	1,16	0,00	0,00	0,00
	LC4 - Wind in -X	-0,14	0,00	-1,16	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,17	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,17	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
205	LC1 - Self- Weight + Secondary structures	0,06	0,58	-2,99	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,10	1,04	-7,66	0,00	0,00	0,00
	LC3 - Wind in +X	-0,10	-2,38	-9,25	0,00	0,00	0,00
	LC4 - Wind in -X	0,11	2,45	9,46	0,00	0,00	0,00
	LC5 - Wind in +Y	0,02	3,91	4,84	0,00	0,00	0,00
	LC6 - Wind in -Y	-0,02	-4,31	-5,28	0,00	0,00	0,00
	LC7 - Temperature +25	-0,15	-1,53	-4,16	0,00	0,00	0,00
	LC8 - Temperature -25	0,15	1,53	4,16	0,00	0,00	0,00
206	LC1 - Self- Weight + Secondary structures	0,00	-0,02	-2,28	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	-0,03	-3,99	0,00	0,00	0,00
	LC3 - Wind in +X	0,18	0,08	2,91	0,00	0,00	0,00

	LC4 - Wind in -X	-0,18	-0,08	-2,99	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,04	-4,55	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,03	5,05	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,05	1,87	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	-0,05	-1,87	0,00	0,00	0,00
207	LC1 - Self- Weight + Secondary structures	-1,71	-0,01	-9,50	0,00	0,00	0,00
	LC2 - Live + Technical Load	-3,10	-0,01	-25,14	0,00	0,00	0,00
	LC3 - Wind in +X	12,45	0,02	5,49	0,00	0,00	0,00
	LC4 - Wind in -X	-12,71	-0,02	-5,54	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,57	0,26	5,59	0,00	0,00	0,00
	LC6 - Wind in -Y	0,46	-0,25	-6,01	0,00	0,00	0,00
	LC7 - Temperature +25	4,44	0,04	0,52	0,00	0,00	0,00
	LC8 - Temperature -25	-4,44	-0,04	-0,52	0,00	0,00	0,00
208	LC1 - Self- Weight + Secondary structures	0,00	0,48	-5,28	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,86	-9,49	0,00	0,00	0,00
	LC3 - Wind in +X	0,30	-0,72	-0,70	0,00	0,00	0,00
	LC4 - Wind in -X	-0,30	0,79	0,62	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	5,16	-5,91	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-5,47	6,28	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	-1,50	1,78	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	1,50	-1,78	0,00	0,00	0,00
209	LC1 - Self- Weight + Secondary structures	-0,01	0,00	-8,99	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,01	0,00	-11,91	0,00	0,00	0,00
	LC3 - Wind in +X	0,16	0,00	0,91	0,00	0,00	0,00
	LC4 - Wind in -X	-0,16	0,00	-0,91	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,17	1,12	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,17	-1,12	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
210	LC1 - Self- Weight + Secondary structures	-0,39	-1,11	-21,31	0,00	0,00	0,00
	LC2 - Live + Technical Load	-1,38	-2,01	-19,50	0,00	0,00	0,00
	LC3 - Wind in +X	3,01	3,63	-6,20	0,00	0,00	0,00
	LC4 - Wind in -X	-3,05	-3,75	6,21	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,22	8,55	3,63	0,00	0,00	0,00
	LC6 - Wind in -Y	0,22	-8,39	-3,57	0,00	0,00	0,00
	LC7 - Temperature +25	-1,07	-2,39	1,00	0,00	0,00	0,00
	LC8 - Temperature -25	1,07	2,39	-1,00	0,00	0,00	0,00
211	LC1 - Self- Weight + Secondary structures	2,08	0,00	-12,34	0,00	0,00	0,00
	LC2 - Live + Technical Load	4,52	0,00	-23,54	0,00	0,00	0,00
	LC3 - Wind in +X	-0,03	0,00	7,94	0,00	0,00	0,00
	LC4 - Wind in -X	0,00	0,00	-7,99	0,00	0,00	0,00
	LC5 - Wind in +Y	0,12	0,30	0,29	0,00	0,00	0,00
	LC6 - Wind in -Y	-0,12	-0,30	-0,29	0,00	0,00	0,00
	LC7 - Temperature +25	-1,70	0,00	-1,72	0,00	0,00	0,00
	LC8 - Temperature -25	1,70	0,00	1,72	0,00	0,00	0,00

212	LC1 - Self- Weight + Secondary structures	-9,64	0,04	-14,90	0,00	0,00	0,00
	LC2 - Live + Technical Load	-11,12	0,07	-8,54	0,00	0,00	0,00
	LC3 - Wind in +X	17,91	-0,12	-15,19	0,00	0,00	0,00
	LC4 - Wind in -X	-16,85	0,13	14,57	0,00	0,00	0,00
	LC5 - Wind in +Y	1,04	-0,11	-5,12	0,00	0,00	0,00
	LC6 - Wind in -Y	-0,95	0,11	5,01	0,00	0,00	0,00
	LC7 - Temperature +25	-0,78	0,08	1,79	0,00	0,00	0,00
	LC8 - Temperature -25	0,78	-0,08	-1,79	0,00	0,00	0,00
213	LC1 - Self- Weight + Secondary structures	9,72	0,05	-37,38	0,00	0,00	0,00
	LC2 - Live + Technical Load	11,11	0,32	-57,01	0,00	0,00	0,00
	LC3 - Wind in +X	-7,98	-0,27	14,03	0,00	0,00	0,00
	LC4 - Wind in -X	7,97	0,24	-13,37	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,30	12,38	3,86	0,00	0,00	0,00
	LC6 - Wind in -Y	0,30	-12,90	-3,96	0,00	0,00	0,00
	LC7 - Temperature +25	-0,87	2,01	-0,45	0,00	0,00	0,00
	LC8 - Temperature -25	0,87	-2,01	0,45	0,00	0,00	0,00
214	LC1 - Self- Weight + Secondary structures	0,00	-0,06	-6,81	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	-0,04	-5,26	0,00	0,00	0,00
	LC3 - Wind in +X	0,93	-0,02	-0,26	0,00	0,00	0,00
	LC4 - Wind in -X	-0,93	0,02	0,26	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,91	0,11	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,91	-0,11	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	-0,03	-0,01	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,03	0,01	0,00	0,00	0,00
215	LC1 - Self- Weight + Secondary structures	-0,02	-0,03	-12,82	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,01	-0,15	-21,36	0,00	0,00	0,00
	LC3 - Wind in +X	0,41	-0,23	-40,02	0,00	0,00	0,00
	LC4 - Wind in -X	-0,42	0,17	41,42	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	-0,52	14,57	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	0,54	-14,36	0,00	0,00	0,00
	LC7 - Temperature +25	0,85	-2,26	-0,94	0,00	0,00	0,00
	LC8 - Temperature -25	-0,85	2,26	0,94	0,00	0,00	0,00
216	LC1 - Self- Weight + Secondary structures	-5,02	-0,63	-23,98	0,00	0,00	0,00
	LC2 - Live + Technical Load	-4,63	-1,37	-37,86	0,00	0,00	0,00
	LC3 - Wind in +X	21,26	0,23	38,01	0,00	0,00	0,00
	LC4 - Wind in -X	-22,30	-0,24	-39,72	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,76	0,73	18,09	0,00	0,00	0,00
	LC6 - Wind in -Y	0,93	-0,71	-17,86	0,00	0,00	0,00
	LC7 - Temperature +25	-0,09	-3,80	0,37	0,00	0,00	0,00
	LC8 - Temperature -25	0,09	3,80	-0,37	0,00	0,00	0,00
217	LC1 - Self- Weight + Secondary structures	10,06	0,00	-33,21	0,00	0,00	0,00
	LC2 - Live + Technical Load	9,17	-0,01	-60,51	0,00	0,00	0,00
	LC3 - Wind in +X	-9,05	0,01	0,17	0,00	0,00	0,00
	LC4 - Wind in -X	9,07	-0,01	-0,19	0,00	0,00	0,00
	LC5 - Wind in +Y	0,24	-0,11	-3,76	0,00	0,00	0,00

	LC6 - Wind in -Y	-0,26	0,13	3,92	0,00	0,00	0,00
	LC7 - Temperature +25	-1,00	-0,07	-0,96	0,00	0,00	0,00
	LC8 - Temperature -25	1,00	0,07	0,96	0,00	0,00	0,00
218	LC1 - Self- Weight + Secondary structures	-0,16	1,46	-22,49	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,29	1,55	-44,63	0,00	0,00	0,00
	LC3 - Wind in +X	0,58	-3,89	-34,26	0,00	0,00	0,00
	LC4 - Wind in -X	-0,59	3,48	34,14	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,03	17,97	-16,74	0,00	0,00	0,00
	LC6 - Wind in -Y	0,03	-16,99	16,38	0,00	0,00	0,00
	LC7 - Temperature +25	0,77	-1,16	0,63	0,00	0,00	0,00
	LC8 - Temperature -25	-0,77	1,16	-0,63	0,00	0,00	0,00
219	LC1 - Self- Weight + Secondary structures	-5,25	-0,92	-32,74	0,00	0,00	0,00
	LC2 - Live + Technical Load	-4,39	-0,78	-62,76	0,00	0,00	0,00
	LC3 - Wind in +X	24,03	3,47	39,19	0,00	0,00	0,00
	LC4 - Wind in -X	-24,29	-3,51	-38,57	0,00	0,00	0,00
	LC5 - Wind in +Y	0,44	17,56	-9,50	0,00	0,00	0,00
	LC6 - Wind in -Y	-0,40	-17,04	9,28	0,00	0,00	0,00
	LC7 - Temperature +25	-1,33	1,37	0,47	0,00	0,00	0,00
	LC8 - Temperature -25	1,33	-1,37	-0,47	0,00	0,00	0,00
220	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,67	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,01	0,00	-6,49	0,00	0,00	0,00
	LC3 - Wind in +X	0,17	0,00	0,00	0,00	0,00	0,00
	LC4 - Wind in -X	-0,17	0,00	0,00	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,17	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,17	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
221	LC1 - Self- Weight + Secondary structures	0,17	-0,01	-9,83	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,11	-0,05	-31,61	0,00	0,00	0,00
	LC3 - Wind in +X	-1,46	-0,22	-11,75	0,00	0,00	0,00
	LC4 - Wind in -X	1,19	0,20	11,15	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,09	-0,09	-2,71	0,00	0,00	0,00
	LC6 - Wind in -Y	0,09	0,13	2,75	0,00	0,00	0,00
	LC7 - Temperature +25	0,61	1,92	0,31	0,00	0,00	0,00
	LC8 - Temperature -25	-0,61	-1,92	-0,31	0,00	0,00	0,00
222	LC1 - Self- Weight + Secondary structures	-7,05	0,21	-19,57	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,34	0,57	-40,90	0,00	0,00	0,00
	LC3 - Wind in +X	19,87	0,09	9,03	0,00	0,00	0,00
	LC4 - Wind in -X	-19,14	-0,10	-8,62	0,00	0,00	0,00
	LC5 - Wind in +Y	4,88	0,49	-3,92	0,00	0,00	0,00
	LC6 - Wind in -Y	-4,96	-0,47	4,02	0,00	0,00	0,00
	LC7 - Temperature +25	-0,77	2,23	-0,08	0,00	0,00	0,00
	LC8 - Temperature -25	0,77	-2,23	0,08	0,00	0,00	0,00
223	LC1 - Self- Weight + Secondary structures	7,19	0,00	-30,60	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,28	0,00	-63,23	0,00	0,00	0,00

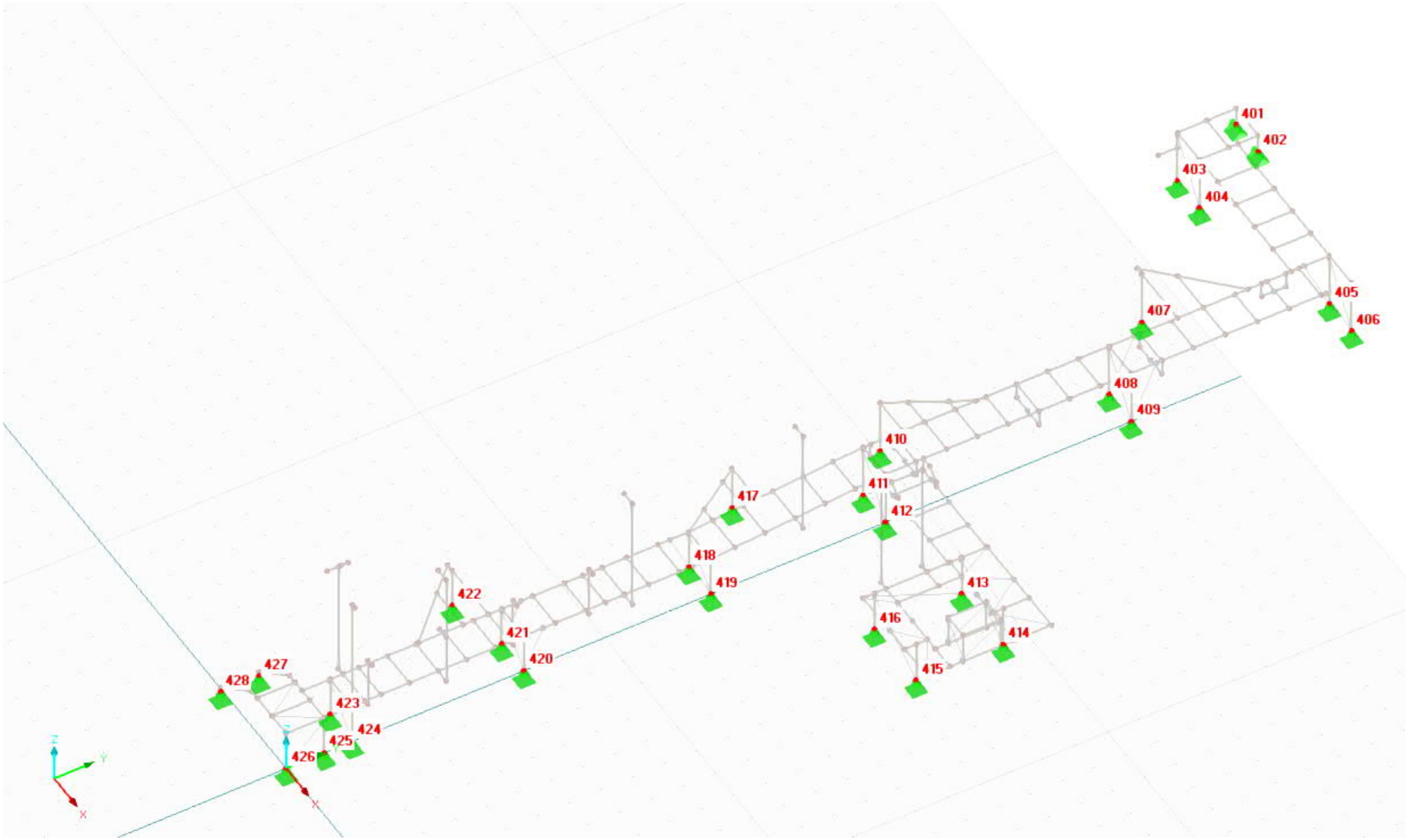
	LC3 - Wind in +X	-9,20	0,00	-0,07	0,00	0,00	0,00
	LC4 - Wind in -X	9,24	0,00	0,09	0,00	0,00	0,00
	LC5 - Wind in +Y	-4,86	0,30	6,36	0,00	0,00	0,00
	LC6 - Wind in -Y	4,86	-0,30	-6,36	0,00	0,00	0,00
	LC7 - Temperature +25	-1,05	0,00	-0,39	0,00	0,00	0,00
	LC8 - Temperature -25	1,05	0,00	0,39	0,00	0,00	0,00
224	LC1 - Self- Weight + Secondary structures	0,22	-0,18	-7,54	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,11	-0,49	-34,10	0,00	0,00	0,00
	LC3 - Wind in +X	-0,07	0,34	-6,16	0,00	0,00	0,00
	LC4 - Wind in -X	0,06	0,04	6,25	0,00	0,00	0,00
	LC5 - Wind in +Y	0,17	3,28	3,84	0,00	0,00	-0,01
	LC6 - Wind in -Y	-0,17	-2,78	-3,51	0,00	0,00	0,01
	LC7 - Temperature +25	-0,10	-3,23	-2,95	0,00	0,00	0,00
	LC8 - Temperature -25	0,10	3,23	2,95	0,00	0,00	0,00
225	LC1 - Self- Weight + Secondary structures	-6,47	-0,02	-12,93	0,00	0,00	0,00
	LC2 - Live + Technical Load	3,33	-0,01	-32,74	0,00	0,00	0,00
	LC3 - Wind in +X	14,07	0,07	6,40	0,00	0,00	0,00
	LC4 - Wind in -X	-13,87	0,00	-6,27	0,00	0,00	0,00
	LC5 - Wind in +Y	-5,06	1,86	-1,55	0,00	0,00	0,00
	LC6 - Wind in -Y	4,88	-2,68	1,07	0,00	0,00	0,00
	LC7 - Temperature +25	2,99	1,06	2,27	0,00	0,00	0,00
	LC8 - Temperature -25	-2,99	-1,06	-2,27	0,00	0,00	0,00
226	LC1 - Self- Weight + Secondary structures	6,21	0,00	-26,37	0,00	0,00	0,00
	LC2 - Live + Technical Load	-3,31	0,00	-50,55	0,00	0,00	0,00
	LC3 - Wind in +X	-6,98	0,00	-0,02	0,00	0,00	0,00
	LC4 - Wind in -X	6,96	0,00	0,02	0,00	0,00	0,00
	LC5 - Wind in +Y	4,82	0,30	-6,27	0,00	0,00	0,00
	LC6 - Wind in -Y	-4,76	-0,30	6,15	0,00	0,00	0,00
	LC7 - Temperature +25	-0,94	0,00	-0,39	0,00	0,00	0,00
	LC8 - Temperature -25	0,94	0,00	0,39	0,00	0,00	0,00
227	LC1 - Self- Weight + Secondary structures	-0,12	0,01	-10,45	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,32	0,02	-30,57	0,00	0,00	0,00
	LC3 - Wind in +X	3,47	-0,01	-1,68	0,00	0,00	0,00
	LC4 - Wind in -X	-3,99	0,00	1,72	0,00	0,00	0,00
	LC5 - Wind in +Y	-1,15	0,06	-0,94	0,00	0,00	-0,01
	LC6 - Wind in -Y	1,18	-0,08	0,70	0,00	0,00	0,01
	LC7 - Temperature +25	-4,71	0,11	3,30	0,00	0,00	0,00
	LC8 - Temperature -25	4,71	-0,11	-3,30	0,00	0,00	0,00
228	LC1 - Self- Weight + Secondary structures	0,00	0,00	-7,94	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,01	0,00	-24,55	0,00	0,00	0,00
	LC3 - Wind in +X	0,06	0,00	1,40	0,00	0,00	0,00
	LC4 - Wind in -X	-0,05	0,00	-1,67	0,00	0,00	0,00
	LC5 - Wind in +Y	0,04	0,12	-1,08	0,00	0,00	0,00
	LC6 - Wind in -Y	-0,04	-0,09	1,47	0,00	0,00	0,00
	LC7 - Temperature +25	0,16	-0,03	-2,76	0,00	0,00	0,00

	LC8 - Temperature -25	-0,16	0,03	2,76	0,00	0,00	0,00
229	LC1 - Self- Weight + Secondary structures	0,00	0,00	-8,00	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-19,91	0,00	0,00	0,00
	LC3 - Wind in +X	0,17	0,00	0,09	0,00	0,00	0,00
	LC4 - Wind in -X	-0,17	0,00	-0,10	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,17	-0,12	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,17	0,26	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
230	LC1 - Self- Weight + Secondary structures	0,00	0,00	-8,11	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-22,74	0,00	0,00	0,00
	LC3 - Wind in +X	0,20	0,00	0,00	0,00	0,00	0,00
	LC4 - Wind in -X	-0,20	0,00	0,00	0,00	0,00	0,00
	LC5 - Wind in +Y	-0,01	0,20	-0,27	0,00	0,00	0,00
	LC6 - Wind in -Y	0,01	-0,20	0,27	0,00	0,00	0,00
	LC7 - Temperature +25	-0,03	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,03	0,00	0,00	0,00	0,00	0,00
231	LC1 - Self- Weight + Secondary structures	0,00	0,00	-3,08	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-5,80	0,00	0,00	0,00
	LC3 - Wind in +X	0,20	0,00	0,00	0,00	0,00	-0,01
	LC4 - Wind in -X	-0,20	0,00	0,00	0,00	0,00	0,01
	LC5 - Wind in +Y	0,00	0,20	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,20	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
232	LC1 - Self- Weight + Secondary structures	0,00	0,00	-5,51	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-14,07	0,00	0,00	0,00
	LC3 - Wind in +X	0,20	0,00	0,00	0,00	0,00	0,00
	LC4 - Wind in -X	-0,20	0,00	0,00	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,20	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,20	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,00
233	LC1 - Self- Weight + Secondary structures	0,00	0,00	-1,93	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-3,51	0,00	0,00	0,00
	LC3 - Wind in +X	0,16	0,00	1,43	0,00	0,00	0,00
	LC4 - Wind in -X	-0,13	0,00	-2,53	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,21	-0,01	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,21	0,01	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,01	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	-0,01	0,00	0,00	0,00	0,00
234	LC1 - Self- Weight + Secondary structures	0,00	0,00	-5,59	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-14,99	0,00	0,00	0,00
	LC3 - Wind in +X	1,22	0,00	-1,43	0,00	0,00	0,00
	LC4 - Wind in -X	-2,00	0,00	2,53	0,00	0,00	0,00

	LC5 - Wind in +Y	-0,01	0,35	0,01	0,00	0,00	0,00
	LC6 - Wind in -Y	0,01	-0,35	-0,01	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	-0,01	0,00	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,01	0,00	0,00	0,00	0,00
235	LC1 - Self- Weight + Secondary structures	0,00	0,00	-4,31	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-10,87	0,00	0,00	0,00
	LC3 - Wind in +X	0,20	0,00	0,03	0,00	0,00	0,00
	LC4 - Wind in -X	-0,20	0,00	-0,03	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,20	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,20	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	-1,02	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	1,02	0,00	0,00	0,00
236	LC1 - Self- Weight + Secondary structures	0,02	0,00	-2,04	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,05	0,00	-4,07	0,00	0,00	0,00
	LC3 - Wind in +X	0,13	0,00	0,62	0,00	0,00	0,00
	LC4 - Wind in -X	-0,13	0,00	-0,62	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,14	0,01	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,14	-0,01	0,00	0,00	0,00
	LC7 - Temperature +25	-0,43	0,00	1,91	0,00	0,00	0,00
	LC8 - Temperature -25	0,43	0,00	-1,91	0,00	0,00	0,00
237	LC1 - Self- Weight + Secondary structures	-0,01	0,00	-1,53	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,04	0,00	-2,74	0,00	0,00	0,00
	LC3 - Wind in +X	0,77	0,00	-0,64	0,00	0,00	0,00
	LC4 - Wind in -X	-0,77	0,00	0,65	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	0,15	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,15	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,86	0,00	-0,91	0,00	0,00	0,00
	LC8 - Temperature -25	-0,86	0,00	0,91	0,00	0,00	0,00
238	LC1 - Self- Weight + Secondary structures	0,00	0,00	-0,92	0,00	0,00	0,00
	LC2 - Live + Technical Load	0,00	0,00	-2,77	0,00	0,00	-0,01
	LC3 - Wind in +X	0,00	0,00	0,00	0,00	0,00	0,07
	LC4 - Wind in -X	0,00	0,00	0,00	0,00	0,00	-0,09
	LC5 - Wind in +Y	0,00	0,42	0,00	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,42	0,00	0,00	0,00	0,00
	LC7 - Temperature +25	0,00	0,00	0,00	0,00	0,00	-0,37
	LC8 - Temperature -25	0,00	0,00	0,00	0,00	0,00	0,37
239	LC1 - Self- Weight + Secondary structures	0,00	0,00	-1,14	0,00	0,00	0,00
	LC2 - Live + Technical Load	-0,01	0,01	-3,30	0,00	0,00	-0,01
	LC3 - Wind in +X	0,49	0,00	0,00	0,00	0,00	0,14
	LC4 - Wind in -X	-0,53	0,00	0,00	0,00	0,00	-0,12
	LC5 - Wind in +Y	0,00	0,51	-0,01	0,00	0,00	0,00
	LC6 - Wind in -Y	0,00	-0,51	0,01	0,00	0,00	0,00
	LC7 - Temperature +25	-0,43	-0,03	0,01	0,00	0,00	-0,42
	LC8 - Temperature -25	0,43	0,03	-0,01	0,00	0,00	0,42
240	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,11	0,00	0,00	0,00

	LC2 - Live + Technical Load	0,00	-0,01	-5,28	0,00	0,00	0,00
	LC3 - Wind in +X	0,00	0,00	0,00	0,00	0,00	0,00
	LC4 - Wind in -X	0,00	0,00	0,00	0,00	0,00	0,00
	LC5 - Wind in +Y	0,00	1,39	0,00	-0,61	0,00	0,00
	LC6 - Wind in -Y	0,00	-1,39	0,00	0,61	0,00	0,00
	LC7 - Temperature +25	0,00	0,03	0,00	-0,01	0,00	0,00
	LC8 - Temperature -25	0,00	-0,03	0,00	0,01	0,00	0,00

WALKWAY EAST



Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
401	LC1 - Self- Weight + Secondary structures	0,00	-1,10	-0,86
	LC2 - Live + Technical Load	0,00	-0,72	-2,20
	LC3 - Wind in +X	0,50	0,00	0,00
	LC4 - Wind in -X	-0,15	0,00	0,00
	LC5 - Wind in +Y	0,00	0,68	-0,21
	LC6 - Wind in -Y	0,00	-0,51	0,14
	LC7 - Temperature +25	-0,41	0,00	0,00
	LC8 - Temperature -25	0,41	0,00	0,00
402	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,93
	LC2 - Live + Technical Load	0,00	0,00	-6,70
	LC3 - Wind in +X	0,31	0,00	0,00
	LC4 - Wind in -X	-0,14	0,00	0,00
	LC5 - Wind in +Y	0,00	2,07	-0,67
	LC6 - Wind in -Y	0,00	-1,88	0,61
	LC7 - Temperature +25	0,41	0,00	0,00
	LC8 - Temperature -25	-0,41	0,00	0,00
403	LC1 - Self- Weight + Secondary structures	0,00	1,10	-5,80
	LC2 - Live + Technical Load	0,01	0,72	-5,09
	LC3 - Wind in +X	-0,01	0,00	8,58
	LC4 - Wind in -X	-0,01	0,00	-7,47
	LC5 - Wind in +Y	-0,02	0,22	0,85
	LC6 - Wind in -Y	0,02	-0,22	-0,84
	LC7 - Temperature +25	0,10	0,00	-3,58
	LC8 - Temperature -25	-0,10	0,00	3,58
404	LC1 - Self- Weight + Secondary structures	-0,09	0,00	-5,24
	LC2 - Live + Technical Load	-0,14	0,00	-11,09
	LC3 - Wind in +X	6,03	0,00	-8,58
	LC4 - Wind in -X	-5,28	0,00	7,47
	LC5 - Wind in +Y	0,44	0,22	0,02
	LC6 - Wind in -Y	-0,47	-0,22	0,09
	LC7 - Temperature +25	-2,42	0,00	3,58
	LC8 - Temperature -25	2,42	0,00	-3,58
405	LC1 - Self- Weight + Secondary structures	0,00	6,58	-13,67
	LC2 - Live + Technical Load	0,00	12,96	-26,20
	LC3 - Wind in +X	0,36	0,26	-2,37
	LC4 - Wind in -X	-0,34	-0,22	1,50
	LC5 - Wind in +Y	0,00	2,90	-1,43
	LC6 - Wind in -Y	0,00	-3,62	1,71
	LC7 - Temperature +25	-0,08	0,61	2,77
	LC8 - Temperature -25	0,08	-0,61	-2,77
406	LC1 - Self- Weight + Secondary structures	0,04	2,89	-4,97
	LC2 - Live + Technical Load	0,07	6,51	-9,38
	LC3 - Wind in +X	-1,29	-0,25	2,37

	LC4 - Wind in -X	0,72	0,22	-1,50
	LC5 - Wind in +Y	-0,06	1,58	-0,64
	LC6 - Wind in -Y	-0,01	-1,93	0,91
	LC7 - Temperature +25	1,95	1,18	-3,31
	LC8 - Temperature -25	-1,95	-1,18	3,31
407	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,68
	LC2 - Live + Technical Load	0,00	0,00	-4,00
	LC3 - Wind in +X	0,22	0,00	0,00
	LC4 - Wind in -X	-0,22	0,00	0,00
	LC5 - Wind in +Y	0,00	0,22	0,00
	LC6 - Wind in -Y	0,00	-0,22	0,00
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
408	LC1 - Self- Weight + Secondary structures	0,00	-6,02	-14,97
	LC2 - Live + Technical Load	0,00	-12,11	-26,97
	LC3 - Wind in +X	-0,03	0,15	9,21
	LC4 - Wind in -X	0,01	-0,11	-8,27
	LC5 - Wind in +Y	0,01	2,76	1,01
	LC6 - Wind in -Y	-0,01	-3,44	-1,35
	LC7 - Temperature +25	-0,02	-0,11	0,66
	LC8 - Temperature -25	0,02	0,11	-0,66
409	LC1 - Self- Weight + Secondary structures	-0,01	-3,04	-12,31
	LC2 - Live + Technical Load	-0,01	-6,51	-21,69
	LC3 - Wind in +X	6,69	-0,14	-9,20
	LC4 - Wind in -X	-6,08	0,11	8,27
	LC5 - Wind in +Y	-0,25	1,67	1,18
	LC6 - Wind in -Y	0,28	-2,02	-1,41
	LC7 - Temperature +25	0,34	0,34	0,00
	LC8 - Temperature -25	-0,34	-0,34	0,00
410	LC1 - Self- Weight + Secondary structures	0,00	-0,01	-1,84
	LC2 - Live + Technical Load	0,00	-0,02	-2,11
	LC3 - Wind in +X	0,23	0,00	0,00
	LC4 - Wind in -X	-0,23	0,00	0,00
	LC5 - Wind in +Y	0,00	0,23	0,00
	LC6 - Wind in -Y	0,00	-0,23	0,00
	LC7 - Temperature +25	0,00	-0,01	0,00
	LC8 - Temperature -25	0,00	0,01	0,00
411	LC1 - Self- Weight + Secondary structures	0,00	0,00	-12,25
	LC2 - Live + Technical Load	0,00	0,00	-22,42
	LC3 - Wind in +X	-0,07	0,00	12,60
	LC4 - Wind in -X	0,03	0,00	-11,23
	LC5 - Wind in +Y	0,01	0,22	-0,29
	LC6 - Wind in -Y	-0,01	-0,22	0,56
	LC7 - Temperature +25	0,02	0,00	-0,63
	LC8 - Temperature -25	-0,02	0,00	0,63

412	LC1 - Self- Weight + Secondary structures	0,08	-0,10	-18,19
	LC2 - Live + Technical Load	0,08	-0,21	-31,58
	LC3 - Wind in +X	7,39	-0,01	-12,61
	LC4 - Wind in -X	-6,52	0,00	11,23
	LC5 - Wind in +Y	-0,20	0,39	-0,58
	LC6 - Wind in -Y	0,39	-0,34	0,32
	LC7 - Temperature +25	-0,57	-0,50	0,65
	LC8 - Temperature -25	0,57	0,50	-0,65
413	LC1 - Self- Weight + Secondary structures	0,03	1,06	-9,34
	LC2 - Live + Technical Load	0,08	1,53	-16,85
	LC3 - Wind in +X	-0,20	0,17	4,16
	LC4 - Wind in -X	0,18	-0,14	-4,00
	LC5 - Wind in +Y	0,14	8,59	-8,73
	LC6 - Wind in -Y	-0,14	-8,59	8,73
	LC7 - Temperature +25	-0,06	1,19	0,19
	LC8 - Temperature -25	0,06	-1,19	-0,19
414	LC1 - Self- Weight + Secondary structures	-0,40	-0,18	-11,33
	LC2 - Live + Technical Load	-1,01	-0,56	-21,70
	LC3 - Wind in +X	5,08	0,14	-4,16
	LC4 - Wind in -X	-4,80	-0,15	4,00
	LC5 - Wind in +Y	-1,93	0,74	0,29
	LC6 - Wind in -Y	1,91	-0,74	-0,29
	LC7 - Temperature +25	0,67	-0,03	-0,19
	LC8 - Temperature -25	-0,67	0,03	0,19
415	LC1 - Self- Weight + Secondary structures	0,39	-0,26	-9,11
	LC2 - Live + Technical Load	1,03	-0,62	-16,38
	LC3 - Wind in +X	5,57	-0,94	-6,80
	LC4 - Wind in -X	-5,62	0,92	6,82
	LC5 - Wind in +Y	2,10	0,45	-0,39
	LC6 - Wind in -Y	-2,10	-0,45	0,39
	LC7 - Temperature +25	0,12	-0,32	-0,24
	LC8 - Temperature -25	-0,12	0,32	0,24
416	LC1 - Self- Weight + Secondary structures	-0,02	-0,63	-6,02
	LC2 - Live + Technical Load	-0,08	-0,35	-2,64
	LC3 - Wind in +X	-0,24	0,62	6,80
	LC4 - Wind in -X	0,25	-0,62	-6,82
	LC5 - Wind in +Y	-0,15	-0,34	8,84
	LC6 - Wind in -Y	0,15	0,34	-8,84
	LC7 - Temperature +25	-0,03	-0,85	0,24
	LC8 - Temperature -25	0,03	0,85	-0,24
417	LC1 - Self- Weight + Secondary structures	0,00	0,00	-1,75
	LC2 - Live + Technical Load	0,00	-0,01	-2,13
	LC3 - Wind in +X	0,19	0,01	0,00
	LC4 - Wind in -X	-0,19	-0,01	0,00
	LC5 - Wind in +Y	0,00	0,19	-0,08

	LC6 - Wind in -Y	0,00	-0,19	0,06
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
418	LC1 - Self- Weight + Secondary structures	0,00	2,23	-8,10
	LC2 - Live + Technical Load	0,00	5,84	-18,37
	LC3 - Wind in +X	-0,19	-2,74	10,95
	LC4 - Wind in -X	0,18	2,74	-10,75
	LC5 - Wind in +Y	0,01	3,74	-1,77
	LC6 - Wind in -Y	-0,01	-3,45	1,57
	LC7 - Temperature +25	0,00	0,39	0,13
	LC8 - Temperature -25	0,00	-0,39	-0,13
419	LC1 - Self- Weight + Secondary structures	0,00	6,25	-13,26
	LC2 - Live + Technical Load	0,01	7,41	-17,80
	LC3 - Wind in +X	6,13	2,71	-10,93
	LC4 - Wind in -X	-5,94	-2,72	10,75
	LC5 - Wind in +Y	-0,28	4,87	-1,65
	LC6 - Wind in -Y	0,19	-4,65	1,66
	LC7 - Temperature +25	0,01	0,24	0,07
	LC8 - Temperature -25	-0,01	-0,24	-0,07
420	LC1 - Self- Weight + Secondary structures	0,02	-6,26	-13,48
	LC2 - Live + Technical Load	0,03	-7,69	-17,50
	LC3 - Wind in +X	7,65	-3,27	-13,54
	LC4 - Wind in -X	-7,27	3,26	13,15
	LC5 - Wind in +Y	0,25	4,85	1,53
	LC6 - Wind in -Y	-0,09	-4,63	-1,61
	LC7 - Temperature +25	0,06	-1,14	-0,26
	LC8 - Temperature -25	-0,06	1,14	0,26
421	LC1 - Self- Weight + Secondary structures	0,00	-2,53	-8,83
	LC2 - Live + Technical Load	0,00	-6,19	-19,71
	LC3 - Wind in +X	-0,26	3,27	13,53
	LC4 - Wind in -X	0,24	-3,27	-13,13
	LC5 - Wind in +Y	-0,01	3,74	1,55
	LC6 - Wind in -Y	0,00	-3,45	-1,25
	LC7 - Temperature +25	0,00	-1,00	-0,08
	LC8 - Temperature -25	0,00	1,00	0,08
422	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,51
	LC2 - Live + Technical Load	0,00	0,00	-4,00
	LC3 - Wind in +X	0,17	0,00	0,00
	LC4 - Wind in -X	-0,17	0,00	0,00
	LC5 - Wind in +Y	0,00	0,17	0,00
	LC6 - Wind in -Y	0,00	-0,17	0,00
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
423	LC1 - Self- Weight + Secondary structures	-0,02	0,00	-7,66
	LC2 - Live + Technical Load	0,00	0,00	-11,23

	LC3 - Wind in +X	-0,52	0,00	17,15
	LC4 - Wind in -X	0,51	0,00	-16,97
	LC5 - Wind in +Y	0,01	0,17	0,25
	LC6 - Wind in -Y	0,00	-0,17	-0,45
	LC7 - Temperature +25	0,00	0,00	-0,09
	LC8 - Temperature -25	0,00	0,00	0,09
424	LC1 - Self- Weight + Secondary structures	-0,01	0,00	-10,47
	LC2 - Live + Technical Load	-0,06	0,00	-10,82
	LC3 - Wind in +X	7,47	0,00	-17,15
	LC4 - Wind in -X	-7,30	0,00	16,97
	LC5 - Wind in +Y	-0,09	-2,18	0,91
	LC6 - Wind in -Y	-0,09	2,18	-0,73
	LC7 - Temperature +25	-0,08	0,00	0,09
	LC8 - Temperature -25	0,08	0,00	-0,09
425	LC1 - Self- Weight + Secondary structures	0,00	0,00	-3,18
	LC2 - Live + Technical Load	0,00	0,00	-6,88
	LC3 - Wind in +X	0,15	0,00	-0,40
	LC4 - Wind in -X	-0,15	0,00	0,40
	LC5 - Wind in +Y	0,00	-0,03	-3,53
	LC6 - Wind in -Y	0,00	0,03	3,53
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
426	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,62
	LC2 - Live + Technical Load	0,00	0,00	-4,96
	LC3 - Wind in +X	0,17	0,00	-0,32
	LC4 - Wind in -X	-0,17	0,00	0,32
	LC5 - Wind in +Y	0,00	2,70	3,53
	LC6 - Wind in -Y	0,00	-2,70	-3,53
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
427	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,21
	LC2 - Live + Technical Load	-0,02	0,00	-5,98
	LC3 - Wind in +X	1,23	0,00	0,40
	LC4 - Wind in -X	-1,23	0,00	-0,40
	LC5 - Wind in +Y	2,18	0,36	0,90
	LC6 - Wind in -Y	-2,18	-0,36	-0,90
	LC7 - Temperature +25	0,00	0,00	0,00
	LC8 - Temperature -25	0,00	0,00	0,00
428	LC1 - Self- Weight + Secondary structures	0,00	0,00	-2,09
	LC2 - Live + Technical Load	0,01	0,00	-5,58
	LC3 - Wind in +X	1,05	0,00	0,32
	LC4 - Wind in -X	-1,05	0,00	-0,32
	LC5 - Wind in +Y	-2,18	0,36	-0,90
	LC6 - Wind in -Y	2,18	-0,36	0,90
	LC7 - Temperature +25	0,00	0,00	0,00

	LC8 - Temperature -25	0,00	0,00	0,00

SYC
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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

PRIMARY CYANIDE DESTRUCTION

DESIGN CALCULATION OF STAIRBLOCK

S BUILT

MD-216-3670-EG-ST-CAL-2305

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	27-11-2023	AS BUILT	SGI	SYC	PRI	
C01	13-04-2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	SGI	SYC	PRI	
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
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Checked	SYC	SYC	SYC			Checked	SYC	SYC	SYC	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
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Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	MansourahMassarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

1. INTENT

The document is prepared for the design calculation of stair supporting structures in Primary Cyanide Destruction of Plant.
The design calculation ensures the safety and serviceability of structural members.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah&Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

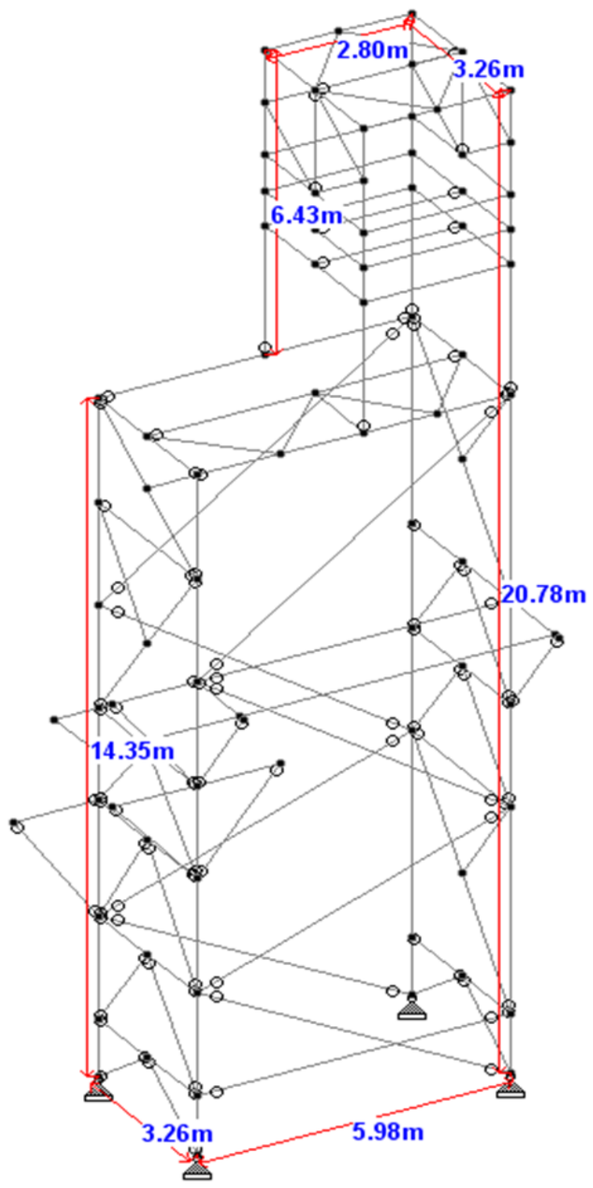
The design document is prepared for stair supporting structures in Primary Cyanide Destruction of Plant. Structures are modelled in Staad-Pro and analysis & design has been done for stair supporting structure in Staad

a) MD-216-1000-EG-ST-CRT-1001 _Civil, Structural & Architectural Design Criteria

4. DESIGN OF PLATFORM SUPPORTING STRUCTURE

4.1 Structural Geometry of Staircase

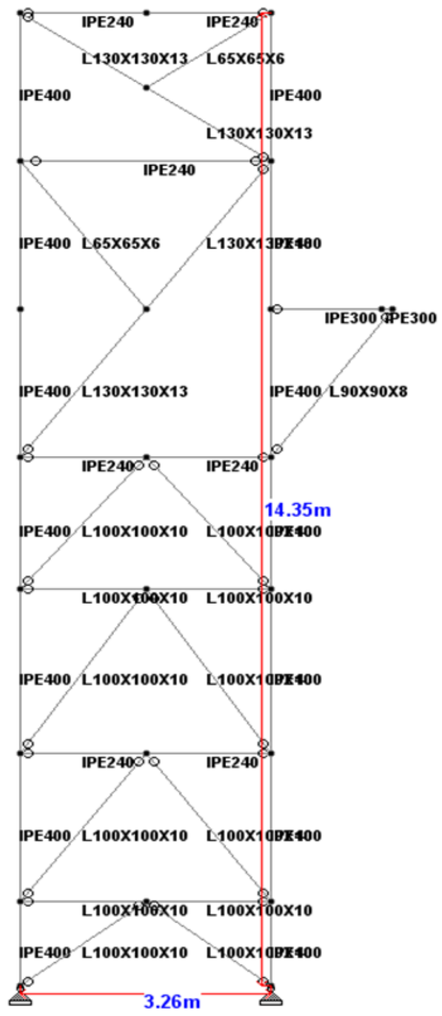
Three Dimensional view of staircase supporting structure from Staad Model is shown below:

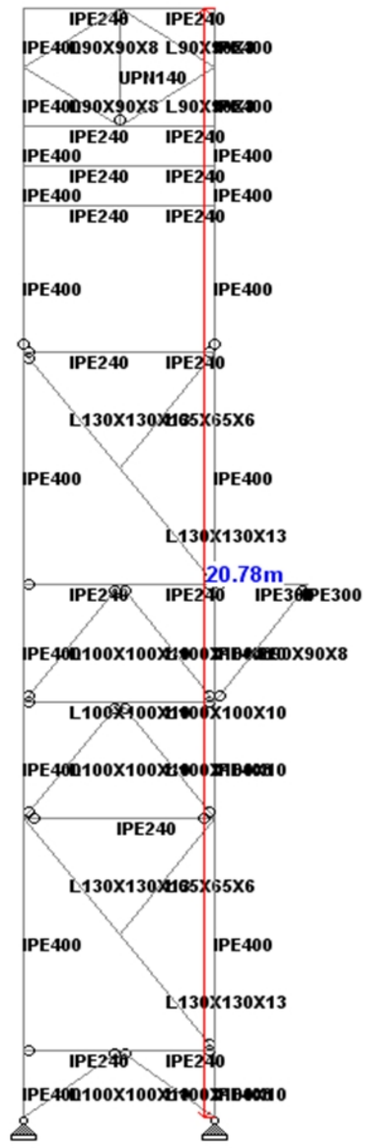


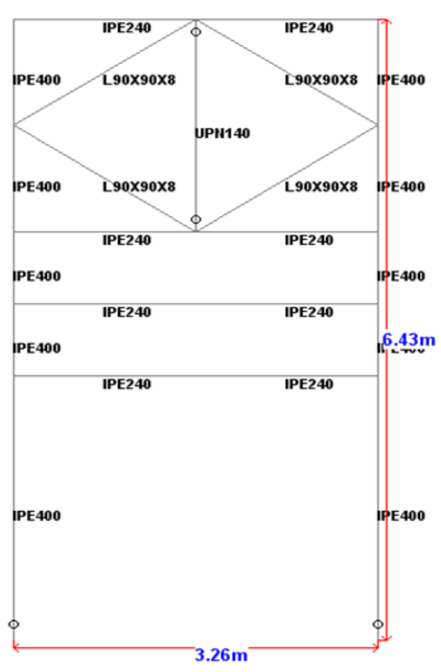
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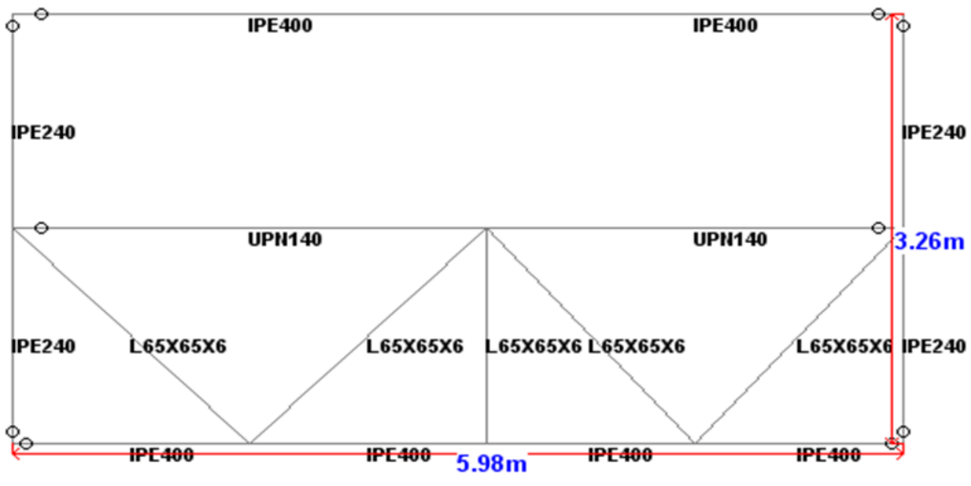
4.2 Section Profiles of Staircase
For Staircase supporting structure

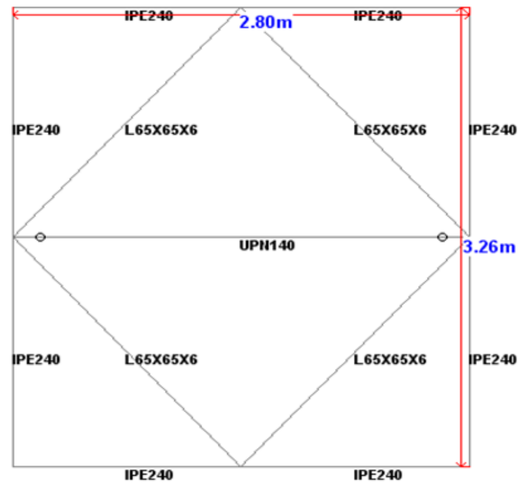






CENTRAL CROSS SECTION VIEW





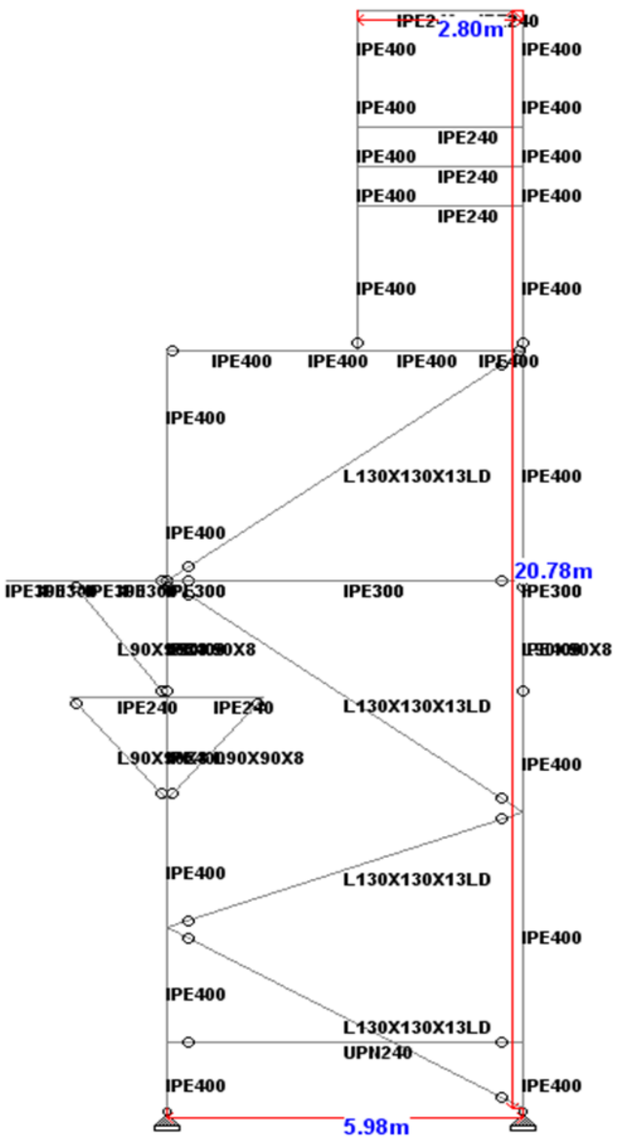
MID LEVEL PLAN

TOP PLAN

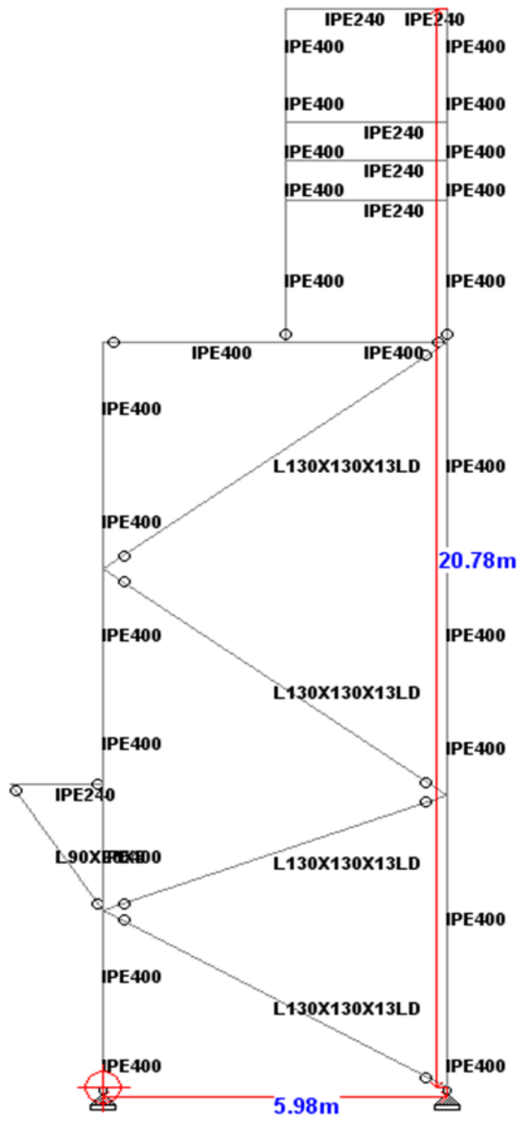
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4.2 Section Profiles of Staircase
For Staircase supporting structure



FRONT ELEVATION

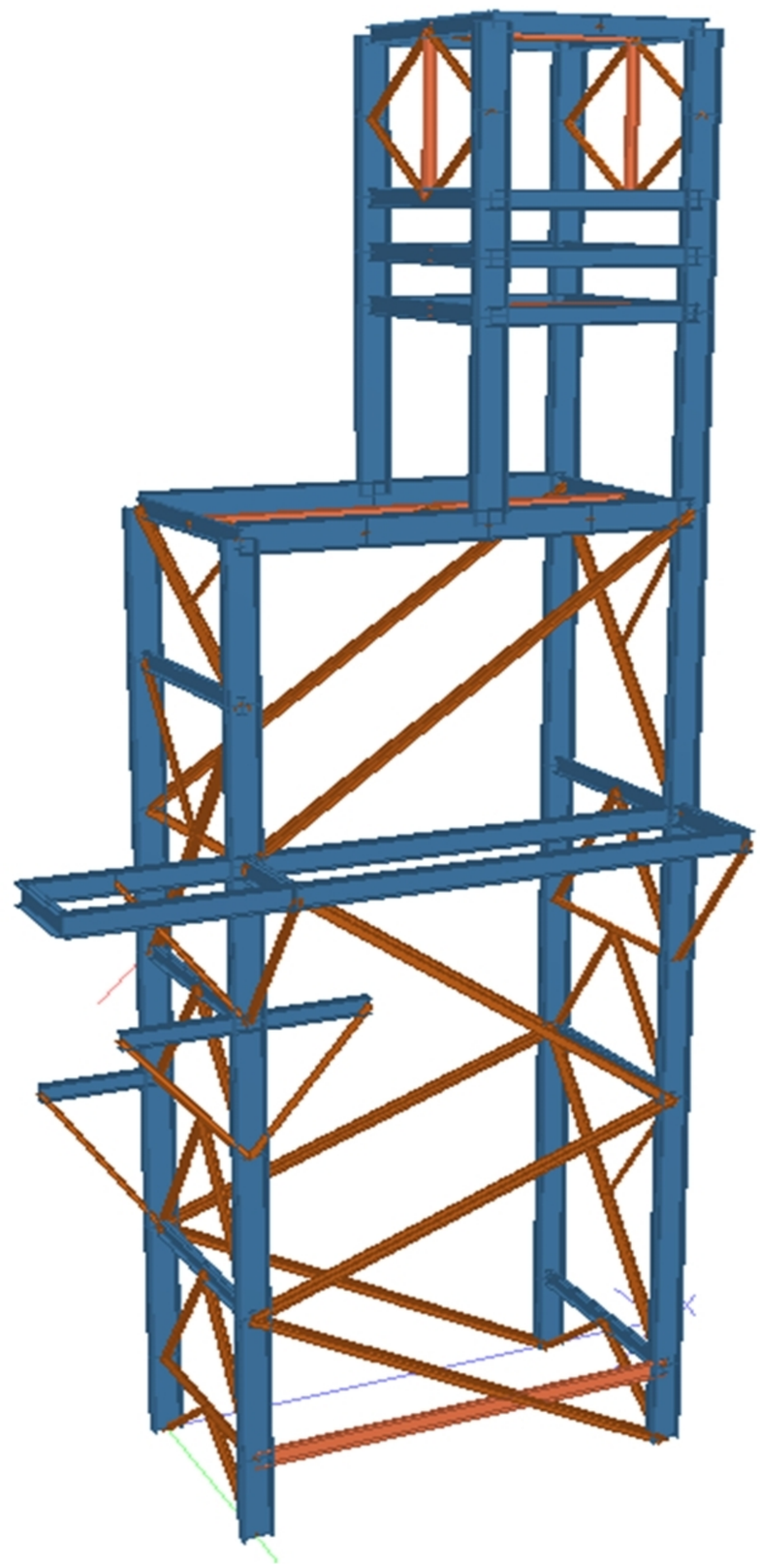


BACK ELEVATION

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4.2a 3D MODEL
For Staircase supporting structure



3D MODEL

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4.3 Load Calculation for Platform Supporting Structure

All the structural members are designed in Staad using AISC LRFD code.
Max. Elevation considered to be 15m above Base Plate Level.
All the structural members are designed in Staad using AISC LRFD code.

L/C 1. Dead Load :

a) Load due to self wt of stringer channel & grating :
Weight of Grating @ 35kg/m2 = 0.35 kN/m2
Self weight of stringer = 0.33 kN/m (UPN 240)
Stair width = 1.0 m
Stringer Length = 6 m
Conc. Load on beam supporting stringer = {(1.2/2 *0.35+0.33+0.15)*6.0/2} kN = 2.07 kN = 0.25 T
Self weight of handrail = 0.15 kN/m
Dead load of out side platform weight including handrail handrail = 0.35*1.5/2+0.15 = 0.05 T/m
Dead load of equipment on out side platform per point = 0.1 T/m

Cable Load

Load on each tray = 120 kg/m
Tray supporting member spacing 1.5 m
Total vertical load on each post = (120*1.5) = 0.18 T
Concentrated Moment = 0.18*0.5 = 0.09 T-m

Pipe vertical Load

At +923502 lev 350 dia pipe SW = 104 kg/m*6 m = 0.75 T
Unused area load on each side beam @ 2.5kPa at level +931372 & +930622 = 250 kg/m^2*3.26 m/2 = 0.5 T/m

At Tank Side

At +932122 & +931372 lev 150 dia pipe SW = 34 kg/m*3 m = 0.15 T
At +931372 & +930622 lev 50 dia pipe SW = 8 kg/m*2 m = 0.02 T
At +930622 lev 100 dia pipe SW = 20 kg/m*3 m = 0.1 T
At +930622 lev 200 dia pipe SW = 60 kg/m*3 m = 0.2 T

At Middle

At +931372 & +930622 lev 50 dia pipe SW = 8 kg/m*2 m = 0.02 T

At Vertical Drop Side

At +932122 & +931372 lev 150 dia pipe SW = 34 kg/m*8 m = 0.3 T
At +931372 & +930622 lev 50 dia pipe SW = 8 kg/m*7 m = 0.06 T
At +930622 lev 100 dia pipe SW = 20 kg/m*8 m = 0.2 T
At +930622 lev 200 dia pipe SW = 60 kg/m*8 m = 0.5 T

L/C 2. Live load :

Live Load on stair @ 400 kg/m2 = 4 kN/m2 (as per Civil, Structural Design Criteria)
Conc. Load on beam supporting stringer = {(1.2/2 *4)*6.0/2} kN = 7.2 kN = 0.72 T
Live load of out side platform weight including handrail handrail = 4*1.5/2 = 3 kN/m

Pipe vertical Load

At +923502 lev 350 dia pipe = 175 kg/m*6 m = 1.2 T

At Tank Side

At +932122 & +931372 lev 150 dia pipe = 40 kg/m*3 m = 0.15 T
At +931372 & +930622 lev 50 dia pipe = 4 kg/m*2 m = 0.01 T
At +930622 lev 100 dia pipe = 15 kg/m*3 m = 0.05 T
At +930622 lev 200 dia pipe = 50 kg/m*3 m = 0.2 T

At Middle

At +931372 & +930622 lev 50 dia pipe = 4 kg/m*2 m = 0.01 T

At Vertical Drop Side

At +932122 & +931372 lev 150 dia pipe = 40 kg/m*8 m = 0.4 T
At +931372 & +930622 lev 50 dia pipe = 4 kg/m*7 m = 0.03 T
At +930622 lev 100 dia pipe = 15 kg/m*8 m = 0.2 T
At +930622 lev 200 dia pipe = 50 kg/m*8 m = 0.5 T

Pipe horizontal Load

Horizontal load taken in staad 30% of vertical load

L/C 3. Wind load : (As per SBC 301 : 2007)

Wind load is calculated using the following parameters:
Basic Wind Speed = 155 Km/hr (as per Civil, Structural Design Criteria)
Exposure Category = C (as per Civil, Structural Design Criteria)
Topographic factor, Kzt = 1 (Section 6.4.3, SBC 301)
Importance Factor, I = 1 (Section 6.5, SBC 301)
Velocity Pressure Co-efficient, Kz = 1.09 (Table 7.2-2 , SBC 301)
Wind Directionality Factor, Kd = 0.85 (Section 6.4.1, SBC 301)
Velocity pressure, qz = 0.0473 x 10-3 x Kz x Kzt x Kd x V2 x I = 1.06 kN/m2

Wind Load in +X Dir.

Sections of the members exposed to Wind :
For Column Section IPE400
Flange width = 180 mm
Beam Depth = 300 mm
Beam Depth = 240 mm
Bracing Depth = 65 mm, 90 mm, 100 mm, 130 mm
Stringer Depth = 240 mm
Force Coeff. , Cf for Flat Section = 2 (Fig. 7.2-17 , SBC 301)
Wind load on Column IPE400 = 1.06*180/1000*2 = 0.040 T/m
Wind load on 300 mm Beam = 1.06*300/1000*2 = 0.065 T/m
Wind load on 240 mm Beam = 1.06*240/1000*2 = 0.052 T/m
Wind load on 65 mm Bracing = 1.06*65/1000*2 = 0.014 T/m
Wind load on 90 mm Bracing = 1.06*90/1000*2 = 0.020 T/m

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Wind load on 100 mm Bracing = 1.06*65/1000*2 = 0.022 T/m
Wind load on 130 mm Bracing = 1.06*65/1000*2 = 0.028 T/m
Wind load on 2Ls-130 mm Bracing = 1.06*260/1000*2 = 0.056 T/m

Wind ON **Cable Tray**
Wind load on each tray = 1.06*150/1000*2 = 0.032 T/m * 1.5 m = 0.05 T

Pipe horizontal Load
At +923502 lev 350 dia pipe = 1.06/10*0.35*6 m*1.2 = 0.27 T

At Tank Side & Vertical Drop Side
At +931372 & +930622 lev total horizontal load = 1.06/10*1.5*(0.5m+3.26m+2m)*1.2 = 1.2 T
Each nodal load = 1.2 T / 4 = 0.3 T

Wind Load in +Z Dir.

Sections of the members exposed to Wind :
For Column Section IPE400,
Column Depth = 400 mm
Beam Depth = 400 mm
Beam Depth = 300 mm
Beam Depth = 240 mm
Bracing Depth = 65 mm, 90 mm, 100 mm, 130 mm
Stringer Depth = 240 mm
Force Coeff. , Cf for Flat Section = 2 (Fig. 7.2-17 , SBC 301)
Wind load on Column IPE400 = 1.06*400/1000*2 = 0.085 T/m
Wind load on 400 mm Beam = 1.06*400/1000*2 = 0.085 T/m
Wind load on 300 mm Beam = 1.06*300/1000*2 = 0.065 T/m
Wind load on 240 mm Beam = 1.06*240/1000*2 = 0.052 T/m
Wind load on 65 mm Bracing = 1.06*65/1000*2 = 0.014 T/m
Wind load on 90 mm Bracing = 1.06*90/1000*2 = 0.020 T/m
Wind load on 100 mm Bracing = 1.06*65/1000*2 = 0.022 T/m
Wind load on 130 mm Bracing = 1.06*65/1000*2 = 0.028 T/m
Wind load on Stringer = 1.06*240/1000*2 = 0.052 T/m
Conc. Load on beam supporting stringer for Stair-1 = 0.052 * 7.5 / 2 = 0.2 T

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4.4 LOAD COMBINATIONS (Refer SBC 301 : 2007 for Strength Design)

100 1.4*DL
101 1.2DL+1.6*LL
102 1.2DL+0.8WLX
103 1.2DL+0.5LL+1.6WLX
104 0.9DL+1.6WLX
105 1.2DL+0.8WLZ
106 1.2DL+0.5LL+1.6WLZ
107 0.9DL+1.6WLZ

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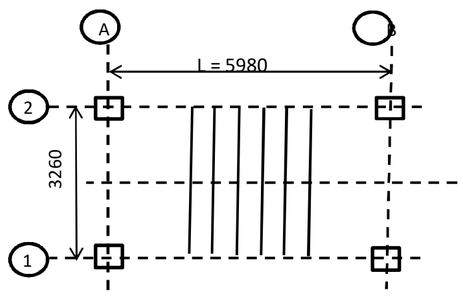
4.5 Foundation Load Data for columns of Primary Cyanide Destruction

(+ve Fy means Vertical Downward Load)

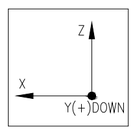
Col. Mkd.	Load Comb.	HORZ. Fx (Mton)	Vertical Fy(Mton)	HORZ. Fz (Mton)
A2	DL	0	7	0
	LL-1	0	5	0
	LL-2	0	9	0
	LL-3	0	3	0
	LL-4	0	10	0
	WL(+)X	0	-14	0
	WL(+)Z	0	-23	-4
B2	DL	0	10	1
	LL-1	0	7	1
	LL-2	1	3	1
	LL-3	0	1	1
	LL-4	0	9	2
	WL(+)X	-6	14	0
	WL(+)Z	0	-34	-5
A1	DL	0	9	0
	LL-1	0	10	0
	LL-2	0	14	0
	LL-3	0	16	0
	LL-4	0	9	0
	WL(+)X	0	-15	0
	WL(+)Z	0	23	-4
B1	DL	0	12	-1
	LL-1	-1	12	-1
	LL-2	1	7	-1
	LL-3	0	13	-2
	LL-4	0	6	-1
	WL(+)X	-7	15	0
	WL(+)Z	0	34	-5

NOTE:

1. ALL LOAD VALUES ARE ACTUAL WITHOUT ANY INCREASE.
2. WL(+)X :- WIND LOAD FROM GRID-A TO GRID-B
3. WL(+)Z :- WIND LOAD FROM LINE-2 TO LINE-1
4. WIND LOAD IN X AND Z DIRECTION WILL NOT ACT SIMULTANEOUSLY.



PRIMARY CYANIDE DESTRUCTION



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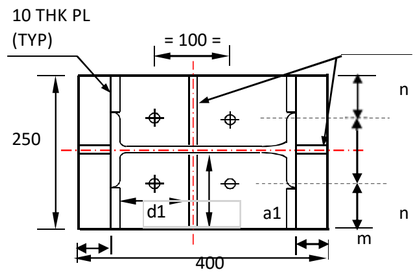
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4.6 Design of Base Plates for Columns

DESIGN OF BASE PLATE FOR STAIR-1

(AS PER SBC - 306 & AISC : 2018 LRFD Specification)

Here, Column Section =	IPE 400			
Grade of Conc. =	35	N/mm2		
Grade of Steel, Fy =	275	N/mm2		
Maximum axial load, Pu =	75.00	Ton	L/C	114
& Minimum axial Load =	-45.30	Ton (Ten.)	(Refer Staad File for	Load Combs)
Length of base plate (N)=	450	mm		
Width of base plate (B) =	250	mm		
Area of base plate, A1 =	0.1125	m2		
Length of concrete pedestal =	500	mm		
Width of concrete pedestal =	500	mm		
Area of concrete pedestal, A2 =	0.25	m2		
Design Bearing strength = 0.85*fc' =	29.75	N/mm2	(As per Section 10.9 of SBC 306)	
Pp = 0.85*fc' *A1*SQRT(A2/A1) =	4989.2	KN	(Eqn. 10.9-2 of SBC 306)	
ϕc * Pp = 2993.5 KN	> 750.02KN - OK		(ϕ_c = 0.6)	



10 THK PL (TYP)

m =	(N - 0.95d)/ 2 =	35	mm
n =	(B - 0.85bf)/2 =	48.5	mm
l =	2 (X) / (1+ (1-X)) =	0.49	
= 100 = l n'=		32.9	mm
l =	max(m, n, ln')	48.5	mm
tmin, l*sqrt((2*Pu)/(0.9*Fy*B*N))=		11.3 mm	(As per AISC-2018 : LRFD)

Actual area of base plate =	1125	cm2		
Uniform base pressure below base plate,fpu =	6.67	N/mm2		
Moment capacity of Base Plate, M_{pu} = $f_{pu} * l^2/2$ =	7845	N-mm/mm width		(As per AISC-2018 : LRFD)
Required thk. of Base Plate, t_{reqd} = $\sqrt{\frac{4 * Mpu}{\phi * F_y}}$ =	11.3	mm	> tmin	ϕ = 0.9

Now, a1 = 12.07 cm and d1 = 18.15 cm
Therefore, a1/d1 = 12.07/18.15 = 0.665
Hence, α_3 = 0.083 as per Mukhanov's chart
Now BM per cm = α3 X q X $d1^2$
= 0.083 X 6.67 X 18.15^2 = 1825.14 kg-cm/cm

Considering 4 nos bolt, tension per bolt = -45.298 / 4 = - 11.33 Ton

Considering 50% of tension will be carried to the web, BM at the face of the web (thickness = 8.6 mm),
=0.5 X 11.33 X 1000 X 4.57 = 25889.05 kg-cm

Moment per unit width =	25889.05 / 9.14 = 2832.5 kg-cm/cm			i.e. > 1825.14 kg-cm/cm
Hence, thickness of baseplate required, treqd = $\sqrt{4*M/(\phi * fy)}$ =		2.140	cm	f = 0.9

Hence, provide 450mm X 250 mm X 32 mm thk. baseplate

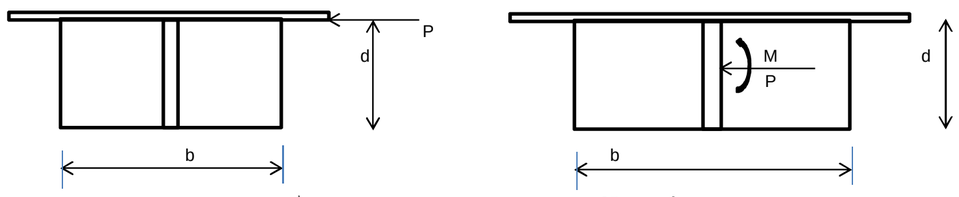
Dia of Bolt =	30	mm		
Ultimate Tensile strength of Bolts, Fu =	410	N/mm2		
Nominal Tensile Capcity of bolt, Rn =	21.74	Ton	= 0.75*f_u * A_b	[AISC Spec]
Design Tensile Capcity of bolt, ϕ * Rn =	16.30	Ton	*> 11.33 Ton, OK*	ϕ = 0.75

Hence, 4 nos 30mm dia. Bolts are required
However provide, 4 nos 32mm dia. Bolts

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Design of Shear Key (As per AISC 2018 Specification - LSD)



Concrete Compressive Strength , f_c' =		35	N/mm^2	
Concrete Pedestal Length =		500	mm	
Concrete Pedestal Width =		500	mm	
Max. Factored shear, P_u =		120	KN	(From STAAD Output - L/C 109)
Equating the load to the bearing capacity of concrete, Area reqd. for shear key,				
$A_{lreqd.}$ =	$P_u/(0.8 * fc')$ =	42.86	cm^2	
Shear Key Width , b =		200	mm	d_{reqd} = 21.43 mm
d_{provd} for shear Key =		75	mm	(excluding Grout thk. = 25mm)
Thickness of shear Key, t =		36	mm	
Projected Depth of Shear Key at Pier edge =		232	mm	
Projected area excl. the area of Shear Key, A_v =		101000	mm^2	
Shear cap. of conc. In front of shear key =		4*f*sqrt(fc')*A_v =	1434 KN	> 120
Using Cantilever Model for Mid Plate,				
Moment @ shear key base, M = V (G + d/2) =		10.5	KN-m	
	Z, $bt^2/4$ =	M/ (f * Fy)		
	t_{reqd} =	30	mm	< t provided - OK
For Wing Plate, Cantilever Moment =		3	KN-m	
	t_{reqd} =	16	mm	< t provided - OK

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
Date	17-03-2020	13-04-2020	27-11-2023			Date	17-03-2020	13-04-2020	27-11-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	AS BUILT	
Prepared By	SGI	SGI	SGI			Prepared By	SGI	SGI	SGI	
Checked	SYC	SYC	SYC			Checked	SYC	SYC	SYC	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

4.7 Design of Staircase Stringer for Stair-1 [As per SBC-Limit State Method]: (SBC 306 : 2007)

Design parameters:

Axial compression, P_u =	0.0	kN	
Axial tension, T =	0.0	kN	
Moment about z-axis, M_{uz} =	25.0	kN-m	
Moment about y-axis, M_{uy} =	5.0	kN-m	(assumed nominal moment)
Shear along y-axis, V_y =	15.0	kN	
Shear along z-axis, V_z =	0.0	kN	
Effective length about z-axis, k_zL_z =	6	m	
Effective length about y-axis, k_yL_y =	2	m	
Eff. lat. torsional buck. length, L_{LT} =	2	m	
Allowable slenderness ratio =	200		

Grade of steel =	S275 (Ult 450)		E =	200000	N/mm^2
Lateral support condition =	Unsupported	("1" for simply supported/"2" for cantilever)	G =	77200	N/mm^2
Loading applied at =	Shear centre		F_u =	450	N/mm^2
Loading direction w.r.t. shear centre =	Towards		f_{yf} =	275	N/mm^2
Beam/column type =	Simply suppo	("1" for simply supported/"2" for cantilever)	F_{yw} =	275	N/mm^2
Provision of tranv. stiff. =	At supports		Fr=	69	N/mm^2
Stiffener spacing, c =	2000 mm		$F_L(F_{yf}-F_r)$ =	206	N/mm^2

Sectional properties:

Section: UPN 240

D =	240	mm	A = 42.3	cm^2
b_{top} =	85	mm	I_y = 248	cm^4
t_{top} =	13	mm	r_z = 9.22	cm
b_{bot} =	85	mm	r_y = 2.42	cm
t_{bot} =	13	mm	Z_{ezT} = 300	cm^3
t_w =	9.5	mm	Z_{ey} = 39.6	cm^3
d =	184	mm	Z_{pz} = 358	cm^3
I_z =	3600	cm^4	Z_{py} = 75.7	cm^3
			J = 19.7	cm^4
			Cw = 2.21E+10	cm^6

Unstiffened Element	Flanges	Compact		*(Table 2.5-1)*		
		λ=	b/t = 6.5			
Compact Limit		λp=	$0.38(E/F_y)^{1/2}$	=	10.2	
Non Compact Limit		λr=	$0.83(E/F_L)^{1/2}$	=	25.9	
Unstiffened Element	Web	Compact				
		λ=	d/t_w = 19.4			
Compact Limit		λp=	$3.76(E/F_y)^{1/2}$	=	101.4	
Non Compact Limit		λr=	$5.70(E/F_y)^{1/2}$	=	153.7	

Hence the section is compact

Design Compressive Strength for Flexural Buckling *(Section 5.2)*

Design Compressive Strength		=	$f_c * P_n$			f_c =	0.85
	λc	=	$(Kl/(rp))*(F_y/E)^{1/2}$	=	1.2		< 1.5
	F_{cr}	=	$(0.658^{lc2})*Fy$	if		λc	< 1.5
		=	$(0.877/lc^2)*F$	if		λc	> 1.5
	F_{cr}	=	147.59	N/mm^2			
	P_n	=	624.33	KN			
	$\phi_c * P_n$	=	530.68	KN			

Check Bending about the Z-Z Axis *(Section 6.1)*

Actual Unbraced Length		L_b =	2500	mm		
Limit. lat. unbraced length,	L_p, 1.76 *ry sqrt(E/Fy* ϕ) =		1149	mm	<	L_b
Limiting Buckling moment =		$M_r = F_L * Z_{EZ}$ =	61.8	KN-m		
		X_1 =	26547.3			
		X_2 =	13.870			
	Lr = (r_y* X_1/F_L)*SQRT(1+SQRT(1+ X_2 F_L^2)) =		86437.89	mm	>	L_b
Plastic moment of section =		M_p=	98.45	KN-m	<	1.5*M_y = 123.75
		C_b =	1.00			ϕ_b = 0.9
		M_n =	97.87	KN-m	<	M_p
Flexural Design Strength about the Z-Z Axis = $\varphi_b * M_n$ =				88.1	KN-m	

Check Bending about the Y-Y Axis

	M_p=	20.82	KN-m	>	1.5*M_y = 16.34
	M_n=	16.34	KN-m		
Flexural Design Strength about the Y-Y Axis = $\varphi_b * M_n$ =				14.7015	KN-m

Check For Biaxial Bending with Compression *(Section 8.1)*

$P_u/(\phi*P_n)$ =	0.00	< 0.2	
(Pu/(2*Pn) + Muz/φb * Mnz+ Muy/φb * Mny) =	0.62	< 1	

Check Shear *(Section 6.2)*

Clear distance between stiffeners	a =		0.001	mm	φv =	0.9
	h =		184	mm		
	a/h =		5.43478E-06			
	kv =	5				
	h/t_w =		19.37	< 1.1*(kvE/F_{yw})^0.5 =		66.332496
	V_n =		376.2	KN		
Design shear strength = $\varphi_v * V_n$ =	338.58	KN		>	15.0	

Check for Deflection

UDL on beam, w =	4.5	Kn/m		
Actual Deflection,	d_{act} = 5/384 * 4.5 * 6^4 / (200000 * 3600) =		10.55	mm
Allowable Deflection,	d_{allw} = L/ 360		= 16.67	mm
			> Actual deflection- Ok	

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Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

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Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 STAAD INPUT FOR STAIR SUPPORTING STRUCTURE

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 19-Jun-19
END JOB INFORMATION
INPUT WIDTH 79
SET ECHO OFF
UNIT METER MTON
JOINT COORDINATES
1 0 0 0; 2 5.98 0 0; 3 0 0 3.26; 4 5.98 0 3.26; 5 0 14.35 0; 6 5.98 14.35 0;
7 0 14.35 3.26; 8 5.98 14.35 3.26; 9 0 3.41 3.26; 10 5.98 5.6 3.26;
11 5.98 9.975 3.26; 13 0 9.975 3.26; 15 0 1.225 3.26; 16 5.98 1.225 3.26;
17 5.98 1.225 0; 28 0 3.41 0; 37 5.98 5.6 0; 49 0 7.79 3.26; 50 0 12.165 3.26;
51 0 7.79 0; 56 5.98 9.975 0; 61 0 12.165 0; 95 1.6 7.79 3.26;
97 -1.6 7.79 3.26; 99 0 5.84 3.26; 103 0 5.84 0; 106 -1.6 5.84 0;
116 5.98 9.975 4.835; 117 0 9.975 4.835; 120 5.98 9.975 4.71; 121 0 9.975 4.71;
122 -2.7 9.975 3.26; 123 -2.7 9.975 4.71; 124 -1.6 9.975 3.26;
125 3.18 14.35 0; 126 3.18 14.35 3.26; 127 5.98 18.58 0; 128 5.98 18.58 3.26;
133 3.18 18.58 0; 134 3.18 18.58 3.26; 135 5.98 17.83 0; 136 5.98 17.83 3.26;
137 3.18 17.83 0; 138 3.18 17.83 3.26; 139 5.98 17.08 0; 140 5.98 17.08 3.26;
141 3.18 17.08 0; 142 3.18 17.08 3.26; 143 5.98 18.58 1.63;
144 3.18 18.58 1.63; 145 5.98 20.78 0; 146 5.98 20.78 3.26; 147 3.18 20.78 0;
148 3.18 20.78 3.26; 149 5.98 20.78 1.63; 150 3.18 20.78 1.63; 151 0 9.975 0;
152 5.98 1.225 1.63; 153 0 14.35 1.63; 154 5.98 14.35 1.63;
155 3.18 14.35 1.63; 156 1.59 14.35 3.26; 157 4.58 14.35 3.26;
158 5.98 7.7875 0; 159 5.98 7.7875 3.26; 160 5.98 9.975 1.63;
161 5.98 7.7875 1.63; 162 0 1.225 0; 163 0 3.41 1.63; 164 0 1.225 1.63;
165 3.18 17.83 1.63; 166 5.98 17.83 1.63; 167 3.18 17.08 1.63;
168 5.98 17.08 1.63; 169 0 5.84 1.63; 170 0 7.79 1.63; 171 0 9.9775 1.63;
172 0 13.2575 1.63; 173 5.98 3.4125 1.63; 174 5.98 12.1625 1.63;
175 5.98 19.68 0; 176 5.98 19.68 3.26; 177 3.18 19.68 0; 178 3.18 19.68 3.26;
179 4.58 20.78 0; 180 4.58 20.78 3.26;
MEMBER INCIDENCES
1 1 162; 2 2 17; 3 3 15; 4 4 16; 5 5 125; 6 5 153; 7 7 156; 8 6 154; 9 9 99;
10 10 159; 11 11 8; 13 4 9; 14 9 10; 15 13 50; 16 10 13; 17 13 8; 20 15 9;
21 16 10; 22 15 16; 23 17 37; 24 17 152; 36 28 163; 47 37 158; 48 37 10;
63 50 7; 64 51 151; 65 51 170; 70 56 6; 71 56 160; 76 61 5; 77 61 50; 112 2 28;
113 28 37; 124 9 169; 125 51 171; 126 50 172; 127 16 173; 129 11 174;
138 49 95; 139 99 49; 141 97 99; 142 95 99; 144 103 51; 145 97 49; 148 28 103;
149 106 28; 150 106 103; 161 11 120; 162 13 121; 163 159 11; 164 49 13;
165 116 159; 166 117 49; 167 13 11; 168 120 116; 169 121 117; 170 121 120;
171 13 124; 172 121 123; 173 124 122; 174 124 49; 175 125 6; 176 126 157;
178 6 139; 179 8 140; 184 125 141; 185 126 142; 186 127 143; 196 133 127;
197 134 128; 199 133 144; 200 135 127; 201 136 128; 202 137 133; 203 138 134;
204 135 166; 205 137 135; 206 138 136; 207 137 165; 208 139 135; 209 140 136;
210 141 137; 211 142 138; 212 139 168; 213 141 139; 214 142 140; 215 141 167;
216 143 128; 217 144 134; 218 127 175; 219 128 176; 220 133 177; 221 134 178;
222 143 149; 223 144 150; 224 147 179; 225 150 149; 226 148 180; 227 147 150;
228 150 148; 229 145 149; 230 149 146; 231 151 61; 232 37 151; 233 151 6;
234 152 16; 235 152 2; 236 152 4; 237 153 7; 238 154 8; 239 153 155;
240 155 154; 241 156 126; 243 157 8; 245 153 156; 246 156 155; 247 155 157;
248 157 154; 249 155 126; 250 158 56; 251 158 161; 252 160 11; 253 161 159;
254 160 158; 255 160 159; 256 161 37; 257 161 10; 258 162 28; 259 162 164;
260 163 9; 261 164 15; 262 163 162; 263 163 15; 264 164 1; 265 164 3;
266 122 123; 267 165 138; 268 166 136; 269 167 142; 270 168 140; 271 165 166;
272 167 168; 273 103 169; 274 169 99; 275 169 28; 276 170 49; 277 170 99;
278 170 103; 279 171 50; 280 172 5; 281 173 37; 282 174 6; 283 7 172;
284 61 171; 285 8 174; 286 10 173; 289 175 145; 290 176 146; 291 177 147;
292 178 148; 293 150 177; 294 177 144; 295 149 175; 296 175 143; 297 150 178;
298 178 144; 299 149 176; 300 176 143; 301 179 145; 302 180 146; 303 179 150;
304 179 149; 305 149 180; 306 150 180;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
6 8 24 36 48 65 71 77 138 145 150 186 196 197 199 204 TO 207 212 TO 217 224 -
226 TO 230 234 237 238 252 260 267 TO 270 276 301 302 TABLE ST IPE240
22 TABLE ST UPN240
141 142 149 165 166 174 293 TO 300 TABLE ST L90X90X8
245 TO 249 303 TO 306 TABLE ST L65X65X6
222 223 225 239 240 271 272 TABLE ST UPN140
```

```
MEMBER PROPERTY EUROPEAN
13 14 16 17 112 113 232 233 TABLE LD L130X130X13 SP 0.01
124 235 236 251 253 TO 257 259 261 TO 265 273 TO 275 277 -
278 TABLE ST L100X100X10
125 TO 127 129 279 TO 282 TABLE ST L130X130X13
283 TO 286 TABLE ST L65X65X6
5 7 175 176 241 243 TABLE ST IPE400
161 162 167 TO 173 266 TABLE ST IPE300
1 TO 4 9 TO 11 15 20 21 23 47 63 64 70 76 139 144 148 163 164 178 179 184 -
185 200 TO 203 208 TO 211 218 TO 221 231 250 258 289 TO 291 -
292 TABLE ST IPE400
CONSTANTS
MATERIAL STEEL ALL
SUPPORTS
1 TO 4 PINNED
MEMBER RELEASE
5 TO 8 13 14 16 17 22 24 36 48 65 71 77 112 113 124 TO 127 129 141 142 149 -
161 162 165 TO 167 171 174 178 179 184 185 222 223 225 232 233 235 236 239 -
251 254 TO 257 259 262 TO 265 271 TO 273 275 277 278 START MY MZ
13 14 16 17 22 48 77 112 113 124 141 142 149 150 165 TO 167 174 175 222 223 -
225 232 TO 234 237 238 240 243 252 TO 257 260 TO 265 271 272 274 TO 280 -
282 END MY MZ
MEMBER TRUSS
245 TO 249 293 294 296 TO 300 303 TO 306
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
MEMBER LOAD
22 CON GY -0.25 4.65
22 CON GY -0.25 5.65
8 24 36 48 65 71 77 CON GY -0.25 0.42
8 24 36 48 65 71 77 CON GY -0.25 1.42
234 252 260 CON GY -0.25 0.21
234 252 260 CON GY -0.25 1.21
48 77 CON GY -0.25 1.84
276 CON GY -0.25 0.21
48 77 CON GY -0.25 2.84
276 CON GY -0.25 1.21
145 150 CON GY -0.25 0.3
145 150 CON GY -0.25 1.3
167 170 TO 173 UNI GY -0.05
*equipment load
JOINT LOAD
122 123 FY -0.1
*cable load
MEMBER LOAD
218 TO 223 CON GY -0.18 0.2
218 TO 223 CON GY -0.18 0.65
218 TO 223 CON GY -0.18 1.1
222 223 CON GY -0.18 1.55
292 CON GY -0.18 0.45
291 CON GY -0.18 0.45
290 CON GY -0.18 0.45
289 CON GY -0.18 0.45
220 221 223 CMOM GZ -0.09 0.2
220 221 223 CMOM GZ -0.09 0.65
220 221 223 CMOM GZ -0.09 1.1
223 CMOM GZ 0.09 1.55
292 CMOM GZ 0.09 0.45
291 CMOM GZ 0.09 0.45
218 219 222 CMOM GZ 0.09 0.2
218 219 222 CMOM GZ 0.09 0.65
218 219 222 CMOM GZ 0.09 1.1
222 CMOM GZ 0.09 1.55
290 CMOM GZ 0.09 0.45
289 CMOM GZ 0.09 0.45
*
196 CON GY -0.15 1
196 CON GY -0.15 1.8
205 CON GY -0.15
205 271 CON GY -0.02 0.95
213 272 CON GY -0.02 0.5
213 272 CON GY -0.02 0.85
213 272 CON GY -0.02 1.2
213 CON GY -0.2 1.7
213 272 CON GY -0.02 2.2
213 CON GY -0.1 2.5
*
```

```
197 CON GY -0.3 1
197 CON GY -0.3 1.8
206 CON GY -0.3
206 CON GY -0.06 0.95
214 CON GY -0.06 0.5
214 CON GY -0.06 0.85
214 CON GY -0.06 1.2
214 CON GY -0.5 1.7
214 CON GY -0.06 2.2
214 CON GY -0.2 2.5
*
205 206 213 214 UNI GY -0.5
JOINT LOAD
50 61 FY -0.75
*********************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE-1 LOAD
MEMBER LOAD
22 CON GY -0.72 4.65
22 CON GY -0.72 5.65
8 24 36 48 65 71 77 CON GY -0.72 0.42
8 24 36 48 65 71 77 CON GY -0.72 1.42
234 252 260 CON GY -0.72 0.21
234 252 260 CON GY -0.72 1.21
48 77 CON GY -0.72 1.84
276 CON GY -0.72 0.21
48 77 CON GY -0.72 2.84
276 CON GY -0.72 1.21
145 150 CON GY -0.72 0.3
145 150 CON GY -0.72 1.3
167 170 TO 173 UNI GY -0.3
*
196 CON GY -0.15 1
196 CON GX 0.045 1
196 CON GY -0.15 1.8
196 CON GX 0.045 1.8
205 CON GY -0.15
205 CON GX 0.045
205 271 CON GY -0.01 0.95
205 271 CON GX 0.003 0.95
213 272 CON GY -0.01 0.5
213 272 CON GX 0.003 0.5
213 272 CON GY -0.01 0.85
213 272 CON GX 0.003 0.85
213 272 CON GY -0.01 1.2
213 272 CON GX 0.003 1.2
213 CON GY -0.2 1.7
213 CON GX 0.06 1.7
213 272 CON GY -0.01 2.2
213 272 CON GX 0.003 2.2
213 CON GY -0.05 2.5
213 CON GX 0.015 2.5
*
197 CON GY -0.4 1
197 CON GX 0.12 1
197 CON GY -0.4 1.8
197 CON GX 0.12 1.8
206 CON GY -0.4
206 CON GX 0.12
206 CON GY -0.03 0.95
206 CON GX 0.009 0.95
214 CON GY -0.03 0.5
214 CON GX 0.009 0.5
214 CON GY -0.03 0.85
214 CON GX 0.009 0.85
214 CON GY -0.03 1.2
214 CON GX 0.009 1.2
214 CON GY -0.5 1.7
214 CON GX 0.15 1.7
214 CON GY -0.03 2.2
214 CON GX 0.009 2.2
214 CON GY -0.2 2.5
214 CON GX 0.06 2.5
*
JOINT LOAD
50 61 FX 0.36 FY -1.2
*****************************************************
LOAD 3 LOADTYPE None  TITLE LIVE-2 LOAD
```

```
MEMBER LOAD
22 CON GY -0.72 4.65
22 CON GY -0.72 5.65
8 24 36 48 65 71 77 CON GY -0.72 0.42
8 24 36 48 65 71 77 CON GY -0.72 1.42
234 252 260 CON GY -0.72 0.21
234 252 260 CON GY -0.72 1.21
48 77 CON GY -0.72 1.84
276 CON GY -0.72 0.21
48 77 CON GY -0.72 2.84
276 CON GY -0.72 1.21
145 150 CON GY -0.72 0.3
145 150 CON GY -0.72 1.3
167 170 TO 173 UNI GY -0.3
*
196 CON GY -0.15 1
196 CON GX -0.045 1
196 CON GY -0.15 1.8
196 CON GX -0.045 1.8
205 CON GY -0.15
205 CON GX -0.045
205 271 CON GY -0.01 0.95
205 271 CON GX -0.003 0.95
213 272 CON GY -0.01 0.5
213 272 CON GX -0.003 0.5
213 272 CON GY -0.01 0.85
213 272 CON GX -0.003 0.85
213 272 CON GY -0.01 1.2
213 272 CON GX -0.003 1.2
213 CON GY -0.2 1.7
213 CON GX -0.06 1.7
213 272 CON GY -0.01 2.2
213 272 CON GX -0.003 2.2
213 CON GY -0.05 2.5
213 CON GX -0.015 2.5
*
197 CON GY -0.4 1
197 CON GX -0.12 1
197 CON GY -0.4 1.8
197 CON GX -0.12 1.8
206 CON GY -0.4
206 CON GX -0.12
206 CON GY -0.03 0.95
206 CON GX -0.009 0.95
214 CON GY -0.03 0.5
214 CON GX -0.009 0.5
214 CON GY -0.03 0.85
214 CON GX -0.009 0.85
214 CON GY -0.03 1.2
214 CON GX -0.009 1.2
214 CON GY -0.5 1.7
214 CON GX -0.15 1.7
214 CON GY -0.03 2.2
214 CON GX -0.009 2.2
214 CON GY -0.2 2.5
214 CON GX -0.06 2.5
*
JOINT LOAD
50 61 FX -0.36 FY -1.2
*******************************************************
LOAD 4 LOADTYPE None  TITLE LIVE-3 LOAD
MEMBER LOAD
22 CON GY -0.72 4.65
22 CON GY -0.72 5.65
8 24 36 48 65 71 77 CON GY -0.72 0.42
8 24 36 48 65 71 77 CON GY -0.72 1.42
234 252 260 CON GY -0.72 0.21
234 252 260 CON GY -0.72 1.21
48 77 CON GY -0.72 1.84
276 CON GY -0.72 0.21
48 77 CON GY -0.72 2.84
276 CON GY -0.72 1.21
145 150 CON GY -0.72 0.3
145 150 CON GY -0.72 1.3
167 170 TO 173 UNI GY -0.3
*
196 CON GY -0.15 1
```

```
196 CON GZ 0.045 1
196 CON GY -0.15 1.8
196 CON GZ 0.045 1.8
205 CON GY -0.15
205 CON GZ 0.045
205 271 CON GY -0.01 0.95
205 271 CON GZ 0.003 0.95
213 272 CON GY -0.01 0.5
213 272 CON GZ 0.003 0.5
213 272 CON GY -0.01 0.85
213 272 CON GZ 0.003 0.85
213 272 CON GY -0.01 1.2
213 272 CON GZ 0.003 1.2
213 CON GY -0.2 1.7
213 CON GZ 0.06 1.7
213 272 CON GY -0.01 2.2
213 272 CON GZ 0.003 2.2
213 CON GY -0.05 2.5
213 CON GZ 0.015 2.5
*
197 CON GY -0.4 1
197 CON GZ 0.12 1
197 CON GY -0.4 1.8
197 CON GZ 0.12 1.8
206 CON GY -0.4
206 CON GZ 0.12
206 CON GY -0.03 0.95
206 CON GZ 0.009 0.95
214 CON GY -0.03 0.5
214 CON GZ 0.009 0.5
214 CON GY -0.03 0.85
214 CON GZ 0.009 0.85
214 CON GY -0.03 1.2
214 CON GZ 0.009 1.2
214 CON GY -0.5 1.7
214 CON GZ 0.15 1.7
214 CON GY -0.03 2.2
214 CON GZ 0.009 2.2
214 CON GY -0.2 2.5
214 CON GZ 0.06 2.5
*
JOINT LOAD
50 61 FY -1.2 FZ 0.36
*******************************************************
LOAD 5 LOADTYPE None  TITLE LIVE-4 LOAD
MEMBER LOAD
22 CON GY -0.72 4.65
22 CON GY -0.72 5.65
8 24 36 48 65 71 77 CON GY -0.72 0.42
8 24 36 48 65 71 77 CON GY -0.72 1.42
234 252 260 CON GY -0.72 0.21
234 252 260 CON GY -0.72 1.21
48 77 CON GY -0.72 1.84
276 CON GY -0.72 0.21
48 77 CON GY -0.72 2.84
276 CON GY -0.72 1.21
145 150 CON GY -0.72 0.3
145 150 CON GY -0.72 1.3
167 170 TO 173 UNI GY -0.3
*
196 CON GY -0.15 1
196 CON GZ -0.045 1
196 CON GY -0.15 1.8
196 CON GZ -0.045 1.8
205 CON GY -0.15
205 CON GZ -0.045
205 271 CON GY -0.01 0.95
205 271 CON GZ -0.003 0.95
213 272 CON GY -0.01 0.5
213 272 CON GZ -0.003 0.5
213 272 CON GY -0.01 0.85
213 272 CON GZ -0.003 0.85
213 272 CON GY -0.01 1.2
213 272 CON GZ -0.003 1.2
213 CON GY -0.2 1.7
213 CON GZ -0.06 1.7
213 272 CON GY -0.01 2.2
```

```
213 272 CON GZ -0.003 2.2
213 CON GY -0.05 2.5
213 CON GZ -0.015 2.5
*
197 CON GY -0.4 1
197 CON GZ -0.12 1
197 CON GY -0.4 1.8
197 CON GZ -0.12 1.8
206 CON GY -0.4
206 CON GZ -0.12
206 CON GY -0.03 0.95
206 CON GZ -0.009 0.95
214 CON GY -0.03 0.5
214 CON GZ -0.009 0.5
214 CON GY -0.03 0.85
214 CON GZ -0.009 0.85
214 CON GY -0.03 1.2
214 CON GZ -0.009 1.2
214 CON GY -0.5 1.7
214 CON GZ -0.15 1.7
214 CON GY -0.03 2.2
214 CON GZ -0.009 2.2
214 CON GY -0.2 2.5
214 CON GZ -0.06 2.5
*
JOINT LOAD
50 61 FY -1.2 FZ -0.36
*
LOAD 6 LOADTYPE Wind  TITLE WIND LOAD (X DIR)
MEMBER LOAD
1 TO 4 9 TO 11 15 20 21 23 47 63 64 70 76 139 144 148 163 164 178 179 184 -
185 200 TO 203 208 TO 211 218 TO 221 231 250 258 289 TO 292 UNI GX 0.04
161 162 168 169 266 UNI GX 0.065
6 8 24 36 48 65 71 77 186 199 204 207 212 215 TO 217 227 TO 230 234 237 238 -
252 260 267 TO 270 276 UNI GX 0.052
283 TO 286 UNI GX 0.014
165 166 293 TO 300 UNI GX 0.02
124 222 223 235 236 251 253 TO 257 259 261 TO 265 273 TO 275 277 -
278 UNI GX 0.022
125 TO 127 129 279 TO 282 UNI GX 0.028
MEMBER LOAD
22 CON GX 0.2 4.65
22 CON GX 0.2 5.65
145 150 CON GX 0.2 0.3
145 150 CON GX 0.2 1.3
*
*wind on cable
MEMBER LOAD
218 TO 223 CON GX 0.05 0.2
218 TO 223 CON GX 0.05 0.65
218 TO 223 CON GX 0.05 1.1
222 223 CON GX 0.05 1.55
292 CON GX 0.05 0.45
291 CON GX 0.05 0.45
290 CON GX 0.05 0.45
289 CON GX 0.05 0.45
*
*wind on pipe
*
JOINT LOAD
50 61 FX 0.27
135 TO 142 FX 0.3
*
LOAD 7 LOADTYPE Wind  TITLE WIND LOAD(Z DIR)
MEMBER LOAD
1 TO 5 7 9 TO 11 15 20 21 23 47 63 64 70 76 139 144 148 163 164 175 176 178 -
179 184 185 200 TO 203 208 TO 211 218 TO 221 231 241 243 250 258 289 TO 291 -
292 UNI GZ 0.085
167 170 TO 173 UNI GZ 0.065
22 138 145 150 196 197 205 206 213 214 224 226 239 240 271 272 301 -
302 UNI GZ 0.052
13 14 16 17 112 113 222 223 225 232 233 UNI GZ 0.028
141 142 149 174 UNI GZ 0.02
MEMBER LOAD
8 24 36 48 65 71 77 CON GZ 0.2 0.42
8 24 36 48 65 71 77 CON GZ 0.2 1.42
234 252 260 CON GZ 0.2 0.21
```

```
234 252 260 CON GZ 0.2 1.21
48 77 CON GZ 0.2 1.84
276 CON GZ 0.2 0.21
48 77 CON GZ 0.2 2.84
276 CON GZ 0.2 1.21
*
*******************************************************
*************************
*****************************************************************************
*********LOAD COMBS FOR DESIGN*********************************************
LOAD COMB 100 1.4*DL
1 1.4
LOAD COMB 101 1.2DL+1.6*LL-1
1 1.2 2 1.6
LOAD COMB 102 1.2DL+1.6*LL-2
1 1.2 3 1.6
LOAD COMB 103 1.2DL+1.6*LL-3
1 1.2 4 1.6
LOAD COMB 104 1.2DL+1.6*LL-4
1 1.2 5 1.6
LOAD COMB 105 1.2DL+0.8WLX
1 1.2 6 0.8
LOAD COMB 106 1.2DL+0.5LL-1+1.6WLX
1 1.2 2 0.5 6 1.6
LOAD COMB 107 1.2DL+0.5LL-2+1.6WLX
1 1.2 3 0.5 6 1.6
LOAD COMB 108 1.2DL+0.5LL-3+1.6WLX
1 1.2 4 0.5 6 1.6
LOAD COMB 109 1.2DL+0.5LL-4+1.6WLX
1 1.2 5 0.5 6 1.6
LOAD COMB 110 0.9DL+1.6WLX
1 0.9 6 1.6
LOAD COMB 111 1.2DL+0.8WLZ
1 1.2 7 0.8
LOAD COMB 112 1.2DL+0.5LL-1+1.6WLZ
1 1.2 2 0.5 7 1.6
LOAD COMB 113 1.2DL+0.5LL-2+1.6WLZ
1 1.2 3 0.5 7 1.6
LOAD COMB 114 1.2DL+0.5LL-3+1.6WLZ
1 1.2 4 0.5 7 1.6
LOAD COMB 115 1.2DL+0.5LL-4+1.6WLZ
1 1.2 5 0.5 7 1.6
LOAD COMB 116 0.9DL+1.6WLZ
1 0.9 7 1.6
*****************************************************
********LOAD COMBS FOR DEFLECTION*****************************************
LOAD COMB 201 DL+LL-1
1 1.0 2 1.0
LOAD COMB 202 DL+LL-2
1 1.0 3 1.0
LOAD COMB 203 DL+LL-3
1 1.0 4 1.0
LOAD COMB 204 DL+LL-4
1 1.0 5 1.0
LOAD COMB 205 DL+LL-1+WLX
1 1.0 2 1.0 6 1.0
LOAD COMB 206 DL+LL-2+WLX
1 1.0 3 1.0 6 1.0
LOAD COMB 207 DL+LL-3+WLX
1 1.0 4 1.0 6 1.0
LOAD COMB 208 DL+LL-4+WLX
1 1.0 5 1.0 6 1.0
LOAD COMB 209 DL+WLX
1 1.0 6 1.0
LOAD COMB 210 DL+LL-1+WLZ
1 1.0 2 1.0 7 1.0
LOAD COMB 211 DL+LL-2+WLZ
1 1.0 3 1.0 7 1.0
LOAD COMB 212 DL+LL-3+WLZ
1 1.0 4 1.0 7 1.0
LOAD COMB 213 DL+LL-4+WLZ
1 1.0 5 1.0 7 1.0
LOAD COMB 214 DL+WLZ
1 1.0 7 1.0
*****************************************************
PERFORM ANALYSIS PRINT STATICS CHECK
LOAD LIST 1 TO 7 100 TO 116
```

```
PRINT SUPPORT REACTION LIST 1 TO 11 13 15 TO 17 28 37 49 TO 51 56 61 95 97 -
99 103 106 116 117 120 TO 128 133 TO 169
LOAD LIST 1 TO 7 201 TO 214
PRINT JOINT DISPLACEMENTS LIST 5 TO 8 145 TO 148
LOAD LIST 100 TO 116
PARAMETER 1
CODE LRFD
FYLD 27500 ALL
FU 41000 ALL
MAIN 250 MEMB 13 14 16 17 112 113 124 TO 127 129 141 142 149 165 166 174 232 -
233 235 236 239 240 245 TO 249 251 253 TO 257 259 261 TO 265 273 TO 275 277 -
278 TO 286 293 TO 300 303 TO 306
LY 4.38 MEMB 15 64 164 231
LZ 4.38 MEMB 10 15 47 63 64 76 139 144 163 164 231 250
LZ 3.41 MEMB 1 258
LZ 6 MEMB 5 7 175 176 239 TO 241 243
LY 6 MEMB 5 175
LZ 3.26 MEMB 6 8 186 199 204 207 212 215 TO 217 237 238 267 TO 270
LY 3.26 MEMB 186 199 216 217
KY 2.52 MEMB 125 TO 127 129 251 253 259 261 273 274 279 TO 282
LZ 2.2 MEMB 218 TO 221 289 TO 292
LZ 2.8 MEMB 224 226 301 302
TRACK 0 ALL
CHECK CODE ALL
STEEL TAKE OFF ALL
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	17-03-2020	13-04-2020	27-11-2023				Date	17-03-2020	13-04-2020	27-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	AS BUILT		
Prepared By	SGI	SGI	SGI				Prepared By	SGI	SGI	SGI		
Checked	SYC	SYC	SYC				Checked	SYC	SYC	SYC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 2 STAAD OUTPUT FOR STAIR SUPPORTING STRUCTURE

```
*************************************************
*                                               *
*           STAAD.Pro V8i SELECTseries4         *
*           Version  20.07.09.31                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    MAR 16, 2020               *
*           Time=    21: 9:55                   *
*                                               *
*      USER ID:                                 *
*************************************************
```

```
     1. STAAD SPACE
INPUT FILE: MD-216-3670-EG-ST-CAL-2305-A01.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 19-JUN-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET ECHO OFF
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**86**	**NUMBER OF MEMBERS**	**174**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 504

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.341439359E+01
Y = 0.123288619E+02
Z = 0.182945379E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -38.04
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 69.58 MY= 0.00 MZ= -129.18

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 38.04
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -69.58 MY= 0.00 MZ= 129.18

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X	-9.64290E-01	146
Y	-5.05509E-01	148
Z	2.62023E-01	123
RX	2.13658E-03	122
RY	-7.73194E-04	128
RZ	1.58327E-03	157

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE NONE TITLE LIVE-1 LOAD

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.256628572E+01
Y = 0.153587637E+02
Z = 0.203361892E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.260924594E+01
Y = 0.864691026E+01
Z = 0.209894251E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 1.58
SUMMATION FORCE-Y = -33.50
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 70.31 MY= 3.20 MZ= -111.40

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -1.58
SUMMATION FORCE-Y = 33.50
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -70.31 MY= -3.20 MZ= 111.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	1.52511E-01	148
Y =	-4.67067E-01	123
Z =	5.59682E-01	123
RX=	2.68748E-03	123
RY=	6.38102E-04	124
RZ=	1.34002E-03	120

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE LIVE-2 LOAD

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.256628572E+01
Y = 0.151075223E+02
Z = 0.203361892E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.260924594E+01
Y = 0.864691026E+01
Z = 0.209894251E+01

```
***TOTAL APPLIED LOAD ( MTON METE ) SUMMARY (LOADING     3 )
    SUMMATION FORCE-X =          -1.58
    SUMMATION FORCE-Y =         -33.50
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=          70.31  MY=          -3.20  MZ=         -63.41


***TOTAL REACTION LOAD( MTON METE ) SUMMARY (LOADING     3 )
    SUMMATION FORCE-X =           1.58
    SUMMATION FORCE-Y =          33.50
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=         -70.31  MY=           3.20  MZ=          63.41


MAXIMUM DISPLACEMENTS (   CM   /RADIANS) (LOADING      3)
          MAXIMUMS    AT NODE
   X = -5.90734E-01     146
   Y = -5.02495E-01     123
   Z =  5.68334E-01     123
   RX=  2.57619E-03     123
   RY=  6.41043E-04     124
   RZ=  1.39718E-03     120


       STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.     4
       LOADTYPE NONE  TITLE LIVE-3 LOAD


        CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X =  0.260924594E+01
                    Y =  0.864691026E+01
                    Z =  0.209894251E+01

        CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X =  0.256628572E+01
                    Y =  0.150777421E+02
                    Z =  0.203361892E+01

***TOTAL APPLIED LOAD ( MTON METE ) SUMMARY (LOADING     4 )
    SUMMATION FORCE-X =           0.00
    SUMMATION FORCE-Y =         -33.50
    SUMMATION FORCE-Z =           1.58

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=          94.30  MY=          -4.04  MZ=         -87.40
```

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 33.50
SUMMATION FORCE-Z = -1.58

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -94.30 MY= 4.04 MZ= 87.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-2.52003E-01	146
Y =	-5.11058E-01	123
Z =	1.11910E+00	149
RX=	2.76210E-03	123
RY=	-9.61878E-04	134
RZ=	1.36595E-03	120

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE NONE TITLE LIVE-4 LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.260924594E+01
Y = 0.864691026E+01
Z = 0.209894251E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.256628572E+01
Y = 0.153885440E+02
Z = 0.203361892E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -33.50
SUMMATION FORCE-Z = -1.58

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 46.32 MY= 4.04 MZ= -87.40

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 33.50
SUMMATION FORCE-Z = 1.58

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -46.32 MY= -4.04 MZ= 87.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	-1.86415E-01	146
Y =	-4.58503E-01	123
Z =	-8.72541E-01	150
RX=	2.50157E-03	123
RY=	6.47518E-04	124
RZ=	1.37125E-03	120

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE WIND TITLE WIND LOAD (X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.338114267E+01
Y = 0.124242297E+02
Z = 0.175107368E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 13.90
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 24.34 MZ= -172.71

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -13.90
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -24.34 MZ= 172.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	2.47172E+00	180
Y =	4.39525E-01	147
Z =	4.43246E-01	122
RX=	-9.06000E-05	50
RY=	3.03775E-03	61
RZ=	-3.22893E-03	139

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
LOADTYPE WIND TITLE WIND LOAD(Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.321665108E+01
Y = 0.995977578E+01
Z = 0.179949714E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 18.72

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 186.49 MY= -60.23 MZ= 0.00

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -18.72

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -186.49 MY= 60.23 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	3.97491E-01	116
Y =	-4.28270E-01	123
Z =	3.46932E+02	106
RX=	1.15452E-01	106
RY=	2.16533E+00	106
RZ=	1.30410E-02	164

********** END OF DATA FROM INTERNAL STORAGE **********

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	1	-0.03	7.05	0.06	0.00	0.00	0.00
	2	-0.10	5.11	0.02	0.00	0.00	0.00
	3	-0.09	8.53	0.04	0.00	0.00	0.00
	4	-0.10	3.30	-0.42	0.00	0.00	0.00
	5	-0.09	10.33	0.47	0.00	0.00	0.00
	6	0.03	-13.77	0.00	0.00	0.00	0.00
	7	-0.04	-23.47	-4.30	0.00	0.00	0.00
	100	-0.04	9.88	0.08	0.00	0.00	0.00
	101	-0.20	16.65	0.10	0.00	0.00	0.00
	102	-0.18	22.11	0.13	0.00	0.00	0.00
	103	-0.19	13.75	-0.60	0.00	0.00	0.00
	104	-0.18	25.00	0.83	0.00	0.00	0.00
	105	-0.02	-2.55	0.07	0.00	0.00	0.00
	106	-0.04	-11.01	0.07	0.00	0.00	0.00
	107	-0.04	-9.30	0.08	0.00	0.00	0.00
	108	-0.04	-11.91	-0.15	0.00	0.00	0.00
	109	-0.04	-8.40	0.30	0.00	0.00	0.00
	110	0.02	-15.68	0.05	0.00	0.00	0.00
	111	-0.07	-10.31	-3.37	0.00	0.00	0.00
	112	-0.15	-26.53	-6.80	0.00	0.00	0.00
	113	-0.15	-24.83	-6.78	0.00	0.00	0.00
	114	-0.15	-27.44	-7.01	0.00	0.00	0.00
	115	-0.15	-23.92	-6.57	0.00	0.00	0.00
	116	-0.10	-31.21	-6.82	0.00	0.00	0.00
2	1	0.07	9.64	0.51	0.00	0.00	0.00
	2	-0.48	6.82	1.23	0.00	0.00	0.00
	3	0.77	3.41	1.19	0.00	0.00	0.00
	4	0.17	1.27	0.87	0.00	0.00	0.00
	5	0.13	8.96	1.55	0.00	0.00	0.00
	6	-6.47	13.77	0.05	0.00	0.00	0.00
	7	0.11	-33.73	-5.06	0.00	0.00	0.00
	100	0.10	13.50	0.71	0.00	0.00	0.00
	101	-0.68	22.48	2.57	0.00	0.00	0.00
	102	1.32	17.02	2.51	0.00	0.00	0.00
	103	0.35	13.60	1.99	0.00	0.00	0.00
	104	0.29	25.90	3.09	0.00	0.00	0.00
	105	-5.09	22.58	0.65	0.00	0.00	0.00
	106	-10.51	37.01	1.30	0.00	0.00	0.00
	107	-9.88	35.30	1.28	0.00	0.00	0.00
	108	-10.18	34.23	1.12	0.00	0.00	0.00
	109	-10.20	38.08	1.46	0.00	0.00	0.00
	110	-10.29	30.71	0.54	0.00	0.00	0.00
	111	0.18	-15.42	-3.44	0.00	0.00	0.00
	112	0.03	-39.00	-6.87	0.00	0.00	0.00
	113	0.66	-40.70	-6.89	0.00	0.00	0.00
	114	0.36	-41.77	-7.05	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	115	0.34	-37.93	-6.71	0.00	0.00	0.00
	116	0.25	-45.30	-7.64	0.00	0.00	0.00
3	1	0.00	9.38	-0.04	0.00	0.00	0.00
	2	0.02	9.76	-0.01	0.00	0.00	0.00
	3	0.03	14.37	0.01	0.00	0.00	0.00
	4	0.02	15.58	-0.45	0.00	0.00	0.00
	5	0.03	8.55	0.45	0.00	0.00	0.00
	6	-0.30	-15.11	0.00	0.00	0.00	0.00
	7	-0.04	23.47	-4.32	0.00	0.00	0.00
	100	0.00	13.14	-0.05	0.00	0.00	0.00
	101	0.03	26.87	-0.06	0.00	0.00	0.00
	102	0.05	34.25	-0.03	0.00	0.00	0.00
	103	0.03	36.18	-0.76	0.00	0.00	0.00
	104	0.05	24.94	0.67	0.00	0.00	0.00
	105	-0.25	-0.83	-0.04	0.00	0.00	0.00
	106	-0.48	-8.04	-0.05	0.00	0.00	0.00
	107	-0.47	-5.73	-0.04	0.00	0.00	0.00
	108	-0.48	-5.13	-0.27	0.00	0.00	0.00
	109	-0.47	-8.65	0.18	0.00	0.00	0.00
	110	-0.49	-15.73	-0.03	0.00	0.00	0.00
	111	-0.04	30.04	-3.50	0.00	0.00	0.00
	112	-0.06	53.69	-6.95	0.00	0.00	0.00
	113	-0.06	56.00	-6.94	0.00	0.00	0.00
	114	-0.07	56.60	-7.17	0.00	0.00	0.00
	115	-0.06	53.09	-6.73	0.00	0.00	0.00
	116	-0.07	46.00	-6.94	0.00	0.00	0.00
4	1	-0.04	11.96	-0.53	0.00	0.00	0.00
	2	-1.02	11.81	-1.23	0.00	0.00	0.00
	3	0.86	7.20	-1.24	0.00	0.00	0.00
	4	-0.09	13.35	-1.58	0.00	0.00	0.00
	5	-0.07	5.66	-0.89	0.00	0.00	0.00
	6	-7.16	15.11	-0.05	0.00	0.00	0.00
	7	-0.03	33.73	-5.05	0.00	0.00	0.00
	100	-0.05	16.75	-0.74	0.00	0.00	0.00
	101	-1.67	33.25	-2.61	0.00	0.00	0.00
	102	1.32	25.87	-2.62	0.00	0.00	0.00
	103	-0.19	35.71	-3.16	0.00	0.00	0.00
	104	-0.16	23.41	-2.07	0.00	0.00	0.00
	105	-5.77	26.44	-0.67	0.00	0.00	0.00
	106	-12.00	44.44	-1.32	0.00	0.00	0.00
	107	-11.07	42.13	-1.33	0.00	0.00	0.00
	108	-11.54	45.21	-1.50	0.00	0.00	0.00
	109	-11.53	41.36	-1.15	0.00	0.00	0.00
	110	-11.48	34.94	-0.55	0.00	0.00	0.00
	111	-0.07	41.34	-4.68	0.00	0.00	0.00
	112	-0.60	74.23	-9.34	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	113	0.34	71.93	-9.34	0.00	0.00	0.00
	114	-0.14	75.00	-9.51	0.00	0.00	0.00
	115	-0.13	71.16	-9.17	0.00	0.00	0.00
	116	-0.08	64.74	-8.56	0.00	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
5	1	0.0176	-0.0382	0.0498	0.0000	0.0000	0.0000
	2	0.0182	-0.0218	0.0911	0.0001	-0.0001	0.0001
	3	-0.0635	-0.0373	0.1072	0.0001	-0.0002	0.0000
	4	-0.0309	-0.0191	0.2232	0.0001	-0.0001	0.0001
	5	-0.0143	-0.0400	-0.0248	0.0000	-0.0001	0.0000
	6	0.3295	0.0618	-0.0317	0.0000	0.0028	-0.0002
	7	-0.1035	0.0587	0.7497	0.0005	0.0065	0.0002
	201	0.0359	-0.0601	0.1409	0.0001	-0.0001	0.0000
	202	-0.0459	-0.0755	0.1570	0.0001	-0.0002	-0.0001
	203	-0.0133	-0.0573	0.2729	0.0001	-0.0002	0.0000
	204	0.0033	-0.0782	0.0249	0.0000	-0.0002	0.0000
	205	0.3654	0.0017	0.1092	0.0001	0.0026	-0.0002
	206	0.2837	-0.0137	0.1253	0.0001	0.0026	-0.0003
	207	0.3162	0.0044	0.2413	0.0001	0.0026	-0.0002
	208	0.3328	-0.0165	-0.0067	0.0000	0.0026	-0.0003
	209	0.3471	0.0235	0.0181	0.0000	0.0027	-0.0003
	210	-0.0676	-0.0014	0.8905	0.0006	0.0063	0.0002
	211	-0.1493	-0.0168	0.9067	0.0006	0.0063	0.0001
	212	-0.1168	0.0013	1.0226	0.0007	0.0063	0.0002
	213	-0.1002	-0.0196	0.7746	0.0005	0.0063	0.0001
	214	-0.0859	0.0204	0.7994	0.0005	0.0065	0.0001
6	1	0.0178	-0.0629	-0.0259	0.0000	-0.0001	0.0000
	2	0.0174	-0.0288	0.0285	0.0000	0.0001	0.0000
	3	-0.0628	-0.0147	0.0341	0.0000	-0.0003	0.0001
	4	-0.0310	-0.0014	0.1782	0.0002	-0.0001	0.0000
	5	-0.0144	-0.0422	-0.1157	-0.0001	-0.0001	0.0000
	6	0.3262	-0.0557	-0.0388	0.0000	0.0008	0.0000
	7	-0.1036	0.1491	1.2310	0.0004	0.0017	0.0001
	201	0.0352	-0.0917	0.0026	0.0000	0.0001	-0.0001
	202	-0.0450	-0.0776	0.0082	0.0000	-0.0004	0.0000
	203	-0.0132	-0.0642	0.1523	0.0001	-0.0001	0.0000
	204	0.0034	-0.1050	-0.1416	-0.0002	-0.0002	0.0000
	205	0.3614	-0.1473	-0.0362	0.0000	0.0009	-0.0001
	206	0.2812	-0.1333	-0.0307	0.0000	0.0004	0.0000
	207	0.3130	-0.1199	0.1135	0.0001	0.0007	0.0000
	208	0.3296	-0.1607	-0.1804	-0.0002	0.0006	0.0000
	209	0.3440	-0.1185	-0.0647	-0.0001	0.0008	0.0000
	210	-0.0684	0.0575	1.2336	0.0004	0.0018	0.0001
	211	-0.1487	0.0715	1.2392	0.0004	0.0014	0.0002
	212	-0.1168	0.0849	1.3834	0.0005	0.0016	0.0001
	213	-0.1003	0.0441	1.0894	0.0002	0.0015	0.0001
	214	-0.0859	0.0863	1.2051	0.0004	0.0017	0.0001
7	1	0.0118	-0.0500	0.0492	0.0003	0.0000	-0.0001
	2	0.0117	-0.0465	0.0902	0.0007	0.0000	0.0000
	3	-0.1056	-0.0675	0.1063	0.0007	-0.0001	0.0000
	4	-0.0469	-0.0702	0.2225	0.0008	0.0000	0.0000

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	5	-0.0470	-0.0438	-0.0260	0.0006	-0.0001	0.0000
	6	0.3615	0.0672	-0.0318	0.0000	-0.0008	-0.0002
	7	0.0308	-0.0696	0.7512	0.0010	0.0003	-0.0001
	201	0.0235	-0.0965	0.1394	0.0010	0.0000	-0.0001
	202	-0.0938	-0.1175	0.1554	0.0010	-0.0002	-0.0001
	203	-0.0351	-0.1202	0.2716	0.0010	-0.0001	-0.0001
	204	-0.0352	-0.0938	0.0232	0.0009	-0.0001	-0.0001
	205	0.3850	-0.0293	0.1076	0.0010	-0.0008	-0.0003
	206	0.2677	-0.0502	0.1236	0.0010	-0.0009	-0.0003
	207	0.3264	-0.0530	0.2398	0.0011	-0.0009	-0.0003
	208	0.3263	-0.0265	-0.0086	0.0009	-0.0009	-0.0003
	209	0.3733	0.0172	0.0174	0.0003	-0.0008	-0.0003
	210	0.0543	-0.1661	0.8905	0.0020	0.0002	-0.0001
	211	-0.0630	-0.1871	0.9066	0.0020	0.0001	-0.0002
	212	-0.0043	-0.1898	1.0228	0.0020	0.0002	-0.0002
	213	-0.0044	-0.1634	0.7743	0.0019	0.0001	-0.0002
	214	0.0426	-0.1196	0.8003	0.0013	0.0002	-0.0002
8	1	0.0120	-0.0737	-0.0260	0.0000	0.0000	-0.0001
	2	0.0109	-0.0486	0.0285	0.0000	0.0000	0.0000
	3	-0.1045	-0.0294	0.0342	0.0000	-0.0002	0.0001
	4	-0.0464	-0.0549	0.1795	0.0002	0.0000	0.0000
	5	-0.0472	-0.0231	-0.1168	-0.0001	-0.0002	0.0000
	6	0.3578	-0.0592	-0.0387	0.0000	-0.0023	-0.0002
	7	0.0332	-0.1064	1.2386	0.0004	0.0003	-0.0001
	201	0.0229	-0.1223	0.0025	0.0000	0.0000	-0.0001
	202	-0.0925	-0.1031	0.0082	0.0000	-0.0002	0.0000
	203	-0.0344	-0.1286	0.1535	0.0001	0.0000	-0.0001
	204	-0.0352	-0.0968	-0.1428	-0.0002	-0.0002	0.0000
	205	0.3808	-0.1815	-0.0362	0.0000	-0.0023	-0.0004
	206	0.2653	-0.1623	-0.0305	0.0000	-0.0025	-0.0003
	207	0.3234	-0.1878	0.1148	0.0001	-0.0023	-0.0003
	208	0.3227	-0.1560	-0.1815	-0.0002	-0.0025	-0.0003
	209	0.3698	-0.1329	-0.0647	-0.0001	-0.0023	-0.0003
	210	0.0561	-0.2287	1.2411	0.0003	0.0003	-0.0002
	211	-0.0593	-0.2095	1.2468	0.0003	0.0001	-0.0001
	212	-0.0012	-0.2350	1.3922	0.0005	0.0003	-0.0002
	213	-0.0020	-0.2032	1.0958	0.0002	0.0001	-0.0001
	214	0.0452	-0.1801	1.2126	0.0003	0.0003	-0.0002
145	1	-0.9364	-0.0708	0.0416	0.0001	-0.0001	0.0015
	2	0.0807	-0.0303	0.0667	0.0001	0.0002	0.0000
	3	-0.4177	-0.0141	0.1808	0.0001	-0.0005	0.0005
	4	-0.1515	-0.0005	1.1191	0.0002	-0.0003	0.0002
	5	-0.1855	-0.0439	-0.8716	-0.0001	0.0000	0.0003
	6	2.4445	-0.0652	-0.0413	0.0000	0.0001	-0.0024
	7	-0.0869	0.1558	4.8363	0.0008	0.0002	-0.0001
	201	-0.8556	-0.1010	0.1083	0.0001	0.0001	0.0014

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	202	-1.3541	-0.0848	0.2225	0.0001	-0.0006	0.0019
	203	-1.0879	-0.0712	1.1607	0.0002	-0.0004	0.0017
	204	-1.1219	-0.1146	-0.8299	0.0000	-0.0001	0.0017
	205	1.5888	-0.1662	0.0670	0.0001	0.0003	-0.0009
	206	1.0904	-0.1501	0.1812	0.0001	-0.0005	-0.0004
	207	1.3566	-0.1365	1.1194	0.0003	-0.0002	-0.0007
	208	1.3226	-0.1798	-0.8712	0.0000	0.0001	-0.0006
	209	1.5081	-0.1360	0.0003	0.0001	0.0001	-0.0009
	210	-0.9425	0.0548	4.9446	0.0010	0.0003	0.0014
	211	-1.4410	0.0710	5.0588	0.0010	-0.0004	0.0019
	212	-1.1748	0.0846	5.9970	0.0011	-0.0002	0.0016
	213	-1.2087	0.0412	4.0064	0.0008	0.0001	0.0017
	214	-1.0233	0.0851	4.8779	0.0009	0.0001	0.0014
146	1	-0.9643	-0.0833	0.0416	0.0000	-0.0001	0.0015
	2	0.1523	-0.0524	0.0667	0.0001	0.0002	0.0000
	3	-0.5907	-0.0309	0.1808	0.0001	-0.0005	0.0006
	4	-0.2520	-0.0588	1.1191	0.0002	-0.0003	0.0003
	5	-0.1864	-0.0245	-0.8716	-0.0001	0.0000	0.0002
	6	2.4715	-0.0686	-0.0413	0.0000	0.0000	-0.0024
	7	-0.0418	-0.1131	4.8363	0.0008	0.0002	0.0002
	201	-0.8120	-0.1357	0.1082	0.0001	0.0001	0.0015
	202	-1.5550	-0.1142	0.2224	0.0001	-0.0006	0.0021
	203	-1.2163	-0.1421	1.1606	0.0002	-0.0004	0.0018
	204	-1.1507	-0.1078	-0.8300	0.0000	-0.0001	0.0017
	205	1.6595	-0.2044	0.0670	0.0001	0.0002	-0.0009
	206	0.9165	-0.1828	0.1811	0.0001	-0.0006	-0.0003
	207	1.2552	-0.2107	1.1194	0.0002	-0.0004	-0.0005
	208	1.3208	-0.1764	-0.8713	0.0000	-0.0001	-0.0006
	209	1.5072	-0.1520	0.0003	0.0000	-0.0001	-0.0008
	210	-0.8538	-0.2488	4.9445	0.0009	0.0003	0.0016
	211	-1.5969	-0.2272	5.0587	0.0009	-0.0004	0.0022
	212	-1.2581	-0.2552	5.9970	0.0010	-0.0002	0.0020
	213	-1.1925	-0.2209	4.0063	0.0008	0.0001	0.0019
	214	-1.0061	-0.1964	4.8779	0.0008	0.0001	0.0017
147	1	-0.9359	-0.4736	0.0188	0.0001	-0.0001	0.0015
	2	0.0808	-0.0308	0.1281	0.0001	0.0002	0.0000
	3	-0.4176	-0.1251	0.0352	0.0002	-0.0005	0.0005
	4	-0.1514	-0.0387	1.0359	0.0004	-0.0003	0.0002
	5	-0.1854	-0.1172	-0.8725	-0.0001	0.0000	0.0003
	6	2.4445	0.4395	-0.0182	0.0000	0.0001	-0.0024
	7	-0.0869	0.2069	4.8836	0.0012	0.0001	-0.0001
	201	-0.8550	-0.5044	0.1470	0.0002	0.0001	0.0015
	202	-1.3535	-0.5987	0.0541	0.0003	-0.0006	0.0020
	203	-1.0873	-0.5123	1.0547	0.0005	-0.0004	0.0017
	204	-1.1213	-0.5908	-0.8537	0.0000	-0.0001	0.0018
	205	1.5894	-0.0648	0.1288	0.0002	0.0003	-0.0009

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	206	1.0910	-0.1592	0.0359	0.0003	-0.0005	-0.0004
	207	1.3572	-0.0728	1.0366	0.0005	-0.0002	-0.0007
	208	1.3232	-0.1513	-0.8718	0.0000	0.0001	-0.0006
	209	1.5086	-0.0341	0.0007	0.0001	0.0001	-0.0009
	210	-0.9419	-0.2974	5.0306	0.0014	0.0002	0.0014
	211	-1.4404	-0.3918	4.9377	0.0015	-0.0005	0.0019
	212	-1.1742	-0.3054	5.9384	0.0017	-0.0003	0.0016
	213	-1.2081	-0.3838	4.0299	0.0012	0.0000	0.0017
	214	-1.0227	-0.2667	4.9024	0.0013	0.0000	0.0014
148	1	-0.9637	-0.5055	0.0188	0.0001	-0.0001	0.0015
	2	0.1525	-0.0637	0.1281	0.0001	0.0002	0.0000
	3	-0.5905	-0.1781	0.0352	0.0002	-0.0005	0.0006
	4	-0.2518	-0.1590	1.0359	0.0004	-0.0003	0.0003
	5	-0.1862	-0.0828	-0.8725	-0.0001	0.0000	0.0002
	6	2.4715	0.4360	-0.0182	0.0000	0.0000	-0.0024
	7	-0.0419	-0.1893	4.8836	0.0012	0.0001	0.0002
	201	-0.8112	-0.5692	0.1469	0.0002	0.0001	0.0015
	202	-1.5542	-0.6836	0.0540	0.0002	-0.0006	0.0021
	203	-1.2155	-0.6645	1.0547	0.0004	-0.0004	0.0019
	204	-1.1499	-0.5883	-0.8538	0.0000	-0.0001	0.0018
	205	1.6603	-0.1333	0.1287	0.0002	0.0002	-0.0008
	206	0.9173	-0.2476	0.0358	0.0002	-0.0006	-0.0002
	207	1.2560	-0.2286	1.0365	0.0004	-0.0004	-0.0005
	208	1.3216	-0.1523	-0.8719	0.0000	-0.0001	-0.0006
	209	1.5078	-0.0696	0.0006	0.0001	-0.0001	-0.0008
	210	-0.8530	-0.7586	5.0305	0.0014	0.0003	0.0017
	211	-1.5961	-0.8729	4.9376	0.0014	-0.0005	0.0023
	212	-1.2574	-0.8538	5.9383	0.0017	-0.0003	0.0020
	213	-1.1918	-0.7776	4.0299	0.0012	0.0000	0.0019
	214	-1.0055	-0.6948	4.9024	0.0013	0.0000	0.0017

********** END OF LATEST ANALYSIS RESULT **********

STAAD.Pro CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.081	116
		26.18 T	0.08	-0.13	1.23
2	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.150	116
		39.67 T	0.30	-0.02	1.23
3	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.278	114
		51.15 C	0.01	-0.08	1.23
4	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.412	114
		67.31 C	0.27	0.33	1.23
5	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.576	114
		0.16 C	-2.12	-5.33	3.18
6	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.131	116
		1.23 T	0.20	-0.08	1.63
7	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.192	113
		0.05 T	0.07	-5.78	1.59
8	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.213	104
		0.56 C	-0.06	-1.55	1.49
9	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.253	114
		32.07 C	0.20	0.68	0.00
10	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.259	114
		37.75 C	0.17	-0.19	0.00
11	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.144	114
		16.81 C	0.25	0.37	0.00
13	LD L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.478	106
		13.69 C	0.00	-0.31	3.44
14	LD L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.139	113
		0.90 C	-0.23	-0.29	3.18
15	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.500	114
		17.78 C	-2.15	0.44	0.00
16	LD L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.464	106
		11.26 C	0.00	-0.34	3.70

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
17	LD L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.168	113
		0.49 C	-0.31	-0.34	3.70
20	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.296	114
		40.72 C	0.20	0.68	2.18
21	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.528	114
		47.11 C	0.27	0.33	0.00
22	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.434	103
		0.37 T	0.00	-1.94	4.48
23	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.344	109
		30.14 C	0.05	0.94	4.38
24	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.133	106
		0.06 T	0.17	0.31	1.63
36	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.120	110
		0.05 T	0.19	-0.11	1.63
47	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.094	109
		19.26 C	0.03	0.95	0.00
48	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.389	104
		0.48 C	0.00	-2.72	1.63
63	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.266	113
		4.02 C	1.32	0.36	0.00
64	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.104	113
		5.19 C	-0.38	-0.28	2.19
65	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.123	108
		0.98 C	0.18	0.11	1.63
70	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.120	109
		17.53 C	-0.02	-0.71	0.00
71	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.145	106
		0.02 T	0.19	0.33	1.63
76	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.058	112
		0.73 C	0.26	-0.26	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
77	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.390	103
		0.58 C	0.00	-2.72	1.63
112	LD L130X130X13	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1A-C	0.424	106
		11.89 C	0.00	-0.31	3.44
113	LD L130X130X13	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.136	113
		0.69 C	-0.23	-0.29	3.18
124	ST L100X100X10	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1A-C	0.701	114
		9.49 C	0.00	-0.01	1.46
125	ST L130X130X13	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.094	114
		11.61 T	0.05	0.01	2.73
126	ST L130X130X13	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.043	109
		0.56 T	-0.11	0.01	1.96
127	ST L130X130X13	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1A-C	0.826	114
		22.96 C	-0.01	0.02	2.73
129	ST L130X130X13	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1A-C	0.460	114
		12.75 C	0.00	0.02	2.73
138	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.112	114
		0.10 T	0.17	-0.13	0.00
139	ST IPE400	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.359	113
		23.32 C	1.54	-0.27	1.95
141	ST L90X90X8	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.093	102
		1.74 C	0.00	-0.01	1.26
142	ST L90X90X8	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.054	114
		0.16 C	0.03	-0.01	1.26
144	ST IPE400	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.082	116
		6.15 T	0.36	-0.07	1.95
145	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.106	112
		0.55 T	0.17	0.05	1.60
148	ST IPE400	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.089	104
		16.38 C	0.03	1.01	2.43

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
149 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.232	101
		1.80 C	0.00	-0.01	1.45
150 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.117	112
		0.47 T	-0.17	-0.18	0.27
161 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.086	106
		0.32 T	0.24	-0.07	1.45
162 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.092	112
		2.14 T	0.21	-0.22	1.45
163 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.142	114
		28.41 C	0.25	0.37	2.19
164 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.506	114
		19.30 C	-2.15	0.44	2.19
165 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.097	101
		1.59 C	0.00	-0.01	1.35
166 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.484	102
		4.54 C	0.00	-0.01	1.35
167 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.325	103
		0.20 T	0.00	-2.64	2.99
168 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.040	101
		0.93 T	0.00	0.00	0.13
169 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.116	101
		2.66 T	0.00	0.00	0.13
170 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.408	113
		0.08 C	0.72	1.36	0.00
171 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.082	112
		0.87 T	0.17	0.34	1.60
172 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.269	113
		0.22 C	-0.51	1.36	0.00
173 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.079	112
		0.22 T	0.17	0.34	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
174 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.226	101
		2.03 C	0.00	-0.01	1.35
175 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.569	114
		0.12 T	-2.12	-5.33	0.00
176 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.385	113
		2.49 T	-0.15	-11.33	0.00
178 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.858	114
		1.30 T	-4.60	0.26	2.73
179 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.914	114
		11.43 C	-4.61	-0.86	2.73
184 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.453	114
		3.13 C	-2.27	0.76	2.73
185 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.847	114
		6.96 C	-4.45	-0.16	2.73
186 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.186	109
		0.05 T	0.31	-0.06	1.63
196 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.419	109
		0.28 C	-0.21	2.26	2.80
197 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.454	106
		0.53 C	0.18	2.60	2.80
199 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.097	108
		0.20 C	-0.16	0.05	0.00
200 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.110	114
		0.19 C	-0.58	-0.07	0.75
201 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.139	114
		4.04 C	-0.65	-0.32	0.75
202 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.089	109
		0.34 T	0.07	-2.46	0.75
203 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.083	106
		0.31 T	-0.02	-2.53	0.75

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
204 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.159	114
		0.05 T	-0.01	-1.34	0.00
205 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.460	106
		0.07 C	-0.07	3.18	2.80
206 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.484	106
		0.12 C	0.06	3.42	2.80
207 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.164	114
		1.26 T	-0.11	-0.85	0.00
208 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.227	114
		0.21 C	-1.22	0.07	0.75
209 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.263	114
		6.45 C	-1.26	-0.45	0.75
210 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.198	114
		3.43 C	-0.95	0.43	0.75
211 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.206	114
		3.84 C	-0.98	-0.51	0.00
212 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.496	114
		0.17 T	-0.01	-4.27	0.00
213 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.601	106
		0.49 T	-0.07	4.25	2.80
214 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.654	106
		0.65 T	0.05	4.70	2.80
215 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.375	114
		1.36 C	-0.11	-2.65	0.00
216 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.186	109
		0.02 T	0.31	-0.06	0.00
217 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.107	114
		0.33 C	0.13	0.27	1.63
218 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.044	114
		0.36 C	-0.18	-0.32	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
219 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.045	114
		2.83 C	-0.15	0.36	0.00
220 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.042	106
		0.02 T	0.00	-1.34	1.10
221 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.044	106
		0.14 T	0.00	-1.42	1.10
222 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.088	112
		0.17 T	-0.02	0.11	1.65
223 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.144	106
		0.03 T	0.00	-0.31	1.28
224 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.155	110
		0.02 T	0.01	-1.33	0.00
225 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.095	114
		0.14 C	-0.04	-0.02	1.40
226 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.163	106
		0.06 C	-0.01	-1.40	0.00
227 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.016	108
		0.03 C	-0.02	-0.04	1.63
228 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.016	106
		0.03 C	-0.02	-0.04	0.00
229 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.014	109
		0.01 C	-0.02	-0.03	1.63
230 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.014	107
		0.01 C	-0.02	-0.03	0.00
231 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.098	113
		4.09 C	-0.38	-0.28	0.00
232 LD	L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.395	106
		9.28 C	0.00	-0.34	3.70
233 LD	L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.165	113
		0.36 C	-0.31	-0.34	3.70

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
234	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.134	109
		0.33 C	0.17	0.31	0.00
235	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-T	0.218	116
		9.52 T	-0.02	0.00	2.04
236	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.478	114
		11.87 C	0.00	0.02	2.04
237	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.127	116
		0.30 C	0.20	-0.08	0.00
238	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.202	104
		0.38 C	-0.06	-1.45	0.00
239	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.290	114
		1.20 C	0.09	-0.12	3.18
240	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.281	114
		1.40 C	0.09	-0.12	0.00
241	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.385	113
		2.39 T	-0.16	-11.33	1.59
243	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.193	113
		0.58 T	0.08	-5.75	0.00
245	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.110	114
		2.05 T	0.00	0.00	0.00
246	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.399	114
		1.47 C	0.00	0.00	0.00
247	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.302	114
		1.26 C	0.00	0.00	0.00
248	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.097	114
		1.81 T	0.00	0.00	0.00
249	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.091	114
		1.69 T	0.00	0.00	0.00
250	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.080	109
		17.93 C	-0.02	-0.71	2.19

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
251	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.107	109
		0.70 T	0.13	0.01	1.63
252	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.149	108
		0.78 T	0.19	0.33	0.00
253	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.107	108
		0.70 T	0.12	0.01	0.00
254	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.099	114
		8.06 T	0.00	-0.01	1.36
255	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.666	114
		10.43 C	0.00	-0.01	1.36
256	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.109	114
		9.03 T	0.00	-0.01	1.36
257	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.585	114
		9.14 C	0.00	-0.01	1.36
258	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.087	104
		21.08 C	-0.02	-0.61	2.18
259	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.266	116
		6.50 C	-0.01	0.01	1.63
260	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.120	110
		0.04 T	0.19	-0.11	0.00
261	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.095	114
		7.81 T	0.01	0.01	0.00
262	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.246	104
		3.74 C	0.00	-0.01	1.36
263	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.786	114
		12.38 C	0.00	-0.01	1.36
264	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.103	114
		8.76 T	0.00	-0.01	1.02
265	ST L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.354	114
		8.88 C	0.00	-0.01	1.02

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
266 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.079	116
		0.13 C	-0.24	0.00	1.45
267 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.166	114
		1.15 T	0.11	0.84	1.63
268 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.165	114
		0.06 C	0.00	1.42	1.63
269 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.478	114
		1.48 C	0.13	3.47	1.63
270 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.499	114
		0.05 T	0.00	4.36	1.63
271 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.165	114
		0.01 C	-0.08	-0.03	1.40
272 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.184	114
		0.01 C	-0.08	-0.06	1.40
273 ST	L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.214	116
		5.11 C	-0.01	0.01	1.63
274 ST	L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.106	108
		0.65 T	0.13	0.01	0.00
275 ST	L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.114	114
		9.38 T	0.00	-0.01	1.46
276 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.119	107
		0.19 C	0.18	0.11	0.00
277 ST	L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.548	114
		9.86 C	0.00	-0.01	1.27
278 ST	L100X100X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.099	104
		3.14 C	0.00	-0.01	1.27
279 ST	L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.098	114
		11.60 T	0.06	0.01	0.00
280 ST	L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.043	109
		0.59 T	-0.11	0.01	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
281	ST L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.828	114
		22.98 C	-0.01	0.02	0.00
282	ST L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.462	114
		12.76 C	0.00	0.02	0.00
283	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.048	112
		0.08 T	0.00	0.01	0.00
284	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.337	112
		0.07 T	0.09	0.00	0.00
285	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.066	109
		0.07 T	-0.02	0.00	1.36
286	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.072	109
		0.05 T	-0.02	0.00	1.36
289	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.050	106
		1.09 C	0.02	-1.39	1.10
290	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.053	106
		1.15 C	-0.02	-1.48	1.10
291	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.048	106
		0.84 T	0.03	-1.32	1.10
292	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.050	106
		0.90 T	-0.02	-1.40	1.10
293	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.033	103
		0.56 C	0.00	0.00	1.97
294	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.017	102
		0.58 T	0.00	0.00	0.00
295	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.036	106
		0.37 C	-0.02	0.00	0.98
296	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.043	116
		0.74 C	0.00	0.00	1.97
297	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.026	100
		0.44 C	0.00	0.00	1.97

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
298 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.019	112
		0.66 T	0.00	0.00	0.00
299 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.061	113
		1.04 C	0.00	0.00	1.97
300 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.044	114
		1.53 T	0.00	0.00	0.00
301 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.164	106
		0.34 C	-0.01	1.39	1.40
302 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.175	106
		0.42 C	0.01	1.48	1.40
303 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.057	110
		0.24 C	0.00	0.00	0.00
304 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.135	114
		0.56 C	0.00	0.00	0.00
305 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.026	114
		0.49 T	0.00	0.00	0.00
306 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.078	108
		0.32 C	0.00	0.00	0.00

********** END OF TABULATED RESULT OF DESIGN **********

STEEL TAKE-OFF

	PROFILE	LENGTH(METE)	WEIGHT(MTON)
ST	IPE400	95.08	6.294
ST	IPE240	79.36	2.431
LD	L130X130X13	56.14	2.841
ST	UPN240	5.98	0.198
ST	L100X100X10	45.24	0.680
ST	L130X130X13	20.29	0.513
ST	L90X90X8	31.78	0.346
ST	IPE300	21.96	0.925
ST	UPN140	18.78	0.300
ST	L65X65X6	29.22	0.172
	TOTAL =		14.701

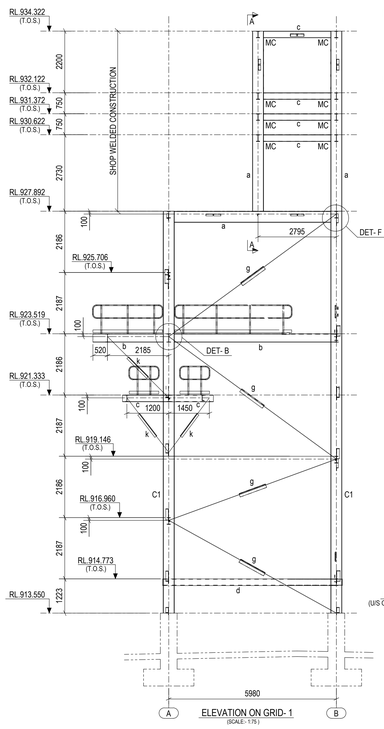
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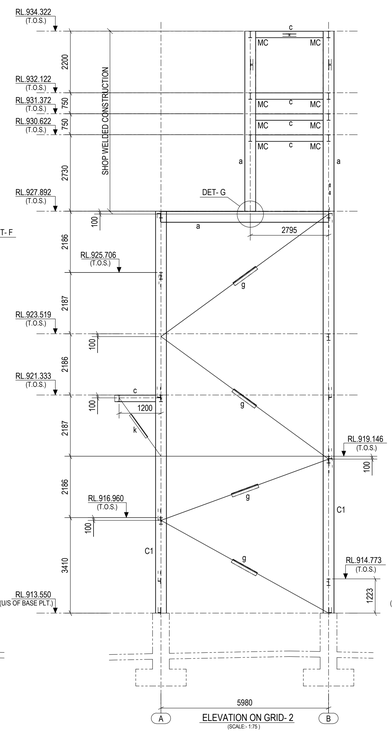
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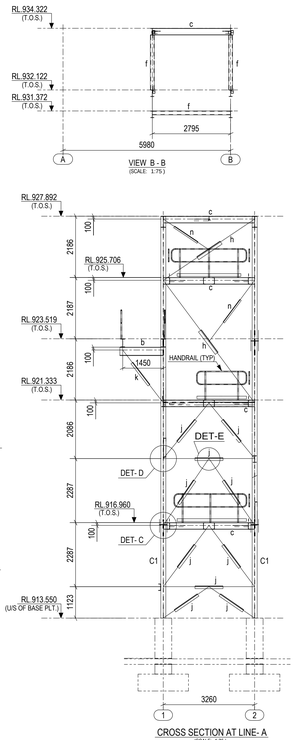
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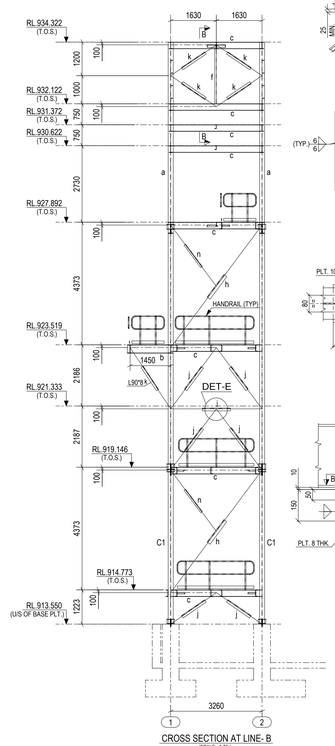
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*************************************************************
*        For questions on STAAD.Pro, please contact         *
*        Bentley Systems or Partner offices                 *
*                                                           *
*            Telephone             Web / Email              *
*  USA       +1  (714) 974-2500                             *
*  UK        +44 (0) 808 101 9246                           *
*  SINGAPORE +65 6225-6158                                  *
*  FRANCE    +33 (0) 1 55238400                             *
*  GERMANY   +49 0931 40468                                 *
*  INDIA     +91 (033) 4006-2021                            *
*  JAPAN     +81 (03)5952-6500    http://www.ctc-g.co.jp    *
*  CHINA     +86 21 6288 4040                               *
*  THAILAND  +66 (0)2645-1018/19 partha.p@reisoftwareth.com*
*                                                           *
*  Worldwide  http://selectservices.bentley.com/en-US/      *
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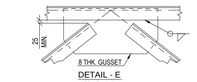
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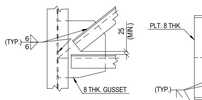


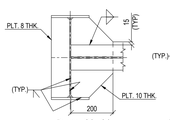


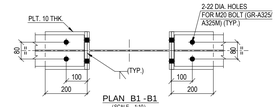


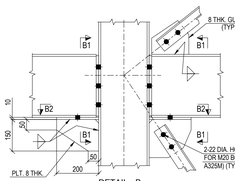


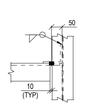




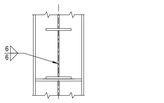


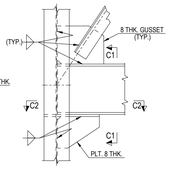


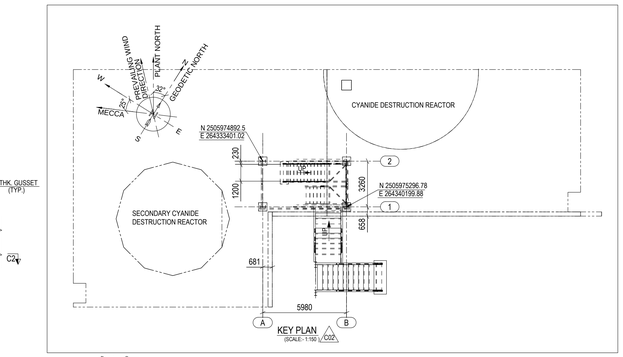


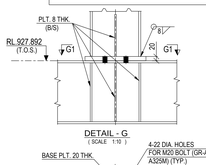


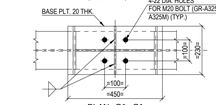




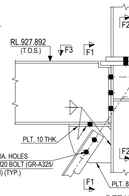


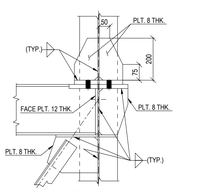


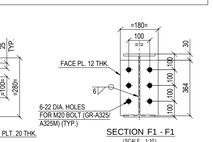


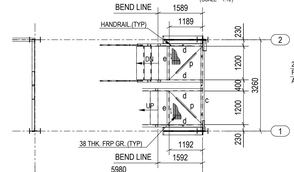


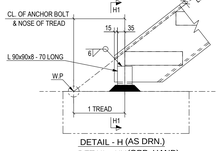


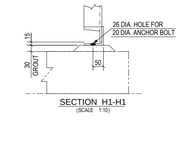


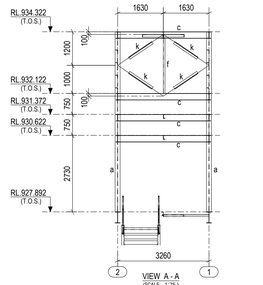


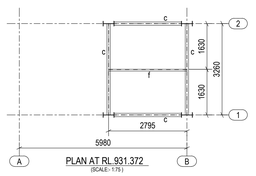


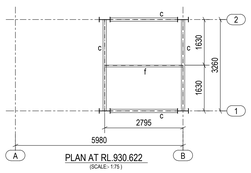


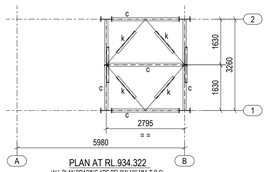


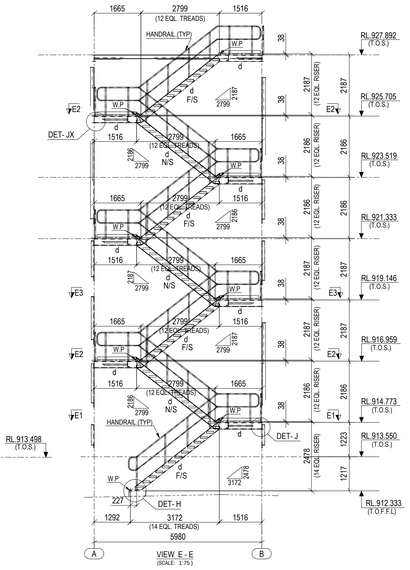


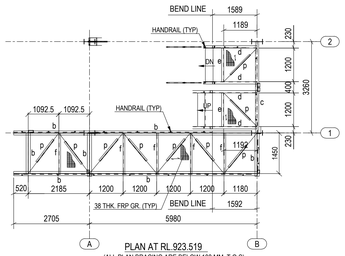


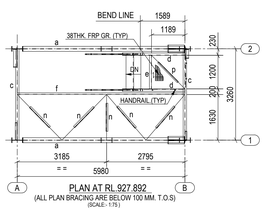


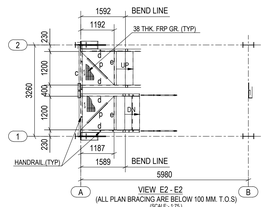


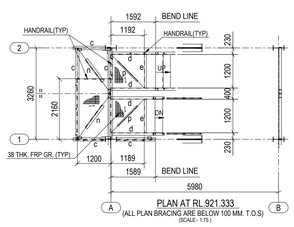


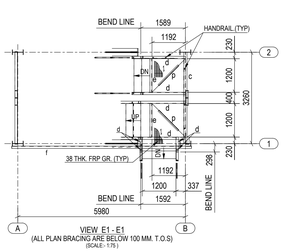


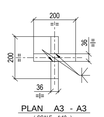


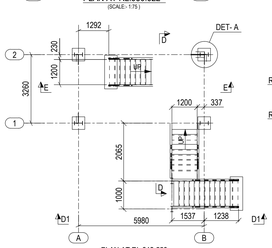


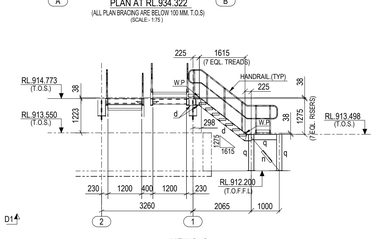


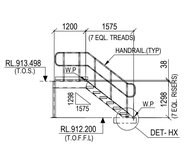


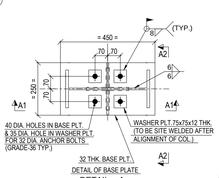


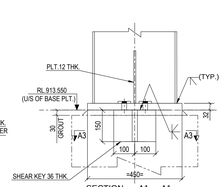


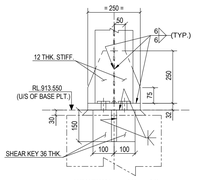












SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(T)	FACTORED SHEAR FORCE(T)	REMARKS
C1	IPE400	I	-	-	MATERIAL GRADE:S275 JR
a	IPE400	I	4.0	5.0	
b	IPE300	I	3.0	4.0	
c	IPE240	I	16.0	5.0	
d	UPN240	[	2.0	4.0	
e	UPN200	[	-	-	
f	UPN140	[	2.0	-	
g	L - 130X130X13	⊓	14.0	-	
h	L - 130X130X13	L	24.0	-	
j	L - 100X100X10	L	13.0	-	
k	L - 90X90X8	L	5.0	-	
n	L - 65X65X6	L	3.0	-	
p	L - 50X50X6	L	-	-	
q	IPE200	I	-	-	

ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION.
FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.

LEGENDS:-

TOS - TOP OF STEEL
U/S - UNDER SIDE
RL - REDUCED LEVEL
T.O.F.F.L - TOP OF FINISHED FLOOR LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN - DOWN
W.P. WORKING POINT
GR. - GRATING
MC. - MOMENT CONNECTION
TYP - TYPICAL
PLT - PLATE
THK. - THICKNESS
UNO - UNLESS NOTED OTHERWISE

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr A36.
3. HANDRAIL PIPE / POST MATERIAL SHALL BE AS PER ASTM A53 Gr.B
4. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
5. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
6. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	27/11/2023	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED D01	SJH	SYC	SGI	PRI	
C02	26/09/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	SJH	SYC	SGI	PRI	
C01	27/04/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C01	JVM	SYC	SGI	PRI	
A01	17/03/2020	ISSUED FOR REFERENCE	JVM	SYC	SGI	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REFERENCE DOCUMENTS
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A01	17/03/2020	ISSUED FOR REFERENCE

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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LARSEN & TOUBRO DRAWING NUMBER	0190-L5_450_S_G_26061_001
REVISION	2
OUTOTEC DRAWING NUMBER	900707__GAU01__RN12__MTA01__36706
DOCUMENT ID	OU100669148
REVISION	2
DRAWN	JVM 12/03/2020
CHECKED	SYC 16/03/2020
ENGINEER	SGI 13/03/2020
APPROVED	PRI 17/03/2020
EMPLOYER	
SCALE	1:750,500
	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CYANIDE DESTRUCTION
PLAN, ELEVATIONS & DETAILS OF STAIRBLOCK

DRAWING NUMBER: MD-216-3670-EG-ST-GAD-2303 | REV: D01

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

PROPRIETARY & CONFIDENTIAL

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START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
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SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
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1102 1100 1131 1134 1128; 1103 1130 1135 1136 1131; 1104 1135 1137 1138 1136;
1105 1137 1139 1140 1138; 1106 1139 1141 1142 1140; 1107 1132 1143 1135 1130;
1108 1143 1144 1137 1135; 1109 1144 1145 1139 1137; 1110 1145 1146 1141 1139;
1111 1133 1147 1143 1132; 1112 1147 1148 1144 1143; 1113 1148 1149 1145 1144;
1114 1149 1150 1146 1145; 1115 1029 1046 1147 1133; 1116 1046 1047 1148 1147;
1117 1047 1048 1149 1148; 1118 1048 1049 1150 1149; 1119 1131 1136 1151 1134;
1120 1136 1138 1152 1151; 1121 1138 1140 1153 1152; 1122 1140 1142 1154 1153;
1123 1141 1155 1156 1142; 1124 1146 1157 1155 1141; 1125 1150 1158 1157 1146;
1126 1049 1054 1158 1150; 1127 1142 1156 1159 1154; 1128 1155 1160 1161 1156;
1129 1160 1162 1163 1161; 1130 1162 1164 1165 1163; 1131 1157 1166 1160 1155;
1132 1166 1167 1162 1160; 1133 1167 1168 1164 1162; 1134 1158 1169 1166 1157;
1135 1169 1170 1167 1166; 1136 1170 1171 1168 1167; 1137 1054 1067 1169 1158;
1138 1067 1068 1170 1169; 1139 1068 1069 1171 1170; 1140 1156 1161 1172 1159;
1141 1161 1163 1173 1172; 1142 1163 1165 1174 1173; 1143 1164 1175 1176 1165;
1144 1168 1177 1175 1164; 1145 1171 1178 1177 1168; 1146 1069 1074 1178 1171;
1147 1165 1176 1179 1174; 1148 1175 1180 1181 1176; 1149 1177 1182 1180 1175;
1150 1178 1183 1182 1177; 1151 1074 1079 1183 1178; 1152 1176 1181 1184 1179;
1153 1180 1185 1186 1181; 1154 1182 1187 1185 1180; 1155 1183 1188 1187 1182;
1156 1079 1084 1188 1183; 1157 1181 1186 1189 1184; 1158 1190 1191 1192 1193;
1159 1191 1194 1195 1192; 1160 1194 1196 1197 1195; 1161 1196 1198 1199 1197;
1162 1198 1200 1201 1199; 1163 1200 1202 1203 1201; 1164 1202 1204 1205 1203;
1165 1206 1190 1193 1207; 1166 1208 1209 1191 1190; 1167 1209 1210 1194 1191;
1168 1210 1211 1196 1194; 1169 1211 1212 1198 1196; 1170 1212 1213 1200 1198;
1171 1213 1214 1202 1200; 1172 1214 1215 1204 1202; 1173 1216 1208 1190 1206;
1174 1217 1218 1209 1208; 1175 1218 1219 1210 1209; 1176 1219 1220 1211 1210;
1177 1220 1221 1212 1211; 1178 1221 1222 1213 1212; 1179 1222 1223 1214 1213;
1180 1223 1224 1215 1214; 1181 1225 1217 1208 1216; 1182 1122 1121 1218 1217;
1183 1121 1123 1219 1218; 1184 1123 1124 1220 1219; 1185 1124 1125 1221 1220;
1186 1125 1126 1222 1221; 1187 1126 1127 1223 1222; 1188 1127 1128 1224 1223;
1189 1129 1122 1217 1225; 1198 1204 1235 1236 1205; 1199 1215 1237 1235 1204;
1200 1224 1238 1237 1215; 1201 1128 1134 1238 1224; 1203 1235 1240 1241 1236;
1204 1240 1242 1243 1241; 1205 1242 1244 1245 1243; 1206 1244 1246 1247 1245;
1207 1237 1248 1240 1235; 1208 1248 1249 1242 1240; 1209 1249 1250 1244 1242;
1210 1250 1251 1246 1244; 1211 1238 1252 1248 1237; 1212 1252 1253 1249 1248;
1213 1253 1254 1250 1249; 1214 1254 1255 1251 1250; 1215 1134 1151 1252 1238;
1216 1151 1152 1253 1252; 1217 1152 1153 1254 1253; 1218 1153 1154 1255 1254;
1223 1246 1260 1261 1247; 1224 1251 1262 1260 1246; 1225 1255 1263 1262 1251;
1226 1154 1159 1263 1255; 1228 1260 1265 1266 1261; 1229 1265 1267 1268 1266;
1230 1267 1269 1270 1268; 1231 1262 1271 1265 1260; 1232 1271 1272 1267 1265;
1233 1272 1273 1269 1267; 1234 1263 1274 1271 1262; 1235 1274 1275 1272 1271;
1236 1275 1276 1273 1272; 1237 1159 1172 1274 1263; 1238 1172 1173 1275 1274;
1239 1173 1174 1276 1275; 1243 1269 1280 1281 1270; 1244 1273 1282 1280 1269;
1245 1276 1283 1282 1273; 1246 1174 1179 1283 1276; 1248 1280 1285 1286 1281;
1249 1282 1287 1285 1280; 1250 1283 1288 1287 1282; 1251 1179 1184 1288 1283;
1253 1285 1290 1291 1286; 1254 1287 1292 1290 1285; 1255 1288 1293 1292 1287;
1256 1184 1189 1293 1288; 1260 758 1297 1298 759; 1261 760 1299 1297 758;
1262 763 1300 1301 764; 1263 765 1302 1300 763; 1264 766 1303 1302 765;
1265 759 1298 1303 766; 1266 764 1301 1304 767; 1267 870 1305 1306 871;
1268 872 1307 1305 870; 1269 873 1308 1307 872; 1270 767 1304 1308 873;
1271 871 1306 1309 874; 1272 975 1310 1311 976; 1273 977 1312 1310 975;
1274 978 1313 1312 977; 1275 874 1309 1313 978; 1276 976 1311 1314 979;
1277 1080 1315 1316 1081; 1278 1082 1317 1315 1080; 1279 1083 1318 1317 1082;
1280 979 1314 1318 1083; 1281 1081 1316 1319 1084; 1282 1185 1320 1321 1186;
1283 1187 1322 1320 1185; 1284 1188 1323 1322 1187; 1285 1084 1319 1323 1188;
1286 1186 1321 1324 1189; 1287 1290 1325 1326 1291; 1288 1292 1327 1325 1290;
1289 1293 1328 1327 1292; 1290 1189 1324 1328 1293;
START GROUP DEFINITION
ELEMENT

```
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 -
412 TO 419 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 -
585 TO 599 604 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 -
685 TO 687 706 707 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 -
756 758 TO 1189 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 -
1244 TO 1246 1248 TO 1251 1253 TO 1256 1260 TO 1290
_RAFT_MOM 326 327 333 334 337 342 TO 349 352 393 TO 399 402 TO 409 -
412 TO 419 422 TO 424 427 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 -
569 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 TO 663 -
670 TO 680 685 TO 687 707 710 TO 712 715 731 732 735 TO 738 740 747 748 751 -
752 TO 754 756 758 TO 814 816 TO 1114 1116 TO 1167 1170 TO 1175 1178 TO 1189 -
1198 TO 1201 1203 TO 1214 1216 TO 1218 1223 TO 1226 1228 TO 1232 1234 1235 -
1237 TO 1239 1243 1246 1248 TO 1251 1253 TO 1256 1260 TO 1290
_RAFT_SHEAR 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 -
604 TO 607 641 685 TO 687 715 740 756 758 TO 1089 1098 TO 1122 1124 TO 1126 -
1131 TO 1139 1144 TO 1146 1149 TO 1151 1154 TO 1156 1198 TO 1201 -
1203 TO 1218 1266 TO 1281 1283 TO 1285
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
1248 TO 1251 1253 TO 1256 1260 TO 1290 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 730 1258 1259 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
1248 TO 1251 1253 TO 1256 1260 TO 1289 -
1290 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
1248 TO 1251 1253 TO 1256 1260 TO 1290 PR GY -16
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
1248 TO 1251 1253 TO 1256 1260 TO 1290 PR GY -20
JOINT LOAD
128 FX -0.01 FY -52.59 FZ 0.88
740 FX -0.66 FY -138.22 FZ 0.26
1295 FX 0.07 FY -74.1 FZ -0.42
1296 FX 0.0001 FY -144 FZ -2.07
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX -0.14 FY -101.7 FZ 2
740 FX -8.89 FY -241.3 FZ 0.81
1295 FX 0.09 FY -139.24 FZ -1.06
1296 FX -0.04 FY -346.9 FZ -10.35
```



```
1296 FX -0.04 FY -346.9 FZ -10.35
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WL1 )
JOINT LOAD
128 FX -2.54 FY 393.02 FZ -0.24
740 FX -114.49 FY -420.8 FZ -0.31
1295 FX -5.79 FY -58.16 FZ 1.21
1296 FX -4.87 FY 263.66 FZ 18
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X ( WL2 )
JOINT LOAD
128 FX 12 FY -390.1 FZ 0.23
740 FX 106.5 FY 416.92 FZ 0.31
1295 FX 4.49 FY -56.45 FZ -1.24
1296 FX 7.87 FY -266.01 FZ -17.58
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Y ( WL3 )
JOINT LOAD
128 FX -0.04 FY 111.7 FZ -16
740 FX -6.47 FY 80.95 FZ -9.04
1295 FX 0.1 FY -85.93 FZ -45.02
1296 FX 0.03 FY -130.38 FZ -73.74
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Y ( WL4 )
JOINT LOAD
128 FX 0.04 FY -109 FZ 16
740 FX 5.78 FY -82.8 FZ 5
1295 FX -0.1 FY 85.55 FZ 45.9
1296 FX -0.03 FY 128.72 FZ 77.13
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX -0.02 FY -11.7 FZ -0.2
740 FX -2.57 FY 14.34 FZ -0.03
1295 FX -0.02 FY 2.37 FZ -0.75
1296 FX -0.02 FY 6.48 FZ -1.7
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX 0.02 FY 11.7 FZ 0.2
740 FX 2.57 FY -14.34 FZ 0.03
1295 FX 0.03 FY -2.37 FZ 0.75
1296 FX 0.02 FY -6.48 FZ 1.7
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 8.9 FY -9.77 FZ 4.05
740 FX -5.15 FY 2 FZ 1.5
1295 FX 0.58 FY -0.3 FZ 3.75
1296 FX 0.08 FY -11.26 FZ -3.07
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -8.9 FY 9.77 FZ -4.05
740 FX 5.15 FY -2 FZ -1.5
1295 FX -0.58 FY -0.3 FZ 3.75
1296 FX 0.08 FY -11.26 FZ -3.07
PERFORM ANALYSIS
CHANGE
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
1248 TO 1251 1253 TO 1256 1260 TO 1289 -
1290 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
```

```
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+0.75L+0.75TL1
REPEAT LOAD
1 1.0 2 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL2
REPEAT LOAD
1 1.0 2 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+WL2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL3
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL4
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+0.75WL1+0.75L
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+0.75WL2+0.75L
REPEAT LOAD
1 1.0 2 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+0.75WL3+0.75L
REPEAT LOAD
1 1.0 2 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+0.75WL4+0.75L
REPEAT LOAD
1 1.0 2 0.75 6 0.75
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL+WL1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 0.6DL+WL2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 0.6DL+WL3
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL+WL4
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 7 0.7
```



```
1 0.6 7 0.7
PERFORM ANALYSIS
CHANGE
LOAD 219 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 8 0.7
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L
REPEAT LOAD
1 1.4 2 1.7 9 1.4
PERFORM ANALYSIS
CHANGE
LOAD 103 1.4DL+1.4TL2+1.7L
REPEAT LOAD
1 1.4 2 1.7 10 1.4
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.0L
REPEAT LOAD
1 1.2 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL1+1.0L
REPEAT LOAD
1 1.2 3 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L
REPEAT LOAD
1 1.2 4 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L
REPEAT LOAD
1 1.2 5 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 7 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 8 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 112 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
```

```
PERFORM ANALYSIS
CHANGE
LOAD 115 0.9DL+1.0SL1
REPEAT LOAD
1 0.9 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 0.9DL+1.0SL2
REPEAT LOAD
1 0.9 8 1.0
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 730 1258 1259
CLB 0.05 MEMB 185 730 1258 1259
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 -
403 TO 409 412 TO 419 422 TO 429 432 497 498 501 TO 504 568 TO 575 -
584 TO 599 632 633 636 TO 639 658 TO 663 670 TO 681 706 707 710 TO 713 730 -
731 TO 732 735 TO 738 747 748 751 TO 754 758 TO 789 798 TO 801 803 TO 818 -
823 TO 826 828 TO 839 843 TO 846 848 TO 851 853 TO 856 858 TO 889 -
898 TO 901 903 TO 918 923 TO 926 928 TO 939 943 TO 946 948 TO 951 -
953 TO 956 958 TO 989 998 TO 1001 1003 TO 1018 1023 TO 1026 1028 TO 1039 -
1043 TO 1046 1048 TO 1051 1053 TO 1056 1058 TO 1089 1098 TO 1101 -
1103 TO 1118 1123 TO 1126 1128 TO 1139 1143 TO 1146 1148 TO 1151 -
1153 TO 1156 1158 TO 1189 1198 TO 1201 1203 TO 1218 1223 TO 1226 -
1228 TO 1239 1243 TO 1246 1248 TO 1251 1253 TO 1256 1258 TO 1265 -
1267 TO 1270 1272 TO 1275 1277 TO 1280 1282 TO 1285 1287 TO 1290
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 730 1258 1259
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 3 STAAD INPUT – FOUNDATION F2

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/09/2020
JOB NAME FDN. MKD. F2
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
492 1.35 911 3.42538; 493 1.35 911 3.175; 494 1.35 911 3.67538;
497 1.35 911 2.42538; 498 1.35 911 2.17538; 499 1.35 911 2.67538;
500 1.35 911 2.92538; 501 1.35 911 1.92538; 569 1.6 911 3.42538;
570 1.6 911 3.175; 571 1.85 911 3.42538; 572 1.85 911 3.175;
573 2.1 911 3.42538; 574 2.1 911 3.175; 575 2.35 911 3.42538;
576 2.35 911 3.175; 577 1.6 911 3.67538; 578 1.85 911 3.67538;
579 2.1 911 3.67538; 580 2.35 911 3.67538; 589 1.6 911 2.42538;
590 1.6 911 2.17538; 591 1.85 911 2.42538; 592 1.85 911 2.17538;
593 2.1 911 2.42538; 594 2.1 911 2.17538; 595 2.35 911 2.42538;
596 2.35 911 2.17538; 597 1.6 911 2.67538; 598 1.85 911 2.67538;
599 2.1 911 2.67538; 600 2.35 911 2.67538; 601 1.6 911 2.92538;
602 1.85 911 2.92538; 603 2.1 911 2.92538; 604 2.35 911 2.92538;
605 1.6 911 1.92538; 607 1.85 911 1.92538; 609 2.1 911 1.92538;
611 2.35 911 1.92538; 637 2.6 911 3.42538; 639 2.6 911 3.67538;
642 2.6 911 2.42538; 643 2.6 911 2.17538; 644 2.6 911 2.67538;
645 2.6 911 2.92538; 646 2.6 911 1.92538; 664 2.85 911 3.42538;
665 2.85 911 3.175; 666 3.1 911 3.42538; 667 3.1 911 3.175;
670 2.85 911 3.67538; 671 3.1 911 3.67538; 679 2.85 911 2.42538;
680 2.85 911 2.17538; 681 3.1 911 2.42538; 682 3.1 911 2.17538;
685 2.85 911 2.67538; 686 3.1 911 2.67538; 688 2.85 911 2.92538;
689 3.1 911 2.92538; 691 2.85 911 1.92538; 693 3.1 911 1.92538;
740 2.6 913.6 3.175; 815 1.35 911 1.17538; 816 1.35 911 0.92538;
817 1.35 911 1.42538; 818 1.35 911 1.67538; 819 1.35 911 0.67538;
820 1.6 911 1.17538; 821 1.6 911 0.92538; 822 1.85 911 1.17538;
823 1.85 911 0.92538; 824 2.1 911 1.17538; 825 2.1 911 0.92538;
826 2.35 911 1.17538; 827 2.35 911 0.92538; 828 1.6 911 1.42538;
829 1.85 911 1.42538; 830 2.1 911 1.42538; 831 2.35 911 1.42538;
832 1.6 911 1.67538; 833 1.85 911 1.67538; 834 2.1 911 1.67538;
835 2.35 911 1.67538; 836 1.6 911 0.67538; 837 1.85 911 0.67538;
838 2.1 911 0.67538; 839 2.35 911 0.67538; 840 2.6 911 1.17538;
842 2.6 911 1.42538; 843 2.6 911 1.67538; 844 2.6 911 0.67538;
845 2.85 911 1.17538; 846 2.85 911 0.92538; 847 3.1 911 1.17538;
848 3.1 911 0.92538; 851 2.85 911 1.42538; 852 3.1 911 1.42538;
854 2.85 911 1.67538; 855 3.1 911 1.67538; 857 2.85 911 0.67538;
858 3.1 911 0.67538; 923 1.35 911 0.42538; 937 1.6 911 0.42538;
938 1.85 911 0.42538; 939 2.1 911 0.42538; 940 2.35 911 0.42538;
948 2.6 911 0.42538; 959 2.85 911 0.42538; 960 3.1 911 0.42538;
1296 2.6 913.6 0.925; 1329 2.6 911 3.175; 1332 2.6 911 0.92538;
1333 3.6 911 3.42538; 1334 3.85 911 3.42538; 1335 3.85 911 3.175;
1336 3.6 911 3.175; 1339 3.6 911 3.67538; 1340 3.85 911 3.67538;
1342 3.6 911 2.42538; 1343 3.85 911 2.42538; 1344 3.85 911 2.17538;
1345 3.6 911 2.17538; 1348 3.6 911 2.67538; 1349 3.85 911 2.67538;
1351 3.6 911 2.92538; 1352 3.85 911 2.92538; 1354 3.85 911 1.92538;
1355 3.6 911 1.92538; 1357 3.6 911 1.17538; 1358 3.85 911 1.17538;
1359 3.85 911 0.92538; 1360 3.6 911 0.92538; 1363 3.6 911 1.42538;
1364 3.85 911 1.42538; 1366 3.6 911 1.67538; 1367 3.85 911 1.67538;
1369 3.85 911 0.67538; 1370 3.6 911 0.67538; 1372 3.85 911 0.42538;
1373 3.6 911 0.42538; 1375 3.35 911 3.42538; 1376 3.35 911 3.175;
1377 3.35 911 3.67538; 1378 3.35 911 2.42538; 1379 3.35 911 2.17538;
1380 3.35 911 2.67538; 1381 3.35 911 2.92538; 1382 3.35 911 1.92538;
1383 3.35 911 1.17538; 1384 3.35 911 0.92538; 1385 3.35 911 1.42538;
1386 3.35 911 1.67538; 1387 3.35 911 0.67538; 1388 3.35 911 0.42538;
1452 1.35 911 0.17538; 1453 1.6 911 0.17538; 1454 1.85 911 0.17538;
1455 2.1 911 0.17538; 1456 2.35 911 0.17538; 1457 2.6 911 0.17538;
1458 2.85 911 0.17538; 1459 3.1 911 0.17538; 1467 1.35 911 -0.07462;
1468 1.6 911 -0.07462; 1469 1.85 911 -0.07462; 1470 2.1 911 -0.07462;
1471 2.35 911 -0.07462; 1472 2.6 911 -0.07462; 1473 2.85 911 -0.07462;
1474 3.1 911 -0.07462; 1475 3.85 911 0.17538; 1476 3.6 911 0.17538;
1478 3.85 911 -0.07462; 1479 3.6 911 -0.07462; 1481 3.35 911 0.17538;
1482 3.35 911 -0.07462; 1505 1.35 911 3.92538; 1506 1.35 911 4.17538;
1507 1.6 911 3.92538; 1508 1.85 911 3.92538; 1509 2.1 911 3.92538;
1510 2.35 911 3.92538; 1511 1.6 911 4.17538; 1512 1.85 911 4.17538;
1513 2.1 911 4.17538; 1514 2.35 911 4.17538; 1515 2.6 911 3.92538;
1516 2.6 911 4.17538; 1517 2.85 911 3.92538; 1518 3.1 911 3.92538;
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1516 2.6 911 4.17538; 1517 2.85 911 3.92538; 1518 3.1 911 3.92538;
1519 2.85 911 4.17538; 1520 3.1 911 4.17538; 1521 3.6 911 3.92538;
1522 3.85 911 3.92538; 1524 3.6 911 4.17538; 1525 3.85 911 4.17538;
1527 3.35 911 3.92538; 1528 3.35 911 4.17538;
MEMBER INCIDENCES
730 1329 740; 1259 1332 1296;
ELEMENT INCIDENCES SHELL
568 492 569 570 493; 569 569 571 572 570; 570 571 573 574 572;
571 573 575 576 574; 572 494 577 569 492; 573 577 578 571 569;
574 578 579 573 571; 575 579 580 575 573; 584 497 589 590 498;
585 589 591 592 590; 586 591 593 594 592; 587 593 595 596 594;
588 499 597 589 497; 589 597 598 591 589; 590 598 599 593 591;
591 599 600 595 593; 592 500 601 597 499; 593 601 602 598 597;
594 602 603 599 598; 595 603 604 600 599; 596 493 570 601 500;
597 570 572 602 601; 598 572 574 603 602; 599 574 576 604 603;
604 498 590 605 501; 605 590 592 607 605; 606 592 594 609 607;
607 594 596 611 609; 632 575 637 1329 576; 633 580 639 637 575;
636 595 642 643 596; 637 600 644 642 595; 638 604 645 644 600;
639 576 1329 645 604; 641 596 643 646 611; 658 637 664 665 1329;
659 664 666 667 665; 661 639 670 664 637; 662 670 671 666 664;
670 642 679 680 643; 671 679 681 682 680; 673 644 685 679 642;
674 685 686 681 679; 676 645 688 685 644; 677 688 689 686 685;
679 1329 665 688 645; 680 665 667 689 688; 685 643 680 691 646;
686 680 682 693 691; 803 815 820 821 816; 804 820 822 823 821;
805 822 824 825 823; 806 824 826 827 825; 807 817 828 820 815;
808 828 829 822 820; 809 829 830 824 822; 810 830 831 826 824;
811 818 832 828 817; 812 832 833 829 828; 813 833 834 830 829;
814 834 835 831 830; 815 501 605 832 818; 816 605 607 833 832;
817 607 609 834 833; 818 609 611 835 834; 819 816 821 836 819;
820 821 823 837 836; 821 823 825 838 837; 822 825 827 839 838;
823 826 840 1332 827; 824 831 842 840 826; 825 835 843 842 831;
826 611 646 843 835; 827 827 1332 844 839; 828 840 845 846 1332;
829 845 847 848 846; 831 842 851 845 840; 832 851 852 847 845;
834 843 854 851 842; 835 854 855 852 851; 837 646 691 854 843;
838 691 693 855 854; 840 1332 846 857 844; 841 846 848 858 857;
915 819 836 937 923; 916 836 837 938 937; 917 837 838 939 938;
918 838 839 940 939; 926 839 844 948 940; 937 844 857 959 948;
938 857 858 960 959; 1260 1333 1334 1335 1336; 1262 1339 1340 1334 1333;
1264 1342 1343 1344 1345; 1266 1348 1349 1343 1342; 1268 1351 1352 1349 1348;
1270 1336 1335 1352 1351; 1272 1345 1344 1354 1355; 1274 1357 1358 1359 1360;
1276 1363 1364 1358 1357; 1278 1366 1367 1364 1363; 1280 1355 1354 1367 1366;
1282 1360 1359 1369 1370; 1284 1370 1369 1372 1373; 1286 666 1375 1376 667;
1287 1375 1333 1336 1376; 1288 671 1377 1375 666; 1289 1377 1339 1333 1375;
1290 681 1378 1379 682; 1291 1378 1342 1345 1379; 1292 686 1380 1378 681;
1293 1380 1348 1342 1378; 1294 689 1381 1380 686; 1295 1381 1351 1348 1380;
1296 667 1376 1381 689; 1297 1376 1336 1351 1381; 1298 682 1379 1382 693;
1299 1379 1345 1355 1382; 1300 847 1383 1384 848; 1301 1383 1357 1360 1384;
1302 852 1385 1383 847; 1303 1385 1363 1357 1383; 1304 855 1386 1385 852;
1305 1386 1366 1363 1385; 1306 693 1382 1386 855; 1307 1382 1355 1366 1386;
1308 848 1384 1387 858; 1309 1384 1360 1370 1387; 1310 858 1387 1388 960;
1311 1387 1370 1373 1388; 1371 923 937 1453 1452; 1372 937 938 1454 1453;
1373 938 939 1455 1454; 1374 939 940 1456 1455; 1375 940 948 1457 1456;
1376 948 959 1458 1457; 1377 959 960 1459 1458; 1385 1452 1453 1468 1467;
1386 1453 1454 1469 1468; 1387 1454 1455 1470 1469; 1388 1455 1456 1471 1470;
1389 1456 1457 1472 1471; 1390 1457 1458 1473 1472; 1391 1458 1459 1474 1473;
1392 1373 1372 1475 1476; 1394 1476 1475 1478 1479; 1396 960 1388 1481 1459;
1397 1388 1373 1476 1481; 1398 1459 1481 1482 1474; 1399 1481 1476 1479 1482;
1422 1505 1507 577 494; 1423 1507 1508 578 577; 1424 1508 1509 579 578;
1425 1509 1510 580 579; 1426 1506 1511 1507 1505; 1427 1511 1512 1508 1507;
1428 1512 1513 1509 1508; 1429 1513 1514 1510 1509; 1430 1510 1515 639 580;
1431 1514 1516 1515 1510; 1432 1515 1517 670 639; 1433 1517 1518 671 670;
1434 1516 1519 1517 1515; 1435 1519 1520 1518 1517; 1436 1521 1522 1340 1339;
1438 1524 1525 1522 1521; 1440 1518 1527 1377 671; 1441 1527 1521 1339 1377;
1442 1520 1528 1527 1518; 1443 1528 1524 1521 1527;
START GROUP DEFINITION
ELEMENT
_RAFT 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 -
662 670 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 -
838 840 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 -
1276 1278 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 -
1397 TO 1399 1422 TO 1436 1438 1440 TO 1443
_RAFT_MOM 568 TO 575 584 TO 599 604 TO 607 633 636 TO 638 641 659 661 662 -
670 671 673 674 676 677 680 685 686 803 TO 822 824 TO 826 829 831 832 834 -
835 837 838 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 -
1274 1276 1278 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 -
1396 TO 1399 1422 TO 1436 1438 1440 TO 1443
_RAFT_SHEAR 568 572 584 588 592 596 604 TO 607 641 685 686 803 807 811 815 -
819 915 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 1280 1282 1284 -
1298 1299 1371 1385 1392 1394 1422 1426 1436 1438
```

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1298 1299 1371 1385 1392 1394 1422 1426 1436 1438
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
1423 TO 1436 1438 1440 TO 1443 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
730 1259 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
1423 TO 1436 1438 1440 TO 1443 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
1423 TO 1436 1438 1440 TO 1443 PR GY -16
* LOAD DUE TO SURCHARGE
568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
1423 TO 1436 1438 1440 TO 1443 PR GY -20
JOINT LOAD
740 FX -0.35 FY -90 FZ 0.39
1296 FX -2.17 FY -89.83 FZ 0.4
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
740 FX 0.18 FY -42.94 FZ 0.04
1296 FX -0.09 FY -5.87 FZ 2.22
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WL1 )
JOINT LOAD
740 FX -3.82 FY 19.37 FZ 0.98
1296 FX 3.87 FY -276.8 FZ -6.86
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X ( WL2 )
JOINT LOAD
740 FX 3.85 FY 19.37 FZ -0.97
1296 FX -3.87 FY 275.48 FZ 6.67
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Y ( WL3 )
JOINT LOAD
740 FX 0.35 FY 260 FZ -0.45
1296 FX 0.61 FY -288.67 FZ -60.96
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Y ( WL4 )
JOINT LOAD
740 FX -0.34 FY -255.87 FZ 0.47
1296 FX -0.59 FY 284.46 FZ 60.73
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
740 FX -0.05 FY 7.65 FZ -0.03
1296 FX -0.21 FY 8.1 FZ -1.02
*********************************************************************************************
```



```
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
740 FX 0.05 FY -7.65 FZ 0.03
1296 FX 0.21 FY -8.1 FZ 1.02
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
740 FX -0.35 FY -33.04 FZ 2.39
1296 FX -1.03 FY 32.98 FZ 2.29
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
740 FX 0.35 FY 33.04 FZ -2.39
1296 FX 1.03 FY -32.98 FZ -2.29
PERFORM ANALYSIS
CHANGE
SUPPORTS
568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
1423 TO 1436 1438 1440 TO 1443 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+0.75L+0.75TL1
REPEAT LOAD
1 1.0 2 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL2
REPEAT LOAD
1 1.0 2 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+WL2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL3
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL4
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+0.75WL1+0.75L
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+0.75WL2+0.75L
REPEAT LOAD
1 1.0 2 0.75 4 0.75
PERFORM ANALYSIS
```

```
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+0.75WL3+0.75L
REPEAT LOAD
1 1.0 2 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+0.75WL4+0.75L
REPEAT LOAD
1 1.0 2 0.75 6 0.75
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL+WL1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 0.6DL+WL2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 0.6DL+WL3
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL+WL4
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 7 0.7
PERFORM ANALYSIS
CHANGE
LOAD 219 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 8 0.7
PERFORM ANALYSIS
CHANGE
********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L
REPEAT LOAD
1 1.4 2 1.7 9 1.4
PERFORM ANALYSIS
CHANGE
LOAD 103 1.4DL+1.4TL2+1.7L
REPEAT LOAD
1 1.4 2 1.7 10 1.4
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.0L
REPEAT LOAD
1 1.2 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL1+1.0L
REPEAT LOAD
1 1.2 3 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L
REPEAT LOAD
1 1.2 4 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L
REPEAT LOAD
1 1.2 5 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
```



```
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 7 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 8 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 112 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 115 0.9DL+1.0SL1
REPEAT LOAD
1 0.9 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 0.9DL+1.0SL2
REPEAT LOAD
1 0.9 8 1.0
PERFORM ANALYSIS
CHANGE
********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 730 1259
CLB 0.05 MEMB 730 1259
CLS 0.05 ALL
CLT 0.05 MEMB 568 TO 575 584 TO 599 632 633 636 TO 639 658 659 661 662 670 -
671 673 674 676 677 679 680 730 803 TO 818 823 TO 826 828 829 831 832 834 -
835 837 838 915 TO 918 926 937 938 1259 1260 1262 1264 1266 1268 1270 1274 -
1276 1278 1280 1284 1286 TO 1297 1300 TO 1307 1310 1311 1385 TO 1391 1394 -
1398 1399 1422 TO 1436 1438 1440 TO 1443
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 730 1259
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 4 STAAD INPUT – FOUNDATION F3

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/11/2020
JOB NAME FDN. MKD. F3
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911 3.175; 6 0.75 911 3.175; 34 -0.5 911 3.175; 35 -0.25 911 3.175;
41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175; 302 -0.5 911 3.42538;
304 -0.25 911 3.42538; 305 0 911 3.42538; 306 0.25 911 3.42538;
307 0.5 911 3.42538; 308 0.75 911 3.42538; 309 1 911 3.175; 313 1 911 3.42538;
322 -0.5 911 3.67538; 323 -0.25 911 3.67538; 324 0 911 3.67538;
325 0.25 911 3.67538; 326 0.5 911 3.67538; 327 0.75 911 3.67538;
328 1 911 3.67538; 366 -0.5 911 2.17538; 369 -0.25 911 2.17538;
371 0 911 2.17538; 373 0.25 911 2.17538; 375 0.5 911 2.17538;
377 0.75 911 2.17538; 379 1 911 2.17538; 388 -0.5 911 2.42538;
389 -0.25 911 2.42538; 390 0 911 2.42538; 391 0.25 911 2.42538;
392 0.5 911 2.42538; 393 0.75 911 2.42538; 394 1 911 2.42538;
399 -0.5 911 2.67538; 400 -0.25 911 2.67538; 401 0 911 2.67538;
402 0.25 911 2.67538; 403 0.5 911 2.67538; 404 0.75 911 2.67538;
405 1 911 2.67538; 410 -0.5 911 2.92538; 411 -0.25 911 2.92538;
412 0 911 2.92538; 413 0.25 911 2.92538; 414 0.5 911 2.92538;
415 0.75 911 2.92538; 416 1 911 2.92538; 421 -0.5 911 1.92538;
424 -0.25 911 1.92538; 426 0 911 1.92538; 428 0.25 911 1.92538;
430 0.5 911 1.92538; 432 0.75 911 1.92538; 434 1 911 1.92538;
492 1.25 911 3.42538; 493 1.25 911 3.175; 494 1.25 911 3.67538;
497 1.25 911 2.42538; 498 1.25 911 2.17538; 499 1.25 911 2.67538;
500 1.25 911 2.92538; 501 1.25 911 1.92538; 771 -0.5 911 1.17538;
772 -0.5 911 0.92538; 774 -0.25 911 1.17538; 775 -0.25 911 0.92538;
776 0 911 1.17538; 778 0.25 911 1.17538; 779 0.25 911 0.92538;
780 0.5 911 1.17538; 781 0.5 911 0.92538; 782 0.75 911 1.17538;
783 0.75 911 0.92538; 784 1 911 1.17538; 785 1 911 0.92538;
789 -0.5 911 1.42538; 790 -0.25 911 1.42538; 791 0 911 1.42538;
792 0.25 911 1.42538; 793 0.5 911 1.42538; 794 0.75 911 1.42538;
795 1 911 1.42538; 798 -0.5 911 1.67538; 799 -0.25 911 1.67538;
800 0 911 1.67538; 801 0.25 911 1.67538; 802 0.5 911 1.67538;
803 0.75 911 1.67538; 804 1 911 1.67538; 806 -0.5 911 0.67538;
808 -0.25 911 0.67538; 809 0 911 0.67538; 810 0.25 911 0.67538;
811 0.5 911 0.67538; 812 0.75 911 0.67538; 813 1 911 0.67538;
815 1.25 911 1.17538; 816 1.25 911 0.92538; 817 1.25 911 1.42538;
818 1.25 911 1.67538; 819 1.25 911 0.67538; 903 -0.5 911 0.42538;
904 -0.25 911 0.42538; 905 0 911 0.42538; 906 0.25 911 0.42538;
907 0.5 911 0.42538; 908 0.75 911 0.42538; 909 1 911 0.42538;
923 1.25 911 0.42538; 1295 0 913.6 0.075; 1331 0 911 0.92538;
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1463 0.25 911 -0.32462; 1464 0.5 911 -0.32462; 1465 0.75 911 -0.32462;
1466 1 911 -0.32462; 1467 1.25 911 -0.32462; 1483 -1.25 911 0.07538;
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1530 -1.25 911 3.92538; 1531 -1 911 3.92538; 1532 -0.75 911 3.92538;
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1530 -1.25 911 3.92538; 1531 -1 911 3.92538; 1532 -0.75 911 3.92538;
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1537 0 911 -0.32462; 1538 0 911 0.07538; 1539 0 913.6 -0.625;
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1620 2 911 1.17538; 1621 2 911 0.92538; 1622 2 911 1.42538; 1623 2 911 1.67538;
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MEMBER INCIDENCES
185 2 128; 1258 1538 1295; 1452 1539 1295;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 337 308 313 309 6; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
349 327 328 313 308; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 404 399 400 389 388; 405 400 401 390 389;
406 401 402 391 390; 407 402 403 392 391; 408 403 404 393 392;
409 404 405 394 393; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
419 415 416 405 404; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 442 366 369 424 421;
443 369 371 426 424; 444 371 373 428 426; 445 373 375 430 428;
446 375 377 432 430; 447 377 379 434 432; 497 313 492 493 309;
498 328 494 492 313; 501 394 497 498 379; 502 405 499 497 394;
503 416 500 499 405; 504 309 493 500 416; 506 379 498 501 434;
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767 789 790 774 771; 768 790 791 776 774; 769 791 792 778 776;
770 792 793 780 778; 771 793 794 782 780; 772 794 795 784 782;
775 798 799 790 789; 776 799 800 791 790; 777 800 801 792 791;
778 801 802 793 792; 779 802 803 794 793; 780 803 804 795 794;
783 421 424 799 798; 784 424 426 800 799; 785 426 428 801 800;
786 428 430 802 801; 787 430 432 803 802; 788 432 434 804 803;
791 772 775 808 806; 792 775 1331 809 808; 793 1331 779 810 809;
794 779 781 811 810; 795 781 783 812 811; 796 783 785 813 812;
798 784 815 816 785; 799 795 817 815 784; 800 804 818 817 795;
801 434 501 818 804; 802 785 816 819 813; 883 806 808 904 903;
884 808 809 905 904; 885 809 810 906 905; 886 810 811 907 906;
887 811 812 908 907; 888 812 813 909 908; 901 813 819 923 909;
1313 1390 1393 1394 1391; 1314 1393 1395 1396 1394; 1315 1395 302 34 1396;
1317 1398 1399 1393 1390; 1318 1399 1400 1395 1393; 1319 1400 322 302 1395;
1321 1402 1405 1406 1403; 1322 1405 1407 1408 1406; 1323 1407 388 366 1408;
1325 1410 1411 1405 1402; 1326 1411 1412 1407 1405; 1327 1412 399 388 1407;
1329 1414 1415 1411 1410; 1330 1415 1416 1412 1411; 1331 1416 410 399 1412;
1333 1391 1394 1415 1414; 1334 1394 1396 1416 1415; 1335 1396 34 410 1416;
1337 1403 1406 1419 1417; 1338 1406 1408 1420 1419; 1339 1408 366 421 1420;
1341 1422 1425 1426 1423; 1342 1425 1427 1428 1426; 1343 1427 771 772 1428;
1345 1430 1431 1425 1422; 1346 1431 1432 1427 1425; 1347 1432 789 771 1427;
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1357 1423 1426 1439 1437; 1358 1426 1428 1440 1439; 1359 1428 772 806 1440;
1361 1437 1439 1443 1441; 1362 1439 1440 1444 1443; 1363 1440 806 903 1444;
1364 903 904 1445 1446; 1365 904 905 1538 1445; 1366 905 906 1448 1538;
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1370 909 923 1452 1451; 1378 1446 1445 1460 1461; 1379 1445 1538 1537 1460;
1380 1538 1448 1463 1537; 1381 1448 1449 1464 1463; 1382 1449 1450 1465 1464;
1383 1450 1451 1466 1465; 1384 1451 1452 1467 1466; 1401 1441 1443 1485 1483;
1402 1443 1444 1486 1485; 1403 1444 903 1446 1486; 1405 1483 1485 1489 1487;
1406 1485 1486 1490 1489; 1407 1486 1446 1461 1490; 1408 1491 1492 323 322;
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1412 1495 1496 327 326; 1413 1496 1497 328 327; 1414 1498 1499 1492 1491;
1415 1499 1500 1493 1492; 1416 1500 1501 1494 1493; 1417 1501 1502 1495 1494;
1418 1502 1503 1496 1495; 1419 1503 1504 1497 1496; 1420 1497 1505 494 328;
1421 1504 1506 1505 1497; 1445 1530 1531 1399 1398; 1446 1531 1532 1400 1399;
1447 1532 1491 322 1400; 1449 1534 1535 1531 1530; 1450 1535 1536 1532 1531;
1451 1536 1498 1491 1532; 1453 492 1540 1541 493; 1454 494 1542 1540 492;
1455 497 1543 1544 498; 1456 499 1545 1543 497; 1457 500 1546 1545 499;
1458 493 1541 1546 500; 1459 498 1544 1547 501; 1460 1540 1548 1549 1541;
1461 1542 1550 1548 1540; 1462 1543 1551 1552 1544; 1463 1545 1553 1551 1543;
1464 1546 1554 1553 1545; 1465 1541 1549 1554 1546; 1466 1544 1552 1555 1547;
1467 815 1556 1557 816; 1468 817 1558 1556 815; 1469 818 1559 1558 817;
1470 501 1547 1559 818; 1471 816 1557 1560 819; 1472 1556 1561 1562 1557;
1473 1558 1563 1561 1556; 1474 1559 1564 1563 1558; 1475 1547 1555 1564 1559;
1476 1557 1562 1565 1560; 1477 819 1560 1566 923; 1478 1560 1565 1567 1566;
1479 923 1566 1568 1452; 1480 1566 1567 1569 1568; 1481 1452 1568 1570 1467;
1482 1568 1569 1571 1570; 1483 1505 1572 1542 494; 1484 1506 1573 1572 1505;
1485 1572 1574 1550 1542; 1486 1573 1575 1574 1572; 1487 1576 1577 1578 1579;
1488 1577 1390 1391 1578; 1489 1580 1581 1577 1576; 1490 1581 1398 1390 1577;
1491 1582 1583 1584 1585; 1492 1583 1402 1403 1584; 1493 1586 1587 1583 1582;
1494 1587 1410 1402 1583; 1495 1588 1589 1587 1586; 1496 1589 1414 1410 1587;
1497 1579 1578 1589 1588; 1498 1578 1391 1414 1589; 1499 1585 1584 1590 1591;
1500 1584 1403 1417 1590; 1501 1592 1593 1594 1595; 1502 1593 1422 1423 1594;
1503 1596 1597 1593 1592; 1504 1597 1430 1422 1593; 1505 1598 1599 1597 1596;
1506 1599 1434 1430 1597; 1507 1591 1590 1599 1598; 1508 1590 1417 1434 1599;
1509 1595 1594 1600 1601; 1510 1594 1423 1437 1600; 1511 1601 1600 1602 1603;
1512 1600 1437 1441 1602; 1513 1603 1602 1604 1605; 1514 1602 1441 1483 1604;
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1518 1609 1530 1398 1581; 1519 1610 1611 1609 1608; 1520 1611 1534 1530 1609;
1521 1548 1612 1613 1549; 1522 1550 1614 1612 1548; 1523 1551 1615 1616 1552;
1524 1553 1617 1615 1551; 1525 1554 1618 1617 1553; 1526 1549 1613 1618 1554;
1527 1552 1616 1619 1555; 1528 1561 1620 1621 1562; 1529 1563 1622 1620 1561;
1530 1564 1623 1622 1563; 1531 1555 1619 1623 1564; 1532 1562 1621 1624 1565;
1533 1565 1624 1625 1567; 1534 1567 1625 1626 1569; 1535 1569 1626 1627 1571;
1536 1574 1628 1614 1550; 1537 1575 1629 1628 1574; 1538 1630 1576 1579 1631;
1539 1632 1580 1576 1630; 1540 1633 1582 1585 1634; 1541 1635 1586 1582 1633;
1542 1636 1588 1586 1635; 1543 1631 1579 1588 1636; 1544 1634 1585 1591 1637;
1545 1638 1592 1595 1639; 1546 1640 1596 1592 1638; 1547 1641 1598 1596 1640;
1548 1637 1591 1598 1641; 1549 1639 1595 1601 1642; 1550 1642 1601 1603 1643;
1551 1643 1603 1605 1644; 1552 1644 1605 1607 1645; 1553 1646 1608 1580 1632;
1554 1647 1610 1608 1646;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
425 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
1449 TO 1451 1453 TO 1554
_RAFT_MOM 327 333 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
427 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 786 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 -
1317 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 -
1337 TO 1339 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 1355 1357 TO 1359 -
1361 1363 1364 1367 TO 1370 1378 1381 TO 1384 1401 TO 1403 1405 TO 1421 1445 -
1447 1449 TO 1451 1453 TO 1554
_RAFT_SHEAR 442 TO 447 497 498 501 TO 504 506 767 TO 772 775 TO 780 -
783 TO 788 798 TO 802 901 1313 1317 1321 1325 1329 1333 1337 TO 1339 1341 -
1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 1361 1370 1384 1401 1405 1420 -
1421 1445 1449 1453 TO 1554
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
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775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
1449 TO 1451 1453 TO 1554 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.6 ZD 0.6
MEMBER PROPERTY AMERICAN
1258 PRIS YD 0.8 ZD 0.8
1452 PRIS YD 0.75 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
1449 TO 1451 1453 TO 1554 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
MEMBER OFFSET
1452 START 0 -0.375 0
1452 END 0 -0.375 0
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
1449 TO 1451 1453 TO 1554 PR GY -16
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
1449 TO 1451 1453 TO 1554 PR GY -20
JOINT LOAD
128 FX 0.38 FY -117.3 FZ 5.2
1295 FX -0.725 FY -94.5 FZ -5
1539 FX -0.04 FY -33.15 FZ 0.09
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 10 FY -115.8 FZ 12
1295 FX 4.7 FY -66.8 FZ -12
1539 FX -0.03 FY -13.43 FZ 0.15
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WL1 )
JOINT LOAD
128 FX 70.2 FY -148.2 FZ 0.5
1295 FX 63.5 FY -135 FZ -0.5
1539 FX -0.69 FY -95.18 FZ 0.01
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X ( WL2 )
JOINT LOAD
128 FX -70.2 FY -148.2 FZ -0.5
1295 FX -63.5 FY -135 FZ 0.5
1539 FX 0.69 FY 90.7 FZ -0.01
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Y ( WL3 )
JOINT LOAD
128 FX 0.3 FY -330.8 FZ 49.5
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128 FX 0.3 FY -330.8 FZ 49.5
1295 FX -1.12 FY 330.8 FZ 49.6
1539 FX 0.09 FY 20.68 FZ -6.15
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Y ( WL4 )
JOINT LOAD
128 FX -0.3 FY 330.8 FZ -49.5
1295 FX 1.12 FY -330.8 FZ -49.6
1539 FX -0.09 FY -20.36 FZ 6.15
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
1539 FX -0.02 FY 6.48 FZ -1.7
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
1539 FX 0.02 FY -6.48 FZ 1.7
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
1539 FX -0.28 FY 13.79 FZ 0.07
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
1539 FX 0.28 FY -13.79 FZ -0.07
PERFORM ANALYSIS
CHANGE
SUPPORTS
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
1449 TO 1451 1453 TO 1554 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+0.75L+0.75TL1
REPEAT LOAD
1 1.0 2 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL2
REPEAT LOAD
1 1.0 2 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+WL2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL3
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL4
REPEAT LOAD
```



```
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+0.75WL1+0.75L
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+0.75WL2+0.75L
REPEAT LOAD
1 1.0 2 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+0.75WL3+0.75L
REPEAT LOAD
1 1.0 2 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+0.75WL4+0.75L
REPEAT LOAD
1 1.0 2 0.75 6 0.75
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL+WL1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 0.6DL+WL2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 0.6DL+WL3
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL+WL4
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 7 0.7
PERFORM ANALYSIS
CHANGE
LOAD 219 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 8 0.7
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L
REPEAT LOAD
1 1.4 2 1.7 9 1.4
PERFORM ANALYSIS
CHANGE
LOAD 103 1.4DL+1.4TL2+1.7L
REPEAT LOAD
1 1.4 2 1.7 10 1.4
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.0L
REPEAT LOAD
1 1.2 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL1+1.0L
REPEAT LOAD
1 1.2 3 1.6 2 0.5 2 1.0
```

```
1 1.2 3 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L
REPEAT LOAD
1 1.2 4 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L
REPEAT LOAD
1 1.2 5 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 7 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 8 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 112 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 115 0.9DL+1.0SL1
REPEAT LOAD
1 0.9 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 0.9DL+1.0SL2
REPEAT LOAD
1 0.9 8 1.0
PERFORM ANALYSIS
CHANGE
*****************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 MEMB 185 1258
FC 35000 MEMB 185 1258
CLB 0.05 MEMB 185 1258
CLS 0.05 MEMB 185 1258
CLT 0.05 MEMB 185 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 -
414 TO 419 424 TO 429 497 498 501 TO 504 759 TO 764 767 TO 772 775 TO 780 -
783 TO 788 798 TO 801 883 TO 888 901 1258 1313 TO 1315 1317 TO 1319 1321 -
1322 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1341 TO 1343 1345 TO 1347 -
1349 TO 1351 1353 TO 1355 1361 TO 1363 1378 TO 1384 1405 TO 1421 -
1445 TO 1447 1449 TO 1451 1453 TO 1458 1460 TO 1465 1467 TO 1470 -
1472 TO 1475 1477 1478 1481 TO 1498 1501 TO 1508 1511 1512 1515 TO 1526 1528 -
1529 TO 1531 1533 1535 TO 1543 1545 TO 1548 1550 1552 TO 1554
MAXMAIN 25 MEMB 185 1258
MINMAIN 25 MEMB 185 1258
MINSEC 10 MEMB 185 1258
DESIGN COLUMN 185 1258
```



```
DESIGN COLUMN 185 1258
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 5 STAAD INPUT – FOUNDATION F4

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/11/2020
JOB NAME FDN. MKD. F4
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911 3.175; 6 0.75 911 3.175; 34 -0.5 911 3.175; 35 -0.25 911 3.175;
41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175; 302 -0.5 911 3.42538;
304 -0.25 911 3.42538; 305 0 911 3.42538; 306 0.25 911 3.42538;
307 0.5 911 3.42538; 308 0.75 911 3.42538; 309 1 911 3.175; 313 1 911 3.42538;
322 -0.5 911 3.67538; 323 -0.25 911 3.67538; 324 0 911 3.67538;
325 0.25 911 3.67538; 326 0.5 911 3.67538; 327 0.75 911 3.67538;
328 1 911 3.67538; 366 -0.5 911 2.17538; 369 -0.25 911 2.17538;
371 0 911 2.17538; 373 0.25 911 2.17538; 375 0.5 911 2.17538;
377 0.75 911 2.17538; 379 1 911 2.17538; 388 -0.5 911 2.42538;
389 -0.25 911 2.42538; 390 0 911 2.42538; 391 0.25 911 2.42538;
392 0.5 911 2.42538; 393 0.75 911 2.42538; 394 1 911 2.42538;
399 -0.5 911 2.67538; 400 -0.25 911 2.67538; 401 0 911 2.67538;
402 0.25 911 2.67538; 403 0.5 911 2.67538; 404 0.75 911 2.67538;
405 1 911 2.67538; 410 -0.5 911 2.92538; 411 -0.25 911 2.92538;
412 0 911 2.92538; 413 0.25 911 2.92538; 414 0.5 911 2.92538;
415 0.75 911 2.92538; 416 1 911 2.92538; 421 -0.5 911 1.92538;
424 -0.25 911 1.92538; 426 0 911 1.92538; 428 0.25 911 1.92538;
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492 1.25 911 3.42538; 493 1.25 911 3.175; 494 1.25 911 3.67538;
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818 1.25 911 1.67538; 819 1.25 911 0.67538; 903 -0.5 911 0.42538;
904 -0.25 911 0.42538; 905 0 911 0.42538; 906 0.25 911 0.42538;
907 0.5 911 0.42538; 908 0.75 911 0.42538; 909 1 911 0.42538;
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1466 1 911 -0.07462; 1467 1.25 911 -0.07462; 1483 -1.25 911 0.17538;
1485 -1 911 0.17538; 1486 -0.75 911 0.17538; 1487 -1.25 911 -0.07462;
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1557 1.25 911 -0.57462; 1558 -1 911 -0.57462; 1559 -1.25 911 -0.57462;
1560 -0.75 911 -0.57462; 1561 0 911 -0.07462;
MEMBER INCIDENCES
185 2 128; 1258 1561 1295;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 337 308 313 309 6; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
349 327 328 313 308; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 404 399 400 389 388; 405 400 401 390 389;
406 401 402 391 390; 407 402 403 392 391; 408 403 404 393 392;
409 404 405 394 393; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
419 415 416 405 404; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 442 366 369 424 421;
443 369 371 426 424; 444 371 373 428 426; 445 373 375 430 428;
446 375 377 432 430; 447 377 379 434 432; 497 313 492 493 309;
498 328 494 492 313; 501 394 497 498 379; 502 405 499 497 394;
503 416 500 499 405; 504 309 493 500 416; 506 379 498 501 434;
759 771 774 775 772; 760 774 776 1331 775; 761 776 778 779 1331;
762 778 780 781 779; 763 780 782 783 781; 764 782 784 785 783;
767 789 790 774 771; 768 790 791 776 774; 769 791 792 778 776;
770 792 793 780 778; 771 793 794 782 780; 772 794 795 784 782;
775 798 799 790 789; 776 799 800 791 790; 777 800 801 792 791;
778 801 802 793 792; 779 802 803 794 793; 780 803 804 795 794;
783 421 424 799 798; 784 424 426 800 799; 785 426 428 801 800;
786 428 430 802 801; 787 430 432 803 802; 788 432 434 804 803;
791 772 775 808 806; 792 775 1331 809 808; 793 1331 779 810 809;
794 779 781 811 810; 795 781 783 812 811; 796 783 785 813 812;
798 784 815 816 785; 799 795 817 815 784; 800 804 818 817 795;
801 434 501 818 804; 802 785 816 819 813; 883 806 808 904 903;
884 808 809 905 904; 885 809 810 906 905; 886 810 811 907 906;
887 811 812 908 907; 888 812 813 909 908; 901 813 819 923 909;
1313 1390 1393 1394 1391; 1314 1393 1395 1396 1394; 1315 1395 302 34 1396;
1317 1398 1399 1393 1390; 1318 1399 1400 1395 1393; 1319 1400 322 302 1395;
1321 1402 1405 1406 1403; 1322 1405 1407 1408 1406; 1323 1407 388 366 1408;
1325 1410 1411 1405 1402; 1326 1411 1412 1407 1405; 1327 1412 399 388 1407;
1329 1414 1415 1411 1410; 1330 1415 1416 1412 1411; 1331 1416 410 399 1412;
1333 1391 1394 1415 1414; 1334 1394 1396 1416 1415; 1335 1396 34 410 1416;
1337 1403 1406 1419 1417; 1338 1406 1408 1420 1419; 1339 1408 366 421 1420;
1341 1422 1425 1426 1423; 1342 1425 1427 1428 1426; 1343 1427 771 772 1428;
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1353 1417 1419 1435 1434; 1354 1419 1420 1436 1435; 1355 1420 421 798 1436;
1357 1423 1426 1439 1437; 1358 1426 1428 1440 1439; 1359 1428 772 806 1440;
1361 1437 1439 1443 1441; 1362 1439 1440 1444 1443; 1363 1440 806 903 1444;
1364 903 904 1445 1446; 1365 904 905 1538 1445; 1366 905 906 1448 1538;
1367 906 907 1449 1448; 1368 907 908 1450 1449; 1369 908 909 1451 1450;
1370 909 923 1452 1451; 1378 1446 1445 1460 1461; 1379 1445 1538 1561 1460;
1380 1538 1448 1463 1561; 1381 1448 1449 1464 1463; 1382 1449 1450 1465 1464;
1383 1450 1451 1466 1465; 1384 1451 1452 1467 1466; 1401 1441 1443 1485 1483;
1402 1443 1444 1486 1485; 1403 1444 903 1446 1486; 1405 1483 1485 1489 1487;
1406 1485 1486 1490 1489; 1407 1486 1446 1461 1490; 1408 1461 1460 1539 1540;
1409 1460 1561 1541 1539; 1410 1561 1463 1542 1541; 1411 1463 1464 1543 1542;
1412 1464 1465 1544 1543; 1413 1465 1466 1545 1544; 1414 1466 1467 1546 1545;
1415 1487 1489 1547 1548; 1416 1489 1490 1549 1547; 1417 1490 1461 1540 1549;
1418 1540 1539 1550 1551; 1419 1539 1541 1552 1550; 1420 1541 1542 1553 1552;
1421 1542 1543 1554 1553; 1422 1543 1544 1555 1554; 1423 1544 1545 1556 1555;
1424 1545 1546 1557 1556; 1425 1548 1547 1558 1559; 1426 1547 1549 1560 1558;
1427 1549 1540 1551 1560;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
425 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427
_RAFT_MOM 327 333 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
427 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 886 TO 888 901 1313 TO 1315 -
1317 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 -
1337 TO 1339 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 -
1357 TO 1359 1361 TO 1370 1378 1381 TO 1384 1401 TO 1403 1405 TO 1408 1411 -
1412 TO 1427
_RAFT_SHEAR 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 798 TO 802 901 1313 1317 1321 1325 1329 1333 -
```

```
775 TO 780 783 TO 788 798 TO 802 901 1313 1317 1321 1325 1329 1333 -
1337 TO 1339 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 1361 -
1370 1384 1401 1405 1414 1415 1424 1425
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1258 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1426 -
1427 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1 LIST ALL
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427 PR GY -16
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427 PR GY -20
JOINT LOAD
128 FX 0.03 FY -92.02 FZ 0.35
1295 FX 0.32 FY -69.18 FZ -0.58
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX -0.21 FY -95.7 FZ 0.1
1295 FX 0.97 FY -50.14 FZ -0.18
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WL1 )
JOINT LOAD
128 FX 2.96 FY 148.2 FZ 0.0001
1295 FX -0.28 FY -135.03 FZ 0.04
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X ( WL2 )
JOINT LOAD
128 FX -2.96 FY -148.2 FZ -0.0001
1295 FX 0.28 FY 135.03 FZ -0.04
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Y ( WL3 )
JOINT LOAD
128 FX 0.44 FY 230.2 FZ 42.33
1295 FX 0.41 FY -230.18 FZ 42.14
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Y ( WL4 )
JOINT LOAD
128 FX -0.44 FY -230.2 FZ -42.33
```



```
128 FX -0.44 FY -230.2 FZ -42.33
1295 FX -0.41 FY 230.18 FZ -42.14
*********************************************************************************************
PERFORM ANALYSIS
CHANGE
SUPPORTS
327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1426 -
1427 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+0.75L
REPEAT LOAD
1 1.0 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+WL1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+WL2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL3
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+WL4
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+0.75WL1+0.75L
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+0.75WL2+0.75L
REPEAT LOAD
1 1.0 2 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+0.75WL3+0.75L
REPEAT LOAD
1 1.0 2 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+0.75WL4+0.75L
REPEAT LOAD
1 1.0 2 0.75 6 0.75
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL+WL1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL+WL2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
```

```
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL+WL3
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 0.6DL+WL4
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.7L
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL+1.0L
REPEAT LOAD
1 1.2 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.6WL1+1.0L
REPEAT LOAD
1 1.2 3 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL2+1.0L
REPEAT LOAD
1 1.2 4 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL3+1.0L
REPEAT LOAD
1 1.2 5 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL4+1.0L
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 111
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 1258
CLB 0.05 MEMB 185 1258
CLS 0.05 ALL
CLT 0.05 MEMB 185 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 -
```



```
CLT 0.05 MEMB 185 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 -
414 TO 419 424 TO 429 497 498 501 TO 504 759 TO 764 767 TO 772 775 TO 780 -
783 TO 788 798 TO 801 883 TO 888 901 1258 1313 TO 1315 1317 TO 1319 1321 -
1322 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1341 TO 1343 1345 TO 1347 -
1349 TO 1351 1353 TO 1355 1361 TO 1363 1378 TO 1384 1405 TO 1427
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 1258
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 6 STAAD INPUT – FOUNDATION F5

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/11/2020
JOB NAME FDN. MKD. F5
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
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1539 400 1687 390 389; 1540 411 1686 1687 400; 1541 1641 1685 411 410;
1542 1652 1685 1641; 1543 1685 1686 411; 1544 1687 1688 390;
1545 1689 1690 1618; 1546 1626 1691 1624; 1547 1691 1692 1591;
1548 1693 1694 1603; 1549 371 1689 1618 1562; 1550 1695 1563 1564 1696;
1551 1697 1565 1563 1695; 1552 1698 1566 1565 1697; 1553 1696 1564 1567 1699;
1554 1700 1568 1569 1701; 1555 1702 1570 1568 1700; 1556 1703 1571 1570 1702;
1557 1699 1567 1571 1703; 1558 1701 1569 1572 1704; 1559 1705 1646 1566 1698;
1560 1706 1657 1646 1705; 1561 1704 1572 1668 1707; 1562 1664 1663 1708 1709;
1563 1663 1665 1710 1708; 1564 1667 1666 1711 1712; 1565 1666 1664 1709 1711;
1566 1668 1667 1712 1713; 1567 1670 1669 1714 1715; 1568 1673 1672 1716 1717;
1569 1672 1670 1715 1716; 1570 1665 1673 1717 1710; 1571 1707 1668 1713 1718;
1572 1669 1671 1719 1714; 1573 1709 1708 1720 1721; 1574 1708 1710 1722 1720;
1575 1712 1711 1723 1724; 1576 1711 1709 1721 1723; 1577 1713 1712 1724 1725;
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1596 1732 1734 1746 1744; 1597 1736 1735 1747 1748; 1598 1735 1733 1745 1747;
1599 1737 1736 1748 1749; 1600 1739 1738 1750 1751; 1601 1741 1740 1752 1753;
1602 1740 1739 1751 1752; 1603 1734 1741 1753 1746; 1604 1742 1737 1749 1754;
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1608 1748 1747 1759 1760; 1609 1747 1745 1757 1759; 1610 1749 1748 1760 1761;
1611 1751 1750 1762 1763; 1612 1753 1752 1764 1765; 1613 1752 1751 1763 1764;
1614 1746 1753 1765 1758; 1615 1754 1749 1761 1766; 1616 1750 1755 1767 1762;
1617 1757 1756 1768 1769; 1618 1756 1758 1770 1768; 1619 1760 1759 1771 1772;
1620 1759 1757 1769 1771; 1621 1761 1760 1772 1773; 1622 1763 1762 1774 1775;
1623 1765 1764 1776 1777; 1624 1764 1763 1775 1776; 1625 1758 1765 1777 1770;
1626 1766 1761 1773 1778; 1627 1762 1767 1779 1774; 1628 1769 1768 1780 1781;
1629 1768 1770 1782 1780; 1630 1772 1771 1783 1784; 1631 1771 1769 1781 1783;
1632 1773 1772 1784 1785; 1633 1775 1774 1786 1787; 1634 1777 1776 1788 1789;
1635 1776 1775 1787 1788; 1636 1770 1777 1789 1782; 1637 1778 1773 1785 1790;
1638 1774 1779 1791 1786; 1639 1781 1780 1792 1793; 1640 1780 1782 1794 1792;
1641 1784 1783 1795 1796; 1642 1783 1781 1793 1795; 1643 1785 1784 1796 1797;
1644 1787 1786 1798 1799; 1645 1789 1788 1800 1801; 1646 1788 1787 1799 1800;
1647 1782 1789 1801 1794; 1648 1790 1785 1797 1802; 1649 1786 1791 1803 1798;
1650 1793 1792 1804 1805; 1651 1792 1794 1806 1804; 1652 1796 1795 1807 1808;
1653 1795 1793 1805 1807; 1654 1797 1796 1808 1809; 1655 1799 1798 1810 1811;
1656 1801 1800 1812 1813; 1657 1800 1799 1811 1812; 1658 1794 1801 1813 1806;
1659 1802 1797 1809 1814; 1660 1798 1803 1815 1810; 1661 1805 1804 1816 1817;
1662 1804 1806 1818 1816; 1663 1808 1807 1819 1820; 1664 1807 1805 1817 1819;
1665 1809 1808 1820 1821; 1666 1811 1810 1822 1823; 1667 1813 1812 1824 1825;
1668 1812 1811 1823 1824; 1669 1806 1813 1825 1818; 1670 1814 1809 1821 1826;

1668 1812 1811 1823 1824; 1669 1806 1813 1825 1818; 1670 1814 1809 1821 1826;
1671 1810 1815 1827 1822; 1672 1817 1816 1828 1829; 1673 1816 1818 1830 1828;
1674 1820 1819 1831 1832; 1675 1819 1817 1829 1831; 1676 1821 1820 1832 1833;
1677 1823 1822 1834 1835; 1678 1825 1824 1836 1837; 1679 1824 1823 1835 1836;
1680 1818 1825 1837 1830; 1681 1826 1821 1833 1838; 1682 1822 1827 1839 1834;
1683 1829 1828 1840 1841; 1684 1828 1830 1842 1840; 1685 1832 1831 1843 1844;
1686 1831 1829 1841 1843; 1687 1833 1832 1844 1845; 1688 1835 1834 1846 1847;
1689 1837 1836 1848 1849; 1690 1836 1835 1847 1848; 1691 1830 1837 1849 1842;
1692 1838 1833 1845 1850; 1693 1834 1839 1851 1846; 1694 1841 1840 1852 1853;
1695 1840 1842 1854 1852; 1696 1844 1843 1855 1856; 1697 1843 1841 1853 1855;
1698 1845 1844 1856 1857; 1699 1847 1846 1858 1859; 1700 1849 1848 1860 1861;
1701 1848 1847 1859 1860; 1702 1842 1849 1861 1854; 1703 1850 1845 1857 1862;
1704 1846 1851 1863 1858; 1705 1853 1852 1864 1865; 1706 1852 1854 1866 1864;
1707 1856 1855 1867 1868; 1708 1855 1853 1865 1867; 1709 1857 1856 1868 1869;
1710 1859 1858 1870 1871; 1711 1861 1860 1872 1873; 1712 1860 1859 1871 1872;
1713 1854 1861 1873 1866; 1714 1862 1857 1869 1874; 1715 1858 1863 1875 1870;
1716 1865 1864 1876 1877; 1717 1864 1866 1878 1876; 1718 1868 1867 1879 1880;
1719 1867 1865 1877 1879; 1720 1869 1868 1880 1881; 1721 1871 1870 1882 1883;
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1725 1874 1869 1881 1886; 1726 1870 1875 1887 1882; 1727 1877 1876 1888 1889;
1728 1876 1878 1890 1888; 1729 1880 1879 1891 1892; 1730 1879 1877 1889 1891;
1731 1881 1880 1892 1893; 1732 1883 1882 1894 1895; 1733 1885 1884 1896 1897;
1734 1884 1883 1895 1896; 1735 1878 1885 1897 1890; 1736 1886 1881 1893 1898;
1737 1882 1887 1899 1894; 1738 1889 1888 1900 1901; 1739 1888 1890 1902 1900;
1740 1892 1891 1903 1904; 1741 1891 1889 1901 1903; 1742 1893 1892 1904 1905;
1743 1895 1894 1906 1907; 1744 1897 1896 1908 1909; 1745 1896 1895 1907 1908;
1746 1890 1897 1909 1902; 1747 1898 1893 1905 1910; 1748 1894 1899 1911 1906;
1749 1901 1900 1912 1913; 1750 1900 1902 1914 1912; 1751 1904 1903 1915 1916;
1752 1903 1901 1913 1915; 1753 1905 1904 1916 1917; 1754 1907 1906 1918 1919;
1755 1909 1908 1920 1921; 1756 1908 1907 1919 1920; 1757 1902 1909 1921 1914;
1758 1910 1905 1917 1922; 1759 1906 1911 1923 1918; 1760 1913 1912 1924 1925;
1761 1912 1914 1926 1924; 1762 1916 1915 1927 1928; 1763 1915 1913 1925 1927;
1764 1917 1916 1928 1929; 1765 1919 1918 1930 1931; 1766 1921 1920 1932 1933;
1767 1920 1919 1931 1932; 1768 1914 1921 1933 1926; 1769 1922 1917 1929 1934;
1770 1918 1923 1935 1930; 1771 1925 1924 1936 1937; 1772 1924 1926 1938 1936;
1773 1928 1927 1939 1940; 1774 1927 1925 1937 1939; 1775 1929 1928 1940 1941;
1776 1931 1930 1942 1943; 1777 1933 1932 1944 1945; 1778 1932 1931 1943 1944;
1779 1926 1933 1945 1938; 1780 1934 1929 1941 1946; 1781 1930 1935 1947 1942;
1782 1937 1936 1948 1949; 1783 1936 1938 1950 1948; 1784 1940 1939 1951 1952;
1785 1939 1937 1949 1951; 1786 1941 1940 1952 1953; 1787 1943 1942 1954 1955;
1788 1945 1944 1956 1957; 1789 1944 1943 1955 1956; 1790 1938 1945 1957 1950;
1791 1946 1941 1953 1958; 1792 1942 1947 1959 1954; 1793 1949 1948 1960 1961;
1794 1948 1950 1962 1960; 1795 1952 1951 1963 1964; 1796 1951 1949 1961 1963;
1797 1953 1952 1964 1965; 1798 1955 1954 1966 1967; 1799 1957 1956 1968 1969;
1800 1956 1955 1967 1968; 1801 1950 1957 1969 1962; 1802 1958 1953 1965 1970;
1803 1954 1959 1971 1966; 1804 1961 1960 1972 1973; 1805 1960 1962 1974 1972;
1806 1964 1963 1975 1976; 1807 1963 1961 1973 1975; 1808 1965 1964 1976 1977;
1809 1967 1966 1978 1979; 1810 1969 1968 1980 1981; 1811 1968 1967 1979 1980;
1812 1962 1969 1981 1974; 1813 1970 1965 1977 1982; 1814 1966 1971 1983 1978;
1815 1973 1972 1984 1985; 1816 1972 1974 1986 1984; 1817 1976 1975 1987 1988;
1818 1975 1973 1985 1987; 1819 1977 1976 1988 1989; 1820 1979 1978 1990 1991;
1821 1981 1980 1992 1993; 1822 1980 1979 1991 1992; 1823 1974 1981 1993 1986;
1824 1982 1977 1989 1994; 1825 1978 1983 1995 1990; 1826 1985 1984 1996 1997;
1827 1984 1986 1998 1996; 1828 1988 1987 1999 2000; 1829 1987 1985 1997 1999;
1830 1989 1988 2000 2001; 1831 1991 1990 2002 2003; 1832 1993 1992 2004 2005;
1833 1992 1991 2003 2004; 1834 1986 1993 2005 1998; 1835 1994 1989 2001 2006;
1836 1990 1995 2007 2002; 1837 1997 1996 2008 2009; 1838 1996 1998 2010 2008;
1839 2000 1999 2011 2012; 1840 1999 1997 2009 2011; 1841 2001 2000 2012 2013;
1842 2003 2002 2014 2015; 1843 2005 2004 2016 2017; 1844 2004 2003 2015 2016;
1845 1998 2005 2017 2010; 1846 2006 2001 2013 2018; 1847 2002 2007 2019 2014;
1848 2009 2008 2020 2021; 1849 2008 2010 2022 2020; 1850 2012 2011 2023 2024;
1851 2011 2009 2021 2023; 1852 2013 2012 2024 2025; 1853 2015 2014 2026 2027;
1854 2017 2016 2028 2029; 1855 2016 2015 2027 2028; 1856 2010 2017 2029 2022;
1857 2018 2013 2025 2030; 1858 2014 2019 2031 2026;
START GROUP DEFINITION
ELEMENT
_RAFT 394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 -
1323 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 -
1358 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 -
1492 TO 1494 1500 TO 1505 1507 TO 1509 1550 TO 1858
_RAFT_MOM 394 395 404 414 442 759 760 767 768 775 776 783 791 792 1322 1323 -
1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1474 1476 1482 TO 1484 -
1492 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1548 -
1550 TO 1716 1718 TO 1723 1725 TO 1727 1729 TO 1734 1736 TO 1826 -
1828 TO 1833 1835 TO 1837 1839 TO 1844 1846 TO 1858
_RAFT_SHEAR 414 759 760 791 792 1322 1326 1330 1331 1338 1342 1343 1346 1350 -
1354 1358 1359 1408 TO 1416 1448 1449 1456 1457 1472 1476 1482 TO 1484 1492 -
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1534 1540 TO 1543 1548 1550 -

```
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1534 1540 TO 1543 1548 1550 -
1551 TO 1693 1696 1698 1699 1703 1704 1707 1709 1710 1714 1715 1718 1720 1721 -
1725 1726 1729 1731 1732 1736 1737 1740 1742 1743 1747 1748 1751 1753 1754 -
1758 TO 1803 1806 1808 1809 1813 1814 1817 1819 1820 1824 1825 1828 1830 -
1831 1835 1836 1839 1841 1842 1846 1847 1850 1852 1853 1857 1858
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1857 -
1858 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1258 2002 2003 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1857 -
1858 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1 LIST ALL
ELEMENT LOAD
* LOAD DUE TO BACKFILL
394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1858 PR GY -10
* LOAD DUE TO SURCHARGE
394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1858 PR GY -20
JOINT LOAD
1295 FX 0.5 FY -203.1 FZ 0.02
2188 FX 1.1 FY -148.26 FZ 1.5
2189 FX 2.17 FY -107.92 FZ 0.08
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
1295 FX 0.31 FY -245.85 FZ -0.07
2188 FX 8.9 FY -88.68 FZ 6.42
2189 FX 0.09 FY -29.11 FZ 0.1
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WL1 )
JOINT LOAD
1295 FX -0.73 FY 13.53 FZ 0.07
2188 FX -178.53 FY -279 FZ -12.56
2189 FX -62.6 FY 230.1 FZ -0.02
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X ( WL2 )
JOINT LOAD
1295 FX 0.74 FY -13 FZ -0.07
2188 FX 179.4 FY 280.45 FZ 12.41
2189 FX 62.6 FY -230.07 FZ 0.01
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Y ( WL3 )
JOINT LOAD
1295 FX -0.46 FY 399.87 FZ -1.21
2188 FX 12.72 FY -345.28 FZ -160.55
2189 FX -6.44 FY -2.43 FZ -2.48
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Y ( WL4 )
JOINT LOAD
1295 FX 0.45 FY -394.16 FZ 1.21
```



```
1295 FX 0.45 FY -394.16 FZ 1.21
2188 FX -12 FY 341.08 FZ 159.19
2189 FX 6.44 FY 2.21 FZ 2.5
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
1295 FX -0.03 FY 5.07 FZ 0.0001
2188 FX -3.89 FY 6.33 FZ -1.67
2189 FX -2.17 FY 10.83 FZ -0.02
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
1295 FX 0.03 FY -5.07 FZ -0.0001
2188 FX 3.89 FY -6.33 FZ 1.67
2189 FX 2.17 FY -10.83 FZ 0.02
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
1295 FX 0.06 FY 23.97 FZ 0.18
2188 FX -3.53 FY -27.12 FZ -9.28
2189 FX 1.03 FY -1.17 FZ -0.44
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
1295 FX -0.06 FY -23.97 FZ -0.18
2188 FX 3.53 FY 27.12 FZ 9.28
2189 FX -1.03 FY 1.17 FZ 0.44
*********************************************************************************************
PERFORM ANALYSIS
CHANGE
SUPPORTS
394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1857 -
1858 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L+TL1
REPEAT LOAD
1 1.0 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+L+TL2
REPEAT LOAD
1 1.0 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+0.75L+0.75TL1
REPEAT LOAD
1 1.0 2 0.75 9 0.75
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+0.75L+0.75TL2
REPEAT LOAD
1 1.0 2 0.75 10 0.75
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+WL2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+WL3
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+WL4
```

```
LOAD 209 DL+WL4
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+0.75WL1+0.75L
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+0.75WL2+0.75L
REPEAT LOAD
1 1.0 2 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 212 DL+0.75WL3+0.75L
REPEAT LOAD
1 1.0 2 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 213 DL+0.75WL4+0.75L
REPEAT LOAD
1 1.0 2 0.75 6 0.75
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL+WL1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 0.6DL+WL2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 0.6DL+WL3
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL+WL4
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 7 0.7
PERFORM ANALYSIS
CHANGE
LOAD 219 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 8 0.7
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.4TL1+1.7L
REPEAT LOAD
1 1.4 2 1.7 9 1.4
PERFORM ANALYSIS
CHANGE
LOAD 103 1.4DL+1.4TL2+1.7L
REPEAT LOAD
1 1.4 2 1.7 10 1.4
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.0L
REPEAT LOAD
1 1.2 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL1+1.0L
REPEAT LOAD
```



```
REPEAT LOAD
1 1.2 3 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL2+1.0L
REPEAT LOAD
1 1.2 4 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL3+1.0L
REPEAT LOAD
1 1.2 5 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 1.2DL+1.6WL4+1.0L
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 109 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 7 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 110 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 8 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 112 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 115 0.9DL+1.0SL1
REPEAT LOAD
1 0.9 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 0.9DL+1.0SL2
REPEAT LOAD
1 0.9 8 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 1258 2002 2003
CLB 0.05 MEMB 1258 2002 2003
CLS 0.05 ALL
CLT 0.05 MEMB 394 395 404 414 759 760 767 768 775 776 783 784 1258 1322 1323 -
1326 1327 1330 1331 1342 1343 1346 1347 1350 1351 1354 1355 1408 TO 1410 -
1412 TO 1415 1448 TO 1450 1452 1472 TO 1475 1482 TO 1484 1492 TO 1494 1550 -
1551 TO 1552 1554 TO 1557 1559 1560 2002 2003
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 1258 2002 2003
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 7 STAAD INPUT – FOUNDATION F6

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/11/2020
JOB NAME FDN. MKD. F6
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911 3.175; 6 0.75 911 3.175; 34 -0.5 911 3.175; 35 -0.25 911 3.175;
41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175; 302 -0.5 911 3.47538;
304 -0.25 911 3.47538; 305 0 911 3.47538; 306 0.25 911 3.47538;
307 0.5 911 3.47538; 308 0.75 911 3.47538; 309 1 911 3.175; 313 1 911 3.47538;
366 -0.5 911 2.17538; 369 -0.25 911 2.17538; 371 0 911 2.17538;
373 0.25 911 2.17538; 375 0.5 911 2.17538; 377 0.75 911 2.17538;
379 1 911 2.17538; 388 -0.5 911 2.42538; 389 -0.25 911 2.42538;
390 0 911 2.42538; 391 0.25 911 2.42538; 392 0.5 911 2.42538;
393 0.75 911 2.42538; 394 1 911 2.42538; 399 -0.5 911 2.67538;
400 -0.25 911 2.67538; 401 0 911 2.67538; 402 0.25 911 2.67538;
403 0.5 911 2.67538; 404 0.75 911 2.67538; 405 1 911 2.67538;
410 -0.5 911 2.92538; 411 -0.25 911 2.92538; 412 0 911 2.92538;
413 0.25 911 2.92538; 414 0.5 911 2.92538; 415 0.75 911 2.92538;
416 1 911 2.92538; 421 -0.5 911 1.92538; 424 -0.25 911 1.92538;
426 0 911 1.92538; 428 0.25 911 1.92538; 430 0.5 911 1.92538;
432 0.75 911 1.92538; 434 1 911 1.92538; 771 -0.5 911 1.17538;
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MEMBER INCIDENCES
185 2 128; 1258 1562 1295;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 337 308 313 309 6; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 404 399 400 389 388; 405 400 401 390 389;
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409 404 405 394 393; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
419 415 416 405 404; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 442 366 369 424 421;
443 369 371 426 424; 444 371 373 428 426; 445 373 375 430 428;
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779 802 803 794 793; 780 803 804 795 794; 783 421 424 799 798;
784 424 426 800 799; 785 426 428 801 800; 786 428 430 802 801;
787 430 432 803 802; 788 432 434 804 803; 791 772 775 808 806;
792 775 1331 809 808; 793 1331 779 810 809; 794 779 781 811 810;
795 781 783 812 811; 796 783 785 813 812; 883 806 808 904 903;
884 808 809 905 904; 885 809 810 906 905; 886 810 811 907 906;
887 811 812 908 907; 888 812 813 909 908; 1314 1393 1395 1396 1394;
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1326 1411 1412 1407 1405; 1327 1412 399 388 1407; 1330 1415 1416 1412 1411;
1331 1416 410 399 1412; 1334 1394 1396 1416 1415; 1335 1396 34 410 1416;
1338 1406 1408 1420 1419; 1339 1408 366 421 1420; 1342 1425 1427 1428 1426;
1343 1427 771 772 1428; 1346 1431 1432 1427 1425; 1347 1432 789 771 1427;
1350 1435 1436 1432 1431; 1351 1436 798 789 1432; 1354 1419 1420 1436 1435;
1355 1420 421 798 1436; 1358 1426 1428 1440 1439; 1359 1428 772 806 1440;
1362 1439 1440 1444 1443; 1363 1440 806 903 1444; 1364 903 904 1445 1446;
1365 904 905 1562 1445; 1366 905 906 1448 1562; 1367 906 907 1449 1448;
1368 907 908 1450 1449; 1369 908 909 1451 1450; 1378 1446 1445 1460 1461;
1379 1445 1562 1561 1460; 1380 1562 1448 1463 1561; 1381 1448 1449 1464 1463;
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1403 1444 903 1446 1486; 1406 1485 1486 1490 1489; 1407 1486 1446 1461 1490;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 442 -
443 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 883 TO 888 -
1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 1343 1346 -
1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 1403 1406 -
1407
_RAFT_MOM 327 333 334 337 394 TO 399 404 TO 409 414 TO 419 424 427 TO 429 -
442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1364 1367 TO 1369 1378 -
1381 TO 1383 1402 1403 1406 1407
_RAFT_SHEAR 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 1338 1339 -
1342 1343 1346 1347 1350 1351 1354 1355
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
1403 1406 1407 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1258 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
1403 1406 1407 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
**********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
1403 1406 1407 PR GY -16
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
1403 1406 1407 PR GY -20
```

```
1403 1406 1407 PR GY -20
JOINT LOAD
128 FX -0.21 FY -37.55 FZ 0.42
1295 FX 0.02 FY -72 FZ -0.07
**************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 0.15 FY -28.56 FZ 0.15
1295 FX 0.1 FY -58.49 FZ -0.14
**************************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WL1 )
JOINT LOAD
128 FX -28.45 FY 92.71 FZ 1.01
1295 FX -6.1 FY 31.79 FZ 0.09
**************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X ( WL2 )
JOINT LOAD
128 FX 27.73 FY -88.75 FZ -0.88
1295 FX 6.11 FY -29.6 FZ -0.09
**************************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Y ( WL3 )
JOINT LOAD
128 FX 3.19 FY 102.06 FZ -33.34
1295 FX -0.06 FY -117.78 FZ -0.63
**************************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Y ( WL4 )
JOINT LOAD
128 FX -3.12 FY -102.34 FZ 33.33
1295 FX 0.06 FY 117.92 FZ 0.63
**************************************************************************************************
LOAD 7 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX 3.62 FY -25.58 FZ 2.05
1295 FX 0.1 FY 3.3 FZ -0.29
**************************************************************************************************
LOAD 8 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX -3.62 FY 25.58 FZ -2.05
1295 FX -0.1 FY -3.3 FZ 0.29
PERFORM ANALYSIS
CHANGE
SUPPORTS
327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
1403 1406 1407 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
**************************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+0.75L
REPEAT LOAD
1 1.0 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+WL1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+WL2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL3
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
```



```
CHANGE
LOAD 207 DL+WL4
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+0.75WL1+0.75L
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 DL+0.75WL2+0.75L
REPEAT LOAD
1 1.0 2 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 DL+0.75WL3+0.75L
REPEAT LOAD
1 1.0 2 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 DL+0.75WL4+0.75L
REPEAT LOAD
1 1.0 2 0.75 6 0.75
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL+WL1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL+WL2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL+WL3
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 0.6DL+WL4
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 DL+L+TL1
REPEAT LOAD
1 1.0 2 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 DL+L+TL2
REPEAT LOAD
1 1.0 2 1.0 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 218 DL+0.75L+0.75TL1
REPEAT LOAD
1 1.0 2 0.75 7 0.75
PERFORM ANALYSIS
CHANGE
LOAD 219 DL+0.75L+0.75TL2
REPEAT LOAD
1 1.0 2 0.75 8 0.75
PERFORM ANALYSIS
CHANGE
**************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.7L
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL+1.0L
```

```
LOAD 103 1.2DL+1.0L
REPEAT LOAD
1 1.2 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.6WL1+1.0L
REPEAT LOAD
1 1.2 3 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL2+1.0L
REPEAT LOAD
1 1.2 4 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.6WL3+1.0L
REPEAT LOAD
1 1.2 5 1.6 2 0.5 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.6WL4+1.0L
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL+1.6WL3
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.6WL4
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 112 1.4DL+1.4TL1+1.7L
REPEAT LOAD
1 1.4 2 1.7 7 1.4
PERFORM ANALYSIS
CHANGE
LOAD 113 1.4DL+1.4TL2+1.7L
REPEAT LOAD
1 1.4 2 1.7 8 1.4
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 113
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 1258
CLB 0.05 MEMB 185 1258
CLS 0.05 ALL
CLT 0.05 MEMB 185 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 -
424 TO 429 759 TO 764 767 TO 772 775 TO 780 783 TO 788 883 TO 888 1258 1314 -
1315 1322 1323 1326 1327 1330 1331 1334 1335 1342 1343 1346 1347 1350 1351 -
1354 1355 1362 1363 1378 TO 1383 1406 1407
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 1258
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 8 STAAD OUTPUT – FOUNDATION F1

PROPRIETARY & CONFIDENTIAL

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    NOV  5, 2020                 *
*          Time=    17:10:24                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
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INPUT FILE: F1.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911 3.175; 6 0.75 911 3.175; 10 -0.75 911 3.175; 34 -0.5 911 3.175
   17. 35 -0.25 911 3.175; 41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175
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   46. 590 1.5 911 2.17538; 591 1.75 911 2.42538; 592 1.75 911 2.17538
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   53. 639 2.5 911 3.67538; 642 2.5 911 2.42538; 643 2.5 911 2.17538
   54. 644 2.5 911 2.67538; 645 2.5 911 2.92538; 646 2.5 911 1.92538
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   57. 671 3 911 3.67538; 672 3.25 911 3.67538; 679 2.75 911 2.42538
   58. 680 2.75 911 2.17538; 681 3 911 2.42538; 682 3 911 2.17538
   59. 683 3.25 911 2.42538; 684 3.25 911 2.17538; 685 2.75 911 2.67538
   60. 686 3 911 2.67538; 687 3.25 911 2.67538; 688 2.75 911 2.92538
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   63. 716 3.5 911 3.175; 717 3.5 911 3.67538; 720 3.5 911 2.42538
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468. START GROUP DEFINITION
469. ELEMENT
470. _RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 -
471. 412 TO 419 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 -
472. 585 TO 599 604 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 -
473. 685 TO 687 706 707 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 -
474. 756 758 TO 1189 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 -
475. 1244 TO 1246 1248 TO 1251 1253 TO 1256 1260 TO 1290
476. _RAFT_MOM 326 327 333 334 337 342 TO 349 352 393 TO 399 402 TO 409 -
477. 412 TO 419 422 TO 424 427 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 -
478. 569 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 TO 663 -
479. 670 TO 680 685 TO 687 707 710 TO 712 715 731 732 735 TO 738 740 747 748 751 -
480. 752 TO 754 756 758 TO 814 816 TO 1114 1116 TO 1167 1170 TO 1175 1178 TO 1189 -
481. 1198 TO 1201 1203 TO 1214 1216 TO 1218 1223 TO 1226 1228 TO 1232 1234 1235 -
482. 1237 TO 1239 1243 1246 1248 TO 1251 1253 TO 1256 1260 TO 1290
483. _RAFT_SHEAR 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 -
484. 604 TO 607 641 685 TO 687 715 740 756 758 TO 1089 1098 TO 1122 1124 TO 1126 -
485. 1131 TO 1139 1144 TO 1146 1149 TO 1151 1154 TO 1156 1198 TO 1201 -
486. 1203 TO 1218 1266 TO 1281 1283 TO 1285
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487. MEMBER
488. END GROUP DEFINITION
489. ELEMENT PROPERTY
490. 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
491. 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
492. 605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
493. 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
494. 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
495. 1248 TO 1251 1253 TO 1256 1260 TO 1290 THICKNESS 0.7
496. DEFINE MATERIAL START
497. ISOTROPIC CONCRETE
498. E 2.17185E+007
499. POISSON 0.17
500. DENSITY 25
501. ALPHA 1E-005
502. DAMP 0.05
503. TYPE CONCRETE
504. STRENGTH FCU 27579
505. END DEFINE MATERIAL
506. MEMBER PROPERTY AMERICAN
507. 185 730 1258 1259 PRIS YD 0.6 ZD 0.6
508. CONSTANTS
509. MATERIAL CONCRETE ALL
510. SUPPORTS
511. 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
512. 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
513. 605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
514. 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
515. 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
516. 1248 TO 1251 1253 TO 1256 1260 TO 1289 -
517. 1290 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
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760	0.031250
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1014	0.062500

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1122	0.062500
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1126	0.062500

1127	0.062500
1128	0.062500
1129	0.031250
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1198	0.062500
1199	0.031250
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1207	0.015625
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1235	0.062500
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1245	0.031250
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1263	0.062500
1265	0.062500
1266	0.031250
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1268	0.031250
1269	0.062500
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1272	0.062500
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1297	0.031274
1298	0.031250
1299	0.015625
1300	0.031250
1301	0.031250
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1303	0.031226
1304	0.031250
1305	0.031250
1306	0.031250
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1309	0.031250
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1312	0.031250
1313	0.031250
1314	0.031250
1315	0.031250
1316	0.031250
1317	0.031250

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      1318          0.031250
      1319          0.031250
      1320          0.031250
      1321          0.031250
      1322          0.031250
      1323          0.031250
      1324          0.031250
      1325          0.031250
      1326          0.015625
      1327          0.031250
      1328          0.031250
     TOTAL         40.687500
518. ******************************************************************************
519. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
520. SELFWEIGHT Y -1
521. ELEMENT LOAD
522. * LOAD DUE TO BACKFILL
523. 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
524. 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
525. 605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
526. 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
527. 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
528. 1248 TO 1251 1253 TO 1256 1260 TO 1290 PR GY -16
529. * LOAD DUE TO SURCHARGE
530. 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
531. 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
532. 605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
533. 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
534. 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
535. 1248 TO 1251 1253 TO 1256 1260 TO 1290 PR GY -20
536. JOINT LOAD
537. 128 FX -0.01 FY -52.59 FZ 0.88
538. 740 FX -0.66 FY -138.22 FZ 0.26
539. 1295 FX 0.07 FY -74.1 FZ -0.42
540. 1296 FX 0.0001 FY -144 FZ -2.07
541. ******************************************************************************
542. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
543. JOINT LOAD
544. 128 FX -0.14 FY -101.7 FZ 2
545. 740 FX -8.89 FY -241.3 FZ 0.81
546. 1295 FX 0.09 FY -139.24 FZ -1.06
547. 1296 FX -0.04 FY -346.9 FZ -10.35
548. ******************************************************************************
549. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WL1 )
550. JOINT LOAD
551. 128 FX -2.54 FY 393.02 FZ -0.24
552. 740 FX -114.49 FY -420.8 FZ -0.31
553. 1295 FX -5.79 FY -58.16 FZ 1.21
554. 1296 FX -4.87 FY 263.66 FZ 18
555. ******************************************************************************
556. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X ( WL2 )
557. JOINT LOAD
558. 128 FX 12 FY -390.1 FZ 0.23
559. 740 FX 106.5 FY 416.92 FZ 0.31
560. 1295 FX 4.49 FY -56.45 FZ -1.24
561. 1296 FX 7.87 FY -266.01 FZ -17.58
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562. ******************************************************************************
563. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Y ( WL3 )
564. JOINT LOAD
565. 128 FX -0.04 FY 111.7 FZ -16
566. 740 FX -6.47 FY 80.95 FZ -9.04
567. 1295 FX 0.1 FY -85.93 FZ -45.02
568. 1296 FX 0.03 FY -130.38 FZ -73.74
569. ******************************************************************************
570. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Y ( WL4 )
571. JOINT LOAD
572. 128 FX 0.04 FY -109 FZ 16
573. 740 FX 5.78 FY -82.8 FZ 5
574. 1295 FX -0.1 FY 85.55 FZ 45.9
575. 1296 FX -0.03 FY 128.72 FZ 77.13
576. ******************************************************************************
577. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
578. JOINT LOAD
579. 128 FX -0.02 FY -11.7 FZ -0.2
580. 740 FX -2.57 FY 14.34 FZ -0.03
581. 1295 FX -0.02 FY 2.37 FZ -0.75
582. 1296 FX -0.02 FY 6.48 FZ -1.7
583. ******************************************************************************
584. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
585. JOINT LOAD
586. 128 FX 0.02 FY 11.7 FZ 0.2
587. 740 FX 2.57 FY -14.34 FZ 0.03
588. 1295 FX 0.03 FY -2.37 FZ 0.75
589. 1296 FX 0.02 FY -6.48 FZ 1.7
590. ******************************************************************************
591. LOAD 9 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
592. JOINT LOAD
593. 128 FX 8.9 FY -9.77 FZ 4.05
594. 740 FX -5.15 FY 2 FZ 1.5
595. 1295 FX 0.58 FY -0.3 FZ 3.75
596. 1296 FX 0.08 FY -11.26 FZ -3.07
597. ******************************************************************************
598. LOAD 10 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
599. JOINT LOAD
600. 128 FX -8.9 FY 9.77 FZ -4.05
601. 740 FX 5.15 FY -2 FZ -1.5
602. 1295 FX -0.58 FY -0.3 FZ 3.75
603. 1296 FX 0.08 FY -11.26 FZ -3.07
604. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**708**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**651**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**704**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 519/ 36/ 111 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 2136
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 238 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 18.2/ 142122.0 MB

```
605. CHANGE
606. SUPPORTS
607. 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 TO 409 412 TO 419 -
608. 422 TO 429 432 441 TO 448 497 498 501 TO 504 506 568 TO 575 584 TO 599 604 -
609. 605 TO 607 632 633 636 TO 639 641 658 TO 663 670 TO 681 685 TO 687 706 707 -
610. 710 TO 713 715 731 732 735 TO 738 740 747 748 751 TO 754 756 758 TO 1189 -
611. 1198 TO 1201 1203 TO 1218 1223 TO 1226 1228 TO 1239 1243 TO 1246 -
612. 1248 TO 1251 1253 TO 1256 1260 TO 1289 -
613. 1290 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
614. ********************************************************************************
615. * UNFACTORED LOAD COMBINATIONS
616. LOAD 201 DL
617. REPEAT LOAD
618. 1 1.0
619. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201

620. CHANGE
621. LOAD 202 DL+L+TL1
622. REPEAT LOAD
623. 1 1.0 2 1.0 9 1.0
624. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202

625. CHANGE
626. LOAD 203 DL+L+TL2
627. REPEAT LOAD
628. 1 1.0 2 1.0 10 1.0
629. PERFORM ANALYSIS
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**NOTE-Tension/Compression converged after  1 iterations, Case=  203

630. CHANGE
631. LOAD 204 DL+0.75L+0.75TL1
632. REPEAT LOAD
633. 1 1.0 2 0.75 9 0.75
634. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204

635. CHANGE
636. LOAD 205 DL+0.75L+0.75TL2
637. REPEAT LOAD
638. 1 1.0 2 0.75 10 0.75
639. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  205

640. CHANGE
641. LOAD 206 DL+WL1
642. REPEAT LOAD
643. 1 1.0 3 1.0
644. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  206

645. CHANGE
646. LOAD 207 DL+WL2
647. REPEAT LOAD
648. 1 1.0 4 1.0
649. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207
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```
 650. CHANGE
 651. LOAD 208 DL+WL3
 652. REPEAT LOAD
 653. 1 1.0 5 1.0
 654. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  208



 655. CHANGE
 656. LOAD 209 DL+WL4
 657. REPEAT LOAD
 658. 1 1.0 6 1.0
 659. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  209



 660. CHANGE
 661. LOAD 210 DL+0.75WL1+0.75L
 662. REPEAT LOAD
 663. 1 1.0 2 0.75 3 0.75
 664. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



 665. CHANGE
 666. LOAD 211 DL+0.75WL2+0.75L
 667. REPEAT LOAD
 668. 1 1.0 2 0.75 4 0.75
 669. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



 670. CHANGE
 671. LOAD 212 DL+0.75WL3+0.75L
 672. REPEAT LOAD
 673. 1 1.0 2 0.75 5 0.75
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 674. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



 675. CHANGE
 676. LOAD 213 DL+0.75WL4+0.75L
 677. REPEAT LOAD
 678. 1 1.0 2 0.75 6 0.75
 679. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



 680. CHANGE
 681. LOAD 214 0.6DL+WL1
 682. REPEAT LOAD
 683. 1 0.6 3 1.0
 684. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  214



 685. CHANGE
 686. LOAD 215 0.6DL+WL2
 687. REPEAT LOAD
 688. 1 0.6 4 1.0
 689. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  215



 690. CHANGE
 691. LOAD 216 0.6DL+WL3
 692. REPEAT LOAD
 693. 1 0.6 5 1.0
 694. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

695. CHANGE
696. LOAD 217 0.6DL+WL4
697. REPEAT LOAD
698. 1 0.6 6 1.0
699. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 217

700. CHANGE
701. LOAD 218 0.6DL+0.7SL1
702. REPEAT LOAD
703. 1 0.6 7 0.7
704. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

705. CHANGE
706. LOAD 219 0.6DL+0.7SL2
707. REPEAT LOAD
708. 1 0.6 8 0.7
709. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

710. CHANGE
711. **
712. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
713. LOAD 101 1.4DL
714. REPEAT LOAD
715. 1 1.4
716. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

717. CHANGE
718. LOAD 102 1.4DL+1.4TL1+1.7L
719. REPEAT LOAD
720. 1 1.4 2 1.7 9 1.4
721. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

722. CHANGE
723. LOAD 103 1.4DL+1.4TL2+1.7L
724. REPEAT LOAD
725. 1 1.4 2 1.7 10 1.4
726. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

727. CHANGE
728. LOAD 104 1.2DL+1.0L
729. REPEAT LOAD
730. 1 1.2 2 1.0
731. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

732. CHANGE
733. LOAD 105 1.2DL+1.6WL1+1.0L
734. REPEAT LOAD
735. 1 1.2 3 1.6 2 0.5 2 1.0
736. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

737. CHANGE
738. LOAD 106 1.2DL+1.6WL2+1.0L
739. REPEAT LOAD
740. 1 1.2 4 1.6 2 0.5 2 1.0
741. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

742. CHANGE
743. LOAD 107 1.2DL+1.6WL3+1.0L
744. REPEAT LOAD
745. 1 1.2 5 1.6 2 0.5 2 1.0
746. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

747. CHANGE
748. LOAD 108 1.2DL+1.6WL4+1.0L
749. REPEAT LOAD
750. 1 1.2 6 1.6 2 1.0
751. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

752. CHANGE
753. LOAD 109 1.2DL+1.0SL1+1.0L
754. REPEAT LOAD
755. 1 1.2 7 1.0 2 1.0
756. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

757. CHANGE
758. LOAD 110 1.2DL+1.0SL2+1.0L
759. REPEAT LOAD
760. 1 1.2 8 1.0 2 1.0
761. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

762. CHANGE
763. LOAD 111 0.9DL+1.6WL1
764. REPEAT LOAD
765. 1 0.9 3 1.6
766. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 111

767. CHANGE
768. LOAD 112 0.9DL+1.6WL2
769. REPEAT LOAD
770. 1 0.9 4 1.6
771. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 112

772. CHANGE
773. LOAD 113 0.9DL+1.6WL3
774. REPEAT LOAD
775. 1 0.9 5 1.6
776. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

```
 777. CHANGE
 778. LOAD 114 0.9DL+1.6WL4
 779. REPEAT LOAD
 780. 1 0.9 6 1.6
 781. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  114



 782. CHANGE
 783. LOAD 115 0.9DL+1.0SL1
 784. REPEAT LOAD
 785. 1 0.9 7 1.0
 786. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  115



 787. CHANGE
 788. LOAD 116 0.9DL+1.0SL2
 789. REPEAT LOAD
 790. 1 0.9 8 1.0
 791. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  116



 792. CHANGE
 793. ******************************************************************************
 794. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=    2
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**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6
**START ITERATION NO.          7
**START ITERATION NO.          8
**START ITERATION NO.          9

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG= -1.1837110E-06 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902206E+06 L-MATRIX DIAG= -2.4092849E-05 EQN NO    2136
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.         10

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.3795981E+04 L-MATRIX DIAG=  1.2369128E-10 EQN NO    2134
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG=  2.5393441E-05 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902206E+06 L-MATRIX DIAG=  2.0797364E-05 EQN NO    2136
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
 **START ITERATION NO.         11
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 **START ITERATION NO.          12
 **START ITERATION NO.          13
 **START ITERATION NO.          14
 **START ITERATION NO.          15

 **WARNING-Tension/Compression did not converge after 15 iterations, Case=    3
 ** If results are not satisfactory then enter a
 ** SET ITERLIM i command (i>10) before loads to get more iterations.



 **START ITERATION NO.           2
 **START ITERATION NO.           3
 **START ITERATION NO.           4
 **START ITERATION NO.           5
 **START ITERATION NO.           6
 **START ITERATION NO.           7
 **START ITERATION NO.           8
 **START ITERATION NO.           9
 **START ITERATION NO.          10
 **START ITERATION NO.          11
 **START ITERATION NO.          12

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG=  2.5393441E-05 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          13

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG=  2.5390182E-05 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          14
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***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG=  2.5394838E-05 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          15

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG=  2.5392976E-05 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


 **NOTE-Tension/Compression converged after 15 iterations, Case=    4



 **START ITERATION NO.           2
 **START ITERATION NO.           3
 **START ITERATION NO.           4
 **START ITERATION NO.           5
 **START ITERATION NO.           6

***WARNING -  INSTABILITY AT JOINT  1207   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902247E+06 L-MATRIX DIAG=  2.5396235E-05 EQN NO    2135
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           7
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***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5395770E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5393441E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5397632E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5394838E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5395304E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5393441E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -1.6942155E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 12 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.2938627E-06 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 1.1496712E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.4330955E-07 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.0265579E-06 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 7.1795657E-06 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.8453302E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795981E+04 L-MATRIX DIAG= 1.2369128E-10 EQN NO 2134
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5393441E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.0797364E-05 EQN NO 2136

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 6
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.6140828E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.5253277E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.7504284E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.7174130E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 7.1795657E-06 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.8453302E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795981E+04 L-MATRIX DIAG= 1.2369128E-10 EQN NO 2134
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5393441E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= 2.0797364E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 11 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -2.3992732E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 7 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5392044E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5392976E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5395304E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5395304E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= 2.5393441E-05 EQN NO 2135
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1207 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -1.6942155E-05 EQN NO 2136
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 12 iterations, Case= 10

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 214

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 111

**NOTE-Tension/Compression converged after 3 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

795. LOAD LIST 101 TO 116
796. START CONCRETE DESIGN

```
797. CODE ACI
798. TRACK 2 ALL
799. FC 35000 MEMB 185 730 1258 1259
800. CLB 0.05 MEMB 185 730 1258 1259
801. CLS 0.05 ALL
802. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 393 TO 399 402 -
803. 403 TO 409 412 TO 419 422 TO 429 432 497 498 501 TO 504 568 TO 575 -
804. 584 TO 599 632 633 636 TO 639 658 TO 663 670 TO 681 706 707 710 TO 713 730 -
805. 731 TO 732 735 TO 738 747 748 751 TO 754 758 TO 789 798 TO 801 803 TO 818 -
806. 823 TO 826 828 TO 839 843 TO 846 848 TO 851 853 TO 856 858 TO 889 -
807. 898 TO 901 903 TO 918 923 TO 926 928 TO 939 943 TO 946 948 TO 951 -
808. 953 TO 956 958 TO 989 998 TO 1001 1003 TO 1018 1023 TO 1026 1028 TO 1039 -
809. 1043 TO 1046 1048 TO 1051 1053 TO 1056 1058 TO 1089 1098 TO 1101 -
810. 1103 TO 1118 1123 TO 1126 1128 TO 1139 1143 TO 1146 1148 TO 1151 -
811. 1153 TO 1156 1158 TO 1189 1198 TO 1201 1203 TO 1218 1223 TO 1226 -
812. 1228 TO 1239 1243 TO 1246 1248 TO 1251 1253 TO 1256 1258 TO 1265 -
813. 1267 TO 1270 1272 TO 1275 1277 TO 1280 1282 TO 1285 1287 TO 1290
814. MAXMAIN 25 ALL
815. MINMAIN 25 ALL
816. MINSEC 10 ALL
817. DESIGN COLUMN 185 730 1258 1259
```



```
================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
----------------------------------------------------------

  8 - 25 MM               1.091      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53    113.27      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53    113.27      0.00    0.00000
 ---------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |              9022.30   709.00  4511.15  1061.99
       P0 |*             8270.44   812.98  3759.29  1019.91
          | *            7518.58   897.94  3007.43   956.53
    Pn,max|__*           6766.73   962.48  2255.58   875.37
          |   *          6014.87  1009.20  1503.72   752.23
    Pn    |    *         5263.01  1041.19   751.86   595.66
   NOMINAL|     *          Pn       Mn       Pn       Mn    (@ Y )
     AXIAL|      *       9022.30   709.00  4511.15  1061.99
COMPRESSION|       *      8270.44   812.98  3759.29  1019.91
        Pb|-------*Mb     7518.58   897.94  3007.43   956.53
          |      *       6766.73   962.48  2255.58   875.37
__________|____*_______  6014.87  1009.20  1503.72   752.23
          |  * M0   Mn,  5263.01  1041.19   751.86   595.66
          | *   BENDING
    P-tens|*     MOMENT
          |
```

COLUMN NO. 730 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 6055.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	2.182	112	STA	0.900

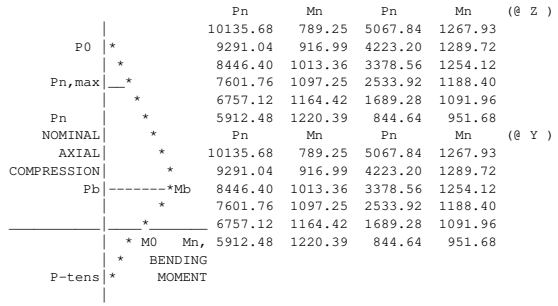
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13725.40	10980.32	4490.13	1297.45	289.0
M0	P-tens.	Des.Pn	Des.Mn	e/h
794.04	-3249.05	-579.57	490.56	-0.32554

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13725.40	10980.32	4490.13	1297.45	289.0
M0	P-tens.	Des.Pn	Des.Mn	e/h
794.04	-3249.05	-579.57	2.11	-0.00140



COLUMN NO. 1258 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

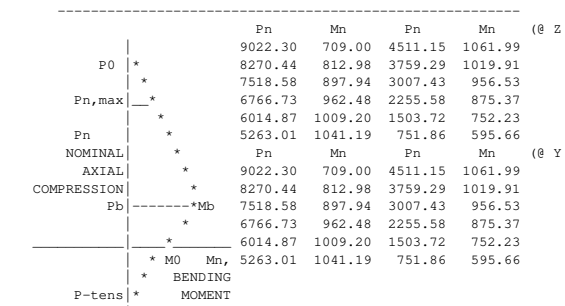
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	159.60	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	159.60	0.00	0.00000



==========

COLUMN NO. 1259 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

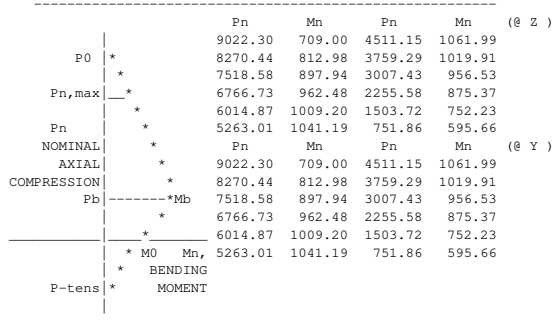
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	310.15	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	310.15	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

818. END CONCRETE DESIGN
819. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 5,2020 TIME= 17:14:17 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 9 STAAD OUTPUT – FOUNDATION F2

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*****************************************************
*                                                   *
*          STAAD.Pro V8i SELECTseries5              *
*          Version  20.07.10.64                     *
*          Proprietary Program of                   *
*          Bentley Systems, Inc.                    *
*          Date=    NOV  6, 2020                    *
*          Time=    14: 5:55                        *
*                                                   *
*     USER ID: LNT ECC DIV                          *
*****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/09/2020
    4. JOB NAME FDN. MKD. F2
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 492 1.35 911 3.42538; 493 1.35 911 3.175; 494 1.35 911 3.67538
   17. 497 1.35 911 2.42538; 498 1.35 911 2.17538; 499 1.35 911 2.67538
   18. 500 1.35 911 2.92538; 501 1.35 911 1.92538; 569 1.6 911 3.42538
   19. 570 1.6 911 3.175; 571 1.85 911 3.42538; 572 1.85 911 3.175
   20. 573 2.1 911 3.42538; 574 2.1 911 3.175; 575 2.35 911 3.42538
   21. 576 2.35 911 3.175; 577 1.6 911 3.67538; 578 1.85 911 3.67538
   22. 579 2.1 911 3.67538; 580 2.35 911 3.67538; 589 1.6 911 2.42538
   23. 590 1.6 911 2.17538; 591 1.85 911 2.42538; 592 1.85 911 2.17538
   24. 593 2.1 911 2.42538; 594 2.1 911 2.17538; 595 2.35 911 2.42538
   25. 596 2.35 911 2.17538; 597 1.6 911 2.67538; 598 1.85 911 2.67538
   26. 599 2.1 911 2.67538; 600 2.35 911 2.67538; 601 1.6 911 2.92538
   27. 602 1.85 911 2.92538; 603 2.1 911 2.92538; 604 2.35 911 2.92538
   28. 605 1.6 911 1.92538; 607 1.85 911 1.92538; 609 2.1 911 1.92538
   29. 611 2.35 911 1.92538; 637 2.6 911 3.42538; 639 2.6 911 3.67538
   30. 642 2.6 911 2.42538; 643 2.6 911 2.17538; 644 2.6 911 2.67538
   31. 645 2.6 911 2.92538; 646 2.6 911 1.92538; 664 2.85 911 3.42538
   32. 665 2.85 911 3.175; 666 3.1 911 3.42538; 667 3.1 911 3.175
   33. 670 2.85 911 3.67538; 671 3.1 911 3.67538; 679 2.85 911 2.42538
   34. 680 2.85 911 2.17538; 681 3.1 911 2.42538; 682 3.1 911 2.17538
   35. 685 2.85 911 2.67538; 686 3.1 911 2.67538; 688 2.85 911 2.92538
   36. 689 3.1 911 2.92538; 691 2.85 911 1.92538; 693 3.1 911 1.92538
   37. 740 2.6 913.6 3.175; 815 1.35 911 1.17538; 816 1.35 911 0.92538
   38. 817 1.35 911 1.42538; 818 1.35 911 1.67538; 819 1.35 911 0.67538
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   39. 820 1.6 911 1.17538; 821 1.6 911 0.92538; 822 1.85 911 1.17538
   40. 823 1.85 911 0.92538; 824 2.1 911 1.17538; 825 2.1 911 0.92538
   41. 826 2.35 911 1.17538; 827 2.35 911 0.92538; 828 1.6 911 1.42538
   42. 829 1.85 911 1.42538; 830 2.1 911 1.42538; 831 2.35 911 1.42538
   43. 832 1.6 911 1.67538; 833 1.85 911 1.67538; 834 2.1 911 1.67538
   44. 835 2.35 911 1.67538; 836 1.6 911 0.67538; 837 1.85 911 0.67538
   45. 838 2.1 911 0.67538; 839 2.35 911 0.67538; 840 2.6 911 1.17538
   46. 842 2.6 911 1.42538; 843 2.6 911 1.67538; 844 2.6 911 0.67538
   47. 845 2.85 911 1.17538; 846 2.85 911 0.92538; 847 3.1 911 1.17538
   48. 848 3.1 911 0.92538; 851 2.85 911 1.42538; 852 3.1 911 1.42538
   49. 854 2.85 911 1.67538; 855 3.1 911 1.67538; 857 2.85 911 0.67538
   50. 858 3.1 911 0.67538; 923 1.35 911 0.42538; 937 1.6 911 0.42538
   51. 938 1.85 911 0.42538; 939 2.1 911 0.42538; 940 2.35 911 0.42538
   52. 948 2.6 911 0.42538; 959 2.85 911 0.42538; 960 3.1 911 0.42538
   53. 1296 2.6 913.6 0.925; 1329 2.6 911 3.175; 1332 2.6 911 0.92538
   54. 1333 3.6 911 3.42538; 1334 3.85 911 3.42538; 1335 3.85 911 3.175
   55. 1336 3.6 911 3.175; 1339 3.6 911 3.67538; 1340 3.85 911 3.67538
   56. 1342 3.6 911 2.42538; 1343 3.85 911 2.42538; 1344 3.85 911 2.17538
   57. 1345 3.6 911 2.17538; 1348 3.6 911 2.67538; 1349 3.85 911 2.67538
   58. 1351 3.6 911 2.92538; 1352 3.85 911 2.92538; 1354 3.85 911 1.92538
   59. 1355 3.6 911 1.92538; 1357 3.6 911 1.17538; 1358 3.85 911 1.17538
   60. 1359 3.85 911 0.92538; 1360 3.6 911 0.92538; 1363 3.6 911 1.42538
   61. 1364 3.85 911 1.42538; 1366 3.6 911 1.67538; 1367 3.85 911 1.67538
   62. 1369 3.85 911 0.67538; 1370 3.6 911 0.67538; 1372 3.85 911 0.42538
   63. 1373 3.6 911 0.42538; 1375 3.35 911 3.42538; 1376 3.35 911 3.175
   64. 1377 3.35 911 3.67538; 1378 3.35 911 2.42538; 1379 3.35 911 2.17538
   65. 1380 3.35 911 2.67538; 1381 3.35 911 2.92538; 1382 3.35 911 1.92538
   66. 1383 3.35 911 1.17538; 1384 3.35 911 0.92538; 1385 3.35 911 1.42538
   67. 1386 3.35 911 1.67538; 1387 3.35 911 0.67538; 1388 3.35 911 0.42538
   68. 1452 1.35 911 0.17538; 1453 1.6 911 0.17538; 1454 1.85 911 0.17538
   69. 1455 2.1 911 0.17538; 1456 2.35 911 0.17538; 1457 2.6 911 0.17538
   70. 1458 2.85 911 0.17538; 1459 3.1 911 0.17538; 1467 1.35 911 -0.07462
   71. 1468 1.6 911 -0.07462; 1469 1.85 911 -0.07462; 1470 2.1 911 -0.07462
   72. 1471 2.35 911 -0.07462; 1472 2.6 911 -0.07462; 1473 2.85 911 -0.07462
   73. 1474 3.1 911 -0.07462; 1475 3.85 911 0.17538; 1476 3.6 911 0.17538
   74. 1478 3.85 911 -0.07462; 1479 3.6 911 -0.07462; 1481 3.35 911 0.17538
   75. 1482 3.35 911 -0.07462; 1505 1.35 911 3.92538; 1506 1.35 911 4.17538
   76. 1507 1.6 911 3.92538; 1508 1.85 911 3.92538; 1509 2.1 911 3.92538
   77. 1510 2.35 911 3.92538; 1511 1.6 911 4.17538; 1512 1.85 911 4.17538
   78. 1513 2.1 911 4.17538; 1514 2.35 911 4.17538; 1515 2.6 911 3.92538
   79. 1516 2.6 911 4.17538; 1517 2.85 911 3.92538; 1518 3.1 911 3.92538
   80. 1519 2.85 911 4.17538; 1520 3.1 911 4.17538; 1521 3.6 911 3.92538
   81. 1522 3.85 911 3.92538; 1524 3.6 911 4.17538; 1525 3.85 911 4.17538
   82. 1527 3.35 911 3.92538; 1528 3.35 911 4.17538
   83. MEMBER INCIDENCES
   84. 730 1329 740; 1259 1332 1296
   85. ELEMENT INCIDENCES SHELL
   86. 568 492 569 570 493; 569 569 571 572 570; 570 571 573 574 572
   87. 571 573 575 576 574; 572 494 577 569 492; 573 577 578 571 569
   88. 574 578 579 573 571; 575 579 580 575 573; 584 497 589 590 498
   89. 585 589 591 592 590; 586 591 593 594 592; 587 593 595 596 594
   90. 588 499 597 589 497; 589 597 598 591 589; 590 598 599 593 591
   91. 591 599 600 595 593; 592 500 601 597 499; 593 601 602 598 597
   92. 594 602 603 599 598; 595 603 604 600 599; 596 493 570 601 500
   93. 597 570 572 602 601; 598 572 574 603 602; 599 574 576 604 603
   94. 604 498 590 605 501; 605 590 592 607 605; 606 592 594 609 607
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 95. 607 594 596 611 609; 632 575 637 1329 576; 633 580 639 637 575
 96. 636 595 642 643 596; 637 600 644 642 595; 638 604 645 644 600
 97. 639 576 1329 645 604; 641 596 643 646 611; 658 637 664 665 1329
 98. 659 664 666 667 665; 661 639 670 664 637; 662 670 671 666 664
 99. 670 642 679 680 643; 671 679 681 682 680; 673 644 685 679 642
100. 674 685 686 681 679; 676 645 688 685 644; 677 688 689 686 685
101. 679 1329 665 688 645; 680 665 667 689 688; 685 643 680 691 646
102. 686 680 682 693 691; 803 815 820 821 816; 804 820 822 823 821
103. 805 822 824 825 823; 806 824 826 827 825; 807 817 828 820 815
104. 808 828 829 822 820; 809 829 830 824 822; 810 830 831 826 824
105. 811 818 832 828 817; 812 832 833 829 828; 813 833 834 830 829
106. 814 834 835 831 830; 815 501 605 832 818; 816 605 607 833 832
107. 817 607 609 834 833; 818 609 611 835 834; 819 816 821 836 819
108. 820 821 823 837 836; 821 823 825 838 837; 822 825 827 839 838
109. 823 826 840 1332 827; 824 831 842 840 826; 825 835 843 842 831
110. 826 611 646 843 835; 827 827 1332 844 839; 828 840 845 846 1332
111. 829 845 847 848 846; 831 842 851 845 840; 832 851 852 847 845
112. 834 843 854 851 842; 835 854 855 852 851; 837 646 691 854 843
113. 838 691 693 855 854; 840 1332 846 857 844; 841 846 848 858 857
114. 915 819 836 937 923; 916 836 837 938 937; 917 837 838 939 938
115. 918 838 839 940 939; 926 839 844 948 940; 937 844 857 959 948
116. 938 857 858 960 959; 1260 1333 1334 1335 1336; 1262 1339 1340 1334 1333
117. 1264 1342 1343 1344 1345; 1266 1348 1349 1343 1342; 1268 1351 1352 1349 1348
118. 1270 1336 1335 1352 1351; 1272 1345 1344 1354 1355; 1274 1357 1358 1359 1360
119. 1276 1363 1364 1358 1357; 1278 1366 1367 1364 1363; 1280 1355 1354 1367 1366
120. 1282 1360 1359 1369 1370; 1284 1370 1369 1372 1373; 1286 666 1375 1376 667
121. 1287 1375 1333 1336 1376; 1288 671 1377 1375 666; 1289 1377 1339 1333 1375
122. 1290 681 1378 1379 682; 1291 1378 1342 1345 1379; 1292 686 1380 1378 681
123. 1293 1380 1348 1342 1378; 1294 689 1381 1380 686; 1295 1381 1351 1348 1380
124. 1296 667 1376 1381 689; 1297 1376 1336 1351 1381; 1298 682 1379 1382 693
125. 1299 1379 1345 1355 1382; 1300 847 1383 1384 848; 1301 1383 1357 1360 1384
126. 1302 852 1385 1383 847; 1303 1385 1363 1357 1383; 1304 855 1386 1385 852
127. 1305 1386 1366 1363 1385; 1306 693 1382 1386 855; 1307 1382 1355 1366 1386
128. 1308 848 1384 1387 858; 1309 1384 1360 1370 1387; 1310 858 1387 1388 960
129. 1311 1387 1370 1373 1388; 1371 923 937 1453 1452; 1372 937 938 1454 1453
130. 1373 938 939 1455 1454; 1374 939 940 1456 1455; 1375 940 948 1457 1456
131. 1376 948 959 1458 1457; 1377 959 960 1459 1458; 1385 1452 1453 1468 1467
132. 1386 1453 1454 1469 1468; 1387 1454 1455 1470 1469; 1388 1455 1456 1471 1470
133. 1389 1456 1457 1472 1471; 1390 1457 1458 1473 1472; 1391 1458 1459 1474 1473
134. 1392 1373 1372 1475 1476; 1394 1476 1475 1478 1479; 1396 960 1388 1481 1459
135. 1397 1388 1373 1476 1481; 1398 1459 1481 1482 1474; 1399 1481 1476 1479 1482
136. 1422 1505 1507 577 494; 1423 1507 1508 578 577; 1424 1508 1509 579 578
137. 1425 1509 1510 580 579; 1426 1506 1511 1507 1505; 1427 1511 1512 1508 1507
138. 1428 1512 1513 1509 1508; 1429 1513 1514 1510 1509; 1430 1510 1515 639 580
139. 1431 1514 1516 1515 1510; 1432 1515 1517 670 639; 1433 1517 1518 671 670
140. 1434 1516 1519 1517 1515; 1435 1519 1520 1518 1517; 1436 1521 1522 1340 1339
141. 1438 1524 1525 1522 1521; 1440 1518 1527 1377 671; 1441 1527 1521 1339 1377
142. 1442 1520 1528 1527 1518; 1443 1528 1524 1521 1527
143. START GROUP DEFINITION
144. ELEMENT
145. _RAFT 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 -
146. 662 670 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 -
147. 838 840 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 -
148. 1276 1278 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 -
149. 1397 TO 1399 1422 TO 1436 1438 1440 TO 1443
150. _RAFT_MOM 568 TO 575 584 TO 599 604 TO 607 633 636 TO 638 641 659 661 662 -
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151. 670 671 673 674 676 677 680 685 686 803 TO 822 824 TO 826 829 831 832 834 -
152. 835 837 838 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 -
153. 1274 1276 1278 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 -
154. 1396 TO 1399 1422 TO 1436 1438 1440 TO 1443
155. _RAFT_SHEAR 568 572 584 588 592 596 604 TO 607 641 685 686 803 807 811 815 -
156. 819 915 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 1280 1282 1284 -
157. 1298 1299 1371 1385 1392 1394 1422 1426 1436 1438
158. MEMBER
159. END GROUP DEFINITION
160. ELEMENT PROPERTY
161. 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
162. 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
163. 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
164. 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
165. 1423 TO 1436 1438 1440 TO 1443 THICKNESS 0.7
166. DEFINE MATERIAL START
167. ISOTROPIC CONCRETE
168. E 2.17185E+007
169. POISSON 0.17
170. DENSITY 25
171. ALPHA 1E-005
172. DAMP 0.05
173. TYPE CONCRETE
174. STRENGTH FCU 27579
175. END DEFINE MATERIAL
176. MEMBER PROPERTY AMERICAN
177. 730 1259 PRIS YD 0.5 ZD 0.5
178. CONSTANTS
179. MATERIAL CONCRETE ALL
180. SUPPORTS
181. 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
182. 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
183. 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
184. 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
185. 1423 TO 1436 1438 1440 TO 1443 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
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JOINT	INFLUENCE AREA
492	0.031274
493	0.031250
494	0.031250
497	0.031250
498	0.031250
499	0.031250
500	0.031226
501	0.031250
569	0.062548
570	0.062500
571	0.062548
572	0.062500
573	0.062547
574	0.062500
575	0.062547
576	0.062500
577	0.062500

578	0.062500
579	0.062500
580	0.062500
589	0.062500
590	0.062500
591	0.062500
592	0.062500
593	0.062500
594	0.062500
595	0.062500
596	0.062500
597	0.062500
598	0.062500
599	0.062500
600	0.062500
601	0.062452
602	0.062452
603	0.062452
604	0.062452
605	0.062500
607	0.062500
609	0.062500
611	0.062500
637	0.062548
639	0.062500
642	0.062500
643	0.062500
644	0.062500
645	0.062452
646	0.062500
664	0.062547
665	0.062500
666	0.062548
667	0.062500
670	0.062500
671	0.062500
679	0.062500
680	0.062500
681	0.062500
682	0.062500
685	0.062500
686	0.062500
688	0.062452
689	0.062452
691	0.062500
693	0.062500
815	0.031250
816	0.031250
817	0.031250
818	0.031250
819	0.031250
820	0.062500
821	0.062500
822	0.062500
823	0.062500
824	0.062500

825	0.062500
826	0.062500
827	0.062500
828	0.062500
829	0.062500
830	0.062500
831	0.062500
832	0.062500
833	0.062500
834	0.062500
835	0.062500
836	0.062500
837	0.062500
838	0.062500
839	0.062500
840	0.062500
842	0.062500
843	0.062500
844	0.062500
845	0.062500
846	0.062500
847	0.062500
848	0.062500
851	0.062500
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854	0.062500
855	0.062500
857	0.062500
858	0.062500
923	0.031250
937	0.062500
938	0.062500
939	0.062500
940	0.062500
948	0.062500
959	0.062500
960	0.062500
1329	0.062500
1332	0.062500
1333	0.062547
1334	0.031274
1335	0.031250
1336	0.062500
1339	0.062500
1340	0.031250
1342	0.062500
1343	0.031250
1344	0.031250
1345	0.062500
1348	0.062500
1349	0.031250
1351	0.062452
1352	0.031226
1354	0.031250
1355	0.062500
1357	0.062500

1358	0.031250
1359	0.031250
1360	0.062500
1363	0.062500
1364	0.031250
1366	0.062500
1367	0.031250
1369	0.031250
1370	0.062500
1372	0.031250
1373	0.062500
1375	0.062548
1376	0.062500
1377	0.062500
1378	0.062500
1379	0.062500
1380	0.062500
1381	0.062452
1382	0.062500
1383	0.062500
1384	0.062500
1385	0.062500
1386	0.062500
1387	0.062500
1388	0.062500
1452	0.031250
1453	0.062500
1454	0.062500
1455	0.062500
1456	0.062500
1457	0.062500
1458	0.062500
1459	0.062500
1467	0.015625
1468	0.031250
1469	0.031250
1470	0.031250
1471	0.031250
1472	0.031250
1473	0.031250
1474	0.031250
1475	0.031250
1476	0.062500
1478	0.015625
1479	0.031250
1481	0.062500
1482	0.031250
1505	0.031250
1506	0.015625
1507	0.062500
1508	0.062500
1509	0.062500
1510	0.062500
1511	0.031250
1512	0.031250
1513	0.031250

1514	0.031250
1515	0.062500
1516	0.031250
1517	0.062500
1518	0.062500
1519	0.031250
1520	0.031250
1521	0.062500
1522	0.031250
1524	0.031250
1525	0.015625
1527	0.062500
1528	0.031250
TOTAL	10.625001

```
186. ********************************************************************************
187. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
188. SELFWEIGHT Y -1
189. ELEMENT LOAD
190. * LOAD DUE TO BACKFILL
191. 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
192. 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
193. 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
194. 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
195. 1423 TO 1436 1438 1440 TO 1443 PR GY -16
196. * LOAD DUE TO SURCHARGE
197. 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
198. 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
199. 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
200. 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
201. 1423 TO 1436 1438 1440 TO 1443 PR GY -20
202. JOINT LOAD
203. 740 FX -0.35 FY -90 FZ 0.39
204. 1296 FX -2.17 FY -89.83 FZ 0.4
205. ********************************************************************************
206. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
207. JOINT LOAD
208. 740 FX 0.18 FY -42.94 FZ 0.04
209. 1296 FX -0.09 FY -5.87 FZ 2.22
210. ********************************************************************************
211. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WL1 )
212. JOINT LOAD
213. 740 FX -3.82 FY 19.37 FZ 0.98
214. 1296 FX 3.87 FY -276.8 FZ -6.86
215. ********************************************************************************
216. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X ( WL2 )
217. JOINT LOAD
218. 740 FX 3.85 FY 19.37 FZ -0.97
219. 1296 FX -3.87 FY 275.48 FZ 6.67
220. ********************************************************************************
221. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Y ( WL3 )
222. JOINT LOAD
223. 740 FX 0.35 FY 260 FZ -0.45
224. 1296 FX 0.61 FY -288.67 FZ -60.96
225. ********************************************************************************
226. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Y ( WL4 )
227. JOINT LOAD
```

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228. 740 FX -0.34 FY -255.87 FZ 0.47
229. 1296 FX -0.59 FY 284.46 FZ 60.73
230. ******************************************************************************
231. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
232. JOINT LOAD
233. 740 FX -0.05 FY 7.65 FZ -0.03
234. 1296 FX -0.21 FY 8.1 FZ -1.02
235. ******************************************************************************
236. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
237. JOINT LOAD
238. 740 FX 0.05 FY -7.65 FZ 0.03
239. 1296 FX 0.21 FY -8.1 FZ 1.02
240. ******************************************************************************
241. LOAD 9 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
242. JOINT LOAD
243. 740 FX -0.35 FY -33.04 FZ 2.39
244. 1296 FX -1.03 FY 32.98 FZ 2.29
245. ******************************************************************************
246. LOAD 10 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
247. JOINT LOAD
248. 740 FX 0.35 FY 33.04 FZ -2.39
249. 1296 FX 1.03 FY -32.98 FZ -2.29
250. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**200**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**170**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**198**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 178/ 18/ 57 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 606
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 35 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.3/ 142099.6 MB

```
251. CHANGE
252. SUPPORTS
253. 568 TO 575 584 TO 599 604 TO 607 632 633 636 TO 639 641 658 659 661 662 670 -
254. 671 673 674 676 677 679 680 685 686 803 TO 829 831 832 834 835 837 838 840 -
255. 841 915 TO 918 926 937 938 1260 1262 1264 1266 1268 1270 1272 1274 1276 1278 -
256. 1280 1282 1284 1286 TO 1311 1371 TO 1377 1385 TO 1392 1394 1396 TO 1399 1422 -
257. 1423 TO 1436 1438 1440 TO 1443 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
258. ******************************************************************************
259. * UNFACTORED LOAD COMBINATIONS
```



```
260. LOAD 201 DL
261. REPEAT LOAD
262. 1 1.0
263. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201


264. CHANGE
265. LOAD 202 DL+L+TL1
266. REPEAT LOAD
267. 1 1.0 2 1.0 9 1.0
268. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202


269. CHANGE
270. LOAD 203 DL+L+TL2
271. REPEAT LOAD
272. 1 1.0 2 1.0 10 1.0
273. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  203


274. CHANGE
275. LOAD 204 DL+0.75L+0.75TL1
276. REPEAT LOAD
277. 1 1.0 2 0.75 9 0.75
278. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204


279. CHANGE
280. LOAD 205 DL+0.75L+0.75TL2
281. REPEAT LOAD
282. 1 1.0 2 0.75 10 0.75
283. PERFORM ANALYSIS
```

```
**NOTE-Tension/Compression converged after  1 iterations, Case=  205


 284. CHANGE
 285. LOAD 206 DL+WL1
 286. REPEAT LOAD
 287. 1 1.0 3 1.0
 288. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  206


 289. CHANGE
 290. LOAD 207 DL+WL2
 291. REPEAT LOAD
 292. 1 1.0 4 1.0
 293. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207


 294. CHANGE
 295. LOAD 208 DL+WL3
 296. REPEAT LOAD
 297. 1 1.0 5 1.0
 298. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  208


 299. CHANGE
 300. LOAD 209 DL+WL4
 301. REPEAT LOAD
 302. 1 1.0 6 1.0
 303. PERFORM ANALYSIS
```



```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  209


 304. CHANGE
 305. LOAD 210 DL+0.75WL1+0.75L
 306. REPEAT LOAD
 307. 1 1.0 2 0.75 3 0.75
 308. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210


 309. CHANGE
 310. LOAD 211 DL+0.75WL2+0.75L
 311. REPEAT LOAD
 312. 1 1.0 2 0.75 4 0.75
 313. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211


 314. CHANGE
 315. LOAD 212 DL+0.75WL3+0.75L
 316. REPEAT LOAD
 317. 1 1.0 2 0.75 5 0.75
 318. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212


 319. CHANGE
 320. LOAD 213 DL+0.75WL4+0.75L
 321. REPEAT LOAD
 322. 1 1.0 2 0.75 6 0.75
 323. PERFORM ANALYSIS
```

```
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  213



 324. CHANGE
 325. LOAD 214 0.6DL+WL1
 326. REPEAT LOAD
 327. 1 0.6 3 1.0
 328. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  214



 329. CHANGE
 330. LOAD 215 0.6DL+WL2
 331. REPEAT LOAD
 332. 1 0.6 4 1.0
 333. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  215



 334. CHANGE
 335. LOAD 216 0.6DL+WL3
 336. REPEAT LOAD
 337. 1 0.6 5 1.0
 338. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
```



```
**NOTE-Tension/Compression converged after  5 iterations, Case=  216



 339. CHANGE
 340. LOAD 217 0.6DL+WL4
 341. REPEAT LOAD
 342. 1 0.6 6 1.0
 343. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  217



 344. CHANGE
 345. LOAD 218 0.6DL+0.7SL1
 346. REPEAT LOAD
 347. 1 0.6 7 0.7
 348. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  218



 349. CHANGE
 350. LOAD 219 0.6DL+0.7SL2
 351. REPEAT LOAD
 352. 1 0.6 8 0.7
 353. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  219



 354. CHANGE
 355. ********************************************************************************
 356. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
 357. LOAD 101 1.4DL
 358. REPEAT LOAD
 359. 1 1.4
 360. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

361. CHANGE
362. LOAD 102 1.4DL+1.4TL1+1.7L
363. REPEAT LOAD
364. 1 1.4 2 1.7 9 1.4
365. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

366. CHANGE
367. LOAD 103 1.4DL+1.4TL2+1.7L
368. REPEAT LOAD
369. 1 1.4 2 1.7 10 1.4
370. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

371. CHANGE
372. LOAD 104 1.2DL+1.0L
373. REPEAT LOAD
374. 1 1.2 2 1.0
375. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

376. CHANGE
377. LOAD 105 1.2DL+1.6WL1+1.0L
378. REPEAT LOAD
379. 1 1.2 3 1.6 2 0.5 2 1.0
380. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

381. CHANGE
382. LOAD 106 1.2DL+1.6WL2+1.0L
383. REPEAT LOAD
384. 1 1.2 4 1.6 2 0.5 2 1.0
385. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 106

386. CHANGE
387. LOAD 107 1.2DL+1.6WL3+1.0L
388. REPEAT LOAD
389. 1 1.2 5 1.6 2 0.5 2 1.0
390. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 107

391. CHANGE
392. LOAD 108 1.2DL+1.6WL4+1.0L
393. REPEAT LOAD
394. 1 1.2 6 1.6 2 1.0
395. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 108

396. CHANGE
397. LOAD 109 1.2DL+1.0SL1+1.0L
398. REPEAT LOAD
399. 1 1.2 7 1.0 2 1.0
400. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

401. CHANGE
402. LOAD 110 1.2DL+1.0SL2+1.0L
403. REPEAT LOAD
404. 1 1.2 8 1.0 2 1.0
405. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

406. CHANGE
407. LOAD 111 0.9DL+1.6WL1
408. REPEAT LOAD
409. 1 0.9 3 1.6
410. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

411. CHANGE
412. LOAD 112 0.9DL+1.6WL2
413. REPEAT LOAD
414. 1 0.9 4 1.6
415. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 112

416. CHANGE
417. LOAD 113 0.9DL+1.6WL3
418. REPEAT LOAD

419. 1 0.9 5 1.6
420. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 113

421. CHANGE
422. LOAD 114 0.9DL+1.6WL4
423. REPEAT LOAD
424. 1 0.9 6 1.6
425. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 114

426. CHANGE
427. LOAD 115 0.9DL+1.0SL1
428. REPEAT LOAD
429. 1 0.9 7 1.0
430. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

431. CHANGE
432. LOAD 116 0.9DL+1.0SL2
433. REPEAT LOAD
434. 1 0.9 8 1.0
435. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

436. CHANGE
437. **
438. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 2

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6205317E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795982E+04 L-MATRIX DIAG= 3.2014214E-10 EQN NO 604
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.5879354E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= -3.9348379E-07 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 9.2643313E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6205317E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795982E+04 L-MATRIX DIAG= 3.2014214E-10 EQN NO 604
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.5879354E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= -3.9348379E-07 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 7 iterations, Case= 6

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795982E+04 L-MATRIX DIAG= 3.2014214E-10 EQN NO 604
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.5879354E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= -3.9348379E-07 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 7

**NOTE-Tension/Compression converged after 1 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6205317E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6112185E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6158751E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6112185E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.6391582E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 2.5853515E-06 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 10 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 9.2643313E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 7.6713040E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 8.4876083E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 6.6324137E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.3369440E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= -3.4179538E-07 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795982E+04 L-MATRIX DIAG= 3.2014214E-10 EQN NO 604
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= 5.5879354E-07 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1525 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902220E+06 L-MATRIX DIAG= -3.9348379E-07 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 10
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

```
**NOTE-Tension/Compression converged after  1 iterations, Case=  206



**NOTE-Tension/Compression converged after  1 iterations, Case=  207


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  208


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  209



**NOTE-Tension/Compression converged after  1 iterations, Case=  210



**NOTE-Tension/Compression converged after  1 iterations, Case=  211



**NOTE-Tension/Compression converged after  1 iterations, Case=  212


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  213



**NOTE-Tension/Compression converged after  1 iterations, Case=  214


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  215


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
```



```
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  216


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  217



**NOTE-Tension/Compression converged after  1 iterations, Case=  218



**NOTE-Tension/Compression converged after  1 iterations, Case=  219



**NOTE-Tension/Compression converged after  1 iterations, Case=  101



**NOTE-Tension/Compression converged after  1 iterations, Case=  102



**NOTE-Tension/Compression converged after  1 iterations, Case=  103



**NOTE-Tension/Compression converged after  1 iterations, Case=  104



**NOTE-Tension/Compression converged after  1 iterations, Case=  105


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  106


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  107
```

```
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  108



**NOTE-Tension/Compression converged after  1 iterations, Case=  109



**NOTE-Tension/Compression converged after  1 iterations, Case=  110



**NOTE-Tension/Compression converged after  1 iterations, Case=  111


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6
**START ITERATION NO.          7

**NOTE-Tension/Compression converged after  7 iterations, Case=  112


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  113


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  114



**NOTE-Tension/Compression converged after  1 iterations, Case=  115



**NOTE-Tension/Compression converged after  1 iterations, Case=  116


 439. LOAD LIST 101 TO 116
 440. START CONCRETE DESIGN
```



```
 441. CODE ACI
 442. TRACK 2 ALL
 443. FC 35000 MEMB 730 1259
 444. CLB 0.05 MEMB 730 1259
 445. CLS 0.05 ALL
 446. CLT 0.05 MEMB 568 TO 575 584 TO 599 632 633 636 TO 639 658 659 661 662 670 -
 447. 671 673 674 676 677 679 680 730 803 TO 818 823 TO 826 828 829 831 832 834 -
 448. 835 837 838 915 TO 918 926 937 938 1259 1260 1262 1264 1266 1268 1270 1274 -
 449. 1276 1278 1280 1284 1286 TO 1297 1300 TO 1307 1310 1311 1385 TO 1391 1394 -
 450. 1398 1399 1422 TO 1436 1438 1440 TO 1443
 451. MAXMAIN 25 ALL
 452. MINMAIN 25 ALL
 453. MINSEC 10 ALL
 454. DESIGN COLUMN 730 1259
```

COLUMN NO. 730 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	101	END	0.650

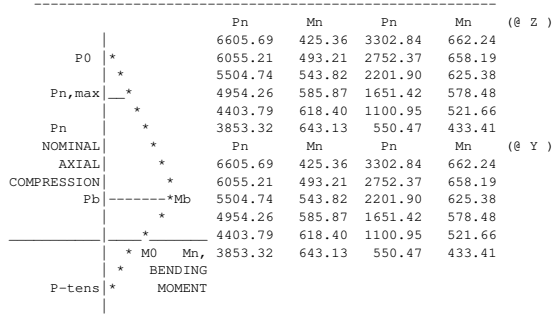
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	193.85	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	193.85	0.00	0.00000



COLUMN NO. 1259 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 4400.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	2.356	108	STA	0.900

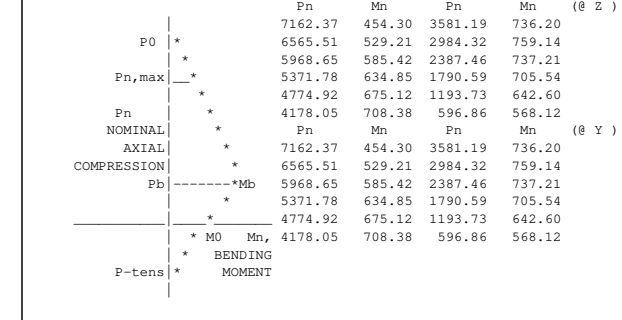
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9699.05	7759.24	3019.55	760.13	251.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
483.46	-2436.79	-357.73	10.51	-0.01130

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9699.05	7759.24	3019.55	760.13	251.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
483.46	-2436.79	-357.73	288.65	-0.31034



**********END OF COLUMN DESIGN RESULTS**********

455. END CONCRETE DESIGN

456. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 6,2020 TIME= 14: 9:35 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 10 STAAD OUTPUT – FOUNDATION F3

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    NOV  5, 2020                 *
*          Time=    16: 1:57                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/11/2020
    4. JOB NAME FDN. MKD. F3
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911 3.175; 6 0.75 911 3.175; 34 -0.5 911 3.175; 35 -0.25 911 3.175
   17. 41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175; 302 -0.5 911 3.42538
   18. 304 -0.25 911 3.42538; 305 0 911 3.42538; 306 0.25 911 3.42538
   19. 307 0.5 911 3.42538; 308 0.75 911 3.42538; 309 1 911 3.175; 313 1 911 3.42538
   20. 322 -0.5 911 3.67538; 323 -0.25 911 3.67538; 324 0 911 3.67538
   21. 325 0.25 911 3.67538; 326 0.5 911 3.67538; 327 0.75 911 3.67538
   22. 328 1 911 3.67538; 366 -0.5 911 2.17538; 369 -0.25 911 2.17538
   23. 371 0 911 2.17538; 373 0.25 911 2.17538; 375 0.5 911 2.17538
   24. 377 0.75 911 2.17538; 379 1 911 2.17538; 388 -0.5 911 2.42538
   25. 389 -0.25 911 2.42538; 390 0 911 2.42538; 391 0.25 911 2.42538
   26. 392 0.5 911 2.42538; 393 0.75 911 2.42538; 394 1 911 2.42538
   27. 399 -0.5 911 2.67538; 400 -0.25 911 2.67538; 401 0 911 2.67538
   28. 402 0.25 911 2.67538; 403 0.5 911 2.67538; 404 0.75 911 2.67538
   29. 405 1 911 2.67538; 410 -0.5 911 2.92538; 411 -0.25 911 2.92538
   30. 412 0 911 2.92538; 413 0.25 911 2.92538; 414 0.5 911 2.92538
   31. 415 0.75 911 2.92538; 416 1 911 2.92538; 421 -0.5 911 1.92538
   32. 424 -0.25 911 1.92538; 426 0 911 1.92538; 428 0.25 911 1.92538
   33. 430 0.5 911 1.92538; 432 0.75 911 1.92538; 434 1 911 1.92538
   34. 492 1.25 911 3.42538; 493 1.25 911 3.175; 494 1.25 911 3.67538
   35. 497 1.25 911 2.42538; 498 1.25 911 2.17538; 499 1.25 911 2.67538
   36. 500 1.25 911 2.92538; 501 1.25 911 1.92538; 771 -0.5 911 1.17538
   37. 772 -0.5 911 0.92538; 774 -0.25 911 1.17538; 775 -0.25 911 0.92538
   38. 776 0 911 1.17538; 778 0.25 911 1.17538; 779 0.25 911 0.92538
```



```
   39. 780 0.5 911 1.17538; 781 0.5 911 0.92538; 782 0.75 911 1.17538
   40. 783 0.75 911 0.92538; 784 1 911 1.17538; 785 1 911 0.92538
   41. 789 -0.5 911 1.42538; 790 -0.25 911 1.42538; 791 0 911 1.42538
   42. 792 0.25 911 1.42538; 793 0.5 911 1.42538; 794 0.75 911 1.42538
   43. 795 1 911 1.42538; 798 -0.5 911 1.67538; 799 -0.25 911 1.67538
   44. 800 0 911 1.67538; 801 0.25 911 1.67538; 802 0.5 911 1.67538
   45. 803 0.75 911 1.67538; 804 1 911 1.67538; 806 -0.5 911 0.67538
   46. 808 -0.25 911 0.67538; 809 0 911 0.67538; 810 0.25 911 0.67538
   47. 811 0.5 911 0.67538; 812 0.75 911 0.67538; 813 1 911 0.67538
   48. 815 1.25 911 1.17538; 816 1.25 911 0.92538; 817 1.25 911 1.42538
   49. 818 1.25 911 1.67538; 819 1.25 911 0.67538; 903 -0.5 911 0.42538
   50. 904 -0.25 911 0.42538; 905 0 911 0.42538; 906 0.25 911 0.42538
   51. 907 0.5 911 0.42538; 908 0.75 911 0.42538; 909 1 911 0.42538
   52. 923 1.25 911 0.42538; 1295 0 913.6 0.075; 1331 0 911 0.92538
   53. 1390 -1.25 911 3.42538; 1391 -1.25 911 3.175; 1393 -1 911 3.42538
   54. 1394 -1 911 3.175; 1395 -0.75 911 3.42538; 1396 -0.75 911 3.175
   55. 1398 -1.25 911 3.67538; 1399 -1 911 3.67538; 1400 -0.75 911 3.67538
   56. 1402 -1.25 911 2.42538; 1403 -1.25 911 2.17538; 1405 -1 911 2.42538
   57. 1406 -1 911 2.17538; 1407 -0.75 911 2.42538; 1408 -0.75 911 2.17538
   58. 1410 -1.25 911 2.67538; 1411 -1 911 2.67538; 1412 -0.75 911 2.67538
   59. 1414 -1.25 911 2.92538; 1415 -1 911 2.92538; 1416 -0.75 911 2.92538
   60. 1417 -1.25 911 1.92538; 1419 -1 911 1.92538; 1420 -0.75 911 1.92538
   61. 1422 -1.25 911 1.17538; 1423 -1.25 911 0.92538; 1425 -1 911 1.17538
   62. 1426 -1 911 0.92538; 1427 -0.75 911 1.17538; 1428 -0.75 911 0.92538
   63. 1430 -1.25 911 1.42538; 1431 -1 911 1.42538; 1432 -0.75 911 1.42538
   64. 1434 -1.25 911 1.67538; 1435 -1 911 1.67538; 1436 -0.75 911 1.67538
   65. 1437 -1.25 911 0.67538; 1439 -1 911 0.67538; 1440 -0.75 911 0.67538
   66. 1441 -1.25 911 0.42538; 1443 -1 911 0.42538; 1444 -0.75 911 0.42538
   67. 1445 -0.25 911 0.07538; 1446 -0.5 911 0.07538; 1448 0.25 911 0.07538
   68. 1449 0.5 911 0.07538; 1450 0.75 911 0.07538; 1451 1 911 0.07538
   69. 1452 1.25 911 0.07538; 1460 -0.25 911 -0.32462; 1461 -0.5 911 -0.32462
   70. 1463 0.25 911 -0.32462; 1464 0.5 911 -0.32462; 1465 0.75 911 -0.32462
   71. 1466 1 911 -0.32462; 1467 1.25 911 -0.32462; 1483 -1.25 911 0.07538
   72. 1485 -1 911 0.07538; 1486 -0.75 911 0.07538; 1487 -1.25 911 -0.32462
   73. 1489 -1 911 -0.32462; 1490 -0.75 911 -0.32462; 1491 -0.5 911 3.92538
   74. 1492 -0.25 911 3.92538; 1493 0 911 3.92538; 1494 0.25 911 3.92538
   75. 1495 0.5 911 3.92538; 1496 0.75 911 3.92538; 1497 1 911 3.92538
   76. 1498 -0.5 911 4.17538; 1499 -0.25 911 4.17538; 1500 0 911 4.17538
   77. 1501 0.25 911 4.17538; 1502 0.5 911 4.17538; 1503 0.75 911 4.17538
   78. 1504 1 911 4.17538; 1505 1.25 911 3.92538; 1506 1.25 911 4.17538
   79. 1530 -1.25 911 3.92538; 1531 -1 911 3.92538; 1532 -0.75 911 3.92538
   80. 1534 -1.25 911 4.17538; 1535 -1 911 4.17538; 1536 -0.75 911 4.17538
   81. 1537 0 911 -0.32462; 1538 0 911 0.07538; 1539 0 913.6 -0.625
   82. 1540 1.5 911 3.42538; 1541 1.5 911 3.175; 1542 1.5 911 3.67538
   83. 1543 1.5 911 2.42538; 1544 1.5 911 2.17538; 1545 1.5 911 2.67538
   84. 1546 1.5 911 2.92538; 1547 1.5 911 1.92538; 1548 1.75 911 3.42538
   85. 1549 1.75 911 3.175; 1550 1.75 911 3.67538; 1551 1.75 911 2.42538
   86. 1552 1.75 911 2.17538; 1553 1.75 911 2.67538; 1554 1.75 911 2.92538
   87. 1555 1.75 911 1.92538; 1556 1.5 911 1.17538; 1557 1.5 911 0.92538
   88. 1558 1.5 911 1.42538; 1559 1.5 911 1.67538; 1560 1.5 911 0.67538
   89. 1561 1.75 911 1.17538; 1562 1.75 911 0.92538; 1563 1.75 911 1.42538
   90. 1564 1.75 911 1.67538; 1565 1.75 911 0.67538; 1566 1.5 911 0.42538
   91. 1567 1.75 911 0.42538; 1568 1.5 911 0.07538; 1569 1.75 911 0.07538
   92. 1570 1.5 911 -0.32462; 1571 1.75 911 -0.32462; 1572 1.5 911 3.92538
   93. 1573 1.5 911 4.17538; 1574 1.75 911 3.92538; 1575 1.75 911 4.17538
   94. 1576 -1.75 911 3.42538; 1577 -1.5 911 3.42538; 1578 -1.5 911 3.175
```

```
 95. 1579 -1.75 911 3.175; 1580 -1.75 911 3.67538; 1581 -1.5 911 3.67538
 96. 1582 -1.75 911 2.42538; 1583 -1.5 911 2.42538; 1584 -1.5 911 2.17538
 97. 1585 -1.75 911 2.17538; 1586 -1.75 911 2.67538; 1587 -1.5 911 2.67538
 98. 1588 -1.75 911 2.92538; 1589 -1.5 911 2.92538; 1590 -1.5 911 1.92538
 99. 1591 -1.75 911 1.92538; 1592 -1.75 911 1.17538; 1593 -1.5 911 1.17538
100. 1594 -1.5 911 0.92538; 1595 -1.75 911 0.92538; 1596 -1.75 911 1.42538
101. 1597 -1.5 911 1.42538; 1598 -1.75 911 1.67538; 1599 -1.5 911 1.67538
102. 1600 -1.5 911 0.67538; 1601 -1.75 911 0.67538; 1602 -1.5 911 0.42538
103. 1603 -1.75 911 0.42538; 1604 -1.5 911 0.07538; 1605 -1.75 911 0.07538
104. 1606 -1.5 911 -0.32462; 1607 -1.75 911 -0.32462; 1608 -1.75 911 3.92538
105. 1609 -1.5 911 3.92538; 1610 -1.75 911 4.17538; 1611 -1.5 911 4.17538
106. 1612 2 911 3.42538; 1613 2 911 3.175; 1614 2 911 3.67538; 1615 2 911 2.42538
107. 1616 2 911 2.17538; 1617 2 911 2.67538; 1618 2 911 2.92538; 1619 2 911 1.92538
108. 1620 2 911 1.17538; 1621 2 911 0.92538; 1622 2 911 1.42538; 1623 2 911 1.67538
109. 1624 2 911 0.67538; 1625 2 911 0.42538; 1626 2 911 0.07538
110. 1627 2 911 -0.32462; 1628 2 911 3.92538; 1629 2 911 4.17538
111. 1630 -2 911 3.42538; 1631 -2 911 3.175; 1632 -2 911 3.67538
112. 1633 -2 911 2.42538; 1634 -2 911 2.17538; 1635 -2 911 2.67538
113. 1636 -2 911 2.92538; 1637 -2 911 1.92538; 1638 -2 911 1.17538
114. 1639 -2 911 0.92538; 1640 -2 911 1.42538; 1641 -2 911 1.67538
115. 1642 -2 911 0.67538; 1643 -2 911 0.42538; 1644 -2 911 0.07538
116. 1645 -2 911 -0.32462; 1646 -2 911 3.92538; 1647 -2 911 4.17538
117. MEMBER INCIDENCES
118. 185 2 128; 1258 1538 1295; 1452 1539 1295
119. ELEMENT INCIDENCES SHELL
120. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
121. 334 307 308 6 42; 337 308 313 309 6; 344 322 323 304 302; 345 323 324 305 304
122. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
123. 349 327 328 313 308; 394 388 389 369 366; 395 389 390 371 369
124. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
125. 399 393 394 379 377; 404 399 400 389 388; 405 400 401 390 389
126. 406 401 402 391 390; 407 402 403 392 391; 408 403 404 393 392
127. 409 404 405 394 393; 414 410 411 400 399; 415 411 412 401 400
128. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
129. 419 415 416 405 404; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
130. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 442 366 369 424 421
131. 443 369 371 426 424; 444 371 373 428 426; 445 373 375 430 428
132. 446 375 377 432 430; 447 377 379 434 432; 497 313 492 493 309
133. 498 328 494 492 313; 501 394 497 498 379; 502 405 499 497 394
134. 503 416 500 499 405; 504 309 493 500 416; 506 379 498 501 434
135. 759 771 774 775 772; 760 774 776 1331 775; 761 776 778 779 1331
136. 762 778 780 781 779; 763 780 782 783 781; 764 782 784 785 783
137. 767 789 790 774 771; 768 790 791 776 774; 769 791 792 778 776
138. 770 792 793 780 778; 771 793 794 782 780; 772 794 795 784 782
139. 775 798 799 790 789; 776 799 800 791 790; 777 800 801 792 791
140. 778 801 802 793 792; 779 802 803 794 793; 780 803 804 795 794
141. 783 421 424 799 798; 784 424 426 800 799; 785 426 428 801 800
142. 786 428 430 802 801; 787 430 432 803 802; 788 432 434 804 803
143. 791 772 775 808 806; 792 775 1331 809 808; 793 1331 779 810 809
144. 794 779 781 811 810; 795 781 783 812 811; 796 783 785 813 812
145. 798 784 815 816 785; 799 795 817 815 784; 800 804 818 817 795
146. 801 434 501 818 804; 802 785 816 819 813; 883 806 808 904 903
147. 884 808 809 905 904; 885 809 810 906 905; 886 810 811 907 906
148. 887 811 812 908 907; 888 812 813 909 908; 901 813 819 923 909
149. 1313 1390 1393 1394 1391; 1314 1393 1395 1396 1394; 1315 1395 302 34 1396
150. 1317 1398 1399 1393 1390; 1318 1399 1400 1395 1393; 1319 1400 322 302 1395
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152. 1325 1410 1411 1405 1402; 1326 1411 1412 1407 1405; 1327 1412 399 388 1407
153. 1329 1414 1415 1411 1410; 1330 1415 1416 1412 1411; 1331 1416 410 399 1412
154. 1333 1391 1394 1415 1414; 1334 1394 1396 1416 1415; 1335 1396 34 410 1416
155. 1337 1403 1406 1419 1417; 1338 1406 1408 1420 1419; 1339 1408 366 421 1420
156. 1341 1422 1425 1426 1423; 1342 1425 1427 1428 1426; 1343 1427 771 772 1428
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161. 1361 1437 1439 1443 1441; 1362 1439 1440 1444 1443; 1363 1440 806 903 1444
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167. 1402 1443 1444 1486 1485; 1403 1444 903 1446 1486; 1405 1483 1485 1489 1487
168. 1406 1485 1486 1490 1489; 1407 1486 1446 1461 1490; 1408 1491 1492 323 322
169. 1409 1492 1493 324 323; 1410 1493 1494 325 324; 1411 1494 1495 326 325
170. 1412 1495 1496 327 326; 1413 1496 1497 328 327; 1414 1498 1499 1492 1491
171. 1415 1499 1500 1493 1492; 1416 1500 1501 1494 1493; 1417 1501 1502 1495 1494
172. 1418 1502 1503 1496 1495; 1419 1503 1504 1497 1496; 1420 1497 1505 494 328
173. 1421 1504 1506 1505 1497; 1445 1530 1531 1399 1398; 1446 1531 1532 1400 1399
174. 1447 1532 1491 322 1400; 1449 1534 1535 1531 1530; 1450 1535 1536 1532 1531
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183. 1476 1557 1562 1565 1560; 1477 819 1560 1566 923; 1478 1560 1565 1567 1566
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191. 1500 1584 1403 1417 1590; 1501 1592 1593 1594 1595; 1502 1593 1422 1423 1594
192. 1503 1596 1597 1593 1592; 1504 1597 1430 1422 1593; 1505 1598 1599 1597 1596
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206. 1545 1638 1592 1595 1639; 1546 1640 1596 1592 1638; 1547 1641 1598 1596 1640
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207. 1548 1637 1591 1598 1641; 1549 1639 1595 1601 1642; 1550 1642 1601 1603 1643
208. 1551 1643 1603 1605 1644; 1552 1644 1605 1607 1645; 1553 1646 1608 1580 1632
209. 1554 1647 1610 1608 1646
210. START GROUP DEFINITION
211. ELEMENT
212. _RAFT 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
213. 425 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
214. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
215. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
216. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
217. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
218. 1449 TO 1451 1453 TO 1554
219. _RAFT_MOM 327 333 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
220. 427 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
221. 775 TO 780 783 786 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 -
222. 1317 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 -
223. 1337 TO 1339 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 1355 1357 TO 1359 -
224. 1361 1363 1364 1367 TO 1370 1378 1381 TO 1384 1401 TO 1403 1405 TO 1421 1445 -
225. 1447 1449 TO 1451 1453 TO 1554
226. _RAFT_SHEAR 442 TO 447 497 498 501 TO 504 506 767 TO 772 775 TO 780 -
227. 783 TO 788 798 TO 802 901 1313 1317 1321 1325 1329 1333 1337 TO 1339 1341 -
228. 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 1361 1370 1384 1401 1405 1420 -
229. 1421 1445 1449 1453 TO 1554
230. MEMBER
231. END GROUP DEFINITION
232. ELEMENT PROPERTY
233. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
234. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
235. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
236. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
237. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
238. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
239. 1449 TO 1451 1453 TO 1554 THICKNESS 0.7
240. DEFINE MATERIAL START
241. ISOTROPIC CONCRETE
242. E 2.17185E+007
243. POISSON 0.17
244. DENSITY 25
245. ALPHA 1E-005
246. DAMP 0.05
247. TYPE CONCRETE
248. STRENGTH FCU 27579
249. END DEFINE MATERIAL
250. MEMBER PROPERTY AMERICAN
251. 185 PRIS YD 0.6 ZD 0.6
252. MEMBER PROPERTY AMERICAN
253. 1258 PRIS YD 0.8 ZD 0.8
254. 1452 PRIS YD 0.75 ZD 0.8
255. CONSTANTS
256. MATERIAL CONCRETE ALL
257. SUPPORTS
258. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
259. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
260. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
261. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
262. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
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263. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
264. 1449 TO 1451 1453 TO 1554 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
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41	0.062500
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304	0.062547
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313	0.062547
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328	0.062500
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400	0.062500
401	0.062500
402	0.062500
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404	0.062500
405	0.062500
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412	0.062452
413	0.062452
414	0.062452
415	0.062452
416	0.062452
421	0.062500

424	0.062500
426	0.062500
428	0.062500
430	0.062500
432	0.062500
434	0.062500
492	0.062547
493	0.062500
494	0.062500
497	0.062500
498	0.062500
499	0.062500
500	0.062452
501	0.062500
771	0.062500
772	0.062500
774	0.062500
775	0.062500
776	0.062500
778	0.062500
779	0.062500
780	0.062500
781	0.062500
782	0.062500
783	0.062500
784	0.062500
785	0.062500
789	0.062500
790	0.062500
791	0.062500
792	0.062500
793	0.062500
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795	0.062500
798	0.062500
799	0.062500
800	0.062500
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809	0.062500
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812	0.062500
813	0.062500
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818	0.062500
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904	0.075000
905	0.075000

906	0.075000
907	0.075000
908	0.075000
909	0.075000
923	0.075000
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1466	0.050000
1467	0.050000
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1531	0.062500
1532	0.062500
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1535	0.031250
1536	0.031250
1537	0.050000
1538	0.093750
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1541	0.062500
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1543	0.062500
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1545	0.062500
1546	0.062452
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1559	0.062500

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1567	0.075000
1568	0.093750
1569	0.093750
1570	0.050000
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1572	0.062500
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1575	0.031250
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1603	0.075000
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1605	0.093750
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1608	0.062500
1609	0.062500
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1611	0.031250
1612	0.031274
1613	0.031250
1614	0.031250
1615	0.031250

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      1617          0.031250
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      1623          0.031250
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      1625          0.037500
      1626          0.046875
      1627          0.025000
      1628          0.031250
      1629          0.015625
      1630          0.031274
      1631          0.031250
      1632          0.031250
      1633          0.031250
      1634          0.031250
      1635          0.031250
      1636          0.031226
      1637          0.031250
      1638          0.031250
      1639          0.031250
      1640          0.031250
      1641          0.031250
      1642          0.031250
      1643          0.037500
      1644          0.046875
      1645          0.025000
      1646          0.031250
      1647          0.015625
    TOTAL          18.000002
265. MEMBER OFFSET
266. 1452 START 0 -0.375 0
267. 1452 END 0 -0.375 0
268. ******************************************************************************
269. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
270. SELFWEIGHT Y -1
271. ELEMENT LOAD
272. * LOAD DUE TO BACKFILL
273. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
274. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
275. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
276. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
277. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
278. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
279. 1449 TO 1451 1453 TO 1554 PR GY -16
280. * LOAD DUE TO SURCHARGE
281. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
282. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
283. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
284. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
285. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
286. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
287. 1449 TO 1451 1453 TO 1554 PR GY -20
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288. JOINT LOAD
289. 128 FX 0.38 FY -117.3 FZ 5.2
290. 1295 FX -0.725 FY -94.5 FZ -5
291. 1539 FX -0.04 FY -33.15 FZ 0.09
292. ******************************************************************************
293. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
294. JOINT LOAD
295. 128 FX 10 FY -115.8 FZ 12
296. 1295 FX 4.7 FY -66.8 FZ -12
297. 1539 FX -0.03 FY -13.43 FZ 0.15
298. ******************************************************************************
299. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WL1 )
300. JOINT LOAD
301. 128 FX 70.2 FY -148.2 FZ 0.5
302. 1295 FX 63.5 FY -135 FZ -0.5
303. 1539 FX -0.69 FY -95.18 FZ 0.01
304. ******************************************************************************
305. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X ( WL2 )
306. JOINT LOAD
307. 128 FX -70.2 FY -148.2 FZ -0.5
308. 1295 FX -63.5 FY -135 FZ 0.5
309. 1539 FX 0.69 FY 90.7 FZ -0.01
310. ******************************************************************************
311. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Y ( WL3 )
312. JOINT LOAD
313. 128 FX 0.3 FY -330.8 FZ 49.5
314. 1295 FX -1.12 FY 330.8 FZ 49.6
315. 1539 FX 0.09 FY 20.68 FZ -6.15
316. ******************************************************************************
317. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Y ( WL4 )
318. JOINT LOAD
319. 128 FX -0.3 FY 330.8 FZ -49.5
320. 1295 FX 1.12 FY -330.8 FZ -49.6
321. 1539 FX -0.09 FY -20.36 FZ 6.15
322. ******************************************************************************
323. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
324. JOINT LOAD
325. 1539 FX -0.02 FY 6.48 FZ -1.7
326. ******************************************************************************
327. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
328. JOINT LOAD
329. 1539 FX 0.02 FY -6.48 FZ 1.7
330. ******************************************************************************
331. LOAD 9 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
332. JOINT LOAD
333. 1539 FX -0.28 FY 13.79 FZ 0.07
334. ******************************************************************************
335. LOAD 10 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
336. JOINT LOAD
337. 1539 FX 0.28 FY -13.79 FZ -0.07
338. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 309 NUMBER OF MEMBERS 3
NUMBER OF PLATES 272 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 306

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 179/ 28/ 87 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 936
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 82 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.4/ 142125.1 MB

```
339. CHANGE
340. SUPPORTS
341. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
342. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
343. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
344. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
345. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
346. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1421 1445 TO 1447 -
347. 1449 TO 1451 1453 TO 1554 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
348. ********************************************************************************
349. * UNFACTORED LOAD COMBINATIONS
350. LOAD 201 DL
351. REPEAT LOAD
352. 1 1.0
353. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

354. CHANGE
355. LOAD 202 DL+L+TL1
356. REPEAT LOAD
357. 1 1.0 2 1.0 9 1.0
358. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202
```



```
359. CHANGE
360. LOAD 203 DL+L+TL2
361. REPEAT LOAD
362. 1 1.0 2 1.0 10 1.0
363. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

364. CHANGE
365. LOAD 204 DL+0.75L+0.75TL1
366. REPEAT LOAD
367. 1 1.0 2 0.75 9 0.75
368. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

369. CHANGE
370. LOAD 205 DL+0.75L+0.75TL2
371. REPEAT LOAD
372. 1 1.0 2 0.75 10 0.75
373. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

374. CHANGE
375. LOAD 206 DL+WL1
376. REPEAT LOAD
377. 1 1.0 3 1.0
378. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

379. CHANGE
380. LOAD 207 DL+WL2
381. REPEAT LOAD
```

```
 382. 1 1.0 4 1.0
 383. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207



 384. CHANGE
 385. LOAD 208 DL+WL3
 386. REPEAT LOAD
 387. 1 1.0 5 1.0
 388. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  208



 389. CHANGE
 390. LOAD 209 DL+WL4
 391. REPEAT LOAD
 392. 1 1.0 6 1.0
 393. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  209



 394. CHANGE
 395. LOAD 210 DL+0.75WL1+0.75L
 396. REPEAT LOAD
 397. 1 1.0 2 0.75 3 0.75
 398. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



 399. CHANGE
 400. LOAD 211 DL+0.75WL2+0.75L
 401. REPEAT LOAD
```



```
 402. 1 1.0 2 0.75 4 0.75
 403. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



 404. CHANGE
 405. LOAD 212 DL+0.75WL3+0.75L
 406. REPEAT LOAD
 407. 1 1.0 2 0.75 5 0.75
 408. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



 409. CHANGE
 410. LOAD 213 DL+0.75WL4+0.75L
 411. REPEAT LOAD
 412. 1 1.0 2 0.75 6 0.75
 413. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  213



 414. CHANGE
 415. LOAD 214 0.6DL+WL1
 416. REPEAT LOAD
 417. 1 0.6 3 1.0
 418. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  214



 419. CHANGE
 420. LOAD 215 0.6DL+WL2
 421. REPEAT LOAD
 422. 1 0.6 4 1.0
 423. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

424. CHANGE
425. LOAD 216 0.6DL+WL3
426. REPEAT LOAD
427. 1 0.6 5 1.0
428. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 216

429. CHANGE
430. LOAD 217 0.6DL+WL4
431. REPEAT LOAD
432. 1 0.6 6 1.0
433. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 217

434. CHANGE
435. LOAD 218 0.6DL+0.7SL1
436. REPEAT LOAD
437. 1 0.6 7 0.7
438. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

439. CHANGE
440. LOAD 219 0.6DL+0.7SL2
441. REPEAT LOAD
442. 1 0.6 8 0.7
443. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

444. CHANGE
445. **
446. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
447. LOAD 101 1.4DL
448. REPEAT LOAD
449. 1 1.4
450. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

451. CHANGE
452. LOAD 102 1.4DL+1.4TL1+1.7L
453. REPEAT LOAD
454. 1 1.4 2 1.7 9 1.4
455. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

456. CHANGE
457. LOAD 103 1.4DL+1.4TL2+1.7L
458. REPEAT LOAD
459. 1 1.4 2 1.7 10 1.4
460. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

461. CHANGE
462. LOAD 104 1.2DL+1.0L

```
 463. REPEAT LOAD
 464. 1 1.2 2 1.0
 465. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104


 466. CHANGE
 467. LOAD 105 1.2DL+1.6WL1+1.0L
 468. REPEAT LOAD
 469. 1 1.2 3 1.6 2 0.5 2 1.0
 470. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105


 471. CHANGE
 472. LOAD 106 1.2DL+1.6WL2+1.0L
 473. REPEAT LOAD
 474. 1 1.2 4 1.6 2 0.5 2 1.0
 475. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106


 476. CHANGE
 477. LOAD 107 1.2DL+1.6WL3+1.0L
 478. REPEAT LOAD
 479. 1 1.2 5 1.6 2 0.5 2 1.0
 480. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  107


 481. CHANGE
 482. LOAD 108 1.2DL+1.6WL4+1.0L
 483. REPEAT LOAD
 484. 1 1.2 6 1.6 2 1.0
```



```
 485. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  108


 486. CHANGE
 487. LOAD 109 1.2DL+1.0SL1+1.0L
 488. REPEAT LOAD
 489. 1 1.2 7 1.0 2 1.0
 490. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  109


 491. CHANGE
 492. LOAD 110 1.2DL+1.0SL2+1.0L
 493. REPEAT LOAD
 494. 1 1.2 8 1.0 2 1.0
 495. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110


 496. CHANGE
 497. LOAD 111 0.9DL+1.6WL1
 498. REPEAT LOAD
 499. 1 0.9 3 1.6
 500. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  111


 501. CHANGE
 502. LOAD 112 0.9DL+1.6WL2
 503. REPEAT LOAD
 504. 1 0.9 4 1.6
 505. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

506. CHANGE
507. LOAD 113 0.9DL+1.6WL3
508. REPEAT LOAD
509. 1 0.9 5 1.6
510. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 113

511. CHANGE
512. LOAD 114 0.9DL+1.6WL4
513. REPEAT LOAD
514. 1 0.9 6 1.6
515. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 114

516. CHANGE
517. LOAD 115 0.9DL+1.0SL1
518. REPEAT LOAD
519. 1 0.9 7 1.0
520. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

521. CHANGE
522. LOAD 116 0.9DL+1.0SL2
523. REPEAT LOAD
524. 1 0.9 8 1.0
525. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

526. CHANGE
527. **
528. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8

**NOTE-Tension/Compression converged after 8 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 2.2426248E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9331611E+04 L-MATRIX DIAG= 2.2919266E-10 EQN NO 932
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 7.3909760E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2491692E+07 L-MATRIX DIAG= 5.8487058E-07 EQN NO 936
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11
**START ITERATION NO. 12
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 2.2426248E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9331611E+04 L-MATRIX DIAG= 2.2919266E-10 EQN NO 932
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 7.3909760E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2491692E+07 L-MATRIX DIAG= 5.8487058E-07 EQN NO 936
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 5
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 6.1541796E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 5 iterations, Case= 6

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 2.2426248E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9331611E+04 L-MATRIX DIAG= 2.2919266E-10 EQN NO 932
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 7.3909760E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2491692E+07 L-MATRIX DIAG= 5.8487058E-07 EQN NO 936
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 2.2426248E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9331611E+04 L-MATRIX DIAG= 2.2919266E-10 EQN NO 932
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 7.3909760E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2491692E+07 L-MATRIX DIAG= 5.8487058E-07 EQN NO 936
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1539 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.6462644E+07 L-MATRIX DIAG= 6.1541796E-06 EQN NO 934
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 10

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 216

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 113

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

529. LOAD LIST 101 TO 116
530. START CONCRETE DESIGN

```
531. CODE ACI
532. TRACK 2 MEMB 185 1258
533. FC 35000 MEMB 185 1258
534. CLB 0.05 MEMB 185 1258
535. CLS 0.05 MEMB 185 1258
536. CLT 0.05 MEMB 185 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 -
537. 414 TO 419 424 TO 429 497 498 501 TO 504 759 TO 764 767 TO 772 775 TO 780 -
538. 783 TO 788 798 TO 801 883 TO 888 901 1258 1313 TO 1315 1317 TO 1319 1321 -
539. 1322 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1341 TO 1343 1345 TO 1347 -
540. 1349 TO 1351 1353 TO 1355 1361 TO 1363 1378 TO 1384 1405 TO 1421 -
541. 1445 TO 1447 1449 TO 1451 1453 TO 1458 1460 TO 1465 1467 TO 1470 -
542. 1472 TO 1475 1477 1478 1481 TO 1498 1501 TO 1508 1511 1512 1515 TO 1526 1528 -
543. 1529 TO 1531 1533 1535 TO 1543 1545 TO 1548 1550 1552 TO 1554
544. MAXMAIN 25 MEMB 185 1258
545. MINMAIN 25 MEMB 185 1258
546. MINSEC 10 MEMB 185 1258
547. DESIGN COLUMN 185 1258
```



```
================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
----------------------------------------------------------

  8 - 25 MM               1.091      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ----------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53    252.65      0.00    0.00000
 ----------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ----------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53    252.65      0.00    0.00000
 ----------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |              9022.30   709.00  4511.15  1061.99
       P0 |*             8270.44   812.98  3759.29  1019.91
          | *            7518.58   897.94  3007.43   956.53
   Pn,max |__*           6766.73   962.48  2255.58   875.37
          |   *          6014.87  1009.20  1503.72   752.23
    Pn    |    *         5263.01  1041.19   751.86   595.66
  NOMINAL |     *          Pn       Mn       Pn       Mn    (@ Y )
    AXIAL |      *       9022.30   709.00  4511.15  1061.99
COMPRESSION|       *     8270.44   812.98  3759.29  1019.91
        Pb|-------*Mb    7518.58   897.94  3007.43   956.53
          |      *       6766.73   962.48  2255.58   875.37
__________|____*_______  6014.87  1009.20  1503.72   752.23
          |  * M0   Mn,  5263.01  1041.19   751.86   595.66
          | *   BENDING
    P-tens|*     MOMENT
          |
```

==========

COLUMN NO. 1258 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 800.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 6400.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	1.227	101	END	0.650

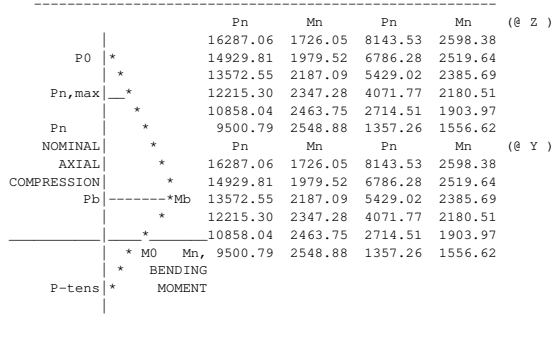
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
22055.40	17644.32	8287.15	2603.84	314.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
1140.95	-3249.05	297.55	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
22055.40	17644.32	8287.15	2603.84	314.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
1140.95	-3249.05	297.55	57.90	0.07483



**********END OF COLUMN DESIGN RESULTS**********

548. END CONCRETE DESIGN
549. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 5,2020 TIME= 16: 6: 3 ****

For technical assistance on STAAD.Pro, please visit
http://selectservices.bentley.com/en-US/

Details about additional assistance from
Bentley and Partners can be found at program menu
Help->Technical Support

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http://www.bentley.com

ATTACHMENT 11 STAAD OUTPUT – FOUNDATION F4

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    NOV  5, 2020                   *
*           Time=    16:23:34                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F4.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/11/2020
    4. JOB NAME FDN. MKD. F4
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911 3.175; 6 0.75 911 3.175; 34 -0.5 911 3.175; 35 -0.25 911 3.175
   17. 41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175; 302 -0.5 911 3.42538
   18. 304 -0.25 911 3.42538; 305 0 911 3.42538; 306 0.25 911 3.42538
   19. 307 0.5 911 3.42538; 308 0.75 911 3.42538; 309 1 911 3.175; 313 1 911 3.42538
   20. 322 -0.5 911 3.67538; 323 -0.25 911 3.67538; 324 0 911 3.67538
   21. 325 0.25 911 3.67538; 326 0.5 911 3.67538; 327 0.75 911 3.67538
   22. 328 1 911 3.67538; 366 -0.5 911 2.17538; 369 -0.25 911 2.17538
   23. 371 0 911 2.17538; 373 0.25 911 2.17538; 375 0.5 911 2.17538
   24. 377 0.75 911 2.17538; 379 1 911 2.17538; 388 -0.5 911 2.42538
   25. 389 -0.25 911 2.42538; 390 0 911 2.42538; 391 0.25 911 2.42538
   26. 392 0.5 911 2.42538; 393 0.75 911 2.42538; 394 1 911 2.42538
   27. 399 -0.5 911 2.67538; 400 -0.25 911 2.67538; 401 0 911 2.67538
   28. 402 0.25 911 2.67538; 403 0.5 911 2.67538; 404 0.75 911 2.67538
   29. 405 1 911 2.67538; 410 -0.5 911 2.92538; 411 -0.25 911 2.92538
   30. 412 0 911 2.92538; 413 0.25 911 2.92538; 414 0.5 911 2.92538
   31. 415 0.75 911 2.92538; 416 1 911 2.92538; 421 -0.5 911 1.92538
   32. 424 -0.25 911 1.92538; 426 0 911 1.92538; 428 0.25 911 1.92538
   33. 430 0.5 911 1.92538; 432 0.75 911 1.92538; 434 1 911 1.92538
   34. 492 1.25 911 3.42538; 493 1.25 911 3.175; 494 1.25 911 3.67538
   35. 497 1.25 911 2.42538; 498 1.25 911 2.17538; 499 1.25 911 2.67538
   36. 500 1.25 911 2.92538; 501 1.25 911 1.92538; 771 -0.5 911 1.17538
   37. 772 -0.5 911 0.92538; 774 -0.25 911 1.17538; 775 -0.25 911 0.92538
   38. 776 0 911 1.17538; 778 0.25 911 1.17538; 779 0.25 911 0.92538
```



```
   39. 780 0.5 911 1.17538; 781 0.5 911 0.92538; 782 0.75 911 1.17538
   40. 783 0.75 911 0.92538; 784 1 911 1.17538; 785 1 911 0.92538
   41. 789 -0.5 911 1.42538; 790 -0.25 911 1.42538; 791 0 911 1.42538
   42. 792 0.25 911 1.42538; 793 0.5 911 1.42538; 794 0.75 911 1.42538
   43. 795 1 911 1.42538; 798 -0.5 911 1.67538; 799 -0.25 911 1.67538
   44. 800 0 911 1.67538; 801 0.25 911 1.67538; 802 0.5 911 1.67538
   45. 803 0.75 911 1.67538; 804 1 911 1.67538; 806 -0.5 911 0.67538
   46. 808 -0.25 911 0.67538; 809 0 911 0.67538; 810 0.25 911 0.67538
   47. 811 0.5 911 0.67538; 812 0.75 911 0.67538; 813 1 911 0.67538
   48. 815 1.25 911 1.17538; 816 1.25 911 0.92538; 817 1.25 911 1.42538
   49. 818 1.25 911 1.67538; 819 1.25 911 0.67538; 903 -0.5 911 0.42538
   50. 904 -0.25 911 0.42538; 905 0 911 0.42538; 906 0.25 911 0.42538
   51. 907 0.5 911 0.42538; 908 0.75 911 0.42538; 909 1 911 0.42538
   52. 923 1.25 911 0.42538; 1295 0 913.6 -0.075; 1331 0 911 0.92538
   53. 1390 -1.25 911 3.42538; 1391 -1.25 911 3.175; 1393 -1 911 3.42538
   54. 1394 -1 911 3.175; 1395 -0.75 911 3.42538; 1396 -0.75 911 3.175
   55. 1398 -1.25 911 3.67538; 1399 -1 911 3.67538; 1400 -0.75 911 3.67538
   56. 1402 -1.25 911 2.42538; 1403 -1.25 911 2.17538; 1405 -1 911 2.42538
   57. 1406 -1 911 2.17538; 1407 -0.75 911 2.42538; 1408 -0.75 911 2.17538
   58. 1410 -1.25 911 2.67538; 1411 -1 911 2.67538; 1412 -0.75 911 2.67538
   59. 1414 -1.25 911 2.92538; 1415 -1 911 2.92538; 1416 -0.75 911 2.92538
   60. 1417 -1.25 911 1.92538; 1419 -1 911 1.92538; 1420 -0.75 911 1.92538
   61. 1422 -1.25 911 1.17538; 1423 -1.25 911 0.92538; 1425 -1 911 1.17538
   62. 1426 -1 911 0.92538; 1427 -0.75 911 1.17538; 1428 -0.75 911 0.92538
   63. 1430 -1.25 911 1.42538; 1431 -1 911 1.42538; 1432 -0.75 911 1.42538
   64. 1434 -1.25 911 1.67538; 1435 -1 911 1.67538; 1436 -0.75 911 1.67538
   65. 1437 -1.25 911 0.67538; 1439 -1 911 0.67538; 1440 -0.75 911 0.67538
   66. 1441 -1.25 911 0.42538; 1443 -1 911 0.42538; 1444 -0.75 911 0.42538
   67. 1445 -0.25 911 0.17538; 1446 -0.5 911 0.17538; 1448 0.25 911 0.17538
   68. 1449 0.5 911 0.17538; 1450 0.75 911 0.17538; 1451 1 911 0.17538
   69. 1452 1.25 911 0.17538; 1460 -0.25 911 -0.07462; 1461 -0.5 911 -0.07462
   70. 1463 0.25 911 -0.07462; 1464 0.5 911 -0.07462; 1465 0.75 911 -0.07462
   71. 1466 1 911 -0.07462; 1467 1.25 911 -0.07462; 1483 -1.25 911 0.17538
   72. 1485 -1 911 0.17538; 1486 -0.75 911 0.17538; 1487 -1.25 911 -0.07462
   73. 1489 -1 911 -0.07462; 1490 -0.75 911 -0.07462; 1538 0 911 0.17538
   74. 1539 -0.25 911 -0.32462; 1540 -0.5 911 -0.32462; 1541 0 911 -0.32462
   75. 1542 0.25 911 -0.32462; 1543 0.5 911 -0.32462; 1544 0.75 911 -0.32462
   76. 1545 1 911 -0.32462; 1546 1.25 911 -0.32462; 1547 -1 911 -0.32462
   77. 1548 -1.25 911 -0.32462; 1549 -0.75 911 -0.32462; 1550 -0.25 911 -0.57462
   78. 1551 -0.5 911 -0.57462; 1552 0 911 -0.57462; 1553 0.25 911 -0.57462
   79. 1554 0.5 911 -0.57462; 1555 0.75 911 -0.57462; 1556 1 911 -0.57462
   80. 1557 1.25 911 -0.57462; 1558 -1 911 -0.57462; 1559 -1.25 911 -0.57462
   81. 1560 -0.75 911 -0.57462; 1561 0 911 -0.07462
   82. MEMBER INCIDENCES
   83. 185 2 128; 1258 1561 1295
   84. ELEMENT INCIDENCES SHELL
   85. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   86. 334 307 308 6 42; 337 308 313 309 6; 344 322 323 304 302; 345 323 324 305 304
   87. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   88. 349 327 328 313 308; 394 388 389 369 366; 395 389 390 371 369
   89. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
   90. 399 393 394 379 377; 404 399 400 389 388; 405 400 401 390 389
   91. 406 401 402 391 390; 407 402 403 392 391; 408 403 404 393 392
   92. 409 404 405 394 393; 414 410 411 400 399; 415 411 412 401 400
   93. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
   94. 419 415 416 405 404; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
```

```
 95. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 442 366 369 424 421
 96. 443 369 371 426 424; 444 371 373 428 426; 445 373 375 430 428
 97. 446 375 377 432 430; 447 377 379 434 432; 497 313 492 493 309
 98. 498 328 494 492 313; 501 394 497 498 379; 502 405 499 497 394
 99. 503 416 500 499 405; 504 309 493 500 416; 506 379 498 501 434
100. 759 771 774 775 772; 760 774 776 1331 775; 761 776 778 779 1331
101. 762 778 780 781 779; 763 780 782 783 781; 764 782 784 785 783
102. 767 789 790 774 771; 768 790 791 776 774; 769 791 792 778 776
103. 770 792 793 780 778; 771 793 794 782 780; 772 794 795 784 782
104. 775 798 799 790 789; 776 799 800 791 790; 777 800 801 792 791
105. 778 801 802 793 792; 779 802 803 794 793; 780 803 804 795 794
106. 783 421 424 799 798; 784 424 426 800 799; 785 426 428 801 800
107. 786 428 430 802 801; 787 430 432 803 802; 788 432 434 804 803
108. 791 772 775 808 806; 792 775 1331 809 808; 793 1331 779 810 809
109. 794 779 781 811 810; 795 781 783 812 811; 796 783 785 813 812
110. 798 784 815 816 785; 799 795 817 815 784; 800 804 818 817 795
111. 801 434 501 818 804; 802 785 816 819 813; 883 806 808 904 903
112. 884 808 809 905 904; 885 809 810 906 905; 886 810 811 907 906
113. 887 811 812 908 907; 888 812 813 909 908; 901 813 819 923 909
114. 1313 1390 1393 1394 1391; 1314 1393 1395 1396 1394; 1315 1395 302 34 1396
115. 1317 1398 1399 1393 1390; 1318 1399 1400 1395 1393; 1319 1400 322 302 1395
116. 1321 1402 1405 1406 1403; 1322 1405 1407 1408 1406; 1323 1407 388 366 1408
117. 1325 1410 1411 1405 1402; 1326 1411 1412 1407 1405; 1327 1412 399 388 1407
118. 1329 1414 1415 1411 1410; 1330 1415 1416 1412 1411; 1331 1416 410 399 1412
119. 1333 1391 1394 1415 1414; 1334 1394 1396 1416 1415; 1335 1396 34 410 1416
120. 1337 1403 1406 1419 1417; 1338 1406 1408 1420 1419; 1339 1408 366 421 1420
121. 1341 1422 1425 1426 1423; 1342 1425 1427 1428 1426; 1343 1427 771 772 1428
122. 1345 1430 1431 1425 1422; 1346 1431 1432 1427 1425; 1347 1432 789 771 1427
123. 1349 1434 1435 1431 1430; 1350 1435 1436 1432 1431; 1351 1436 798 789 1432
124. 1353 1417 1419 1435 1434; 1354 1419 1420 1436 1435; 1355 1420 421 798 1436
125. 1357 1423 1426 1439 1437; 1358 1426 1428 1440 1439; 1359 1428 772 806 1440
126. 1361 1437 1439 1443 1441; 1362 1439 1440 1444 1443; 1363 1440 806 903 1444
127. 1364 903 904 1445 1446; 1365 904 905 1538 1445; 1366 905 906 1448 1538
128. 1367 906 907 1449 1448; 1368 907 908 1450 1449; 1369 908 909 1451 1450
129. 1370 909 923 1452 1451; 1378 1446 1445 1460 1461; 1379 1445 1538 1561 1460
130. 1380 1538 1448 1463 1561; 1381 1448 1449 1464 1463; 1382 1449 1450 1465 1464
131. 1383 1450 1451 1466 1465; 1384 1451 1452 1467 1466; 1401 1441 1443 1485 1483
132. 1402 1443 1444 1486 1485; 1403 1444 903 1446 1486; 1405 1483 1485 1489 1487
133. 1406 1485 1486 1490 1489; 1407 1486 1446 1461 1490; 1408 1461 1460 1539 1540
134. 1409 1460 1561 1541 1539; 1410 1561 1463 1542 1541; 1411 1463 1464 1543 1542
135. 1412 1464 1465 1544 1543; 1413 1465 1466 1545 1544; 1414 1466 1467 1546 1545
136. 1415 1487 1489 1547 1548; 1416 1489 1490 1549 1547; 1417 1490 1461 1540 1549
137. 1418 1540 1539 1550 1551; 1419 1539 1541 1552 1550; 1420 1541 1542 1553 1552
138. 1421 1542 1543 1554 1553; 1422 1543 1544 1555 1554; 1423 1544 1545 1556 1555
139. 1424 1545 1546 1557 1556; 1425 1548 1547 1558 1559; 1426 1547 1549 1560 1558
140. 1427 1549 1540 1551 1560
141. START GROUP DEFINITION
142. ELEMENT
143. _RAFT 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
144. 425 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
145. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
146. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
147. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
148. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427
149. _RAFT_MOM 327 333 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 424 -
150. 427 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
```



```
151. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 886 TO 888 901 1313 TO 1315 -
152. 1317 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 -
153. 1337 TO 1339 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 -
154. 1357 TO 1359 1361 TO 1370 1378 1381 TO 1384 1401 TO 1403 1405 TO 1408 1411 -
155. 1412 TO 1427
156. _RAFT_SHEAR 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
157. 775 TO 780 783 TO 788 798 TO 802 901 1313 1317 1321 1325 1329 1333 -
158. 1337 TO 1339 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 1361 -
159. 1370 1384 1401 1405 1414 1415 1424 1425
160. MEMBER
161. END GROUP DEFINITION
162. ELEMENT PROPERTY
163. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
164. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
165. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
166. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
167. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
168. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427 THICKNESS 0.7
169. DEFINE MATERIAL START
170. ISOTROPIC CONCRETE
171. E 2.17185E+007
172. POISSON 0.17
173. DENSITY 25
174. ALPHA 1E-005
175. DAMP 0.05
176. TYPE CONCRETE
177. STRENGTH FCU 27579
178. END DEFINE MATERIAL
179. MEMBER PROPERTY AMERICAN
180. 185 1258 PRIS YD 0.6 ZD 0.6
181. CONSTANTS
182. MATERIAL CONCRETE ALL
183. SUPPORTS
184. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
185. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
186. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
187. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
188. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
189. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1426 -
190. 1427 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062547
304	0.062547
305	0.062547
306	0.062547
307	0.062547
308	0.062547

309	0.062500
313	0.062547
322	0.031250
323	0.031250
324	0.031250
325	0.031250
326	0.031250
327	0.031250
328	0.031250
366	0.062500
369	0.062500
371	0.062500
373	0.062500
375	0.062500
377	0.062500
379	0.062500
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
394	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
405	0.062500
410	0.062452
411	0.062452
412	0.062452
413	0.062452
414	0.062452
415	0.062452
416	0.062452
421	0.062500
424	0.062500
426	0.062500
428	0.062500
430	0.062500
432	0.062500
434	0.062500
492	0.031274
493	0.031250
494	0.015625
497	0.031250
498	0.031250
499	0.031250
500	0.031226
501	0.031250
771	0.062500
772	0.062500
774	0.062500
775	0.062500

776	0.062500
778	0.062500
779	0.062500
780	0.062500
781	0.062500
782	0.062500
783	0.062500
784	0.062500
785	0.062500
789	0.062500
790	0.062500
791	0.062500
792	0.062500
793	0.062500
794	0.062500
795	0.062500
798	0.062500
799	0.062500
800	0.062500
801	0.062500
802	0.062500
803	0.062500
804	0.062500
806	0.062500
808	0.062500
809	0.062500
810	0.062500
811	0.062500
812	0.062500
813	0.062500
815	0.031250
816	0.031250
817	0.031250
818	0.031250
819	0.031250
903	0.062500
904	0.062500
905	0.062500
906	0.062500
907	0.062500
908	0.062500
909	0.062500
923	0.031250
1331	0.062500
1390	0.031274
1391	0.031250
1393	0.062547
1394	0.062500
1395	0.062547
1396	0.062500
1398	0.015625
1399	0.031250
1400	0.031250
1402	0.031250
1403	0.031250
1405	0.062500

1406	0.062500
1407	0.062500
1408	0.062500
1410	0.031250
1411	0.062500
1412	0.062500
1414	0.031226
1415	0.062452
1416	0.062452
1417	0.031250
1419	0.062500
1420	0.062500
1422	0.031250
1423	0.031250
1425	0.062500
1426	0.062500
1427	0.062500
1428	0.062500
1430	0.031250
1431	0.062500
1432	0.062500
1434	0.031250
1435	0.062500
1436	0.062500
1437	0.031250
1439	0.062500
1440	0.062500
1441	0.031250
1443	0.062500
1444	0.062500
1445	0.062500
1446	0.062500
1448	0.062500
1449	0.062500
1450	0.062500
1451	0.062500
1452	0.031250
1460	0.062500
1461	0.062500
1463	0.062500
1464	0.062500
1465	0.062500
1466	0.062500
1467	0.031250
1483	0.031250
1485	0.062500
1486	0.062500
1487	0.031250
1489	0.062500
1490	0.062500
1538	0.062500
1539	0.062500
1540	0.062500
1541	0.062500
1542	0.062500
1543	0.062500

1544	0.062500
1545	0.062500
1546	0.031250
1547	0.062500
1548	0.031250
1549	0.062500
1550	0.031250
1551	0.031250
1552	0.031250
1553	0.031250
1554	0.031250
1555	0.031250
1556	0.031250
1557	0.015625
1558	0.031250
1559	0.015625
1560	0.031250
1561	0.062500
TOTAL	10.625000

```
191. ********************************************************************************
192. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
193. SELFWEIGHT Y -1 LIST ALL
194. ELEMENT LOAD
195. * LOAD DUE TO BACKFILL
196. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
197. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
198. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
199. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
200. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
201. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427 PR GY -16
202. * LOAD DUE TO SURCHARGE
203. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
204. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
205. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
206. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
207. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
208. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1427 PR GY -20
209. JOINT LOAD
210. 128 FX 0.03 FY -92.02 FZ 0.35
211. 1295 FX 0.32 FY -69.18 FZ -0.58
212. ********************************************************************************
213. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
214. JOINT LOAD
215. 128 FX -0.21 FY -95.7 FZ 0.1
216. 1295 FX 0.97 FY -50.14 FZ -0.18
217. ********************************************************************************
218. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WL1 )
219. JOINT LOAD
220. 128 FX 2.96 FY 148.2 FZ 0.0001
221. 1295 FX -0.28 FY -135.03 FZ 0.04
222. ********************************************************************************
223. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X ( WL2 )
224. JOINT LOAD
225. 128 FX -2.96 FY -148.2 FZ -0.0001
226. 1295 FX 0.28 FY 135.03 FZ -0.04
227. ********************************************************************************
```

```
228. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Y ( WL3 )
229. JOINT LOAD
230. 128 FX 0.44 FY 230.2 FZ 42.33
231. 1295 FX 0.41 FY -230.18 FZ 42.14
232. ******************************************************************************
233. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Y ( WL4 )
234. JOINT LOAD
235. 128 FX -0.44 FY -230.2 FZ -42.33
236. 1295 FX -0.41 FY 230.18 FZ -42.14
237. ******************************************************************************
238. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 200 NUMBER OF MEMBERS 2
NUMBER OF PLATES 170 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 198

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 132/ 19/ 60 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 606
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 37 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.3/ 142120.6 MB

```
239. CHANGE
240. SUPPORTS
241. 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 414 TO 419 -
242. 424 TO 429 442 TO 447 497 498 501 TO 504 506 759 TO 764 767 TO 772 -
243. 775 TO 780 783 TO 788 791 TO 796 798 TO 802 883 TO 888 901 1313 TO 1315 1317 -
244. 1318 TO 1319 1321 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1337 TO 1339 -
245. 1341 TO 1343 1345 TO 1347 1349 TO 1351 1353 TO 1355 1357 TO 1359 -
246. 1361 TO 1370 1378 TO 1384 1401 TO 1403 1405 TO 1426 -
247. 1427 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
248. ******************************************************************************
249. * UNFACTORED LOAD COMBINATIONS
250. LOAD 201 DL
251. REPEAT LOAD
252. 1 1.0
253. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  201


254. CHANGE
255. LOAD 202 DL+L
256. REPEAT LOAD
257. 1 1.0 2 1.0
258. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202


259. CHANGE
260. LOAD 203 DL+0.75L
261. REPEAT LOAD
262. 1 1.0 2 0.75
263. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  203


264. CHANGE
265. LOAD 204 DL+WL1
266. REPEAT LOAD
267. 1 1.0 3 1.0
268. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204


269. CHANGE
270. LOAD 205 DL+WL2
271. REPEAT LOAD
272. 1 1.0 4 1.0
273. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  205
```

```
 274. CHANGE
 275. LOAD 206 DL+WL3
 276. REPEAT LOAD
 277. 1 1.0 5 1.0
 278. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  206



 279. CHANGE
 280. LOAD 207 DL+WL4
 281. REPEAT LOAD
 282. 1 1.0 6 1.0
 283. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207



 284. CHANGE
 285. LOAD 208 DL+0.75WL1+0.75L
 286. REPEAT LOAD
 287. 1 1.0 2 0.75 3 0.75
 288. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  208



 289. CHANGE
 290. LOAD 209 DL+0.75WL2+0.75L
 291. REPEAT LOAD
 292. 1 1.0 2 0.75 4 0.75
 293. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  209



 294. CHANGE
 295. LOAD 210 DL+0.75WL3+0.75L
 296. REPEAT LOAD
 297. 1 1.0 2 0.75 5 0.75
```



```
 298. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



 299. CHANGE
 300. LOAD 211 DL+0.75WL4+0.75L
 301. REPEAT LOAD
 302. 1 1.0 2 0.75 6 0.75
 303. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



 304. CHANGE
 305. LOAD 212 0.6DL+WL1
 306. REPEAT LOAD
 307. 1 0.6 3 1.0
 308. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  212



 309. CHANGE
 310. LOAD 213 0.6DL+WL2
 311. REPEAT LOAD
 312. 1 0.6 4 1.0
 313. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  213



 314. CHANGE
 315. LOAD 214 0.6DL+WL3
 316. REPEAT LOAD
```

```
 317. 1 0.6 5 1.0
 318. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  214



 319. CHANGE
 320. LOAD 215 0.6DL+WL4
 321. REPEAT LOAD
 322. 1 0.6 6 1.0
 323. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  215



 324. CHANGE
 325. ******************************************************************************
 326. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
 327. LOAD 101 1.4DL
 328. REPEAT LOAD
 329. 1 1.4
 330. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101



 331. CHANGE
 332. LOAD 102 1.4DL+1.7L
 333. REPEAT LOAD
 334. 1 1.4 2 1.7
 335. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102
```



```
 336. CHANGE
 337. LOAD 103 1.2DL+1.0L
 338. REPEAT LOAD
 339. 1 1.2 2 1.0
 340. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 341. CHANGE
 342. LOAD 104 1.2DL+1.6WL1+1.0L
 343. REPEAT LOAD
 344. 1 1.2 3 1.6 2 0.5 2 1.0
 345. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104



 346. CHANGE
 347. LOAD 105 1.2DL+1.6WL2+1.0L
 348. REPEAT LOAD
 349. 1 1.2 4 1.6 2 0.5 2 1.0
 350. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  105



 351. CHANGE
 352. LOAD 106 1.2DL+1.6WL3+1.0L
 353. REPEAT LOAD
 354. 1 1.2 5 1.6 2 0.5 2 1.0
 355. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106



 356. CHANGE
 357. LOAD 107 1.2DL+1.6WL4+1.0L
 358. REPEAT LOAD
```

```
 359. 1 1.2 6 1.6 2 1.0
 360. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  107



 361. CHANGE
 362. LOAD 108 0.9DL+1.6WL1
 363. REPEAT LOAD
 364. 1 0.9 3 1.6
 365. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  108



 366. CHANGE
 367. LOAD 109 0.9DL+1.6WL2
 368. REPEAT LOAD
 369. 1 0.9 4 1.6
 370. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  109



 371. CHANGE
 372. LOAD 110 0.9DL+1.6WL3
 373. REPEAT LOAD
 374. 1 0.9 5 1.6
 375. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
```



```
**NOTE-Tension/Compression converged after  4 iterations, Case=  110



 376. CHANGE
 377. LOAD 111 0.9DL+1.6WL4
 378. REPEAT LOAD
 379. 1 0.9 6 1.6
 380. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  111



 381. CHANGE
 382. ******************************************************************************
 383. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

***WARNING -  INSTABILITY AT JOINT  1557   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902233E+06 L-MATRIX DIAG=  3.8072467E-06 EQN NO     605
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

  **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
           DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
           BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
           STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
           IS   1.000E-09

           THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
           ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO.          7
```

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8091093E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -1.0263175E-06 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795979E+04 L-MATRIX DIAG= -2.0372681E-10 EQN NO 604
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8091093E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -1.0263175E-06 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -4.9727969E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -6.9485977E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 7 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8072467E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8072467E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8091093E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8091093E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8091093E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -1.0263175E-06 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795979E+04 L-MATRIX DIAG= -2.0372681E-10 EQN NO 604
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 3.8091093E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -1.0263175E-06 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 11 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -4.9727969E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -5.5902638E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -5.8822334E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -4.4708140E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= -4.2682514E-06 EQN NO 605
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1557 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902233E+06 L-MATRIX DIAG= 9.5600262E-07 EQN NO 606
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 10 iterations, Case= 6

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 212

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 213

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 214

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 108

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 109

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 110

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 111

384. LOAD LIST 101 TO 111

```
385. START CONCRETE DESIGN
```



```
386. CODE ACI
387. TRACK 2 ALL
388. FC 35000 MEMB 185 1258
389. CLB 0.05 MEMB 185 1258
390. CLS 0.05 ALL
391. CLT 0.05 MEMB 185 327 328 332 TO 334 337 344 TO 349 394 TO 399 404 TO 409 -
392. 414 TO 419 424 TO 429 497 498 501 TO 504 759 TO 764 767 TO 772 775 TO 780 -
393. 783 TO 788 798 TO 801 883 TO 888 901 1258 1313 TO 1315 1317 TO 1319 1321 -
394. 1322 TO 1323 1325 TO 1327 1329 TO 1331 1333 TO 1335 1341 TO 1343 1345 TO 1347 -
395. 1349 TO 1351 1353 TO 1355 1361 TO 1363 1378 TO 1384 1405 TO 1427
396. MAXMAIN 25 ALL
397. MINMAIN 25 ALL
398. MINSEC 10 ALL
399. DESIGN COLUMN 185 1258
```

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

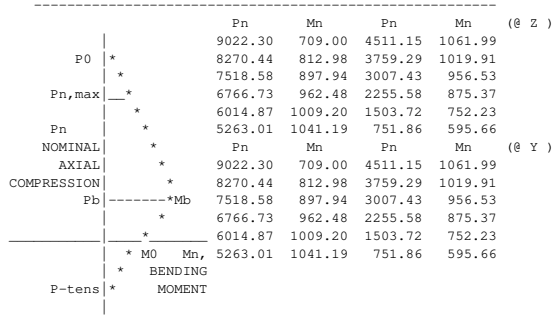
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	198.20	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	198.20	0.00	0.00000



COLUMN NO. 1258 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	149.00	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	149.00	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

```
400. END CONCRETE DESIGN
401. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= NOV  5,2020   TIME= 16:26:12 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
```

ATTACHMENT 12 STAAD OUTPUT – FOUNDATION F5

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    NOV  6, 2020                   *
*           Time=    13:10:14                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F5.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/11/2020
    4. JOB NAME FDN. MKD. F5
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 366 -0.5 911 2.17538; 369 -0.25 911 2.17538; 371 0 911 2.17538
   17. 388 -0.5 911 2.42538; 389 -0.25 911 2.42538; 390 0 911 2.42538
   18. 399 -0.5 911 2.67538; 400 -0.25 911 2.67538; 410 -0.5 911 2.92538
   19. 411 -0.25 911 2.92538; 421 -0.5 911 1.92538; 424 -0.25 911 1.92538
   20. 771 -0.5 911 1.17538; 772 -0.5 911 0.92538; 774 -0.25 911 1.17538
   21. 775 -0.25 911 0.92538; 776 0 911 1.17538; 789 -0.5 911 1.42538
   22. 790 -0.25 911 1.42538; 791 0 911 1.42538; 798 -0.5 911 1.67538
   23. 799 -0.25 911 1.67538; 800 0 911 1.67538; 806 -0.5 911 0.67538
   24. 808 -0.25 911 0.67538; 809 0 911 0.67538; 1295 0 913.6 1.925
   25. 1331 0 911 0.92538; 1405 -1 911 2.42538; 1406 -1 911 2.17538
   26. 1407 -0.75 911 2.42538; 1408 -0.75 911 2.17538; 1411 -1 911 2.67538
   27. 1412 -0.75 911 2.67538; 1415 -1 911 2.92538; 1416 -0.75 911 2.92538
   28. 1419 -1 911 1.92538; 1420 -0.75 911 1.92538; 1425 -1 911 1.17538
   29. 1426 -1 911 0.92538; 1427 -0.75 911 1.17538; 1428 -0.75 911 0.92538
   30. 1431 -1 911 1.42538; 1432 -0.75 911 1.42538; 1435 -1 911 1.67538
   31. 1436 -0.75 911 1.67538; 1439 -1 911 0.67538; 1440 -0.75 911 0.67538
   32. 1562 0 911 1.92538; 1563 -1.25 911 2.42538; 1564 -1.25 911 2.17538
   33. 1565 -1.25 911 2.67538; 1566 -1.25 911 2.92538; 1567 -1.25 911 1.92538
   34. 1568 -1.25 911 1.17538; 1569 -1.25 911 0.92538; 1570 -1.25 911 1.42538
   35. 1571 -1.25 911 1.67538; 1572 -1.25 911 0.67538; 1591 0.75 911 1.17538
   36. 1594 0.75 911 0.92538; 1603 1 911 0.67538; 1604 0.75 911 0.67538
   37. 1618 0.25 911 1.92538; 1619 0.25 911 1.17538; 1620 0.5 911 1.17538
   38. 1621 0.5 911 0.92538; 1622 0.25 911 0.92538; 1623 0.25 911 1.42538
```



```
   39. 1624 0.5 911 1.42538; 1625 0.25 911 1.67538; 1626 0.5 911 1.67538
   40. 1627 0.5 911 0.67538; 1628 0.25 911 0.67538; 1641 -0.5 911 3.17538
   41. 1644 -1 911 3.17538; 1645 -0.75 911 3.17538; 1646 -1.25 911 3.17538
   42. 1652 -0.5 911 3.42538; 1655 -1 911 3.42538; 1656 -0.75 911 3.42538
   43. 1657 -1.25 911 3.42538; 1663 -0.25 911 0.42538; 1664 -0.5 911 0.42538
   44. 1665 0 911 0.42538; 1666 -0.75 911 0.42538; 1667 -1 911 0.42538
   45. 1668 -1.25 911 0.42538; 1669 1 911 0.42538; 1670 0.75 911 0.42538
   46. 1671 1.25 911 0.42538; 1672 0.5 911 0.42538; 1673 0.25 911 0.42538
   47. 1685 -0.354167 911 3.17538; 1686 -0.208333 911 2.92538
   48. 1687 -0.0625 911 2.67538; 1688 0.0833333 911 2.42538
   49. 1689 0.229167 911 2.17538; 1690 0.375 911 1.92538; 1691 0.666667 911 1.42538
   50. 1692 0.8125 911 1.17538; 1693 0.958333 911 0.92538; 1694 1.10417 911 0.67538
   51. 1695 -1.5 911 2.42538; 1696 -1.5 911 2.17538; 1697 -1.5 911 2.67538
   52. 1698 -1.5 911 2.92538; 1699 -1.5 911 1.92538; 1700 -1.5 911 1.17538
   53. 1701 -1.5 911 0.92538; 1702 -1.5 911 1.42538; 1703 -1.5 911 1.67538
   54. 1704 -1.5 911 0.67538; 1705 -1.5 911 3.17538; 1706 -1.5 911 3.42538
   55. 1707 -1.5 911 0.42538; 1708 -0.25 911 0.17538; 1709 -0.5 911 0.17538
   56. 1710 0 911 0.17538; 1711 -0.75 911 0.17538; 1712 -1 911 0.17538
   57. 1713 -1.25 911 0.17538; 1714 1 911 0.17538; 1715 0.75 911 0.17538
   58. 1716 0.5 911 0.17538; 1717 0.25 911 0.17538; 1718 -1.5 911 0.17538
   59. 1719 1.25 911 0.17538; 1720 -0.25 911 -0.07462; 1721 -0.5 911 -0.07462
   60. 1722 0 911 -0.07462; 1723 -0.75 911 -0.07462; 1724 -1 911 -0.07462
   61. 1725 -1.25 911 -0.07462; 1726 1 911 -0.07462; 1727 0.75 911 -0.07462
   62. 1728 0.5 911 -0.07462; 1729 0.25 911 -0.07462; 1730 -1.5 911 -0.07462
   63. 1731 1.25 911 -0.07462; 1732 -0.25 911 -0.32462; 1733 -0.5 911 -0.32462
   64. 1734 0 911 -0.32462; 1735 -0.75 911 -0.32462; 1736 -1 911 -0.32462
   65. 1737 -1.25 911 -0.32462; 1738 1 911 -0.32462; 1739 0.75 911 -0.32462
   66. 1740 0.5 911 -0.32462; 1741 0.25 911 -0.32462; 1742 -1.5 911 -0.32462
   67. 1743 1.25 911 -0.32462; 1744 -0.25 911 -0.57462; 1745 -0.5 911 -0.57462
   68. 1746 0 911 -0.57462; 1747 -0.75 911 -0.57462; 1748 -1 911 -0.57462
   69. 1749 -1.25 911 -0.57462; 1750 1 911 -0.57462; 1751 0.75 911 -0.57462
   70. 1752 0.5 911 -0.57462; 1753 0.25 911 -0.57462; 1754 -1.5 911 -0.57462
   71. 1755 1.25 911 -0.57462; 1756 -0.25 911 -0.82462; 1757 -0.5 911 -0.82462
   72. 1758 0 911 -0.82462; 1759 -0.75 911 -0.82462; 1760 -1 911 -0.82462
   73. 1761 -1.25 911 -0.82462; 1762 1 911 -0.82462; 1763 0.75 911 -0.82462
   74. 1764 0.5 911 -0.82462; 1765 0.25 911 -0.82462; 1766 -1.5 911 -0.82462
   75. 1767 1.25 911 -0.82462; 1768 -0.25 911 -1.07462; 1769 -0.5 911 -1.07462
   76. 1770 0 911 -1.07462; 1771 -0.75 911 -1.07462; 1772 -1 911 -1.07462
   77. 1773 -1.25 911 -1.07462; 1774 1 911 -1.07462; 1775 0.75 911 -1.07462
   78. 1776 0.5 911 -1.07462; 1777 0.25 911 -1.07462; 1778 -1.5 911 -1.07462
   79. 1779 1.25 911 -1.07462; 1780 -0.25 911 -1.32462; 1781 -0.5 911 -1.32462
   80. 1782 0 911 -1.32462; 1783 -0.75 911 -1.32462; 1784 -1 911 -1.32462
   81. 1785 -1.25 911 -1.32462; 1786 1 911 -1.32462; 1787 0.75 911 -1.32462
   82. 1788 0.5 911 -1.32462; 1789 0.25 911 -1.32462; 1790 -1.5 911 -1.32462
   83. 1791 1.25 911 -1.32462; 1792 -0.25 911 -1.57462; 1793 -0.5 911 -1.57462
   84. 1794 0 911 -1.57462; 1795 -0.75 911 -1.57462; 1796 -1 911 -1.57462
   85. 1797 -1.25 911 -1.57462; 1798 1 911 -1.57462; 1799 0.75 911 -1.57462
   86. 1800 0.5 911 -1.57462; 1801 0.25 911 -1.57462; 1802 -1.5 911 -1.57462
   87. 1803 1.25 911 -1.57462; 1804 -0.25 911 -1.82462; 1805 -0.5 911 -1.82462
   88. 1806 0 911 -1.82462; 1807 -0.75 911 -1.82462; 1808 -1 911 -1.82462
   89. 1809 -1.25 911 -1.82462; 1810 1 911 -1.82462; 1811 0.75 911 -1.82462
   90. 1812 0.5 911 -1.82462; 1813 0.25 911 -1.82462; 1814 -1.5 911 -1.82462
   91. 1815 1.25 911 -1.82462; 1816 -0.25 911 -2.07462; 1817 -0.5 911 -2.07462
   92. 1818 0 911 -2.07462; 1819 -0.75 911 -2.07462; 1820 -1 911 -2.07462
   93. 1821 -1.25 911 -2.07462; 1822 1 911 -2.07462; 1823 0.75 911 -2.07462
   94. 1824 0.5 911 -2.07462; 1825 0.25 911 -2.07462; 1826 -1.5 911 -2.07462
```

95. 1827 1.25 911 -2.07462; 1828 -0.25 911 -2.32462; 1829 -0.5 911 -2.32462
96. 1830 0 911 -2.32462; 1831 -0.75 911 -2.32462; 1832 -1 911 -2.32462
97. 1833 -1.25 911 -2.32462; 1834 1 911 -2.32462; 1835 0.75 911 -2.32462
98. 1836 0.5 911 -2.32462; 1837 0.25 911 -2.32462; 1838 -1.5 911 -2.32462
99. 1839 1.25 911 -2.32462; 1840 -0.25 911 -2.57462; 1841 -0.5 911 -2.57462
100. 1842 0 911 -2.57462; 1843 -0.75 911 -2.57462; 1844 -1 911 -2.57462
101. 1845 -1.25 911 -2.57462; 1846 1 911 -2.57462; 1847 0.75 911 -2.57462
102. 1848 0.5 911 -2.57462; 1849 0.25 911 -2.57462; 1850 -1.5 911 -2.57462
103. 1851 1.25 911 -2.57462; 1852 -0.25 911 -2.82462; 1853 -0.5 911 -2.82462
104. 1854 0 911 -2.82462; 1855 -0.75 911 -2.82462; 1856 -1 911 -2.82462
105. 1857 -1.25 911 -2.82462; 1858 1 911 -2.82462; 1859 0.75 911 -2.82462
106. 1860 0.5 911 -2.82462; 1861 0.25 911 -2.82462; 1862 -1.5 911 -2.82462
107. 1863 1.25 911 -2.82462; 1864 -0.25 911 -3.07462; 1865 -0.5 911 -3.07462
108. 1866 0 911 -3.07462; 1867 -0.75 911 -3.07462; 1868 -1 911 -3.07462
109. 1869 -1.25 911 -3.07462; 1870 1 911 -3.07462; 1871 0.75 911 -3.07462
110. 1872 0.5 911 -3.07462; 1873 0.25 911 -3.07462; 1874 -1.5 911 -3.07462
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301. _RAFT 394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 -
302. 1323 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 -
303. 1358 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 -
304. 1492 TO 1494 1500 TO 1505 1507 TO 1509 1550 TO 1858
305. _RAFT_MOM 394 395 404 414 442 759 760 767 768 775 776 783 791 792 1322 1323 -
306. 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
307. 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1474 1476 1482 TO 1484 -
308. 1492 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1548 -
309. 1550 TO 1716 1718 TO 1723 1725 TO 1727 1729 TO 1734 1736 TO 1826 -
310. 1828 TO 1833 1835 TO 1837 1839 TO 1844 1846 TO 1858
311. _RAFT_SHEAR 414 759 760 791 792 1322 1326 1330 1331 1338 1342 1343 1346 1350 -
312. 1354 1358 1359 1408 TO 1416 1448 1449 1456 1457 1472 1476 1482 TO 1484 1492 -
313. 1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1534 1540 TO 1543 1548 1550 -
314. 1551 TO 1693 1696 1698 1699 1703 1704 1707 1709 1710 1714 1715 1718 1720 1721 -
315. 1725 1726 1729 1731 1732 1736 1737 1740 1742 1743 1747 1748 1751 1753 1754 -
316. 1758 TO 1803 1806 1808 1809 1813 1814 1817 1819 1820 1824 1825 1828 1830 -
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318. MEMBER
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319. END GROUP DEFINITION
320. ELEMENT PROPERTY
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323. 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
324. 1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1857 -
325. 1858 THICKNESS 0.7
326. DEFINE MATERIAL START
327. ISOTROPIC CONCRETE
328. E 2.17185E+007
329. POISSON 0.17
330. DENSITY 25
331. ALPHA 1E-005
332. DAMP 0.05
333. TYPE CONCRETE
334. STRENGTH FCU 27579
335. END DEFINE MATERIAL
336. MEMBER PROPERTY AMERICAN
337. 1258 2002 2003 PRIS YD 0.5 ZD 0.5
338. CONSTANTS
339. MATERIAL CONCRETE ALL
340. SUPPORTS
341. 394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
342. 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
343. 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
344. 1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1857 -
345. 1858 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
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411	0.036241
421	0.062500
424	0.062500
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772	0.062500
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1698	0.031250
1699	0.031250
1700	0.031250
1701	0.031250
1702	0.031250
1703	0.031250
1704	0.031250
1705	0.031250
1706	0.015625
1707	0.031250
1708	0.062500
1709	0.062500
1710	0.062500
1711	0.062500
1712	0.062500
1713	0.062500
1714	0.062500
1715	0.062500
1716	0.062500
1717	0.062500
1718	0.031250
1719	0.031250
1720	0.062500
1721	0.062500
1722	0.062500
1723	0.062500
1724	0.062500
1725	0.062500
1726	0.062500
1727	0.062500
1728	0.062500
1729	0.062500

1730	0.031250
1731	0.031250
1732	0.062500
1733	0.062500
1734	0.062500
1735	0.062500
1736	0.062500
1737	0.062500
1738	0.062500
1739	0.062500
1740	0.062500
1741	0.062500
1742	0.031250
1743	0.031250
1744	0.062500
1745	0.062500
1746	0.062500
1747	0.062500
1748	0.062500
1749	0.062500
1750	0.062500
1751	0.062500
1752	0.062500
1753	0.062500
1754	0.031250
1755	0.031250
1756	0.062500
1757	0.062500
1758	0.062500
1759	0.062500
1760	0.062500
1761	0.062500
1762	0.062500
1763	0.062500
1764	0.062500
1765	0.062500
1766	0.031250
1767	0.031250
1768	0.062500
1769	0.062500
1770	0.062500
1771	0.062500
1772	0.062500
1773	0.062500
1774	0.062500
1775	0.062500
1776	0.062500
1777	0.062500
1778	0.031250
1779	0.031250
1780	0.062500
1781	0.062500
1782	0.062500
1783	0.062500
1784	0.062500
1785	0.062500

1786	0.062500
1787	0.062500
1788	0.062500
1789	0.062500
1790	0.031250
1791	0.031250
1792	0.062500
1793	0.062500
1794	0.062500
1795	0.062500
1796	0.062500
1797	0.062500
1798	0.062500
1799	0.062500
1800	0.062500
1801	0.062500
1802	0.031250
1803	0.031250
1804	0.062500
1805	0.062500
1806	0.062500
1807	0.062500
1808	0.062500
1809	0.062500
1810	0.062500
1811	0.062500
1812	0.062500
1813	0.062500
1814	0.031250
1815	0.031250
1816	0.062500
1817	0.062500
1818	0.062500
1819	0.062500
1820	0.062500
1821	0.062500
1822	0.062500
1823	0.062500
1824	0.062500
1825	0.062500
1826	0.031250
1827	0.031250
1828	0.062500
1829	0.062500
1830	0.062500
1831	0.062500
1832	0.062500
1833	0.062500
1834	0.062500
1835	0.062500
1836	0.062500
1837	0.062500
1838	0.031250
1839	0.031250
1840	0.062500
1841	0.062500

1842	0.062500
1843	0.062500
1844	0.062500
1845	0.062500
1846	0.062500
1847	0.062500
1848	0.062500
1849	0.062500
1850	0.031250
1851	0.031250
1852	0.062500
1853	0.062500
1854	0.062500
1855	0.062500
1856	0.062500
1857	0.062500
1858	0.062500
1859	0.062500
1860	0.062500
1861	0.062500
1862	0.031250
1863	0.031250
1864	0.078750
1865	0.078750
1866	0.078750
1867	0.078750
1868	0.078750
1869	0.078750
1870	0.078750
1871	0.078750
1872	0.078750
1873	0.078750
1874	0.039375
1875	0.039375
1876	0.078750
1877	0.078750
1878	0.078750
1879	0.078750
1880	0.078750
1881	0.078750
1882	0.078750
1883	0.078750
1884	0.078750
1885	0.078750
1886	0.039375
1887	0.039375
1888	0.062500
1889	0.062500
1890	0.062500
1891	0.062500
1892	0.062500
1893	0.062500
1894	0.062500
1895	0.062500
1896	0.062500
1897	0.062500

1898	0.031250
1899	0.031250
1900	0.062500
1901	0.062500
1902	0.062500
1903	0.062500
1904	0.062500
1905	0.062500
1906	0.062500
1907	0.062500
1908	0.062500
1909	0.062500
1910	0.031250
1911	0.031250
1912	0.062500
1913	0.062500
1914	0.062500
1915	0.062500
1916	0.062500
1917	0.062500
1918	0.062500
1919	0.062500
1920	0.062500
1921	0.062500
1922	0.031250
1923	0.031250
1924	0.062500
1925	0.062500
1926	0.062500
1927	0.062500
1928	0.062500
1929	0.062500
1930	0.062500
1931	0.062500
1932	0.062500
1933	0.062500
1934	0.031250
1935	0.031250
1936	0.062500
1937	0.062500
1938	0.062500
1939	0.062500
1940	0.062500
1941	0.062500
1942	0.062500
1943	0.062500
1944	0.062500
1945	0.062500
1946	0.031250
1947	0.031250
1948	0.062500
1949	0.062500
1950	0.062500
1951	0.062500
1952	0.062500
1953	0.062500

1954	0.062500
1955	0.062500
1956	0.062500
1957	0.062500
1958	0.031250
1959	0.031250
1960	0.062500
1961	0.062500
1962	0.062500
1963	0.062500
1964	0.062500
1965	0.062500
1966	0.062500
1967	0.062500
1968	0.062500
1969	0.062500
1970	0.031250
1971	0.031250
1972	0.062500
1973	0.062500
1974	0.062500
1975	0.062500
1976	0.062500
1977	0.062500
1978	0.062500
1979	0.062500
1980	0.062500
1981	0.062500
1982	0.031250
1983	0.031250
1984	0.062500
1985	0.062500
1986	0.062500
1987	0.062500
1988	0.062500
1989	0.062500
1990	0.062500
1991	0.062500
1992	0.062500
1993	0.062500
1994	0.031250
1995	0.031250
1996	0.062500
1997	0.062500
1998	0.062500
1999	0.062500
2000	0.062500
2001	0.062500
2002	0.062500
2003	0.062500
2004	0.062500
2005	0.062500
2006	0.031250
2007	0.031250
2008	0.062500
2009	0.062500

2010	0.062500
2011	0.062500
2012	0.062500
2013	0.062500
2014	0.062500
2015	0.062500
2016	0.062500
2017	0.062500
2018	0.031250
2019	0.031250
2020	0.031250
2021	0.031250
2022	0.031250
2023	0.031250
2024	0.031250
2025	0.031250
2026	0.031250
2027	0.031250
2028	0.031250
2029	0.031250
2030	0.015625
2031	0.015625
TOTAL	24.539795

```
346. ********************************************************************************
347. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
348. SELFWEIGHT Y -1 LIST ALL
349. ELEMENT LOAD
350. * LOAD DUE TO BACKFILL
351. 394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
352. 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
353. 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
354. 1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1858 PR GY -10
355. * LOAD DUE TO SURCHARGE
356. 394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
357. 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
358. 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
359. 1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1858 PR GY -20
360. JOINT LOAD
361. 1295 FX 0.5 FY -203.1 FZ 0.02
362. 2188 FX 1.1 FY -148.26 FZ 1.5
363. 2189 FX 2.17 FY -107.92 FZ 0.08
364. ********************************************************************************
365. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
366. JOINT LOAD
367. 1295 FX 0.31 FY -245.85 FZ -0.07
368. 2188 FX 8.9 FY -88.68 FZ 6.42
369. 2189 FX 0.09 FY -29.11 FZ 0.1
370. ********************************************************************************
371. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WL1 )
372. JOINT LOAD
373. 1295 FX -0.73 FY 13.53 FZ 0.07
374. 2188 FX -178.53 FY -279 FZ -12.56
375. 2189 FX -62.6 FY 230.1 FZ -0.02
376. ********************************************************************************
377. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X ( WL2 )
378. JOINT LOAD
```



```
379. 1295 FX 0.74 FY -13 FZ -0.07
380. 2188 FX 179.4 FY 280.45 FZ 12.41
381. 2189 FX 62.6 FY -230.07 FZ 0.01
382. ********************************************************************************
383. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Y ( WL3 )
384. JOINT LOAD
385. 1295 FX -0.46 FY 399.87 FZ -1.21
386. 2188 FX 12.72 FY -345.28 FZ -160.55
387. 2189 FX -6.44 FY -2.43 FZ -2.48
388. ********************************************************************************
389. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Y ( WL4 )
390. JOINT LOAD
391. 1295 FX 0.45 FY -394.16 FZ 1.21
392. 2188 FX -12 FY 341.08 FZ 159.19
393. 2189 FX 6.44 FY 2.21 FZ 2.5
394. ********************************************************************************
395. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
396. JOINT LOAD
397. 1295 FX -0.03 FY 5.07 FZ 0.0001
398. 2188 FX -3.89 FY 6.33 FZ -1.67
399. 2189 FX -2.17 FY 10.83 FZ -0.02
400. ********************************************************************************
401. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
402. JOINT LOAD
403. 1295 FX 0.03 FY -5.07 FZ -0.0001
404. 2188 FX 3.89 FY -6.33 FZ 1.67
405. 2189 FX 2.17 FY -10.83 FZ 0.02
406. ********************************************************************************
407. LOAD 9 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
408. JOINT LOAD
409. 1295 FX 0.06 FY 23.97 FZ 0.18
410. 2188 FX -3.53 FY -27.12 FZ -9.28
411. 2189 FX 1.03 FY -1.17 FZ -0.44
412. ********************************************************************************
413. LOAD 10 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
414. JOINT LOAD
415. 1295 FX -0.06 FY -23.97 FZ -0.18
416. 2188 FX 3.53 FY 27.12 FZ 9.28
417. 2189 FX -1.03 FY 1.17 FZ 0.44
418. ********************************************************************************
419. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**442**	**NUMBER OF MEMBERS**	**3**
NUMBER OF PLATES	**395**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**439**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 154/ 15/ 51 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 1335
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 69 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.0/ 142101.9 MB

```
420. CHANGE
421. SUPPORTS
422. 394 395 404 414 442 443 759 760 767 768 775 776 783 784 791 792 1322 1323 -
423. 1326 1327 1330 1331 1338 1339 1342 1343 1346 1347 1350 1351 1354 1355 1358 -
424. 1359 1408 TO 1416 1448 TO 1450 1452 1456 1457 1472 TO 1476 1482 TO 1484 1492 -
425. 1493 TO 1494 1500 TO 1505 1507 TO 1509 1532 TO 1536 1538 TO 1857 -
426. 1858 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
427. ******************************************************************************
428. * UNFACTORED LOAD COMBINATIONS
429. LOAD 201 DL
430. REPEAT LOAD
431. 1 1.0
432. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201

433. CHANGE
434. LOAD 202 DL+L+TL1
435. REPEAT LOAD
436. 1 1.0 2 1.0 9 1.0
437. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202

438. CHANGE
439. LOAD 203 DL+L+TL2
440. REPEAT LOAD
441. 1 1.0 2 1.0 10 1.0
442. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  203

443. CHANGE
444. LOAD 204 DL+0.75L+0.75TL1
445. REPEAT LOAD
446. 1 1.0 2 0.75 9 0.75
447. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204

448. CHANGE
449. LOAD 205 DL+0.75L+0.75TL2
450. REPEAT LOAD
451. 1 1.0 2 0.75 10 0.75
452. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  205

453. CHANGE
454. LOAD 206 DL+WL1
455. REPEAT LOAD
456. 1 1.0 3 1.0
457. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  206

458. CHANGE
459. LOAD 207 DL+WL2
460. REPEAT LOAD
461. 1 1.0 4 1.0
462. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
```

**NOTE-Tension/Compression converged after 3 iterations, Case= 207

463. CHANGE
464. LOAD 208 DL+WL3
465. REPEAT LOAD
466. 1 1.0 5 1.0
467. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 208

468. CHANGE
469. LOAD 209 DL+WL4
470. REPEAT LOAD
471. 1 1.0 6 1.0
472. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

473. CHANGE
474. LOAD 210 DL+0.75WL1+0.75L
475. REPEAT LOAD
476. 1 1.0 2 0.75 3 0.75
477. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

478. CHANGE
479. LOAD 211 DL+0.75WL2+0.75L
480. REPEAT LOAD
481. 1 1.0 2 0.75 4 0.75
482. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 211

483. CHANGE
484. LOAD 212 DL+0.75WL3+0.75L
485. REPEAT LOAD
486. 1 1.0 2 0.75 5 0.75
487. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

488. CHANGE
489. LOAD 213 DL+0.75WL4+0.75L
490. REPEAT LOAD
491. 1 1.0 2 0.75 6 0.75
492. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

493. CHANGE
494. LOAD 214 0.6DL+WL1
495. REPEAT LOAD
496. 1 0.6 3 1.0
497. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 214

498. CHANGE
499. LOAD 215 0.6DL+WL2
500. REPEAT LOAD
501. 1 0.6 4 1.0
502. PERFORM ANALYSIS
**START ITERATION NO. 2

```
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  215



 503. CHANGE
 504. LOAD 216 0.6DL+WL3
 505. REPEAT LOAD
 506. 1 0.6 5 1.0
 507. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  216



 508. CHANGE
 509. LOAD 217 0.6DL+WL4
 510. REPEAT LOAD
 511. 1 0.6 6 1.0
 512. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  217



 513. CHANGE
 514. LOAD 218 0.6DL+0.7SL1
 515. REPEAT LOAD
 516. 1 0.6 7 0.7
 517. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  218



 518. CHANGE
 519. LOAD 219 0.6DL+0.7SL2
 520. REPEAT LOAD
```



```
 521. 1 0.6 8 0.7
 522. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  219



 523. CHANGE
 524. ******************************************************************************
 525. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
 526. LOAD 101 1.4DL
 527. REPEAT LOAD
 528. 1 1.4
 529. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101



 530. CHANGE
 531. LOAD 102 1.4DL+1.4TL1+1.7L
 532. REPEAT LOAD
 533. 1 1.4 2 1.7 9 1.4
 534. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102



 535. CHANGE
 536. LOAD 103 1.4DL+1.4TL2+1.7L
 537. REPEAT LOAD
 538. 1 1.4 2 1.7 10 1.4
 539. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 540. CHANGE
 541. LOAD 104 1.2DL+1.0L
 542. REPEAT LOAD
 543. 1 1.2 2 1.0
 544. PERFORM ANALYSIS
```

```
**NOTE-Tension/Compression converged after  1 iterations, Case=  104


 545. CHANGE
 546. LOAD 105 1.2DL+1.6WL1+1.0L
 547. REPEAT LOAD
 548. 1 1.2 3 1.6 2 0.5 2 1.0
 549. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  105


 550. CHANGE
 551. LOAD 106 1.2DL+1.6WL2+1.0L
 552. REPEAT LOAD
 553. 1 1.2 4 1.6 2 0.5 2 1.0
 554. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  106


 555. CHANGE
 556. LOAD 107 1.2DL+1.6WL3+1.0L
 557. REPEAT LOAD
 558. 1 1.2 5 1.6 2 0.5 2 1.0
 559. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  107


 560. CHANGE
 561. LOAD 108 1.2DL+1.6WL4+1.0L
 562. REPEAT LOAD
 563. 1 1.2 6 1.6 2 1.0
```



```
 564. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  108


 565. CHANGE
 566. LOAD 109 1.2DL+1.0SL1+1.0L
 567. REPEAT LOAD
 568. 1 1.2 7 1.0 2 1.0
 569. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  109


 570. CHANGE
 571. LOAD 110 1.2DL+1.0SL2+1.0L
 572. REPEAT LOAD
 573. 1 1.2 8 1.0 2 1.0
 574. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110


 575. CHANGE
 576. LOAD 111 0.9DL+1.6WL1
 577. REPEAT LOAD
 578. 1 0.9 3 1.6
 579. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=  111


 580. CHANGE
 581. LOAD 112 0.9DL+1.6WL2
```

```
 582. REPEAT LOAD
 583. 1 0.9 4 1.6
 584. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  112



 585. CHANGE
 586. LOAD 113 0.9DL+1.6WL3
 587. REPEAT LOAD
 588. 1 0.9 5 1.6
 589. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  113



 590. CHANGE
 591. LOAD 114 0.9DL+1.6WL4
 592. REPEAT LOAD
 593. 1 0.9 6 1.6
 594. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  114



 595. CHANGE
 596. LOAD 115 0.9DL+1.0SL1
 597. REPEAT LOAD
 598. 1 0.9 7 1.0
 599. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  115



 600. CHANGE
 601. LOAD 116 0.9DL+1.0SL2
 602. REPEAT LOAD
 603. 1 0.9 8 1.0
 604. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  116



 605. CHANGE
 606. ********************************************************************************
 607. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=    2



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

***WARNING -  INSTABILITY AT JOINT  2031   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  2.8902206E+06 L-MATRIX DIAG= -5.9963204E-06 EQN NO    1335
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
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NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -6.0382299E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -4.0796585E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -4.9402006E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -4.5243651E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.0691888E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -4.4447370E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -7.8200828E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -4.2435713E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 12 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -3.6838464E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.0384551E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -6.1257742E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5618584E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5734999E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.4622069E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1428488E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5814162E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795981E+04 L-MATRIX DIAG= -1.3205863E-09 EQN NO 1333
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1428488E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5814162E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 13 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1429419E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795981E+04 L-MATRIX DIAG= -1.3205863E-09 EQN NO 1333
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS Wwell AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1428488E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5814162E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -2.7855393E-05 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -3.2242388E-05 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -2.7307775E-05 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -2.8656796E-05 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -9.2305243E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -1.6218983E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 13 iterations, Case= 6

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.9963204E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795981E+04 L-MATRIX DIAG= -1.3205863E-09 EQN NO 1333
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1428488E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5814162E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1428022E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1427091E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1427556E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1429885E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1431282E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 12 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -3.2242388E-05 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -2.7946662E-05 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -9.2305243E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -1.6218983E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.3795981E+04 L-MATRIX DIAG= -1.3205863E-09 EQN NO 1333
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902247E+06 L-MATRIX DIAG= -4.1428488E-05 EQN NO 1334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 2031 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8902206E+06 L-MATRIX DIAG= -5.5814162E-06 EQN NO 1335
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 10
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 206

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 207

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 214

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 215

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 216

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 105

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 111

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 112

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 113

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

608. LOAD LIST 101 TO 116
609. START CONCRETE DESIGN

610. CODE ACI
611. TRACK 2 ALL
612. FC 35000 MEMB 1258 2002 2003
613. CLB 0.05 MEMB 1258 2002 2003
614. CLS 0.05 ALL
615. CLT 0.05 MEMB 394 395 404 414 759 760 767 768 775 776 783 784 1258 1322 1323 -
616. 1326 1327 1330 1331 1342 1343 1346 1347 1350 1351 1354 1355 1408 TO 1410 -
617. 1412 TO 1415 1448 TO 1450 1452 1472 TO 1475 1482 TO 1484 1492 TO 1494 1550 -
618. 1551 TO 1552 1554 TO 1557 1559 1560 2002 2003
619. MAXMAIN 25 ALL
620. MINMAIN 25 ALL
621. MINSEC 10 ALL
622. DESIGN COLUMN 1258 2002 2003

COLUMN NO. 1258 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	101	END	0.650

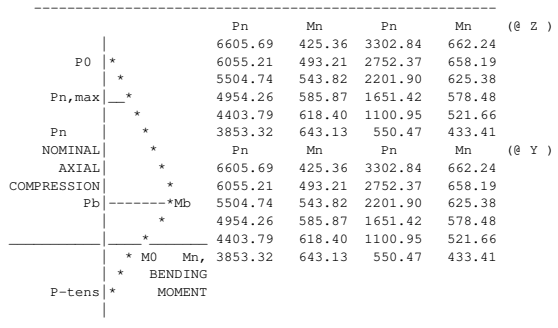
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	437.45	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	437.45	0.00	0.00000



COLUMN NO. 2002 DESIGN PER ACI 318-08 - AXIAL + BENDING

*** NO PROPER BAR ARRANGEMENT IS POSSIBLE.

AREA OF STEEL REQUIRED = 13955.0 SQ. MM

COLUMN NO. 2003 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 4010.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	2.356	111	STA	0.900

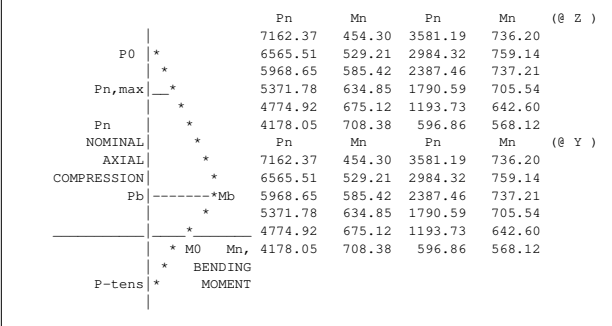
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9699.05	7759.24	3019.55	760.13	251.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
483.46	-2436.79	-284.90	283.71	-0.38301

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9699.05	7759.24	3019.55	760.13	251.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
483.46	-2436.79	-284.90	0.23	-0.00031



**********END OF COLUMN DESIGN RESULTS**********

623. END CONCRETE DESIGN
624. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 6,2020 TIME= 13:14:48 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 13 STAAD OUTPUT – FOUNDATION F6

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```
            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    NOV  7, 2020                   *
            *           Time=    16:27:14                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F6.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/11/2020
    4. JOB NAME FDN. MKD. F6
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911 3.175; 6 0.75 911 3.175; 34 -0.5 911 3.175; 35 -0.25 911 3.175
   17. 41 0.25 911 3.175; 42 0.5 911 3.175; 128 0 913.6 3.175; 302 -0.5 911 3.47538
   18. 304 -0.25 911 3.47538; 305 0 911 3.47538; 306 0.25 911 3.47538
   19. 307 0.5 911 3.47538; 308 0.75 911 3.47538; 309 1 911 3.175; 313 1 911 3.47538
   20. 366 -0.5 911 2.17538; 369 -0.25 911 2.17538; 371 0 911 2.17538
   21. 373 0.25 911 2.17538; 375 0.5 911 2.17538; 377 0.75 911 2.17538
   22. 379 1 911 2.17538; 388 -0.5 911 2.42538; 389 -0.25 911 2.42538
   23. 390 0 911 2.42538; 391 0.25 911 2.42538; 392 0.5 911 2.42538
   24. 393 0.75 911 2.42538; 394 1 911 2.42538; 399 -0.5 911 2.67538
   25. 400 -0.25 911 2.67538; 401 0 911 2.67538; 402 0.25 911 2.67538
   26. 403 0.5 911 2.67538; 404 0.75 911 2.67538; 405 1 911 2.67538
   27. 410 -0.5 911 2.92538; 411 -0.25 911 2.92538; 412 0 911 2.92538
   28. 413 0.25 911 2.92538; 414 0.5 911 2.92538; 415 0.75 911 2.92538
   29. 416 1 911 2.92538; 421 -0.5 911 1.92538; 424 -0.25 911 1.92538
   30. 426 0 911 1.92538; 428 0.25 911 1.92538; 430 0.5 911 1.92538
   31. 432 0.75 911 1.92538; 434 1 911 1.92538; 771 -0.5 911 1.17538
   32. 772 -0.5 911 0.92538; 774 -0.25 911 1.17538; 775 -0.25 911 0.92538
   33. 776 0 911 1.17538; 778 0.25 911 1.17538; 779 0.25 911 0.92538
   34. 780 0.5 911 1.17538; 781 0.5 911 0.92538; 782 0.75 911 1.17538
   35. 783 0.75 911 0.92538; 784 1 911 1.17538; 785 1 911 0.92538
   36. 789 -0.5 911 1.42538; 790 -0.25 911 1.42538; 791 0 911 1.42538
   37. 792 0.25 911 1.42538; 793 0.5 911 1.42538; 794 0.75 911 1.42538
   38. 795 1 911 1.42538; 798 -0.5 911 1.67538; 799 -0.25 911 1.67538
```



```
   39. 800 0 911 1.67538; 801 0.25 911 1.67538; 802 0.5 911 1.67538
   40. 803 0.75 911 1.67538; 804 1 911 1.67538; 806 -0.5 911 0.67538
   41. 808 -0.25 911 0.67538; 809 0 911 0.67538; 810 0.25 911 0.67538
   42. 811 0.5 911 0.67538; 812 0.75 911 0.67538; 813 1 911 0.67538
   43. 903 -0.5 911 0.42538; 904 -0.25 911 0.42538; 905 0 911 0.42538
   44. 906 0.25 911 0.42538; 907 0.5 911 0.42538; 908 0.75 911 0.42538
   45. 909 1 911 0.42538; 1295 0 913.6 0.175; 1331 0 911 0.92538; 1393 -1 911 3.47538
   46. 1394 -1 911 3.175; 1395 -0.75 911 3.47538; 1396 -0.75 911 3.175
   47. 1405 -1 911 2.42538; 1406 -1 911 2.17538; 1407 -0.75 911 2.42538
   48. 1408 -0.75 911 2.17538; 1411 -1 911 2.67538; 1412 -0.75 911 2.67538
   49. 1415 -1 911 2.92538; 1416 -0.75 911 2.92538; 1419 -1 911 1.92538
   50. 1420 -0.75 911 1.92538; 1425 -1 911 1.17538; 1426 -1 911 0.92538
   51. 1427 -0.75 911 1.17538; 1428 -0.75 911 0.92538; 1431 -1 911 1.42538
   52. 1432 -0.75 911 1.42538; 1435 -1 911 1.67538; 1436 -0.75 911 1.67538
   53. 1439 -1 911 0.67538; 1440 -0.75 911 0.67538; 1443 -1 911 0.42538
   54. 1444 -0.75 911 0.42538; 1445 -0.25 911 0.17538; 1446 -0.5 911 0.17538
   55. 1448 0.25 911 0.17538; 1449 0.5 911 0.17538; 1450 0.75 911 0.17538
   56. 1451 1 911 0.17538; 1460 -0.25 911 -0.12462; 1461 -0.5 911 -0.12462
   57. 1463 0.25 911 -0.12462; 1464 0.5 911 -0.12462; 1465 0.75 911 -0.12462
   58. 1466 1 911 -0.12462; 1485 -1 911 0.17538; 1486 -0.75 911 0.17538
   59. 1489 -1 911 -0.12462; 1490 -0.75 911 -0.12462; 1561 0 911 -0.12462
   60. 1562 0 911 0.17538
   61. MEMBER INCIDENCES
   62. 185 2 128; 1258 1562 1295
   63. ELEMENT INCIDENCES SHELL
   64. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   65. 334 307 308 6 42; 337 308 313 309 6; 394 388 389 369 366; 395 389 390 371 369
   66. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
   67. 399 393 394 379 377; 404 399 400 389 388; 405 400 401 390 389
   68. 406 401 402 391 390; 407 402 403 392 391; 408 403 404 393 392
   69. 409 404 405 394 393; 414 410 411 400 399; 415 411 412 401 400
   70. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
   71. 419 415 416 405 404; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
   72. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 442 366 369 424 421
   73. 443 369 371 426 424; 444 371 373 428 426; 445 373 375 430 428
   74. 446 375 377 432 430; 447 377 379 434 432; 759 771 774 775 772
   75. 760 774 776 1331 775; 761 776 778 779 1331; 762 778 780 781 779
   76. 763 780 782 783 781; 764 782 784 785 783; 767 789 790 774 771
   77. 768 790 791 776 774; 769 791 792 778 776; 770 792 793 780 778
   78. 771 793 794 782 780; 772 794 795 784 782; 775 798 799 790 789
   79. 776 799 800 791 790; 777 800 801 792 791; 778 801 802 793 792
   80. 779 802 803 794 793; 780 803 804 795 794; 783 421 424 799 798
   81. 784 424 426 800 799; 785 426 428 801 800; 786 428 430 802 801
   82. 787 430 432 803 802; 788 432 434 804 803; 791 772 775 808 806
   83. 792 775 1331 809 808; 793 1331 779 810 809; 794 779 781 811 810
   84. 795 781 783 812 811; 796 783 785 813 812; 883 806 808 904 903
   85. 884 808 809 905 904; 885 809 810 906 905; 886 810 811 907 906
   86. 887 811 812 908 907; 888 812 813 909 908; 1314 1393 1395 1396 1394
   87. 1315 1395 302 34 1396; 1322 1405 1407 1408 1406; 1323 1407 388 366 1408
   88. 1326 1411 1412 1407 1405; 1327 1412 399 388 1407; 1330 1415 1416 1412 1411
   89. 1331 1416 410 399 1412; 1334 1394 1396 1416 1415; 1335 1396 34 410 1416
   90. 1338 1406 1408 1420 1419; 1339 1408 366 421 1420; 1342 1425 1427 1428 1426
   91. 1343 1427 771 772 1428; 1346 1431 1432 1427 1425; 1347 1432 789 771 1427
   92. 1350 1435 1436 1432 1431; 1351 1436 798 789 1432; 1354 1419 1420 1436 1435
   93. 1355 1420 421 798 1436; 1358 1426 1428 1440 1439; 1359 1428 772 806 1440
   94. 1362 1439 1440 1444 1443; 1363 1440 806 903 1444; 1364 903 904 1445 1446
```

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 95. 1365 904 905 1562 1445; 1366 905 906 1448 1562; 1367 906 907 1449 1448
 96. 1368 907 908 1450 1449; 1369 908 909 1451 1450; 1378 1446 1445 1460 1461
 97. 1379 1445 1562 1561 1460; 1380 1562 1448 1463 1561; 1381 1448 1449 1464 1463
 98. 1382 1449 1450 1465 1464; 1383 1450 1451 1466 1465; 1402 1443 1444 1486 1485
 99. 1403 1444 903 1446 1486; 1406 1485 1486 1490 1489; 1407 1486 1446 1461 1490
100. START GROUP DEFINITION
101. ELEMENT
102. _RAFT 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 442 -
103. 443 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 883 TO 888 -
104. 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 1343 1346 -
105. 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 1403 1406 -
106. 1407
107. _RAFT_MOM 327 333 334 337 394 TO 399 404 TO 409 414 TO 419 424 427 TO 429 -
108. 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
109. 883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
110. 1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1364 1367 TO 1369 1378 -
111. 1381 TO 1383 1402 1403 1406 1407
112. _RAFT_SHEAR 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 1338 1339 -
113. 1342 1343 1346 1347 1350 1351 1354 1355
114. MEMBER
115. END GROUP DEFINITION
116. ELEMENT PROPERTY
117. 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
118. 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
119. 883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
120. 1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
121. 1403 1406 1407 THICKNESS 0.7
122. DEFINE MATERIAL START
123. ISOTROPIC CONCRETE
124. E 2.17185E+007
125. POISSON 0.17
126. DENSITY 25
127. ALPHA 1E-005
128. DAMP 0.05
129. TYPE CONCRETE
130. STRENGTH FCU 27579
131. END DEFINE MATERIAL
132. MEMBER PROPERTY AMERICAN
133. 185 1258 PRIS YD 0.5 ZD 0.5
134. CONSTANTS
135. MATERIAL CONCRETE ALL
136. SUPPORTS
137. 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
138. 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
139. 883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
140. 1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
141. 1403 1406 1407 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.068750
6	0.068750
34	0.068750
35	0.068750
41	0.068750

42	0.068750
302	0.037548
304	0.037548
305	0.037548
306	0.037548
307	0.037548
308	0.037548
309	0.034375
313	0.018774
366	0.062500
369	0.062500
371	0.062500
373	0.062500
375	0.062500
377	0.062500
379	0.031250
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
394	0.031250
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
405	0.031250
410	0.062452
411	0.062452
412	0.062452
413	0.062452
414	0.062452
415	0.062452
416	0.031226
421	0.062500
424	0.062500
426	0.062500
428	0.062500
430	0.062500
432	0.062500
434	0.031250
771	0.062500
772	0.062500
774	0.062500
775	0.062500
776	0.062500
778	0.062500
779	0.062500
780	0.062500
781	0.062500
782	0.062500
783	0.062500
784	0.031250

785	0.031250
789	0.062500
790	0.062500
791	0.062500
792	0.062500
793	0.062500
794	0.062500
795	0.031250
798	0.062500
799	0.062500
800	0.062500
801	0.062500
802	0.062500
803	0.062500
804	0.031250
806	0.062500
808	0.062500
809	0.062500
810	0.062500
811	0.062500
812	0.062500
813	0.031250
903	0.062500
904	0.062500
905	0.062500
906	0.062500
907	0.062500
908	0.062500
909	0.031250
1331	0.062500
1393	0.018774
1394	0.034375
1395	0.037548
1396	0.068750
1405	0.031250
1406	0.031250
1407	0.062500
1408	0.062500
1411	0.031250
1412	0.062500
1415	0.031226
1416	0.062452
1419	0.031250
1420	0.062500
1425	0.031250
1426	0.031250
1427	0.062500
1428	0.062500
1431	0.031250
1432	0.062500
1435	0.031250
1436	0.062500
1439	0.031250
1440	0.062500
1443	0.031250
1444	0.062500

1445	0.068750
1446	0.068750
1448	0.068750
1449	0.068750
1450	0.068750
1451	0.034375
1460	0.037500
1461	0.037500
1463	0.037500
1464	0.037500
1465	0.037500
1466	0.018750
1485	0.034375
1486	0.068750
1489	0.018750
1490	0.037500
1561	0.037500
1562	0.068750
TOTAL	7.199999

```
142. ********************************************************************************
143. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
144. SELFWEIGHT Y -1
145. ELEMENT LOAD
146. * LOAD DUE TO BACKFILL
147. 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
148. 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
149. 883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
150. 1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
151. 1403 1406 1407 PR GY -16
152. * LOAD DUE TO SURCHARGE
153. 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
154. 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
155. 883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
156. 1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
157. 1403 1406 1407 PR GY -20
158. JOINT LOAD
159. 128 FX -0.21 FY -37.55 FZ 0.42
160. 1295 FX 0.02 FY -72 FZ -0.07
161. ********************************************************************************
162. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
163. JOINT LOAD
164. 128 FX 0.15 FY -28.56 FZ 0.15
165. 1295 FX 0.1 FY -58.49 FZ -0.14
166. ********************************************************************************
167. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WL1 )
168. JOINT LOAD
169. 128 FX -28.45 FY 92.71 FZ 1.01
170. 1295 FX -6.1 FY 31.79 FZ 0.09
171. ********************************************************************************
172. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X ( WL2 )
173. JOINT LOAD
174. 128 FX 27.73 FY -88.75 FZ -0.88
175. 1295 FX 6.11 FY -29.6 FZ -0.09
176. ********************************************************************************
177. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Y ( WL3 )
178. JOINT LOAD
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179. 128 FX 3.19 FY 102.06 FZ -33.34
180. 1295 FX -0.06 FY -117.78 FZ -0.63
181. ******************************************************************************
182. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Y ( WL4 )
183. JOINT LOAD
184. 128 FX -3.12 FY -102.34 FZ 33.33
185. 1295 FX 0.06 FY 117.92 FZ 0.63
186. ******************************************************************************
187. LOAD 7 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
188. JOINT LOAD
189. 128 FX 3.62 FY -25.58 FZ 2.05
190. 1295 FX 0.1 FY 3.3 FZ -0.29
191. ******************************************************************************
192. LOAD 8 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
193. JOINT LOAD
194. 128 FX -3.62 FY 25.58 FZ -2.05
195. 1295 FX -0.1 FY -3.3 FZ 0.29
196. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 137 NUMBER OF MEMBERS 2
NUMBER OF PLATES 112 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 135

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 102/ 16/ 51 DOF
TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 417
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 22 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.9/ 142096.8 MB

```
197. CHANGE
198. SUPPORTS
199. 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 424 TO 429 -
200. 442 TO 447 759 TO 764 767 TO 772 775 TO 780 783 TO 788 791 TO 796 -
201. 883 TO 888 1314 1315 1322 1323 1326 1327 1330 1331 1334 1335 1338 1339 1342 -
202. 1343 1346 1347 1350 1351 1354 1355 1358 1359 1362 TO 1369 1378 TO 1383 1402 -
203. 1403 1406 1407 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
204. ******************************************************************************
205. * UNFACTORED LOAD COMBINATIONS
206. LOAD 201 DL
207. REPEAT LOAD
208. 1 1.0
209. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  201


210. CHANGE
211. LOAD 202 DL+L
212. REPEAT LOAD
213. 1 1.0 2 1.0
214. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202


215. CHANGE
216. LOAD 203 DL+0.75L
217. REPEAT LOAD
218. 1 1.0 2 0.75
219. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  203


220. CHANGE
221. LOAD 204 DL+WL1
222. REPEAT LOAD
223. 1 1.0 3 1.0
224. PERFORM ANALYSIS
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  204


225. CHANGE
226. LOAD 205 DL+WL2
227. REPEAT LOAD
228. 1 1.0 4 1.0
229. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

230. CHANGE
231. LOAD 206 DL+WL3
232. REPEAT LOAD
233. 1 1.0 5 1.0
234. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 206

235. CHANGE
236. LOAD 207 DL+WL4
237. REPEAT LOAD
238. 1 1.0 6 1.0
239. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 207

240. CHANGE
241. LOAD 208 DL+0.75WL1+0.75L
242. REPEAT LOAD
243. 1 1.0 2 0.75 3 0.75
244. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

245. CHANGE
246. LOAD 209 DL+0.75WL2+0.75L
247. REPEAT LOAD
248. 1 1.0 2 0.75 4 0.75
249. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

250. CHANGE
251. LOAD 210 DL+0.75WL3+0.75L
252. REPEAT LOAD
253. 1 1.0 2 0.75 5 0.75
254. PERFORM ANALYSIS
**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 210

255. CHANGE
256. LOAD 211 DL+0.75WL4+0.75L
257. REPEAT LOAD
258. 1 1.0 2 0.75 6 0.75
259. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

260. CHANGE
261. LOAD 212 0.6DL+WL1
262. REPEAT LOAD
263. 1 0.6 3 1.0
264. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 212

265. CHANGE
266. LOAD 213 0.6DL+WL2
267. REPEAT LOAD
268. 1 0.6 4 1.0
269. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

270. CHANGE
271. LOAD 214 0.6DL+WL3
272. REPEAT LOAD
273. 1 0.6 5 1.0
274. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 214

275. CHANGE
276. LOAD 215 0.6DL+WL4
277. REPEAT LOAD
278. 1 0.6 6 1.0
279. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 215

280. CHANGE
281. LOAD 216 DL+L+TL1
282. REPEAT LOAD
283. 1 1.0 2 1.0 7 1.0
284. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

285. CHANGE

286. LOAD 217 DL+L+TL2
287. REPEAT LOAD
288. 1 1.0 2 1.0 8 1.0
289. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

290. CHANGE
291. LOAD 218 DL+0.75L+0.75TL1
292. REPEAT LOAD
293. 1 1.0 2 0.75 7 0.75
294. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

295. CHANGE
296. LOAD 219 DL+0.75L+0.75TL2
297. REPEAT LOAD
298. 1 1.0 2 0.75 8 0.75
299. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

300. CHANGE
301. **
302. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
303. LOAD 101 1.4DL
304. REPEAT LOAD
305. 1 1.4
306. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

307. CHANGE
308. LOAD 102 1.4DL+1.7L
309. REPEAT LOAD

```
 310. 1 1.4 2 1.7
 311. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102



 312. CHANGE
 313. LOAD 103 1.2DL+1.0L
 314. REPEAT LOAD
 315. 1 1.2 2 1.0
 316. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 317. CHANGE
 318. LOAD 104 1.2DL+1.6WL1+1.0L
 319. REPEAT LOAD
 320. 1 1.2 3 1.6 2 0.5 2 1.0
 321. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  104



 322. CHANGE
 323. LOAD 105 1.2DL+1.6WL2+1.0L
 324. REPEAT LOAD
 325. 1 1.2 4 1.6 2 0.5 2 1.0
 326. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105



 327. CHANGE
 328. LOAD 106 1.2DL+1.6WL3+1.0L
 329. REPEAT LOAD
 330. 1 1.2 5 1.6 2 0.5 2 1.0
```



```
 331. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  106



 332. CHANGE
 333. LOAD 107 1.2DL+1.6WL4+1.0L
 334. REPEAT LOAD
 335. 1 1.2 6 1.6 2 1.0
 336. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  107



 337. CHANGE
 338. LOAD 108 0.9DL+1.6WL1
 339. REPEAT LOAD
 340. 1 0.9 3 1.6
 341. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=  108



 342. CHANGE
 343. LOAD 109 0.9DL+1.6WL2
 344. REPEAT LOAD
 345. 1 0.9 4 1.6
 346. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  109
```

```
 347. CHANGE
 348. LOAD 110 0.9DL+1.6WL3
 349. REPEAT LOAD
 350. 1 0.9 5 1.6
 351. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  110



 352. CHANGE
 353. LOAD 111 0.9DL+1.6WL4
 354. REPEAT LOAD
 355. 1 0.9 6 1.6
 356. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  111



 357. CHANGE
 358. LOAD 112 1.4DL+1.4TL1+1.7L
 359. REPEAT LOAD
 360. 1 1.4 2 1.7 7 1.4
 361. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  112



 362. CHANGE
 363. LOAD 113 1.4DL+1.4TL2+1.7L
 364. REPEAT LOAD
 365. 1 1.4 2 1.7 8 1.4
 366. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  113



 367. CHANGE
 368. ******************************************************************************
 369. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

***WARNING -  INSTABILITY AT JOINT  1466   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.1846387E+06 L-MATRIX DIAG= -3.7532300E-07 EQN NO     417
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          6

***WARNING -  INSTABILITY AT JOINT  1466   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  3.1846387E+06 L-MATRIX DIAG= -2.0489097E-08 EQN NO     417
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
```

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1427328E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -7.8696758E-08 EQN NO 417
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4035293E+04 L-MATRIX DIAG= -1.5643309E-10 EQN NO 415
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1436641E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -4.0279701E-07 EQN NO 417

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1413358E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1399388E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1427328E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 7 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -3.8831495E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -4.3422915E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -2.4861656E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -1.9418076E-07 EQN NO 417
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4035293E+04 L-MATRIX DIAG= -1.5643309E-10 EQN NO 415
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1436641E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -4.0279701E-07 EQN NO 417
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11
**START ITERATION NO. 12
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -3.8831495E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -4.3422915E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 15

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -2.4861656E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -1.9418076E-07 EQN NO 417
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 6
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -1.4281832E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= -2.7143396E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 8 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1413358E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1422671E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1427328E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -7.8696758E-08 EQN NO 417
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.4035293E+04 L-MATRIX DIAG= -1.5643309E-10 EQN NO 415
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.8187503E+06 L-MATRIX DIAG= 1.1436641E-06 EQN NO 416
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1466 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.1846387E+06 L-MATRIX DIAG= -4.0279701E-07 EQN NO 417
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 8

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 206

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 214

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 106

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  110



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  111



**NOTE-Tension/Compression converged after  1 iterations, Case=  112



**NOTE-Tension/Compression converged after  1 iterations, Case=  113


370. LOAD LIST 101 TO 113
371. START CONCRETE DESIGN
```



```
372. CODE ACI
373. TRACK 2 ALL
374. FC 35000 MEMB 185 1258
375. CLB 0.05 MEMB 185 1258
376. CLS 0.05 ALL
377. CLT 0.05 MEMB 185 327 328 332 TO 334 337 394 TO 399 404 TO 409 414 TO 419 -
378. 424 TO 429 759 TO 764 767 TO 772 775 TO 780 783 TO 788 883 TO 888 1258 1314 -
379. 1315 1322 1323 1326 1327 1330 1331 1334 1335 1342 1343 1346 1347 1350 1351 -
380. 1354 1355 1362 1363 1378 TO 1383 1406 1407
381. MAXMAIN 25 ALL
382. MINMAIN 25 ALL
383. MINSEC 10 ALL
384. DESIGN COLUMN 185 1258
```

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	101	END	0.650

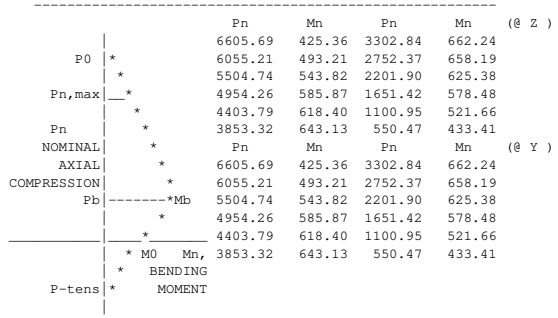
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	80.88	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	80.88	0.00	0.00000



COLUMN NO. 1258 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	155.08	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	155.08	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

385. END CONCRETE DESIGN
386. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 7,2020 TIME= 16:30:43 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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